Analysis on Manifolds

Lecture Notes for Introduction to Analysis II, following Munkres (Ch. 1–9)

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Chapter 1. The Algebra and Topology of $\mathbb{R}^n$

Overview

This chapter collects the linear algebra and the topology of $\mathbb{R}^n$ used throughout. Sections 1 and 3 are review; Sections 2 and 4 contain material that may be new (elementary matrices, the axiomatic determinant, the ϵ-neighborhood theorem, compactness of rectangles). Results that Munkres states without proof are given short proofs here, marked [Supplement]. Elements of $\mathbb{R}^n$ are written as column matrices unless stated otherwise.

§1. Review of Linear Algebra

Motivation. Multivariable calculus is linear algebra applied locally: a derivative is a linear map, a change of variables multiplies volume by a determinant, and differential forms are alternating multilinear maps. We therefore begin by fixing the linear-algebra facts the rest of the book uses.

1.1 Vector spaces

Definition 1.1 (vector space)

A set V with an operation $\mathbf{x}+\mathbf{y}$ (vector addition) and an operation $c\mathbf{x}$ (scalar multiplication by $c \in \mathbb{R}$) is a **vector space** (or **linear space**) if for all $\mathbf{x}, \mathbf{y}, \mathbf{z} \in V$ and $c, d \in \mathbb{R}$:

(1) $\mathbf{x}+\mathbf{y}=\mathbf{y}+\mathbf{x}$;

(2) $\mathbf{x}+(\mathbf{y}+\mathbf{z})=(\mathbf{x}+\mathbf{y})+\mathbf{z}$;

(3) there is a unique vector $\mathbf{0}$ with $\mathbf{x}+\mathbf{0}=\mathbf{x}$ for all $\mathbf{x}$;

(4) $\mathbf{x}+(-1)\mathbf{x}=\mathbf{0}$;

(5) $1\mathbf{x}=\mathbf{x}$;

(6) $c(d\mathbf{x})=(cd)\mathbf{x}$;

(7) $(c+d)\mathbf{x}=c\mathbf{x}+d\mathbf{x}$;

(8) $c(\mathbf{x}+\mathbf{y})=c\mathbf{x}+c\mathbf{y}$.

The basic example is $\mathbb{R}^n$, the set of n-tuples $\mathbf{x}=(x_1,\dots,x_n)$ with componentwise operations $\mathbf{x}+\mathbf{y}=(x_1+y_1,\dots,x_n+y_n)$, $c\mathbf{x}=(cx_1,\dots,cx_n)$.

A subset $W \subset V$ is a **linear subspace** if $\mathbf{x}+\mathbf{y} \in W$ and $c\mathbf{x} \in W$ whenever $\mathbf{x}, \mathbf{y} \in W$, $c \in \mathbb{R}$; then W is a vector space in its own right. In the first part of the book only $\mathbb{R}^n$ and its subspaces occur.

Definition 1.2 (span, independence, basis, dimension)

Let $\mathbf{a}_1,\dots,\mathbf{a}_m \in V$.

- They **span** V if every $\mathbf{x} \in V$ can be written in *at least* one way as a **linear combination** $\mathbf{x}=c_1\mathbf{a}_1+\cdots+c_m\mathbf{a}_m$.
- They are **independent** if every $\mathbf{x} \in V$ can be written in *at most* one way as such a combination; equivalently, $d_1\mathbf{a}_1+\cdots+d_m\mathbf{a}_m=\mathbf{0}$ forces $d_1=\cdots=d_m=0$.
- They form a **basis** if they both span V and are independent.

If V has a basis of m vectors, m is the **dimension** of V (by Theorem 1.1 this is well defined);

the zero space has dimension 0.

Theorem 1.1

Suppose V has a basis consisting of m vectors. Then any set of vectors that spans V has at least m vectors, and any set of vectors of V that is independent has at most m vectors. In particular, any basis for V has exactly m vectors.

Proof [Supplement]. It suffices to show: if $\mathbf{w}_1, \dots, \mathbf{w}_k$ span V and $\mathbf{v}_1, \dots, \mathbf{v}_l$ are independent, then $l \le k$ (apply it with the basis as the spanning set, and with the basis as the independent set). Write $\mathbf{v}_j = \sum_{i=1}^{k} a_{ij}\mathbf{w}_i$. If $l > k$, the homogeneous linear system $\sum_{j=1}^{l} a_{ij}x_j = 0$ $(i = 1, \dots, k)$ has more unknowns than equations, hence (by Gauss–Jordan reduction: an echelon form has at most k corner entries, so some unknown is free) a nonzero solution $(x_1, \dots, x_l)$. Then $\sum_j x_j\mathbf{v}_j = \sum_i(\sum_j a_{ij}x_j)\mathbf{w}_i = \mathbf{0}$, contradicting independence. ■

$\mathbb{R}^n$ has dimension n: the **standard basis** is $\mathbf{e}_1 = (1, 0, \dots, 0)$, $\mathbf{e}_2 = (0, 1, 0, \dots, 0)$, $\dots$, $\mathbf{e}_n = (0, \dots, 0, 1)$. Every basis of $\mathbb{R}^n$ has exactly n vectors. (Spanning, independence and bases make sense for infinite sets of vectors too, but we shall not need this.)

Theorem 1.2

Let V be a vector space of dimension m. If W is a linear subspace of V different from V, then W has dimension less than m. Furthermore, any basis $\mathbf{a}_1, \dots, \mathbf{a}_k$ for W may be extended to a basis $\mathbf{a}_1, \dots, \mathbf{a}_k, \mathbf{a}_{k+1}, \dots, \mathbf{a}_m$ for V.

Proof [Supplement]. First, W has a finite basis: choose independent vectors of W one at a time as long as possible; by Theorem 1.1 the process stops after at most m steps, and a maximal independent subset of W spans W (if $\mathbf{x} \in W$ were not a combination of them, adjoining $\mathbf{x}$ would keep independence). Now let $\mathbf{a}_1, \dots, \mathbf{a}_k$ be a basis of W. If they do not span V, pick $\mathbf{a}_{k+1} \in V$ not in their span; then $\mathbf{a}_1, \dots, \mathbf{a}_{k+1}$ are independent (a relation with nonzero coefficient of $\mathbf{a}_{k+1}$ would put $\mathbf{a}_{k+1}$ in the span; one with zero coefficient contradicts independence of the $\mathbf{a}_i$). Repeat; by Theorem 1.1 the process stops when we have an independent spanning set of m vectors. If $W \neq V$, at least one vector was added, so $k < m$. ■

1.2 Inner products and norms

Definition 1.3 (inner product)

An **inner product** on V assigns to $\mathbf{x}, \mathbf{y} \in V$ a real number $\langle \mathbf{x}, \mathbf{y} \rangle$ such that for all $\mathbf{x}, \mathbf{y}, \mathbf{z}$ and scalars c:
(1) $\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle$;
(2) $\langle \mathbf{x} + \mathbf{y}, \mathbf{z} \rangle = \langle \mathbf{x}, \mathbf{z} \rangle + \langle \mathbf{y}, \mathbf{z} \rangle$;
(3) $\langle c\mathbf{x}, \mathbf{y} \rangle = c\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{x}, c\mathbf{y} \rangle$;
(4) $\langle \mathbf{x}, \mathbf{x} \rangle > 0$ if $\mathbf{x} \neq \mathbf{0}$. V together with an inner product is an **inner product space**.

On $\mathbb{R}^n$ we use the **dot product** $\langle \mathbf{x}, \mathbf{y} \rangle = x_1y_1 + \cdots + x_ny_n$ (written $\langle \mathbf{x}, \mathbf{y} \rangle$ rather than $\mathbf{x} \cdot \mathbf{y}$ to avoid confusion with the matrix product).

Definition 1.4 (norm)

The **length** (or norm) in an inner product space is $\|\mathbf{x}\| = \langle \mathbf{x}, \mathbf{x} \rangle^{1/2}$. It satisfies (1) $\|\mathbf{x}\| > 0$ if $\mathbf{x} \neq \mathbf{0}$;

(2) $\|c\mathbf{x}\| = |c|\,\|\mathbf{x}\|$;
(3) $\|\mathbf{x}+\mathbf{y}\| \le \|\mathbf{x}\| + \|\mathbf{y}\|$ (**triangle inequality**); equivalently (3′) $\|\mathbf{x}-\mathbf{y}\| \ge \|\mathbf{x}\| - \|\mathbf{y}\|$. Any function $V \to \mathbb{R}$ with properties (1)–(3) is called a **norm** on V.

Proof of (3) [Supplement]. *Cauchy–Schwarz*: $|\langle \mathbf{x},\mathbf{y}\rangle| \le \|\mathbf{x}\|\|\mathbf{y}\|$. This is trivial if $\mathbf{x}$ or $\mathbf{y}$ is $\mathbf{0}$; otherwise put $c = 1/\|\mathbf{x}\|$, $d = 1/\|\mathbf{y}\|$; then $0 \le \|c\mathbf{x} \pm d\mathbf{y}\|^2 = 2 \pm 2cd\langle \mathbf{x},\mathbf{y}\rangle$, so $|\langle \mathbf{x},\mathbf{y}\rangle| \le 1/(cd)$. Hence $\|\mathbf{x}+\mathbf{y}\|^2 = \|\mathbf{x}\|^2 + 2\langle \mathbf{x},\mathbf{y}\rangle + \|\mathbf{y}\|^2 \le (\|\mathbf{x}\| + \|\mathbf{y}\|)^2$. Applying (3) to $\mathbf{x} = (\mathbf{x}-\mathbf{y}) + \mathbf{y}$ gives (3′). ■

On $\mathbb{R}^n$ we have the **euclidean norm** $\|\mathbf{x}\|$ coming from the dot product, and the **sup norm**

$$|\mathbf{x}| = \max\{|x_1|, \dots, |x_n|\}.$$

The sup norm is often more convenient. The two are related by

$$|\mathbf{x}| \le \|\mathbf{x}\| \le \sqrt{n}\,|\mathbf{x}|, \tag{1.1}$$

since $\max_i x_i^2 \le \sum_i x_i^2 \le n \max_i x_i^2$.

1.3 Matrices

An n by m matrix A has entries a_{ij} (i = row index, j = column index). With entrywise addition and scalar multiplication the n by m matrices form a vector space (essentially $\mathbb{R}^{nm}$). If A is n by m and B is m by p, the **product** $C = A \cdot B$ is the n by p matrix with

$$c_{ij} = \sum_{k=1}^{m} a_{ik} b_{kj}.$$

It satisfies $A \cdot (B \cdot C) = (A \cdot B) \cdot C$, $A \cdot (B + C) = A \cdot B + A \cdot C$, $(A + B) \cdot C = A \cdot C + B \cdot C$, $(cA) \cdot B = c(A \cdot B) = A \cdot (cB)$, and $I_n \cdot A = A = A \cdot I_m$ for A of size n by m, where the **identity matrix** I_k has entries δ_{ij} ($\delta_{ij} = 1$ if $i = j$, 0 otherwise).

The sup norm extends to matrices: $|A| = \max\{|a_{ij}| : i = 1, \dots, n,\ j = 1, \dots, m\}$; it is a norm.

Theorem 1.3

If A has size n by m and B has size m by p, then $|A \cdot B| \le m|A|\,|B|$.

Proof [Supplement]. $|c_{ij}| = |\sum_{k=1}^m a_{ik}b_{kj}| \le \sum_{k=1}^m |A||B| = m|A||B|$ for every i, j. ■

1.4 Linear transformations

Definition 1.5 (linear transformation)

A function $T : V \to W$ between vector spaces is a **linear transformation** if $T(\mathbf{x}+\mathbf{y}) = T(\mathbf{x}) + T(\mathbf{y})$ and $T(c\mathbf{x}) = cT(\mathbf{x})$. If moreover T is a bijection, T is a **linear isomorphism**.

Composites of linear transformations are linear, and the inverse of a linear isomorphism is linear.

Theorem 1.4

Let V be a vector space with basis $\mathbf{a}_1, \dots, \mathbf{a}_m$. Let W be a vector space. Given any m vectors $\mathbf{b}_1, \dots, \mathbf{b}_m$ in W, there is exactly one linear transformation $T : V \to W$ such that $T(\mathbf{a}_i) = \mathbf{b}_i$ for all i.

Proof [Supplement]. Each $\mathbf{x} \in V$ has a unique expression $\mathbf{x} = \sum c_i \mathbf{a}_i$; define $T(\mathbf{x}) = \sum c_i \mathbf{b}_i$. Uniqueness of the coefficients makes T well defined, and since the coefficients of $\mathbf{x} + \mathbf{y}$ and $c\mathbf{x}$ are the sums and multiples of those of $\mathbf{x}, \mathbf{y}$, T is linear. Any linear T' with $T'(\mathbf{a}_i) = \mathbf{b}_i$ satisfies $T'(\sum c_i \mathbf{a}_i) = \sum c_i \mathbf{b}_i$, so $T' = T$. ■

The maps $(x_1, \dots, x_n) \mapsto [x_1 \ \cdots \ x_n]$ (**row matrix**) and $(x_1, \dots, x_n) \mapsto$ the n by 1 **column matrix** with entries x_i are linear isomorphisms. Representing elements of $\mathbb{R}^m$ as column matrices, a fixed n by m matrix A defines a linear map $T : \mathbb{R}^m \to \mathbb{R}^n$, $T(\mathbf{x}) = A \cdot \mathbf{x}$. Conversely every linear $T : \mathbb{R}^m \to \mathbb{R}^n$ has this form: let $\mathbf{b}_j = T(\mathbf{e}_j)$ and $A = [\mathbf{b}_1 \ \cdots \ \mathbf{b}_m]$; then $A \cdot \mathbf{e}_j = \mathbf{b}_j = T(\mathbf{e}_j)$ for all j, so $A \cdot \mathbf{x} = T(\mathbf{x})$ by Theorem 1.4.

Convention

Throughout, the elements of $\mathbb{R}^n$ are represented by column matrices, unless specifically stated otherwise.

1.5 Rank of a matrix

For an n by m matrix A, the **column space** is the subspace of $\mathbb{R}^n$ spanned by the columns; its dimension is the **column rank** ($\leq \min(m, n)$). The **row space** is the subspace of $\mathbb{R}^m$ spanned by the rows; its dimension is the **row rank**.

Theorem 1.5

For any matrix A, the row rank of A equals the column rank of A.

Proof [Supplement]. Let r be the column rank and let $\mathbf{c}_1, \dots, \mathbf{c}_r \in \mathbb{R}^n$ be a basis of the column space; let $C = [\mathbf{c}_1 \ \cdots \ \mathbf{c}_r]$ (n by r). Each column of A is a combination of the $\mathbf{c}_k$, so $A = C \cdot D$ for some r by m matrix D. Then each row of A is a combination of the r rows of D (row i of A is $\sum_k c_{ik} \cdot$(row k of D)), so the row rank of A is $\leq r$. Applying this to A^{tr} (whose columns are the rows of A) gives the reverse inequality. ■

The common value is the **rank** of A.

Elementary row operations on a matrix are:

(1) exchange rows $i_1 \neq i_2$;

(2) replace row i_1 by itself plus c times row i_2 ($i_1 \neq i_2$);

(3) multiply row i by a scalar $\lambda \neq 0$. Each is invertible, and its inverse is an elementary operation of the same type.

Theorem 1.6

If B is the matrix obtained by applying an elementary row operation to A, then $\operatorname{rank} B = \operatorname{rank} A$.

Proof [Supplement]. Each row of B is a linear combination of rows of A, so the row space of B is contained in that of A. Since A is obtained from B by the inverse elementary operation, the reverse inclusion holds too. The row spaces coincide, so the row ranks agree. ■

Gauss–Jordan reduction applies elementary operations to bring A to **echelon form** (stairstep form): below the "stairsteps" all entries are 0 and the corner entries (pivots) are nonzero. Operations of types (1) and (2) suffice. The nonzero rows of a matrix in echelon form are independent, so its rank is the number of nonzero rows. Using further operations of types (2) and (3) one reaches the **reduced echelon form**, in which each corner entry is 1 and all entries directly above it are 0.

1.6 Transpose

The **transpose** of the n by m matrix A is the m by n matrix A^{tr} with entries $d_{ij} = a_{ji}$. Then $(A^{\text{tr}})^{\text{tr}} = A$, $(A+B)^{\text{tr}} = A^{\text{tr}} + B^{\text{tr}}$, $(A \cdot C)^{\text{tr}} = C^{\text{tr}} \cdot A^{\text{tr}}$, and $\operatorname{rank} A^{\text{tr}} = \operatorname{rank} A$ (the row rank of A^{tr} is the column rank of A).

§2. Matrix Inversion and Determinants

Motivation. The determinant will appear as the volume-scaling factor in the change of variables theorem and as the model for alternating tensors. Here we characterize it by its row properties, which gives the product formula, the inverse formula and cofactor expansion in one framework.

2.1 Elementary matrices

Definition 2.1 (elementary matrix)

An **elementary matrix** of size n by n is the matrix obtained by applying one of the elementary row operations to I_n.

There are three types: E (rows i_1, i_2 of I_n exchanged), E' (I_n with an extra entry c in row i_1, column i_2), and E'' (I_n with the (i,i) entry replaced by $\lambda \neq 0$).

Theorem 2.1

Let A be an n by m matrix. Any elementary row operation on A may be carried out by premultiplying A by the corresponding elementary matrix.

Proof. Direct computation: multiplying A on the left by E interchanges rows i_1 and i_2 of A; by E' replaces row i_1 by itself plus c times row i_2; by E'' multiplies row i by λ. ■

2.2 The inverse of a matrix

Definition 2.2 (left and right inverse)

Let A be n by m, and B, C be m by n. B is a **left inverse** for A if $B \cdot A = I_m$; C is a **right inverse** for A if $A \cdot C = I_n$.

Theorem 2.2

If A has both a left inverse B and a right inverse C, then they are unique and equal.

Proof. $C = I_m \cdot C = (B \cdot A) \cdot C = B \cdot (A \cdot C) = B \cdot I_n = B$. If B_1 is another left inverse, the same computation gives $C = B_1$, so $B_1 = B$; similarly C is unique. ■

Definition 2.3 (invertible)

If A has both a left and a right inverse, A is **invertible**; the unique two-sided inverse is denoted A^{-1}.

Theorem 2.3

Let A be a matrix of size n by m. If A is invertible, then $n = m = \operatorname{rank} A$.

Proof.

Step 1. For any k by n matrix D, $\operatorname{rank}(D \cdot A) \leq \operatorname{rank} A$. Indeed, if R is a 1 by n row matrix, $R \cdot A$ is a linear combination of the rows of A. Each row of $D \cdot A$ is (row of D)$\cdot A$, so the row space of $D \cdot A$ lies in the row space of A.

Step 2. If A has a left inverse B, then $\operatorname{rank} A = m$ (the number of columns). Indeed $m = \operatorname{rank} I_m = \operatorname{rank}(B \cdot A) \leq \operatorname{rank} A$ by Step 1, while the row space of A is a subspace of $\mathbb{R}^m$, so $\operatorname{rank} A \leq m$.

Step 3. Let $B = A^{-1}$. Since B is a left inverse, $\operatorname{rank} A = m$ by Step 2. Since B is a right inverse, $B^{\mathrm{tr}} \cdot A^{\mathrm{tr}} = I_n^{\mathrm{tr}} = I_n$, so B^{tr} is a left inverse of the m by n matrix A^{tr} and Step 2 gives $\operatorname{rank} A = \operatorname{rank} A^{\mathrm{tr}} = n$. ■

Theorem 2.4

Let A be a matrix of size n by m with $n = m = \operatorname{rank} A$. Then A is invertible; furthermore, A equals a product of elementary matrices.

Proof.

Step 1. Every elementary matrix is invertible with elementary inverse: E is its own inverse, E' with c replaced by $-c$ inverts E', and E'' with λ replaced by $1/\lambda$ inverts E''.

Step 2. Reduce A to reduced echelon form C by elementary operations. C is square of rank n, so it has n nonzero rows; hence $C = I_n$. By Theorem 2.1 there are elementary matrices $E_1, \ldots, E_k$ with $E_k \cdots E_2 E_1 A = I_n$. Multiplying on the left successively by $E_k^{-1}, \ldots, E_1^{-1}$ gives

$$A = E_1^{-1} E_2^{-1} \cdots E_k^{-1},$$

a product of elementary matrices, and $B = E_k E_{k-1} \cdots E_1$ is both a left and a right inverse for A. ■

Theorem 2.5

If A is a square matrix and B is a left inverse for A, then B is also a right inverse for A.

Proof. By Step 2 of the proof of Theorem 2.3, $\operatorname{rank} A$ equals the number of columns, which (as A is square) equals the number of rows. By Theorem 2.4, A has an inverse, which equals B by Theorem 2.2. ■

An n by n matrix is **singular** if $\operatorname{rank} A < n$, **non-singular** otherwise. By the above, A is invertible iff it is non-singular.

2.3 Determinants

Since $|A|$ denotes the sup norm, the determinant is written $\det A$. We characterize it by axioms; the existence of such a function for all n is proved in Chapter 6 (§27).

Definition 2.4 (determinant function)

A function assigning to each n by n matrix A a real number $\det A$ is a **determinant function** if:

(1) if B is obtained from A by exchanging two rows, then $\det B = -\det A$;

(2) for each i, $\det A$ is linear as a function of the ith row alone: if $A_i(\mathbf{x})$ denotes A with row i replaced by $\mathbf{x}$, then $\det A_i(a\mathbf{x}+b\mathbf{y}) = a\det A_i(\mathbf{x}) + b\det A_i(\mathbf{y})$;

(3) $\det I_n = 1$.

Example 2.1

For 1 by 1 matrices $\det[a] = a$; for 2 by 2 matrices $\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$; for 3 by 3,

$$\det \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix} = a_1b_2c_3 + a_2b_3c_1 + a_3b_1c_2 - a_3b_2c_1 - a_1b_3c_2 - a_2b_1c_3.$$

These satisfy the axioms. For larger matrices a direct formula has $n!$ terms (24 for $n = 4$, 120 for $n = 5$), so a less direct approach is needed.

Theorem 2.6

Let A be an n by n matrix.

(a) If E is the elementary matrix exchanging rows i_1 and i_2, then $\det(E \cdot A) = -\det A$.

(b) If E' replaces row i_1 of A by itself plus c times row i_2, then $\det(E' \cdot A) = \det A$.

(c) If E'' multiplies row i of A by $\lambda \neq 0$, then $\det(E'' \cdot A) = \lambda \det A$.

(d) If $A = I_n$, then $\det A = 1$.

Proof. (a) is Axiom 1 and (d) is Axiom 3;

(c) is the special case $\det A_i(\lambda\mathbf{x}) = \lambda \det A_i(\mathbf{x})$ of Axiom 2. For (b): if A has two equal rows, exchanging them does not change A but changes the sign of $\det A$, so $\det A = 0$. Let $i = i_1$, $\mathbf{x} =$ row i_1, $\mathbf{y} =$ row i_2. Then

$$\det(E' \cdot A) = \det A_i(\mathbf{x} + c\mathbf{y}) = \det A_i(\mathbf{x}) + c\det A_i(\mathbf{y}) = \det A_i(\mathbf{x}) = \det A,$$

since $A_i(\mathbf{y})$ has two equal rows. ■

Taking $A = I_n$: $\det E = -1$, $\det E' = 1$, $\det E'' = \lambda$.

Theorem 2.7

Let A be a square matrix. If the rows of A are independent, then $\det A \neq 0$; if the rows are dependent, then $\det A = 0$. Thus an n by n matrix A has rank n if and only if $\det A \neq 0$.

Proof. If row i of A is zero, then $\det A = 0$: multiplying that row by 2 leaves A unchanged but doubles $\det A$. An elementary operation multiplies the determinant by -1, 1 or $\lambda \neq 0$, so it does not affect whether $\det A$ vanishes. Reduce A to echelon form B (types (1), (2)). If the rows of A are dependent, $\operatorname{rank} B = \operatorname{rank} A < n$, so B has a zero row, $\det B = 0$, and $\det A = 0$. If they are independent, reduce further to reduced echelon form C; being square of rank n, $C = I_n$, so $\det C = 1 \neq 0$ and $\det A \neq 0$. ■

Theorem 2.8

Given a square matrix A, reduce it to echelon form B by elementary row operations of types (1) and (2). If B has a zero row, then $\det A = 0$. Otherwise, let k be the number of row exchanges involved; then $\det A = (-1)^k$ times the product of the diagonal entries of B.

Proof. If B has a zero row, $\operatorname{rank} A < n$ and $\det A = 0$. Otherwise $\det A = (-1)^k \det B$ by Theorem 2.6(a),(b), and B is upper triangular with nonzero diagonal entries $b_{11}, \ldots, b_{nn}$. Using type (2) operations, clear the entries above the diagonal; the diagonal is unaffected, giving a diagonal matrix C with $\det C = \det B$. Multiplying row i of C by $1/b_{ii}$ for each i yields I_n, so by Theorem 2.6(c),(d), $1 = \det I_n = (1/b_{11}) \cdots (1/b_{nn}) \det C$. Hence $\det B = \det C = b_{11} \cdots b_{nn}$. ■

Corollary 2.9

The determinant function is uniquely characterized by its three axioms. It is also characterized by the four properties listed in Theorem 2.6.

Proof. The calculation of $\det A$ in Theorem 2.8 uses only properties (a)–(d) of Theorem 2.6, which follow from the axioms. ■

Theorem 2.10

Let A and B be n by n matrices. Then $\det(A \cdot B) = (\det A) \cdot (\det B)$.

Proof.

Step 1. The theorem holds when A is elementary: $\det(E \cdot B) = -\det B = (\det E)(\det B)$, $\det(E' \cdot B) = \det B = (\det E')(\det B)$, $\det(E'' \cdot B) = \lambda \det B = (\det E'')(\det B)$.

Step 2. If $\operatorname{rank} A = n$, $A = E_1 \cdots E_k$ is a product of elementary matrices (Theorem 2.4), and repeated use of Step 1 gives $\det(A \cdot B) = (\det E_1) \cdots (\det E_k)(\det B)$ for all B. With $B = I_n$ this reads $\det A = (\det E_1) \cdots (\det E_k)$, and the theorem follows.

Step 3. If $\operatorname{rank} A < n$, then $\operatorname{rank}(A \cdot B) = \operatorname{rank}(B^{\mathrm{tr}} \cdot A^{\mathrm{tr}}) \leq \operatorname{rank} A^{\mathrm{tr}} < n$ by Step 1 of Theorem 2.3, so both sides vanish by Theorem 2.7. ■

Theorem 2.11

$\det A^{\mathrm{tr}} = \det A$.

Proof.

Step 1. For elementary matrices: $E^{\mathrm{tr}} = E$, $(E'')^{\mathrm{tr}} = E''$, and $(E')^{\mathrm{tr}}$ is again of type (2), so both have determinant 1.

Step 2. If $\operatorname{rank} A = n$, write $A = E_1 \cdots E_k$. Then by Theorem 2.10 and Step 1,

$$\det A^{\mathrm{tr}} = \det(E_k^{\mathrm{tr}} \cdots E_1^{\mathrm{tr}}) = (\det E_k^{\mathrm{tr}}) \cdots (\det E_1^{\mathrm{tr}}) = (\det E_1) \cdots (\det E_k) = \det A.$$

Step 3. If $\operatorname{rank} A < n$, then $\operatorname{rank} A^{\mathrm{tr}} < n$ and both determinants are 0. ■

2.4 A formula for A^{-1}

Definition 2.5 (minor, cofactor)

For an n by n matrix A, the $(n-1)$ by $(n-1)$ matrix A_{ij} obtained by deleting row i and column j is the (i,j)**-minor**; the number $(-1)^{i+j}\det A_{ij}$ is the (i,j)**-cofactor** of A.

Lemma 2.12

Let A be an n by n matrix and let b denote its entry in row i and column j.

(a) If all the entries in row i other than b vanish, then $\det A = b(-1)^{i+j}\det A_{ij}$.

(b) The same holds if all the entries in column j other than b vanish.

Proof.

Step 1. Fix $b, a_2, \dots, a_n$. For an $(n-1)$ by $(n-1)$ matrix D let $A(D)$ be the n by n matrix with first row $(b, a_2, \dots, a_n)$, first column $(b, 0, \dots, 0)^{\mathrm{tr}}$ and lower right block D. We show $\det A(D) = b\det D$. If $b = 0$, $\operatorname{rank} A(D) < n$ and both sides vanish. If $b \neq 0$, put $f(D) = (1/b)\det A(D)$. Exchanging two rows of D, adding a multiple of one row of D to another, or multiplying a row of D by λ has the same effect on the corresponding rows $(i+1)$ of $A(D)$, so f changes by -1, 1, λ, respectively. If $D = I_{n-1}$, $A(D)$ is in echelon form and $\det A(D) = b$ by Theorem 2.8, so $f(I_{n-1}) = 1$. By Corollary 2.9, $f(D) = \det D$.

Step 2. Taking transposes (Theorem 2.11), the same formula holds for the matrix with first row $(b, 0, \dots, 0)$, first column $(b, a_2, \dots, a_n)^{\mathrm{tr}}$ and block D.

Step 3. In the general case, $i-1$ exchanges of adjacent rows bring row i to the top without changing the order of the other rows; then $j-1$ exchanges of adjacent columns bring column j to the left edge. The resulting matrix C is of the form in Step 1 or 2, and its $(1,1)$-minor C_{11} equals A_{ij}. Each row exchange, and (by Theorem 2.11) each column exchange, changes the sign of the determinant, so

$$\det A = (-1)^{(i-1)+(j-1)}\det C = (-1)^{i+j} b \det C_{11} = (-1)^{i+j} b \det A_{ij}.$$

■

Theorem 2.13 (Cramer's rule)

Let A be an n by n matrix with successive columns $\mathbf{a}_1, \dots, \mathbf{a}_n$. Let $\mathbf{x} = (x_1, \dots, x_n)$ and $\mathbf{c} = (c_1, \dots, c_n)$ be column matrices. If $A \cdot \mathbf{x} = \mathbf{c}$, then

$$(\det A) \cdot x_i = \det[\mathbf{a}_1 \cdots \mathbf{a}_{i-1}\ \mathbf{c}\ \mathbf{a}_{i+1} \cdots \mathbf{a}_n].$$

Proof. Let $C = [\mathbf{e}_1 \cdots \mathbf{e}_{i-1}\ \mathbf{x}\ \mathbf{e}_{i+1} \cdots \mathbf{e}_n]$. Since $A \cdot \mathbf{e}_j = \mathbf{a}_j$ and $A \cdot \mathbf{x} = \mathbf{c}$, we have $A \cdot C = [\mathbf{a}_1 \cdots \mathbf{a}_{i-1}\ \mathbf{c}\ \mathbf{a}_{i+1} \cdots \mathbf{a}_n]$, so by Theorem 2.10, $(\det A)(\det C)$ equals the right-hand side. C is I_n except for column i, which is $\mathbf{x}$; expanding by Lemma 2.12 along row i (whose only nonzero entry may be x_i in column i) gives $\det C = x_i(-1)^{i+i}\det I_{n-1} = x_i$. ■

Theorem 2.14

Let A be an n by n matrix of rank n; let $B = A^{-1}$. Then

$$b_{ij} = \frac{(-1)^{j+i}\det A_{ji}}{\det A}.$$

Proof. Fix j and let $\mathbf{x}$ be the jth column of B. Then $A \cdot B = I_n$ gives $A \cdot \mathbf{x} = \mathbf{e}_j$, and Cramer's rule gives $(\det A)x_i = \det[\mathbf{a}_1 \cdots \mathbf{a}_{i-1}\ \mathbf{e}_j\ \mathbf{a}_{i+1} \cdots \mathbf{a}_n]$. Column i of this matrix has a single nonzero entry 1 in row j, so by Lemma 2.12(b) the determinant is $1 \cdot (-1)^{j+i} \det A_{ji}$. Since $x_i = b_{ij}$, the theorem follows. ∎

This yields the algorithm:

(1) form the **matrix of cofactors**, with (i, j) entry $(-1)^{i+j} \det A_{ij}$;

(2) transpose it;

(3) divide each entry by $\det A$. Its importance is theoretical (for instance it shows that the entries of A^{-1} are continuous functions of those of A); for computation, Gauss–Jordan reduction is more efficient.

2.5 Expansion by cofactors

Theorem 2.15

Let A be an n by n matrix and fix i. Then

$$\det A = \sum_{k=1}^{n} (-1)^{i+k} a_{ik} \det A_{ik}.$$

Proof. Let $A_i(\mathbf{x})$ be A with row i replaced by the row matrix $\mathbf{x}$. Row i of A is $\sum_k a_{ik}\mathbf{e}_k$, so by linearity (Axiom 2) and Lemma 2.12,

$$\det A = \sum_{k=1}^{n} a_{ik} \det A_i(\mathbf{e}_k) = \sum_{k=1}^{n} a_{ik}(-1)^{i+k} \det A_{ik}.$$

∎

This is the expansion of the determinant by cofactors of the ith row; taking transposes gives expansion by cofactors of the jth column.

§3. Review of Topology in $\mathbb{R}^n$

Motivation. Differentiability and integrability are defined through limits, so we need open sets, closed sets and continuity in $\mathbb{R}^n$; the general metric-space language costs nothing extra and is reused later for manifolds.

3.1 Metric spaces

Definition 3.1 (metric space)

A **metric** on a set X is a function $d : X \times X \to \mathbb{R}$ such that for all $x, y, z \in X$:
(1) $d(x, y) = d(y, x)$;
(2) $d(x, y) \geq 0$, with equality iff $x = y$;
(3) $d(x, z) \leq d(x, y) + d(y, z)$. A **metric space** is a set with a specific metric. A subset $Y \subset X$ with the restricted metric is a **subspace** of X.

On $\mathbb{R}^n$ we have the **euclidean metric** $\|\mathbf{x} - \mathbf{y}\|$ and the **sup metric** $|\mathbf{x} - \mathbf{y}|$. $\mathbb{R}^n$ is called n-dimensional **euclidean space**.

Definition 3.2 (open and closed sets)

For $x_0 \in X$ and $\epsilon > 0$, the **ϵ-neighborhood** of x_0 is $U(x_0;\epsilon) = \{x : d(x,x_0) < \epsilon\}$. A set $U \subset X$ is **open** in X if for each $x_0 \in U$ some $U(x_0;\epsilon) \subset U$; a set C is **closed** in X if $X - C$ is open. Any open set containing x_0 is a **neighborhood** of x_0.

By the triangle inequality, an ϵ-neighborhood is open: if $d(x,x_0) < \epsilon$, then $U(x;\epsilon - d(x,x_0)) \subset U(x_0;\epsilon)$.

Theorem 3.1

Let (X,d) be a metric space. Then finite intersections and arbitrary unions of open sets of X are open in X. Similarly, finite unions and arbitrary intersections of closed sets of X are closed in X.

Proof [Supplement]. If x lies in a union of open sets, it lies in one of them, U_α, and some $U(x;\epsilon) \subset U_\alpha$ is contained in the union. If $x \in U_1 \cap \cdots \cap U_k$, choose ϵ_i with $U(x;\epsilon_i) \subset U_i$ and let $\epsilon = \min \epsilon_i > 0$. The statements about closed sets follow by De Morgan's laws. ■

Theorem 3.2

Let X be a metric space and Y a subspace. A subset A of Y is open in Y if and only if $A = U \cap Y$ for some U open in X. Similarly, A is closed in Y if and only if $A = C \cap Y$ for some C closed in X.

Proof [Supplement]. The ϵ-neighborhood of $y \in Y$ in Y is $U_X(y;\epsilon) \cap Y$. If U is open in X and $y \in U \cap Y$, then $U_X(y;\epsilon) \subset U$ for some ϵ, so $U_Y(y;\epsilon) \subset U \cap Y$. Conversely, if A is open in Y, choose for each $y \in A$ an ϵ_y with $U_X(y;\epsilon_y) \cap Y \subset A$ and put $U = \bigcup_{y \in A} U_X(y;\epsilon_y)$; then U is open in X and $U \cap Y = A$. For closed sets take complements: $Y - A = (X - C) \cap Y$. ■

Hence if A is open in Y and Y is open in X, then A is open in X; likewise for closed sets.

Definition 3.3 (limit point, closure)

$x_0 \in X$ is a **limit point** of $A \subset X$ if every ϵ-neighborhood of x_0 meets A in a point other than x_0 (equivalently, in infinitely many points of A). The **closure** $\bar{A}$ of A is A together with all its limit points.

Theorem 3.3

If A is a subset of X, then $\bar{A}$ is a closed set of X. A subset of X is closed if and only if it contains all its limit points.

Proof [Supplement]. A set C is closed iff every $x \notin C$ has a neighborhood disjoint from C, i.e. iff no point outside C is a limit point of C; this is the second statement. For the first: if $x \notin \bar{A}$, some $U(x;\epsilon)$ misses A. This open set also misses every limit point y of A: a point $y \in U(x;\epsilon)$ has a neighborhood inside $U(x;\epsilon)$, which contains no point of A. So $U(x;\epsilon) \cap \bar{A} = \emptyset$ and $X - \bar{A}$ is open. ■

In $\mathbb{R}^n$, the ϵ-neighborhood of $\mathbf{a}$ in the euclidean metric is the **open ball** $B(\mathbf{a};\epsilon)$; in the sup metric it is the **open cube** $C(\mathbf{a};\epsilon)$. By (1.1),

$$B(\mathbf{a};\epsilon) \subset C(\mathbf{a};\epsilon) \subset B(\mathbf{a};\sqrt{n}\,\epsilon).$$

Theorem 3.4

If X is a subspace of $\mathbb{R}^n$, the collection of open sets of X is the same whether one uses the euclidean metric or the sup metric on X. The same is true for the collection of closed sets of X.

Proof [Supplement]. If A is open for the sup metric and $\mathbf{x} \in A$, then $C(\mathbf{x};\epsilon) \cap X \subset A$ for some ϵ, hence $B(\mathbf{x};\epsilon) \cap X \subset A$. If A is open for the euclidean metric, $B(\mathbf{x};\epsilon) \cap X \subset A$, hence $C(\mathbf{x};\epsilon/\sqrt{n}) \cap X \subset A$. Closed sets are the complements. ■

A property of X depending only on its collection of open sets is a **topological property**; limits, continuity and compactness are examples.

3.2 Limits and continuity

Definition 3.4 (continuity)

Let X, Y be metric spaces. $f : X \to Y$ is **continuous at** x_0 if for each open $V \ni f(x_0)$ there is an open $U \ni x_0$ with $f(U) \subset V$; equivalently, for each $\epsilon > 0$ there is $\delta > 0$ with $d_Y(f(x), f(x_0)) < \epsilon$ whenever $d_X(x, x_0) < \delta$. f is **continuous** if it is continuous at every point; equivalently, $f^{-1}(V)$ is open in X for each open V of Y (or $f^{-1}(D)$ is closed for each closed D).

If some δ-neighborhood of x_0 is $\{x_0\}$, then x_0 is an **isolated point** of X, and every f is continuous at x_0. Constant maps and the identity are continuous.

Theorem 3.5

(a) Let $x_0 \in A$, where A is a subspace of X. If $f : X \to Y$ is continuous at x_0, then $f|A : A \to Y$ is continuous at x_0.
(b) Let $f : X \to Y$ and $g : Y \to Z$. If f is continuous at x_0 and g is continuous at $y_0 = f(x_0)$, then $g \circ f : X \to Z$ is continuous at x_0.

Proof [Supplement]. (a) The same δ works for points of A.

(b) Given ϵ, choose η with $d_Z(g(y), g(y_0)) < \epsilon$ for $d_Y(y, y_0) < \eta$, then δ with $d_Y(f(x), y_0) < \eta$ for $d_X(x, x_0) < \delta$. ■

Theorem 3.6

(a) Let X be a metric space and $f : X \to \mathbb{R}^n$, $f(x) = (f_1(x), \ldots, f_n(x))$. Then f is continuous at x_0 if and only if each **component function** $f_i : X \to \mathbb{R}$ is continuous at x_0.
(b) Let $f, g : X \to \mathbb{R}$ be continuous at x_0. Then $f + g$, $f - g$ and $f \cdot g$ are continuous at x_0, and f/g is continuous at x_0 if $g(x_0) \neq 0$.
(c) The projection $\pi_i : \mathbb{R}^n \to \mathbb{R}$, $\pi_i(\mathbf{x}) = x_i$, is continuous.

Proof [Supplement]. (a) Use the sup metric: $|f(x) - f(x_0)| < \epsilon$ iff $|f_i(x) - f_i(x_0)| < \epsilon$ for all i; take the minimum of the n values of δ.

(c) $|\pi_i(\mathbf{x}) - \pi_i(\mathbf{y})| \leq |\mathbf{x} - \mathbf{y}|$.

(b) Addition, subtraction and multiplication $\mathbb{R}^2 \to \mathbb{R}$ are continuous, and so is division on $\{(s,t) : t \neq 0\}$, by the usual estimates, e.g. $|st - s_0t_0| \leq |s||t - t_0| + |t_0||s - s_0|$. Compose with $x \mapsto (f(x), g(x))$, continuous by (a), and use Theorem 3.5. ■

Thus functions built from the familiar continuous functions by algebraic operations and composition are continuous, e.g. $f(s,t,u,v) = \sin(s+t)/e^{uv}$ on $\mathbb{R}^4$.

Definition 3.5 (limit)

Let $A \subset X$, $f : A \to Y$, and let x_0 be a limit point of A (not necessarily in A). We say $f(x) \to y_0$ as $x \to x_0$, or $\lim_{x\to x_0} f(x) = y_0$, if for each open $V \ni y_0$ there is an open $U \ni x_0$ with $f(x) \in V$ for all $x \in U \cap A$, $x \neq x_0$. Equivalently: for each $\epsilon > 0$ there is $\delta > 0$ with $d_Y(f(x), y_0) < \epsilon$ whenever $x \in A$ and $0 < d_X(x, x_0) < \delta$.

We do not attempt to define the limit if x_0 is not a limit point of the domain; the value of f at x_0 plays no role.

Theorem 3.7

Let $f : X \to Y$. If x_0 is an isolated point of X, then f is continuous at x_0. Otherwise, f is continuous at x_0 if and only if $f(x) \to f(x_0)$ as $x \to x_0$.

Proof [Supplement]. Compare the ϵ-δ conditions: they differ only in whether $x = x_0$ is allowed, and at $x = x_0$ the condition $d_Y(f(x), f(x_0)) < \epsilon$ holds trivially. ■

Theorem 3.8

(a) Let $A \subset X$ and $f : A \to \mathbb{R}^n$, $f = (f_1, \dots, f_n)$; let $\mathbf{a} = (a_1, \dots, a_n)$. Then $f(x) \to \mathbf{a}$ as $x \to x_0$ if and only if $f_i(x) \to a_i$ for each i.
(b) Let $f, g : A \to \mathbb{R}$. If $f(x) \to a$ and $g(x) \to b$ as $x \to x_0$, then $f+g \to a+b$, $f-g \to a-b$, $f \cdot g \to a \cdot b$, and $f/g \to a/b$ if $b \neq 0$.

Proof [Supplement]. Extend f, g to $A \cup \{x_0\}$ by $f(x_0) = \mathbf{a}$ (resp. a, b). The limit statements become continuity at x_0 of the extended functions (Theorem 3.7), and Theorem 3.6 applies. ■

3.3 Interior and exterior

Definition 3.6 (interior, exterior, boundary)

For $A \subset \mathbb{R}^n$: $\operatorname{Int} A$ is the union of all open sets of $\mathbb{R}^n$ contained in A; $\operatorname{Ext} A$ is the union of all open sets disjoint from A; $\operatorname{Bd} A$ consists of the points in neither $\operatorname{Int} A$ nor $\operatorname{Ext} A$.

A point $\mathbf{x}$ is in $\operatorname{Bd} A$ iff every open set containing $\mathbf{x}$ meets both A and $\mathbb{R}^n - A$. $\mathbb{R}^n$ is the disjoint union of $\operatorname{Int} A$, $\operatorname{Ext} A$ and $\operatorname{Bd} A$; the first two are open and $\operatorname{Bd} A$ is closed.

For a **rectangle** $Q = [a_1, b_1] \times \cdots \times [a_n, b_n]$ one checks $\operatorname{Int} Q = (a_1, b_1) \times \cdots \times (a_n, b_n)$ (an **open rectangle**), $\operatorname{Ext} Q = \mathbb{R}^n - Q$, $\operatorname{Bd} Q = Q - \operatorname{Int} Q$. The open cube $C(\mathbf{a}; \epsilon) = (a_1 - \epsilon, a_1 + \epsilon) \times \cdots \times (a_n - \epsilon, a_n + \epsilon)$ is an open rectangle; the corresponding closed rectangle $[a_1 - \epsilon, a_1 + \epsilon] \times \cdots \times [a_n - \epsilon, a_n + \epsilon]$ is the **closed cube** (or simply cube) centered at $\mathbf{a}$.

§4. Compact Subspaces and Connected Subspaces of $\mathbb{R}^n$

Motivation. Compactness is what turns pointwise statements into uniform ones (uniform continuity, existence of maxima, Lebesgue numbers). These facts are used repeatedly in the existence of the integral and in the proof of the change of variables theorem.

4.1 Compact spaces

Definition 4.1 (compact)

Let $X \subset \mathbb{R}^n$. A **covering** of X is a collection of subsets of $\mathbb{R}^n$ whose union contains X; it is an **open covering** if each member is open in $\mathbb{R}^n$. X is **compact** if every open covering of X contains a finite subcollection that also covers X.

Theorem 4.1

A subspace X of $\mathbb{R}^n$ is compact if and only if for every collection of sets open in X whose union is X, there is a finite subcollection whose union equals X.

Proof. Suppose X is compact and $\{A_\alpha\}$ are open in X with union X. By Theorem 3.2 choose U_α open in $\mathbb{R}^n$ with $A_\alpha = U_\alpha \cap X$. The U_α cover X, so finitely many, $U_{\alpha_1}, \ldots, U_{\alpha_k}$, do; then $A_{\alpha_1}, \ldots, A_{\alpha_k}$ have union X. Conversely, given an open covering $\{U_\alpha\}$ of X by open sets of $\mathbb{R}^n$, the sets $U_\alpha \cap X$ are open in X with union X; a finite subfamily with union X gives finitely many U_α covering X. ■

Theorem 4.2

The subspace $[a, b]$ of $\mathbb{R}$ is compact.

Proof [Supplement]. Let $\mathcal{A}$ be an open covering of $[a, b]$ and let S be the set of $x \in [a, b]$ such that $[a, x]$ is covered by finitely many members of $\mathcal{A}$. Then $a \in S$; let $c = \sup S$. Some $U \in \mathcal{A}$ contains c, and with it an interval $(c - \delta, c + \delta)$. There is $x \in S$ with $x > c - \delta$; adding U to the finite family covering $[a, x]$ covers $[a, \min(c + \delta/2, b)]$. Hence $c \in S$, and if $c < b$ we would get a point of S larger than c. So $c = b \in S$. ■

Definition 4.2 (bounded)

$X \subset \mathbb{R}^n$ is **bounded** if there is M with $|\mathbf{x}| \leq M$ for all $\mathbf{x} \in X$.

Theorem 4.3

If X is a compact subspace of $\mathbb{R}^n$, then X is closed and bounded.

Proof.

Step 1. Bounded: the open cubes $U_N = C(\mathbf{0}; N)$, $N = 1, 2, \ldots$, are open, increasing, and cover $\mathbb{R}^n$, hence X. Finitely many, $U_{N_1}, \ldots, U_{N_k}$, cover X; with $M = \max N_i$, $X \subset U_M$.

Step 2. Closed: let $\mathbf{a} \notin X$; we find a neighborhood of $\mathbf{a}$ disjoint from X. Let $C_N = \{\mathbf{x} : |\mathbf{x} - \mathbf{a}| \leq 1/N\}$ and $V_N = \mathbb{R}^n - C_N$. The V_N are open, increasing, and cover $\mathbb{R}^n - \{\mathbf{a}\} \supset X$. Finitely many cover X, so $X \subset V_M$ for some M; then C_M is disjoint from X, and in particular the open cube $C(\mathbf{a}; 1/M)$ lies in $\mathbb{R}^n - X$. ■

Corollary 4.4

Let X be a compact subspace of $\mathbb{R}$. Then X has a largest element and a smallest element.

Proof. X is bounded, so it has a supremum and an infimum; X is closed, so these (being limit points of X or points of X) belong to X. ■

Theorem 4.5 (Extreme-value theorem)

Let X be a compact subspace of $\mathbb{R}^m$. If $f : X \to \mathbb{R}^n$ is continuous, then $f(X)$ is a compact subspace of $\mathbb{R}^n$. In particular, if $\phi : X \to \mathbb{R}$ is continuous, then ϕ has a maximum value and a minimum value.

Proof. Let $\{V_\alpha\}$ be an open covering of $f(X)$. The sets $f^{-1}(V_\alpha)$ are open in X and cover X, so by Theorem 4.1 finitely many, for $\alpha = \alpha_1, \ldots, \alpha_k$, cover X; then $V_{\alpha_1}, \ldots, V_{\alpha_k}$ cover $f(X)$. If $\phi : X \to \mathbb{R}$ is continuous, $\phi(X)$ is compact, and by Corollary 4.4 it has a largest and a smallest element. ■

Definition 4.3 (ϵ-neighborhood of a set)

For $X \subset \mathbb{R}^n$ and $\epsilon > 0$, the union of the balls $B(\mathbf{a}; \epsilon)$, $\mathbf{a} \in X$, is the **ϵ-neighborhood** of X in the euclidean metric; the union of the cubes $C(\mathbf{a}; \epsilon)$ is the ϵ-neighborhood in the sup metric.

Theorem 4.6 (The ϵ-neighborhood theorem)

Let X be a compact subspace of $\mathbb{R}^n$; let U be an open set of $\mathbb{R}^n$ containing X. Then there is an $\epsilon > 0$ such that the ϵ-neighborhood of X (in either metric) is contained in U.

Proof. The euclidean ϵ-neighborhood lies inside the sup-metric one, so it suffices to treat the sup metric.

Step 1. For $C \subset \mathbb{R}^n$ define $d(\mathbf{x}, C) = \inf\{|\mathbf{x} - \mathbf{c}| : \mathbf{c} \in C\}$, the **distance from $\mathbf{x}$ to C**. It is continuous: for $\mathbf{c} \in C$, $d(\mathbf{x}, C) - |\mathbf{x} - \mathbf{y}| \le |\mathbf{x} - \mathbf{c}| - |\mathbf{x} - \mathbf{y}| \le |\mathbf{y} - \mathbf{c}|$; taking the infimum over $\mathbf{c}$, $d(\mathbf{x}, C) - d(\mathbf{y}, C) \le |\mathbf{x} - \mathbf{y}|$, and symmetrically.

Step 2. Let $f(\mathbf{x}) = d(\mathbf{x}, \mathbb{R}^n - U)$ for $\mathbf{x} \in X$. Then f is continuous and positive on X: if $\mathbf{x} \in X$, some $C(\mathbf{x}; \delta) \subset U$, so $f(\mathbf{x}) \ge \delta$. Since X is compact, f has a minimum value $\epsilon > 0$ (Theorem 4.5). Then for $\mathbf{a} \in X$, every point $\mathbf{y}$ with $|\mathbf{y} - \mathbf{a}| < \epsilon$ lies in U; i.e. the ϵ-neighborhood of X lies in U. ■

Some hypothesis on X is needed: if X is the x-axis in $\mathbb{R}^2$ and $U = \{(x, y) : y^2 < 1/(1 + x^2)\}$, no ϵ-neighborhood of X lies in U.

Theorem 4.7 (Uniform continuity)

Let X be a compact subspace of $\mathbb{R}^m$; let $f : X \to \mathbb{R}^n$ be continuous. Given $\epsilon > 0$, there is a $\delta > 0$ such that whenever $\mathbf{x}, \mathbf{y} \in X$,

$$|\mathbf{x} - \mathbf{y}| < \delta \quad \text{implies} \quad |f(\mathbf{x}) - f(\mathbf{y})| < \epsilon.$$

This result also holds if one uses the euclidean metric instead of the sup metric.

The conclusion is the condition of **uniform continuity**.

Proof. Consider $X \times X \subset \mathbb{R}^m \times \mathbb{R}^m$ and the **diagonal** $\Delta = \{(\mathbf{x}, \mathbf{x}) : \mathbf{x} \in X\}$, which is compact as the image of X under the continuous map $\mathbf{x} \mapsto (\mathbf{x}, \mathbf{x})$. First use the euclidean metric. The function $g : X \times X \to \mathbb{R}$, $g(\mathbf{x}, \mathbf{y}) = \|f(\mathbf{x}) - f(\mathbf{y})\|$, is continuous, so $\{(\mathbf{x}, \mathbf{y}) : g(\mathbf{x}, \mathbf{y}) < \epsilon\}$ is open in $X \times X$ and contains Δ; hence it equals $U \cap (X \times X)$ for some open $U \subset \mathbb{R}^{2m}$ containing Δ (Theorem 3.2). By Theorem 4.6, some δ-neighborhood of Δ lies in U. If $\mathbf{x}, \mathbf{y} \in X$ with $\|\mathbf{x} - \mathbf{y}\| < \delta$, then $\|(\mathbf{x}, \mathbf{y}) - (\mathbf{y}, \mathbf{y})\| = \|(\mathbf{x} - \mathbf{y}, \mathbf{0})\| = \|\mathbf{x} - \mathbf{y}\| < \delta$, so $(\mathbf{x}, \mathbf{y})$ lies in the δ-neighborhood of Δ, hence in U, and $g(\mathbf{x}, \mathbf{y}) < \epsilon$.

For the sup metric: if $|\mathbf{x} - \mathbf{y}| < \delta/\sqrt{m}$ then $\|\mathbf{x} - \mathbf{y}\| < \delta$, whence $|f(\mathbf{x}) - f(\mathbf{y})| \le \|f(\mathbf{x}) - f(\mathbf{y})\| < \epsilon$. ■

Lemma 4.8

The rectangle $Q = [a_1, b_1] \times \cdots \times [a_n, b_n]$ in $\mathbb{R}^n$ is compact.

Proof. Induction on n; the case $n = 1$ is Theorem 4.2. Write $Q = X \times [a_n, b_n]$ with X a rectangle in $\mathbb{R}^{n-1}$, compact by the induction hypothesis. Let $\mathcal{A}$ be an open covering of Q.

Step 1. Given $t \in [a_n, b_n]$, there is $\epsilon > 0$ such that $X \times (t - \epsilon, t + \epsilon)$ is covered by finitely many elements of $\mathcal{A}$. Indeed, $X \times t$ is compact (the image of X under $\mathbf{x} \mapsto (\mathbf{x}, t)$), so it is covered by finitely many $A_1, \ldots, A_k \in \mathcal{A}$. Their union U is open and contains $X \times t$; by Theorem 4.6 some ϵ-neighborhood of $X \times t$ lies in U, and in particular $X \times (t - \epsilon, t + \epsilon) \subset U$.

Step 2. By Step 1 choose for each t an open interval $V_t \ni t$ such that $X \times V_t$ is covered by finitely many elements of $\mathcal{A}$. The V_t cover $[a_n, b_n]$, so finitely many, $V_{t_1}, \ldots, V_{t_m}$, do. Then Q is contained in $\bigcup_i X \times V_{t_i}$, each covered by finitely many elements of $\mathcal{A}$; so Q is covered by finitely many. ■

Theorem 4.9

If X is a closed and bounded subspace of $\mathbb{R}^n$, then X is compact.

Proof. Let $\mathcal{A}$ be an open covering of X. Adjoin the open set $\mathbb{R}^n - X$ to get an open covering of $\mathbb{R}^n$. Since X is bounded, it lies in a rectangle Q, which is compact (Lemma 4.8); finitely many members of the enlarged collection cover Q. Discarding $\mathbb{R}^n - X$ if it occurs leaves finitely many members of $\mathcal{A}$; they may not cover Q, but they cover X, since $\mathbb{R}^n - X$ contains no point of X. ■

All theorems of this section hold for arbitrary metric spaces in place of $\mathbb{R}^n, \mathbb{R}^m$, *except* Theorem 4.9.

4.2 Connected spaces

Definition 4.4 (connected)

A metric space X is **connected** if X cannot be written as the union of two disjoint non-empty sets A and B, each of which is open in X.

Theorem 4.10

The closed interval $[a, b]$ of $\mathbb{R}$ is connected.

Proof [Supplement]. Suppose $[a, b] = A \cup B$ with A, B disjoint, non-empty and open in $[a, b]$ (hence also closed in $[a, b]$); say $a \in A$. Let $c = \sup\{x \in [a, b] : [a, x] \subset A\}$. Since A is closed, $c \in A$. If $c < b$, openness of A gives $[c, c + \delta) \subset A$ for some $\delta > 0$, so $[a, c + \delta/2] \subset A$, contradicting the definition of c. So $c = b$ and $[a, b] \subset A$, so $B = \emptyset$, a contradiction. ■

Theorem 4.11 (Intermediate-value theorem)

Let X be connected. If $f : X \to Y$ is continuous, then $f(X)$ is a connected subspace of Y. In particular, if $\phi : X \to \mathbb{R}$ is continuous and $f(x_0) < r < f(x_1)$ for some points x_0, x_1 of X, then $f(x) = r$ for some point x of X.

Proof. If $f(X) = A \cup B$ with A, B disjoint and open in $f(X)$, then $f^{-1}(A)$ and $f^{-1}(B)$ are disjoint, open in X, with union X; connectedness of X forces one of them, hence one of A, B, to be empty. For ϕ: let $A = \{y \in \mathbb{R} : y < r\}$, $B = \{y : y > r\}$, open in $\mathbb{R}$. If $r \notin \phi(X)$, then $\phi(X)$ is the union of the disjoint sets $\phi(X) \cap A$, $\phi(X) \cap B$, open in $\phi(X)$ and both non-empty, contradicting connectedness of $\phi(X)$. ■

Definition 4.5 (line segment, convex)

The **line segment** joining $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$ is the set of points $\mathbf{x} = \mathbf{a} + t(\mathbf{b} - \mathbf{a})$, $0 \leq t \leq 1$. A set $A \subset \mathbb{R}^n$ is **convex** if for every pair $\mathbf{a}, \mathbf{b} \in A$ the segment joining them lies in A.

A line segment is connected, being the image of $[0, 1]$ under $t \mapsto \mathbf{a} + t(\mathbf{b} - \mathbf{a})$. Every convex set A is connected: if $A = U \cup V$ with U, V disjoint, non-empty and open in A, pick $\mathbf{a} \in U$, $\mathbf{b} \in V$ and let L be the segment joining them; then $U \cap L$ and $V \cap L$ are disjoint, non-empty, open in L, with union L, a contradiction. Hence open balls, open cubes and rectangles in $\mathbb{R}^n$ are connected.

Chapter 2. Differentiation

Overview

We define the derivative of a function $f : \mathbb{R}^m \to \mathbb{R}^n$ as a matrix that gives the best linear approximation to the increment $f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a})$. The two major results are the *inverse function theorem*, which gives conditions under which a differentiable map $\mathbb{R}^n \to \mathbb{R}^n$ has a differentiable inverse, and the *implicit function theorem*, the theoretical basis of implicit differentiation. Elements of $\mathbb{R}^m$ and $\mathbb{R}^n$ are column matrices; $|\cdot|$ is the sup norm (Chapter 1).

§5. The Derivative

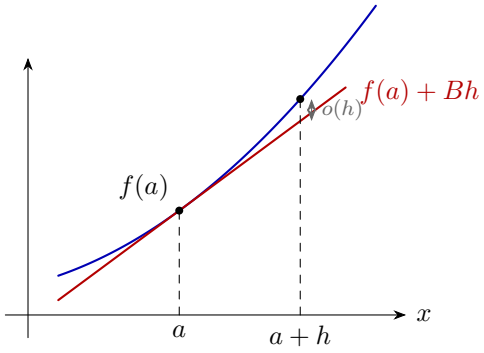


Figure. *The derivative $Df(\mathbf{a}) = B$ is the linear map for which $f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a}) - B\mathbf{h}$ is small compared with $|\mathbf{h}|$: the graph of f is approximated near $\mathbf{a}$ by the affine map $\mathbf{h} \mapsto f(\mathbf{a}) + B\mathbf{h}$.*

For $\phi : A \to \mathbb{R}$ with $A \subset \mathbb{R}$ containing a neighborhood of a, the derivative is $\phi'(a) = \lim_{t\to 0}(\phi(a+t) - \phi(a))/t$. From it one gets (1) differentiable functions are continuous, (2) composites of differentiable functions are differentiable. For $f : \mathbb{R}^m \to \mathbb{R}^n$ we cannot divide by a vector, so we must reformulate.

Definition 5.1 (directional derivative)

Let $A \subset \mathbb{R}^m$, $f : A \to \mathbb{R}^n$, and suppose A contains a neighborhood of $\mathbf{a}$. Given $\mathbf{u} \in \mathbb{R}^m$, $\mathbf{u} \neq \mathbf{0}$, define

$$f'(\mathbf{a};\mathbf{u}) = \lim_{t\to 0} \frac{f(\mathbf{a}+t\mathbf{u}) - f(\mathbf{a})}{t},$$

provided the limit exists. It is the **directional derivative** of f at $\mathbf{a}$ with respect to $\mathbf{u}$ ($\mathbf{u}$ need not be a unit vector).

Example 5.1

Let $f : \mathbb{R}^2 \to \mathbb{R}$, $f(x_1, x_2) = x_1 x_2$, and $\mathbf{a} = (a_1, a_2)$. For $\mathbf{u} = (1, 0)$, $f'(\mathbf{a};\mathbf{u}) = \lim_{t\to 0} \frac{(a_1+t)a_2 - a_1 a_2}{t} = a_2$; for $\mathbf{v} = (1, 2)$, $f'(\mathbf{a};\mathbf{v}) = \lim_{t\to 0} \frac{(a_1+t)(a_2+2t) - a_1 a_2}{t} = a_2 + 2a_1$.

The existence of all directional derivatives is *not* a good definition of differentiability: it does not imply continuity (Example 5.3), nor that composites are differentiable.

Reformulate the one-variable case: ϕ is differentiable at a iff there is a number λ with

$$\frac{\phi(a+t) - \phi(a) - \lambda t}{t} \to 0 \quad \text{as } t \to 0;$$

then $\lambda = \phi'(a)$, and the linear function λt is the "first-order" (linear) approximation to the increment $\phi(a+t) - \phi(a)$. This generalizes.

Definition 5.2 (derivative)

Let $A \subset \mathbb{R}^m$, $f : A \to \mathbb{R}^n$, and suppose A contains a neighborhood of $\mathbf{a}$. f is **differentiable** at $\mathbf{a}$ if there is an n by m matrix B such that

$$\frac{f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a}) - B \cdot \mathbf{h}}{|\mathbf{h}|} \to \mathbf{0} \quad \text{as } \mathbf{h} \to \mathbf{0}.$$

The matrix B is unique; it is the **derivative** of f at $\mathbf{a}$, denoted $Df(\mathbf{a})$.

The quotient is defined for $\mathbf{h}$ in a deleted neighborhood of $\mathbf{0}$. Using $\|\mathbf{h}\|$ instead of $|\mathbf{h}|$ gives an equivalent definition. *Uniqueness*: if C also satisfies the condition, subtracting gives $(C-B)\cdot\mathbf{h}/|\mathbf{h}| \to \mathbf{0}$. Fix $\mathbf{u}$, put $\mathbf{h} = t\mathbf{u}$ and let $t \to 0$: $(C-B)\cdot\mathbf{u}/|\mathbf{u}| = \pm(C-B)\cdot t\mathbf{u}/|t\mathbf{u}| \to \mathbf{0}$, so $(C-B)\cdot\mathbf{u} = \mathbf{0}$ for all $\mathbf{u}$ and $C = B$.

Example 5.2

If $f(\mathbf{x}) = B\cdot\mathbf{x} + \mathbf{b}$ with B an n by m matrix and $\mathbf{b} \in \mathbb{R}^n$, then $f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a}) = B\cdot\mathbf{h}$, so the quotient vanishes identically: f is differentiable and $Df(\mathbf{x}) = B$.

We verify that this definition is suitable:

(1) differentiable functions are continuous;

(2) composites of differentiable functions are differentiable (§7);

(3) differentiability at $\mathbf{a}$ implies the existence of all directional derivatives at $\mathbf{a}$.

Theorem 5.1

Let $A \subset \mathbb{R}^m$; let $f : A \to \mathbb{R}^n$. If f is differentiable at $\mathbf{a}$, then all the directional derivatives of f at $\mathbf{a}$ exist, and

$$f'(\mathbf{a};\mathbf{u}) = Df(\mathbf{a})\cdot\mathbf{u}.$$

Proof. Let $B = Df(\mathbf{a})$ and set $\mathbf{h} = t\mathbf{u}$ $(t \neq 0)$ in the definition:

$$\frac{f(\mathbf{a}+t\mathbf{u}) - f(\mathbf{a}) - B\cdot t\mathbf{u}}{|t\mathbf{u}|} \to \mathbf{0} \quad \text{as } t \to 0. \tag{$*$}$$

If $t \to 0$ through positive values, multiply $(*)$ by $|\mathbf{u}|$ to get $\frac{f(\mathbf{a}+t\mathbf{u})-f(\mathbf{a})}{t} - B\cdot\mathbf{u} \to \mathbf{0}$. If $t \to 0$ through negative values, multiply by $-|\mathbf{u}|$ for the same conclusion. Thus $f'(\mathbf{a};\mathbf{u}) = B\cdot\mathbf{u}$. ■

Example 5.3

Define $f : \mathbb{R}^2 \to \mathbb{R}$ by $f(\mathbf{0}) = 0$ and $f(x,y) = x^2y/(x^4+y^2)$ for $(x,y) \neq \mathbf{0}$. All directional derivatives exist at $\mathbf{0}$, but f is not differentiable there. For $\mathbf{u} = (h,k) \neq \mathbf{0}$,

$$\frac{f(\mathbf{0}+t\mathbf{u}) - f(\mathbf{0})}{t} = \frac{(th)^2(tk)}{(th)^4+(tk)^2}\cdot\frac{1}{t} = \frac{h^2k}{t^2h^4+k^2}, \qquad \text{so} \qquad f'(\mathbf{0};\mathbf{u}) = \begin{cases} h^2/k & k \neq 0, \\ 0 & k = 0. \end{cases}$$

If f were differentiable at $\mathbf{0}$ with $Df(\mathbf{0}) = [a\ b]$, then $f'(\mathbf{0};\mathbf{u}) = ah + bk$ would be linear in $\mathbf{u}$, which it is not. In fact f is not even continuous at $\mathbf{0}$: $f(\mathbf{0}) = 0$, while $f(t,t^2) = 1/2$ for all $t \neq 0$. (On each straight line through the origin f is differentiable: on $y = mx$, $f = mx/(m^2+x^2)$.)

Theorem 5.2

Let $A \subset \mathbb{R}^m$; let $f : A \to \mathbb{R}^n$. If f is differentiable at $\mathbf{a}$, then f is continuous at $\mathbf{a}$.

Proof. Let $B = Df(\mathbf{a})$. For $\mathbf{h}$ near $\mathbf{0}$, $\mathbf{h} \neq \mathbf{0}$,

$$f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a}) = |\mathbf{h}|\Big[\frac{f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a}) - B\cdot\mathbf{h}}{|\mathbf{h}|}\Big] + B\cdot\mathbf{h}.$$

As $\mathbf{h} \to \mathbf{0}$ the bracket tends to $\mathbf{0}$, $|\mathbf{h}| \to 0$ and $B\cdot\mathbf{h} \to \mathbf{0}$; hence $f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a}) \to \mathbf{0}$. ■

Definition 5.3 (partial derivative)

Let $A \subset \mathbb{R}^m$, $f : A \to \mathbb{R}$. The **jth partial derivative** of f at $\mathbf{a}$ is the directional derivative with respect to $\mathbf{e}_j$, if it exists:

$$D_j f(\mathbf{a}) = \lim_{t\to 0} \frac{f(\mathbf{a}+t\mathbf{e}_j) - f(\mathbf{a})}{t}.$$

If $\phi(t) = f(a_1, \ldots, a_{j-1}, t, a_{j+1}, \ldots, a_m)$, then $D_j f(\mathbf{a}) = \phi'(a_j)$: one treats $x_1, \ldots, x_{j-1}, x_{j+1}, \ldots, x_m$ as constants and differentiates in x_j by the familiar single-variable rules.

Theorem 5.3

Let $A \subset \mathbb{R}^m$; let $f : A \to \mathbb{R}$. If f is differentiable at $\mathbf{a}$, then

$$Df(\mathbf{a}) = [D_1 f(\mathbf{a}) \;\; D_2 f(\mathbf{a}) \;\; \cdots \;\; D_m f(\mathbf{a})].$$

Proof. $Df(\mathbf{a})$ is a 1 by m matrix $[\lambda_1 \;\cdots\; \lambda_m]$. By Theorem 5.1, $D_j f(\mathbf{a}) = f'(\mathbf{a};\mathbf{e}_j) = Df(\mathbf{a})\cdot\mathbf{e}_j = \lambda_j$. ■

Theorem 5.4

Let $A \subset \mathbb{R}^m$; let $f : A \to \mathbb{R}^n$. Suppose A contains a neighborhood of $\mathbf{a}$. Let $f_i : A \to \mathbb{R}$ be the ith component function of f, so that $f(\mathbf{x}) = (f_1(\mathbf{x}), \ldots, f_n(\mathbf{x}))$ (a column matrix).

(a) f is differentiable at $\mathbf{a}$ if and only if each component function f_i is differentiable at $\mathbf{a}$.

(b) If f is differentiable at $\mathbf{a}$, then its derivative is the n by m matrix whose ith row is the derivative of f_i.

Thus $Df(\mathbf{a})$ is the matrix whose entry in row i and column j is $D_j f_i(\mathbf{a})$.

Proof. For an arbitrary n by m matrix B consider $F(\mathbf{h}) = (f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a}) - B\cdot\mathbf{h})/|\mathbf{h}|$, defined for $0 < |\mathbf{h}| < \epsilon$. Its ith entry is

$$F_i(\mathbf{h}) = \frac{f_i(\mathbf{a}+\mathbf{h}) - f_i(\mathbf{a}) - (\text{row } i \text{ of } B)\cdot\mathbf{h}}{|\mathbf{h}|}.$$

$F(\mathbf{h}) \to \mathbf{0}$ iff each $F_i(\mathbf{h}) \to 0$. Hence B satisfies the definition for f iff for each i the ith row of B satisfies it for f_i. ■

Definition 5.4 (Jacobian matrix)

If the partial derivatives of the components f_i of f exist at $\mathbf{a}$, the matrix with entry $D_j f_i(\mathbf{a})$ in row i, column j is the **Jacobian matrix** of f at $\mathbf{a}$.

If f is differentiable at $\mathbf{a}$, the Jacobian matrix equals $Df(\mathbf{a})$; but the Jacobian can exist without f being differentiable (Example 5.3). We do not yet have a convenient test for differentiability; e.g. $\sin(xy)$ and xy^2+ze^{xy} obviously have partial derivatives, but are they differentiable? This is settled in §6.

Remark

If $m=1$ or $n=1$, the definition is a matrix reformulation of familiar notions. For $f:\mathbb{R}^1\to\mathbb{R}^3$, $Df(t)$ is the column matrix $(f_1'(t),f_2'(t),f_3'(t))$, the *velocity vector* of the *parametrized-curve* f. For $g:\mathbb{R}^3\to\mathbb{R}^1$, $Dg(\mathbf{x})=[D_1g\ D_2g\ D_3g]$; the vector field $\operatorname{grad} g=(D_1g)\mathbf{e}_1+(D_2g)\mathbf{e}_2+(D_3g)\mathbf{e}_3$ (also written ∇g) is the *gradient*, and the directional derivative $Dg(\mathbf{x})\cdot\mathbf{u}$ is the dot product of $\operatorname{grad} g$ and $\mathbf{u}$. For general $f:\mathbb{R}^m\to\mathbb{R}^n$ matrix notation is needed.

§6. Continuously Differentiable Functions

Motivation. Partial derivatives can exist without f being differentiable. If they are continuous, however, f is differentiable and its derivative is the matrix of partials, so in practice one checks the class C^1 instead of differentiability itself.

Mere existence of partial derivatives does not imply differentiability; their *continuity* does.

Theorem 6.1 (Mean-value theorem)

If $\phi:[a,b]\to\mathbb{R}$ is continuous at each point of $[a,b]$ and differentiable at each point of (a,b), then there exists a point c of (a,b) such that

$$\phi(b)-\phi(a)=\phi'(c)(b-a).$$

Proof [Supplement]. Let $\psi(t)=\phi(t)-\phi(a)-\frac{\phi(b)-\phi(a)}{b-a}(t-a)$; then $\psi(a)=\psi(b)=0$. By the extreme-value theorem ψ attains a maximum and a minimum on $[a,b]$; if both occur at endpoints, $\psi\equiv 0$ and any c works. Otherwise one occurs at some $c\in(a,b)$, where $\psi'(c)=0$ (the difference quotients have opposite signs on the two sides of a local extremum). So $\phi'(c)=(\phi(b)-\phi(a))/(b-a)$. ■

We usually apply it when ϕ is differentiable on an open interval containing $[a,b]$.

Theorem 6.2

Let A be open in $\mathbb{R}^m$. Suppose that the partial derivatives $D_jf_i(\mathbf{x})$ of the component functions of f exist at each point $\mathbf{x}$ of A and are continuous on A. Then f is differentiable at each point of A.

Definition 6.1 (class C^1)

A function satisfying the hypotheses of Theorem 6.2 is **continuously differentiable**, or **of class C^1**, on A.

Proof. By Theorem 5.4 it suffices to treat a real-valued $f:A\to\mathbb{R}$. Let $\mathbf{a}\in A$; the partials $D_jf(\mathbf{x})$ exist and are continuous for $|\mathbf{x}-\mathbf{a}|<\epsilon$.

Step 1. Let $0<|\mathbf{h}|<\epsilon$ with components $h_1,\dots,h_m$, and set

$$\mathbf{p}_0=\mathbf{a},\quad \mathbf{p}_1=\mathbf{a}+h_1\mathbf{e}_1,\quad \mathbf{p}_2=\mathbf{a}+h_1\mathbf{e}_1+h_2\mathbf{e}_2,\quad \dots,\quad \mathbf{p}_m=\mathbf{a}+\mathbf{h}.$$

All $\mathbf{p}_j$ lie in the closed cube C of radius $|\mathbf{h}|$ centered at $\mathbf{a}$. Write

$$f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a}) = \sum_{j=1}^{m} [f(\mathbf{p}_j) - f(\mathbf{p}_{j-1})]. \tag{$*$}$$

Fix j and let $\phi(t) = f(\mathbf{p}_{j-1} + t\mathbf{e}_j)$. Suppose $h_j \neq 0$. As t ranges over the closed interval I with endpoints 0 and h_j, the point $\mathbf{p}_{j-1} + t\mathbf{e}_j$ ranges over the segment from $\mathbf{p}_{j-1}$ to $\mathbf{p}_j$, which lies in $C \subset A$. Only the jth coordinate varies, so since $D_j f$ exists on A, ϕ is differentiable on an open interval containing I. The mean-value theorem gives $\phi(h_j) - \phi(0) = \phi'(c_j)h_j$ for some c_j between 0 and h_j, i.e.

$$f(\mathbf{p}_j) - f(\mathbf{p}_{j-1}) = D_j f(\mathbf{q}_j) h_j, \tag{$**$}$$

where $\mathbf{q}_j = \mathbf{p}_{j-1} + c_j \mathbf{e}_j \in C$. If $h_j = 0$, $(**)$ holds for any $\mathbf{q}_j \in C$. Hence

$$f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a}) = \sum_{j=1}^{m} D_j f(\mathbf{q}_j) h_j, \qquad \mathbf{q}_j \in C. \tag{$***$}$$

Step 2. Let $B = [D_1 f(\mathbf{a}) \ \cdots \ D_m f(\mathbf{a})]$, so $B \cdot \mathbf{h} = \sum_j D_j f(\mathbf{a}) h_j$. By $(***)$,

$$\frac{f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a}) - B\cdot\mathbf{h}}{|\mathbf{h}|} = \sum_{j=1}^{m} [D_j f(\mathbf{q}_j) - D_j f(\mathbf{a})] \frac{h_j}{|\mathbf{h}|}.$$

As $\mathbf{h} \to \mathbf{0}$, $\mathbf{q}_j \to \mathbf{a}$ (it lies in the cube of radius $|\mathbf{h}|$), so by continuity of the partials each bracket tends to 0, while $|h_j|/|\mathbf{h}| \leq 1$. Hence the expression tends to 0. ■

So the familiar functions of calculus, such as $\sin(xy)$ and $xy^2 + ze^{xy}$, are differentiable: their partials exist and are continuous. In practice we deal only with C^1 functions; differentiable functions that are not C^1 occur rarely.

Definition 6.2 (class C^r, C^∞)

Let $f : A \to \mathbb{R}^n$, A open in $\mathbb{R}^m$. If the partials $D_j f_i$ exist on A, their partials $D_k(D_j f_i)$ are the **second-order partial derivatives**; similarly for order r. f is **of class** C^r on A if all partial derivatives of the f_i of order $\leq r$ exist and are continuous on A; equivalently, each $D_j f_i$ is of class C^{r-1}. f is **of class** C^∞ if the partials of all orders are continuous.

Theorem 6.3

Let A be open in $\mathbb{R}^m$; let $f : A \to \mathbb{R}$ be a function of class C^2. Then for each $\mathbf{a} \in A$,

$$D_k D_j f(\mathbf{a}) = D_j D_k f(\mathbf{a}).$$

Proof. Only x_k and x_j vary, so we may assume A open in $\mathbb{R}^2$ and $f : A \to \mathbb{R}$ of class C^2, and prove $D_2 D_1 f = D_1 D_2 f$.

Step 1. A "second-order" mean-value theorem. Let $Q = [a, a+h] \times [b, b+k]$ be a rectangle in A and

$$\lambda(h,k) = f(a,b) - f(a+h,b) - f(a,b+k) + f(a+h,b+k),$$

the sum, with signs, of the values of f at the four vertices. We show there are $\mathbf{p}, \mathbf{q} \in Q$ with

$$\lambda(h,k) = D_2 D_1 f(\mathbf{p}) \cdot hk \qquad \text{and} \qquad \lambda(h,k) = D_1 D_2 f(\mathbf{q}) \cdot hk.$$

By symmetry we prove the first. Let $\phi(s) = f(s, b+k) - f(s,b)$; then $\phi(a+h) - \phi(a) = \lambda(h,k)$. Since $D_1 f$ exists, ϕ is differentiable on an open interval containing $[a, a+h]$, and the mean-value theorem gives s_0 between a and $a+h$ with

$$\lambda(h,k) = [D_1 f(s_0, b+k) - D_1 f(s_0, b)] \cdot h.$$

Now $t \mapsto D_1 f(s_0, t)$ is differentiable on an open interval about $[b, b+k]$ since $D_2 D_1 f$ exists, so by the mean-value theorem again $D_1 f(s_0, b+k) - D_1 f(s_0, b) = D_2 D_1 f(s_0, t_0) \cdot k$ for some t_0 between b and $b+k$. Take $\mathbf{p} = (s_0, t_0)$.

Step 2. Given $\mathbf{a} = (a, b) \in A$ and $t > 0$ small, $Q_t = [a, a+t] \times [b, b+t] \subset A$, and Step 1 gives $\lambda(t,t) = D_2 D_1 f(\mathbf{p}_t)\, t^2$ with $\mathbf{p}_t \in Q_t$. As $t \to 0$, $\mathbf{p}_t \to \mathbf{a}$, and *because $D_2 D_1 f$ is continuous*, $\lambda(t,t)/t^2 \to D_2 D_1 f(\mathbf{a})$. Similarly, using the other equation of Step 1, $\lambda(t,t)/t^2 \to D_1 D_2 f(\mathbf{a})$. The theorem follows. ■

§7. The Chain Rule

Motivation. Almost every computation in the book composes maps: coordinates with functions, parametrizations with forms. The chain rule says the derivative of a composite is the product of the derivative matrices, which is what makes the linear-algebra viewpoint effective.

Theorem 7.1

Let $A \subset \mathbb{R}^m$; let $B \subset \mathbb{R}^n$. Let $f : A \to \mathbb{R}^n$ and $g : B \to \mathbb{R}^p$, with $f(A) \subset B$. Suppose $f(\mathbf{a}) = \mathbf{b}$. If f is differentiable at $\mathbf{a}$, and if g is differentiable at $\mathbf{b}$, then the composite function $g \circ f$ is differentiable at $\mathbf{a}$. Furthermore,

$$D(g \circ f)(\mathbf{a}) = Dg(\mathbf{b}) \cdot Df(\mathbf{a}),$$

where the indicated product is matrix multiplication.

This is the familiar chain rule in a new guise; writing it out in partial derivatives gives the calculus formula (see below).

Proof. Let $\mathbf{x}$ denote the general point of $\mathbb{R}^m$ and $\mathbf{y}$ that of $\mathbb{R}^n$. Choose ϵ so that $g(\mathbf{y})$ is defined for $|\mathbf{y} - \mathbf{b}| < \epsilon$, and δ so that $f(\mathbf{x})$ is defined and $|f(\mathbf{x}) - \mathbf{b}| < \epsilon$ for $|\mathbf{x} - \mathbf{a}| < \delta$ (possible since f is continuous at $\mathbf{a}$). Then $g(f(\mathbf{x}))$ is defined for $|\mathbf{x} - \mathbf{a}| < \delta$.

Step 1. Let $\Delta(\mathbf{h}) = f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a})$ for $|\mathbf{h}| < \delta$. We show $|\Delta(\mathbf{h})|/|\mathbf{h}|$ is bounded in a deleted neighborhood of $\mathbf{0}$. Define $F(\mathbf{0}) = \mathbf{0}$ and $F(\mathbf{h}) = [\Delta(\mathbf{h}) - Df(\mathbf{a}) \cdot \mathbf{h}]/|\mathbf{h}|$ for $0 < |\mathbf{h}| < \delta$. Since f is differentiable at $\mathbf{a}$, F is continuous at $\mathbf{0}$, and

$$\Delta(\mathbf{h}) = Df(\mathbf{a}) \cdot \mathbf{h} + |\mathbf{h}| F(\mathbf{h}) \qquad (*)$$

for $0 < |\mathbf{h}| < \delta$, and trivially for $\mathbf{h} = \mathbf{0}$. By Theorem 1.3 and the triangle inequality, $|\Delta(\mathbf{h})| \le m|Df(\mathbf{a})||\mathbf{h}| + |\mathbf{h}||F(\mathbf{h})|$, and $|F(\mathbf{h})|$ is bounded near $\mathbf{0}$ (it tends to 0).

Step 2. Similarly define $G(\mathbf{0}) = \mathbf{0}$ and $G(\mathbf{k}) = [g(\mathbf{b}+\mathbf{k}) - g(\mathbf{b}) - Dg(\mathbf{b}) \cdot \mathbf{k}]/|\mathbf{k}|$ for $0 < |\mathbf{k}| < \epsilon$. G is continuous at $\mathbf{0}$ and

$$g(\mathbf{b}+\mathbf{k}) - g(\mathbf{b}) = Dg(\mathbf{b}) \cdot \mathbf{k} + |\mathbf{k}| G(\mathbf{k}) \qquad (|\mathbf{k}| < \epsilon). \qquad (**)$$

Step 3. Let $|\mathbf{h}| < \delta$. Then $|\Delta(\mathbf{h})| < \epsilon$, so we may substitute $\mathbf{k} = \Delta(\mathbf{h})$ in $(**)$; since $\mathbf{b} + \Delta(\mathbf{h}) = f(\mathbf{a}+\mathbf{h})$,

$$g(f(\mathbf{a}+\mathbf{h})) - g(f(\mathbf{a})) = Dg(\mathbf{b}) \cdot \Delta(\mathbf{h}) + |\Delta(\mathbf{h})| G(\Delta(\mathbf{h})).$$

Using $(*)$, for $0 < |\mathbf{h}| < \delta$,

$$\frac{1}{|\mathbf{h}|}\big[g(f(\mathbf{a}+\mathbf{h})) - g(f(\mathbf{a})) - Dg(\mathbf{b}) \cdot Df(\mathbf{a}) \cdot \mathbf{h}\big] = Dg(\mathbf{b}) \cdot F(\mathbf{h}) + \frac{|\Delta(\mathbf{h})|}{|\mathbf{h}|} G(\Delta(\mathbf{h})).$$

As $\mathbf{h} \to \mathbf{0}$: $F(\mathbf{h}) \to \mathbf{0}$; $G(\Delta(\mathbf{h})) \to \mathbf{0}$, being a composite of functions continuous at $\mathbf{0}$ and vanishing there; and $|\Delta(\mathbf{h})|/|\mathbf{h}|$ is bounded by Step 1. So the right side tends to $\mathbf{0}$. ■

Corollary 7.2

Let A be open in $\mathbb{R}^m$; let B be open in $\mathbb{R}^n$. Let $f : A \to \mathbb{R}^n$ and $g : B \to \mathbb{R}^p$, with $f(A) \subset B$. If f and g are of class C^r, so is the composite function $g \circ f$.

Proof. By the chain rule, $D(g \circ f)(\mathbf{x}) = Dg(f(\mathbf{x})) \cdot Df(\mathbf{x})$ for $\mathbf{x} \in A$. If f, g are C^1, the entries of Dg are continuous on B, so the entries of $Dg(f(\mathbf{x}))$ are continuous on A; the entries of $Df(\mathbf{x})$ are continuous; and the entries of the product are algebraic (polynomial) functions of these, hence continuous. So $g \circ f$ is C^1.

Induction: suppose the result holds for C^{r-1}, and let f, g be C^r. The entries of Dg are C^{r-1} on B, and f is C^{r-1}, so by the induction hypothesis $D_j g_i(f(\mathbf{x}))$ is C^{r-1}. The entries of Df are C^{r-1}. Hence the entries of $Dg(f(\mathbf{x})) \cdot Df(\mathbf{x})$ are C^{r-1} and $g \circ f$ is C^r. If f, g are C^∞ they are C^r for all r, hence so is $g \circ f$. ∎

Theorem 7.3 (Mean-value theorem)

Let A be open in $\mathbb{R}^m$; let $f : A \to \mathbb{R}$ be differentiable on A. If A contains the line segment with end points $\mathbf{a}$ and $\mathbf{a} + \mathbf{h}$, then there is a point $\mathbf{c} = \mathbf{a} + t_0\mathbf{h}$ with $0 < t_0 < 1$ of this line segment such that

$$f(\mathbf{a} + \mathbf{h}) - f(\mathbf{a}) = Df(\mathbf{c}) \cdot \mathbf{h}.$$

Proof. Let $\phi(t) = f(\mathbf{a} + t\mathbf{h})$, defined on an open interval about $[0, 1]$. By the chain rule ϕ is differentiable with $\phi'(t) = Df(\mathbf{a}+t\mathbf{h})\cdot\mathbf{h}$. The ordinary mean-value theorem gives $\phi(1)-\phi(0) = \phi'(t_0)$ for some $0 < t_0 < 1$, i.e. $f(\mathbf{a} + \mathbf{h}) - f(\mathbf{a}) = Df(\mathbf{a} + t_0\mathbf{h}) \cdot \mathbf{h}$. ∎

In one variable, if $\phi'(x) > 0$ on an open interval, ϕ has an inverse ψ with $\psi'(y) = 1/\phi'(x)$, $y = \phi(x)$. Here we only compute the derivative of an inverse assuming it exists and is differentiable.

Theorem 7.4

Let A be open in $\mathbb{R}^n$; let $f : A \to \mathbb{R}^n$; let $f(\mathbf{a}) = \mathbf{b}$. Suppose that g maps a neighborhood of $\mathbf{b}$ into $\mathbb{R}^n$, that $g(\mathbf{b}) = \mathbf{a}$, and

$$g(f(\mathbf{x})) = \mathbf{x}$$

for all $\mathbf{x}$ in a neighborhood of $\mathbf{a}$. If f is differentiable at $\mathbf{a}$ and if g is differentiable at $\mathbf{b}$, then

$$Dg(\mathbf{b}) = [Df(\mathbf{a})]^{-1}.$$

Proof. Let $i : \mathbb{R}^n \to \mathbb{R}^n$ be the identity, with $Di = I_n$. Since $g(f(\mathbf{x})) = i(\mathbf{x})$ near $\mathbf{a}$, the chain rule gives $Dg(\mathbf{b}) \cdot Df(\mathbf{a}) = I_n$. So $Dg(\mathbf{b})$ is a left inverse of the square matrix $Df(\mathbf{a})$, hence its inverse (Theorem 2.5). ∎

Thus for a differentiable f to have a differentiable inverse it is *necessary* that Df be non-singular. Remarkably, for C^1 functions this is also *sufficient*, at least locally (§8).

Remark on notation

In Leibniz notation, with $y = \phi(x)$, $z = \psi(y)$, the chain rule reads $\frac{dz}{dx} = \frac{dz}{dy} \cdot \frac{dy}{dx}$ and the inverse rule $\frac{dx}{dy} = 1/\frac{dy}{dx}$—easy to remember, but ambiguous (the letter z denotes different functions on the two sides). For $f : \mathbb{R}^m \to \mathbb{R}^n$, $g : \mathbb{R}^n \to \mathbb{R}$ and $F = g \circ f$, the formula $DF(\mathbf{x}) = Dg(f(\mathbf{x})) \cdot Df(\mathbf{x})$ written out entrywise is

$$D_j F(\mathbf{x}) = \sum_{k=1}^{n} D_k g(f(\mathbf{x}))\, D_j f_k(\mathbf{x}), \qquad \text{i.e.} \qquad \frac{\partial z}{\partial x_j} = \sum_{k=1}^{n} \frac{\partial z}{\partial y_k} \frac{\partial y_k}{\partial x_j}$$

with $\mathbf{y} = f(\mathbf{x})$, $z = g(\mathbf{y})$. For the inverse g of $f : \mathbb{R}^2 \to \mathbb{R}^2$, $Dg(\mathbf{y}) = [Df(\mathbf{x})]^{-1}$ reads

$$\begin{bmatrix} \partial x_1/\partial y_1 & \partial x_1/\partial y_2 \\ \partial x_2/\partial y_1 & \partial x_2/\partial y_2 \end{bmatrix} = \begin{bmatrix} \partial y_1/\partial x_1 & \partial y_1/\partial x_2 \\ \partial y_2/\partial x_1 & \partial y_2/\partial x_2 \end{bmatrix}^{-1},$$

so by the formula for the inverse of a matrix, $\partial x_i/\partial y_j$ is far from the reciprocal of $\partial y_j/\partial x_i$. Matrix notation is the clean way to state these results.

§8. The Inverse Function Theorem

Motivation. A linear map with nonzero determinant is invertible. The inverse function theorem says the same holds locally for a C^r map whose derivative is invertible at a point, and the inverse is again C^r. It is the basic tool for producing coordinate systems and is used to define manifolds.

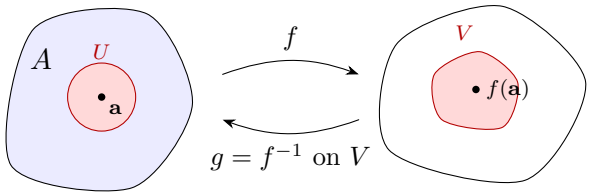


Figure. *The inverse function theorem: if $Df(\mathbf{a})$ is invertible, there is a neighbourhood U of $\mathbf{a}$ that f maps bijectively onto an open set V, and the inverse $g : V \to U$ is again of class C^r. The conclusion is local; f need not be one-to-one on all of A.*

Let A be open in $\mathbb{R}^n$ and $f : A \to \mathbb{R}^n$ of class C^1. We show that non-singularity of Df suffices for f to have a differentiable inverse locally. First, non-singularity implies f is locally one-to-one.

Lemma 8.1

Let A be open in $\mathbb{R}^n$; let $f : A \to \mathbb{R}^n$ be of class C^1. If $Df(\mathbf{a})$ is non-singular, then there exists an $\alpha > 0$ such that the inequality

$$|f(\mathbf{x}_0) - f(\mathbf{x}_1)| \geq \alpha |\mathbf{x}_0 - \mathbf{x}_1|$$

holds for all $\mathbf{x}_0, \mathbf{x}_1$ in some open cube $C(\mathbf{a}; \epsilon)$ centered at $\mathbf{a}$. It follows that f is one-to-one on this open cube.

Proof. Let $E = Df(\mathbf{a})$, non-singular. By Theorem 1.3,

$$|\mathbf{x}_0 - \mathbf{x}_1| = |E^{-1} \cdot (E \cdot \mathbf{x}_0 - E \cdot \mathbf{x}_1)| \leq n|E^{-1}| \cdot |E \cdot \mathbf{x}_0 - E \cdot \mathbf{x}_1|.$$

Setting $2\alpha = 1/(n|E^{-1}|)$, we get $|E \cdot \mathbf{x}_0 - E \cdot \mathbf{x}_1| \geq 2\alpha |\mathbf{x}_0 - \mathbf{x}_1|$ for all $\mathbf{x}_0, \mathbf{x}_1 \in \mathbb{R}^n$.

Let $H(\mathbf{x}) = f(\mathbf{x}) - E \cdot \mathbf{x}$. Then $DH(\mathbf{x}) = Df(\mathbf{x}) - E$, so $DH(\mathbf{a}) = 0$; as H is C^1, choose $\epsilon > 0$ with $|DH(\mathbf{x})| < \alpha/n$ for $\mathbf{x} \in C = C(\mathbf{a}; \epsilon)$. For $\mathbf{x}_0, \mathbf{x}_1 \in C$ (a convex set) the mean-value theorem applied to the ith component of H gives $\mathbf{c} \in C$ with

$$|H_i(\mathbf{x}_0) - H_i(\mathbf{x}_1)| = |DH_i(\mathbf{c}) \cdot (\mathbf{x}_0 - \mathbf{x}_1)| \leq n(\alpha/n)|\mathbf{x}_0 - \mathbf{x}_1|.$$

Hence for $\mathbf{x}_0, \mathbf{x}_1 \in C$,

$$\begin{aligned} \alpha|\mathbf{x}_0 - \mathbf{x}_1| &\geq |H(\mathbf{x}_0) - H(\mathbf{x}_1)| = |f(\mathbf{x}_0) - E \cdot \mathbf{x}_0 - f(\mathbf{x}_1) + E \cdot \mathbf{x}_1| \\ &\geq |E \cdot \mathbf{x}_1 - E \cdot \mathbf{x}_0| - |f(\mathbf{x}_1) - f(\mathbf{x}_0)| \geq 2\alpha|\mathbf{x}_1 - \mathbf{x}_0| - |f(\mathbf{x}_1) - f(\mathbf{x}_0)|, \end{aligned}$$

which is the desired inequality. ∎

Theorem 8.2

Let A be open in $\mathbb{R}^n$; let $f : A \to \mathbb{R}^n$ be of class C^r; let $B = f(A)$. If f is one-to-one on A and if $Df(\mathbf{x})$ is non-singular for $\mathbf{x} \in A$, then the set B is open in $\mathbb{R}^n$ and the inverse function $g : B \to A$ is of class C^r.

Proof.

Step 1. If $\phi : A \to \mathbb{R}$ is differentiable and has a local minimum at $\mathbf{x}_0 \in A$, then $D\phi(\mathbf{x}_0) = 0$. Indeed $\phi(\mathbf{x}_0 + t\mathbf{u}) - \phi(\mathbf{x}_0) \geq 0$ for small t, so the quotient defining $\phi'(\mathbf{x}_0; \mathbf{u})$ is ≥ 0 for $t > 0$ and ≤ 0 for $t < 0$; hence $\phi'(\mathbf{x}_0; \mathbf{u}) = 0$, in particular $D_j\phi(\mathbf{x}_0) = 0$ for all j.

Step 2. B is open. Given $\mathbf{b} \in B$, let $\mathbf{a} = f^{-1}(\mathbf{b})$ and choose a rectangle $Q \subset A$ with $\mathbf{a} \in \operatorname{Int} Q$. $\operatorname{Bd} Q$ is compact, so $f(\operatorname{Bd} Q)$ is compact, hence closed and bounded; since f is one-to-one, $\mathbf{b} \notin f(\operatorname{Bd} Q)$. Choose $\delta > 0$ with $B(\mathbf{b}; 2\delta)$ disjoint from $f(\operatorname{Bd} Q)$. Given $\mathbf{c} \in B(\mathbf{b}; \delta)$ we show $\mathbf{c} = f(\mathbf{x})$ for some $\mathbf{x} \in Q$; then $B(\mathbf{b}; \delta) \subset f(A) = B$.

Let $\phi(\mathbf{x}) = \|f(\mathbf{x}) - \mathbf{c}\|^2$, of class C^r. Since Q is compact, ϕ attains a minimum on Q at some $\mathbf{x}$. Now $\phi(\mathbf{a}) = \|\mathbf{b} - \mathbf{c}\|^2 < \delta^2$, so the minimum is $< \delta^2$. It cannot occur on $\operatorname{Bd} Q$: there $f(\mathbf{x}) \notin B(\mathbf{b}; 2\delta)$, so $\|f(\mathbf{x}) - \mathbf{c}\| \geq \delta$. So $\mathbf{x} \in \operatorname{Int} Q$ and ϕ has a local minimum at $\mathbf{x}$; by Step 1, $D\phi(\mathbf{x}) = 0$. Since $\phi(\mathbf{x}) = \sum_k (f_k(\mathbf{x}) - c_k)^2$, $D_j\phi(\mathbf{x}) = \sum_k 2(f_k(\mathbf{x}) - c_k)D_j f_k(\mathbf{x})$, i.e.

$$2[(f_1(\mathbf{x}) - c_1) \ \cdots \ (f_n(\mathbf{x}) - c_n)] \cdot Df(\mathbf{x}) = 0.$$

$Df(\mathbf{x})$ is non-singular, so multiplying on the right by its inverse gives $f(\mathbf{x}) - \mathbf{c} = \mathbf{0}$.

Step 3. $g : B \to A$ is continuous: for U open in A, $g^{-1}(U) = f(U)$, which is open in $\mathbb{R}^n$ (hence in B) by Step 2 applied to U.

Step 4. g is differentiable at each $\mathbf{b} \in B$. Let $\mathbf{a} = g(\mathbf{b})$, $E = Df(\mathbf{a})$; we show

$$G(\mathbf{k}) = \frac{g(\mathbf{b} + \mathbf{k}) - g(\mathbf{b}) - E^{-1} \cdot \mathbf{k}}{|\mathbf{k}|} \to \mathbf{0} \quad \text{as } \mathbf{k} \to \mathbf{0}.$$

Let $\Delta(\mathbf{k}) = g(\mathbf{b} + \mathbf{k}) - g(\mathbf{b})$. First, $|\Delta(\mathbf{k})|/|\mathbf{k}|$ is bounded for small $\mathbf{k} \neq \mathbf{0}$: by Lemma 8.1 there are a neighborhood C of $\mathbf{a}$ and $\alpha > 0$ with $|f(\mathbf{x}_0) - f(\mathbf{x}_1)| \geq \alpha|\mathbf{x}_0 - \mathbf{x}_1|$ on C; $f(C)$ is a neighborhood of $\mathbf{b}$ (Step 2), so choose ϵ with $\mathbf{b} + \mathbf{k} \in f(C)$ for $|\mathbf{k}| < \epsilon$. Taking $\mathbf{x}_0 = g(\mathbf{b} + \mathbf{k})$, $\mathbf{x}_1 = g(\mathbf{b})$ gives $|\mathbf{k}| \geq \alpha|\Delta(\mathbf{k})|$, i.e. $|\Delta(\mathbf{k})|/|\mathbf{k}| \leq 1/\alpha$. Next,

$$G(\mathbf{k}) = -E^{-1} \cdot \left[\frac{\mathbf{k} - E \cdot \Delta(\mathbf{k})}{|\Delta(\mathbf{k})|}\right] \frac{|\Delta(\mathbf{k})|}{|\mathbf{k}|}$$

(using $\Delta(\mathbf{k}) \neq \mathbf{0}$ for $\mathbf{k} \neq \mathbf{0}$, as g is one-to-one). E^{-1} is constant and $|\Delta(\mathbf{k})|/|\mathbf{k}|$ is bounded. Since $\mathbf{b} + \mathbf{k} = f(g(\mathbf{b} + \mathbf{k})) = f(\mathbf{a} + \Delta(\mathbf{k}))$, the bracket equals

$$\frac{f(\mathbf{a} + \Delta(\mathbf{k})) - f(\mathbf{a}) - E \cdot \Delta(\mathbf{k})}{|\Delta(\mathbf{k})|}.$$

As $\mathbf{k} \to \mathbf{0}$, $\Delta(\mathbf{k}) \to \mathbf{0}$ *because g is continuous*, and since f is differentiable at $\mathbf{a}$ with derivative E, the bracket tends to $\mathbf{0}$.

Step 5. g is of class C^r. By Theorem 7.4, $Dg(\mathbf{y}) = [Df(g(\mathbf{y}))]^{-1}$ for $\mathbf{y} \in B$, so Dg is the composite

$$B \xrightarrow{g} A \xrightarrow{Df} GL(n) \xrightarrow{I} GL(n),$$

where $GL(n)$ is the set of non-singular n by n matrices and $I(C) = C^{-1}$. By Theorem 2.14 the entries of $I(C)$ are rational functions of the entries of C, so I is C^∞. If f is C^1, Df is continuous, g is continuous (even differentiable), so Dg is continuous and g is C^1. If the result holds for C^{r-1} and f is C^r, then g is C^{r-1} by induction, Df is C^{r-1}, so by Corollary 7.2 Dg is C^{r-1} and g is C^r. ■

Remark

Steps 2 and 3 hold without assuming Df non-singular or even f differentiable: if A is open in $\mathbb{R}^n$ and $f : A \to \mathbb{R}^n$ is continuous and one-to-one, then $f(A)$ is open and f^{-1} is continuous (*Brouwer's theorem on invariance of domain*). Its proof requires algebraic topology; we have proved the differentiable version.

Theorem 8.3 (The inverse function theorem)

Let A be open in $\mathbb{R}^n$; let $f : A \to \mathbb{R}^n$ be of class C^r. If $Df(\mathbf{x})$ is non-singular at the point $\mathbf{a}$ of A, there is a neighborhood U of the point $\mathbf{a}$ such that f carries U in a one-to-one fashion onto an open set V of $\mathbb{R}^n$ and the inverse function is of class C^r.

Proof. By Lemma 8.1 there is a neighborhood U_0 of $\mathbf{a}$ on which f is one-to-one. Since $\det Df(\mathbf{x})$ is continuous in $\mathbf{x}$ and $\det Df(\mathbf{a}) \neq 0$, there is a neighborhood U_1 of $\mathbf{a}$ on which $\det Df(\mathbf{x}) \neq 0$. Let $U = U_0 \cap U_1$; the hypotheses of Theorem 8.2 hold for $f : U \to \mathbb{R}^n$. ■

This is the strongest general statement: non-singularity of Df on A makes f locally one-to-one at each point, but not one-to-one on all of A.

Example 8.1

Let $f(r, \theta) = (r\cos\theta, r\sin\theta)$. Then

$$Df(r,\theta) = \begin{bmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{bmatrix}, \qquad \det Df(r,\theta) = r.$$

On the open set $A = (0,1) \times (0,b)$ of the (r,θ)-plane, Df is non-singular at every point, but f is one-to-one on A only if $b \leq 2\pi$.

§9. The Implicit Function Theorem

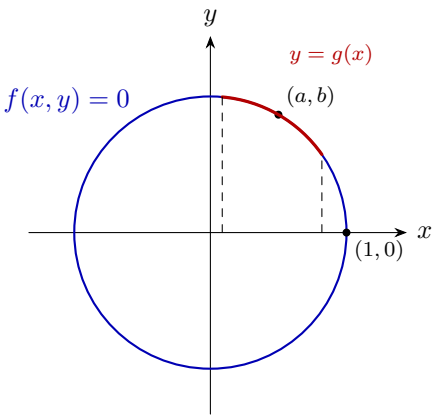


Figure. *Near (a,b) on the circle $x^2+y^2=1$, where $\partial f/\partial y = 2b \neq 0$, the curve is the graph of a C^1 function $y = g(x)$ (red). At $(1,0)$, where $\partial f/\partial y = 0$, no such function exists: every neighbourhood contains two points of the curve over the same x.*

A typical calculus problem: "assume $x^3y + 2e^{xy} = 0$ determines y as a differentiable function of x; find dy/dx." Differentiating, $3x^2y + x^3\,dy/dx + 2e^{xy}(y + x\,dy/dx) = 0$, which one solves for dy/dx. In general, if $f(x,y) = 0$ determines $y = g(x)$ differentiably, then $f(x, g(x)) = 0$ and the chain rule

gives

$$\partial f/\partial x + (\partial f/\partial y)\, g'(x) = 0, \qquad g'(x) = -\frac{\partial f/\partial x}{\partial f/\partial y},$$

evaluated at $(x, g(x))$. This requires $\partial f/\partial y \neq 0$; it turns out this is also *sufficient* for y to be locally a differentiable function of x. More generally, for $f : \mathbb{R}^{k+n} \to \mathbb{R}^n$ of class C^1, the vector equation $f(x_1, \ldots, x_{k+n}) = \mathbf{0}$ is n scalar equations in $k+n$ unknowns; one expects to solve for n unknowns in terms of the other k, with differentiable solutions. There are two problems: computing the derivatives of the implicit functions assuming they exist, and proving existence.

Definition 9.1 (notation for submatrices of Df)

Let A be open in $\mathbb{R}^m$ and $f : A \to \mathbb{R}^n$ differentiable with components $f_1, \ldots, f_n$. We sometimes write

$$Df = \frac{\partial(f_1, \ldots, f_n)}{\partial(x_1, \ldots, x_m)} \qquad \text{or} \qquad Df = \partial f/\partial \mathbf{x}.$$

More generally $\dfrac{\partial(f_{i_1}, \ldots, f_{i_k})}{\partial(x_{j_1}, \ldots, x_{j_l})}$ denotes the k by l matrix of entries of Df in rows $i_1, \ldots, i_k$ and columns $j_1, \ldots, j_l$; its entry in row p, column q is $\partial f_{i_p}/\partial x_{j_q}$.

Theorem 9.1

Let A be open in $\mathbb{R}^{k+n}$; let $f : A \to \mathbb{R}^n$ be differentiable. Write f in the form $f(\mathbf{x}, \mathbf{y})$, for $\mathbf{x} \in \mathbb{R}^k$ and $\mathbf{y} \in \mathbb{R}^n$; then Df has the form

$$Df = [\partial f/\partial \mathbf{x} \quad \partial f/\partial \mathbf{y}].$$

Suppose there is a differentiable function $g : B \to \mathbb{R}^n$ defined on an open set B in $\mathbb{R}^k$, such that $f(\mathbf{x}, g(\mathbf{x})) = \mathbf{0}$ for all $\mathbf{x} \in B$. Then for $\mathbf{x} \in B$,

$$\frac{\partial f}{\partial \mathbf{x}}(\mathbf{x}, g(\mathbf{x})) + \frac{\partial f}{\partial \mathbf{y}}(\mathbf{x}, g(\mathbf{x})) \cdot Dg(\mathbf{x}) = \mathbf{0}.$$

Hence if the n by n matrix $\partial f/\partial \mathbf{y}$ is non-singular at $(\mathbf{x}, g(\mathbf{x}))$,

$$Dg(\mathbf{x}) = -\Big[\frac{\partial f}{\partial \mathbf{y}}(\mathbf{x}, g(\mathbf{x}))\Big]^{-1} \cdot \frac{\partial f}{\partial \mathbf{x}}(\mathbf{x}, g(\mathbf{x})).$$

For $n = k = 1$ this is the formula derived above.

Proof. Let $h : B \to \mathbb{R}^{k+n}$, $h(\mathbf{x}) = (\mathbf{x}, g(\mathbf{x}))$. Then $H(\mathbf{x}) = f(h(\mathbf{x})) = \mathbf{0}$ on B, and the chain rule gives

$$\mathbf{0} = DH(\mathbf{x}) = Df(h(\mathbf{x})) \cdot Dh(\mathbf{x}) = \Big[\frac{\partial f}{\partial \mathbf{x}}(h(\mathbf{x})) \;\; \frac{\partial f}{\partial \mathbf{y}}(h(\mathbf{x}))\Big] \cdot \begin{bmatrix} I_k \\ Dg(\mathbf{x}) \end{bmatrix} = \frac{\partial f}{\partial \mathbf{x}}(h(\mathbf{x})) + \frac{\partial f}{\partial \mathbf{y}}(h(\mathbf{x})) \cdot Dg(\mathbf{x}).$$

■

Theorem 9.2 (Implicit function theorem)

Let A be open in $\mathbb{R}^{k+n}$; let $f : A \to \mathbb{R}^n$ be of class C^r. Write f in the form $f(\mathbf{x}, \mathbf{y})$, for $\mathbf{x} \in \mathbb{R}^k$ and $\mathbf{y} \in \mathbb{R}^n$. Suppose that $(\mathbf{a}, \mathbf{b})$ is a point of A such that $f(\mathbf{a}, \mathbf{b}) = \mathbf{0}$ and

$$\det \frac{\partial f}{\partial \mathbf{y}}(\mathbf{a}, \mathbf{b}) \neq 0.$$

Then there is a neighborhood B of $\mathbf{a}$ in $\mathbb{R}^k$ and a unique continuous function $g : B \to \mathbb{R}^n$

such that $g(\mathbf{a}) = \mathbf{b}$ and $f(\mathbf{x}, g(\mathbf{x})) = \mathbf{0}$ for all $\mathbf{x} \in B$. The function g is in fact of class C^r.

Proof. Define $F : A \to \mathbb{R}^{k+n}$ by $F(\mathbf{x}, \mathbf{y}) = (\mathbf{x}, f(\mathbf{x}, \mathbf{y}))$. Then

$$DF = \begin{bmatrix} I_k & 0 \\ \partial f/\partial \mathbf{x} & \partial f/\partial \mathbf{y} \end{bmatrix},$$

and by repeated application of Lemma 2.12, $\det DF = \det \partial f/\partial \mathbf{y}$, which is nonzero at $(\mathbf{a}, \mathbf{b})$. Also $F(\mathbf{a}, \mathbf{b}) = (\mathbf{a}, \mathbf{0})$. By the inverse function theorem there is an open set $U \times V$ about $(\mathbf{a}, \mathbf{b})$ (U open in $\mathbb{R}^k$, V open in $\mathbb{R}^n$) such that (1) F maps $U \times V$ one-to-one onto an open set W of $\mathbb{R}^{k+n}$ containing $(\mathbf{a}, \mathbf{0})$;

(2) the inverse $G : W \to U \times V$ is of class C^r. Since $F(\mathbf{x}, \mathbf{y}) = (\mathbf{x}, f(\mathbf{x}, \mathbf{y}))$, we have $(\mathbf{x}, \mathbf{y}) = G(\mathbf{x}, f(\mathbf{x}, \mathbf{y}))$, so G preserves the first k coordinates, as F does; write

$$G(\mathbf{x}, \mathbf{z}) = (\mathbf{x}, h(\mathbf{x}, \mathbf{z})), \qquad \mathbf{x} \in \mathbb{R}^k,\ \mathbf{z} \in \mathbb{R}^n,$$

with $h : W \to \mathbb{R}^n$ of class C^r.

Existence. Let B be a connected neighborhood of $\mathbf{a}$ in $\mathbb{R}^k$ so small that $B \times \mathbf{0} \subset W$. For $\mathbf{x} \in B$, $(\mathbf{x}, \mathbf{0}) \in W$ and

$$G(\mathbf{x}, \mathbf{0}) = (\mathbf{x}, h(\mathbf{x}, \mathbf{0})), \qquad (\mathbf{x}, \mathbf{0}) = F(\mathbf{x}, h(\mathbf{x}, \mathbf{0})) = (\mathbf{x}, f(\mathbf{x}, h(\mathbf{x}, \mathbf{0}))),$$

so $f(\mathbf{x}, h(\mathbf{x}, \mathbf{0})) = \mathbf{0}$. Set $g(\mathbf{x}) = h(\mathbf{x}, \mathbf{0})$; g is C^r and $f(\mathbf{x}, g(\mathbf{x})) = \mathbf{0}$. Moreover $(\mathbf{a}, \mathbf{b}) = G(\mathbf{a}, \mathbf{0}) = (\mathbf{a}, h(\mathbf{a}, \mathbf{0}))$, so $g(\mathbf{a}) = \mathbf{b}$.

Uniqueness. Let $g_0 : B \to \mathbb{R}^n$ be continuous with $g_0(\mathbf{a}) = \mathbf{b}$ and $f(\mathbf{x}, g_0(\mathbf{x})) = \mathbf{0}$ on B. If g_0 agrees with g at $\mathbf{a}_0 \in B$, it agrees with g near $\mathbf{a}_0$: since g_0 is continuous and $g_0(\mathbf{a}_0) = g(\mathbf{a}_0) \in V$, there is a neighborhood $B_0 \subset B$ of $\mathbf{a}_0$ with $g_0(B_0) \subset V$. For $\mathbf{x} \in B_0$, $f(\mathbf{x}, g_0(\mathbf{x})) = \mathbf{0}$ gives $F(\mathbf{x}, g_0(\mathbf{x})) = (\mathbf{x}, \mathbf{0})$, and since $(\mathbf{x}, g_0(\mathbf{x})) \in U \times V$, $(\mathbf{x}, g_0(\mathbf{x})) = G(\mathbf{x}, \mathbf{0}) = (\mathbf{x}, h(\mathbf{x}, \mathbf{0}))$, so $g_0 = g$ on B_0. Hence $\{\mathbf{x} \in B : |g(\mathbf{x}) - g_0(\mathbf{x})| = 0\}$ is open in B; so is $\{\mathbf{x} \in B : |g(\mathbf{x}) - g_0(\mathbf{x})| > 0\}$, by continuity. The first contains $\mathbf{a}$, and B is connected, so the second is empty. ∎

Nothing is special about the *last* n coordinates. For example, if $f : A \to \mathbb{R}^2$, A open in $\mathbb{R}^5$, is C^r, $f(\mathbf{a}) = \mathbf{0}$, and $\det \frac{\partial f}{\partial (y,u)}(\mathbf{a}) \neq 0$, then near $\mathbf{a}$ one can solve $f(x, y, z, u, v) = \mathbf{0}$ for $y = \phi(x, z, v)$, $u = \psi(x, z, v)$, with

$$\frac{\partial(\phi, \psi)}{\partial(x, z, v)} = -\Big[\frac{\partial f}{\partial(y, u)}\Big]^{-1} \cdot \Big[\frac{\partial f}{\partial(x, z, v)}\Big].$$

Example 9.1

Let $f(x, y) = x^2 + y^2 - 5$. The point $(1, 2)$ satisfies $f = 0$, and both $\partial f/\partial x$ and $\partial f/\partial y$ are nonzero there, so we can solve locally for either variable in terms of the other. Solving for y gives $y = g(x) = [5 - x^2]^{1/2}$. This is not unique near $x = 1$ unless g is required to be continuous: $h(x) = [5 - x^2]^{1/2}$ for $x \geq 1$, $h(x) = -[5 - x^2]^{1/2}$ for $x < 1$ also satisfies the conditions but is not continuous.

Example 9.2

For the same f, the point $(\sqrt{5}, 0)$ satisfies $f = 0$ but $\partial f/\partial y = 0$ there; indeed no neighborhood B of $\sqrt{5}$ allows solving for y in terms of x.

Example 9.3

Let $f(x,y) = x^2 - y^3$. At $(0,0)$, $\partial f/\partial y = 0$, so we do not expect to solve for y near $(0,0)$. But we can, and the solution $y = x^{2/3}$ is unique; however it is not differentiable at $x = 0$.

Example 9.4

Let $f(x,y) = y^2 - x^4$. At $(0,0)$, $\partial f/\partial y = 0$, yet we can solve for y near $(0,0)$ so that the solution is differentiable ($y = x^2$ or $y = -x^2$); but the solution is not unique. At a point such as $(1,1)$,$\partial f/\partial y \neq 0$, so we can solve for y as a continuous function of x near $x = 1$; one can extend to a larger neighborhood, but if it contains 0 the solution is no longer unique.

Chapter 3. Integration

Overview

We define the (Riemann) integral of a bounded real-valued function of several variables, first over a rectangle (§10), and characterize integrability: f is integrable iff its set of discontinuities has measure zero (§11). Fubini's theorem reduces integrals to iterated integrals (§12). The integral is then extended to bounded sets (§13), which leads to rectifiable sets and volume (§14), and finally to the extended (improper) integral over open sets (§15).

§10. The Integral over a Rectangle

Motivation. To integrate over regions in $\mathbb{R}^n$ we follow the one-variable Riemann approach: partition a rectangle into subrectangles and compare upper and lower sums. The results here are the basic bookkeeping of refinements.

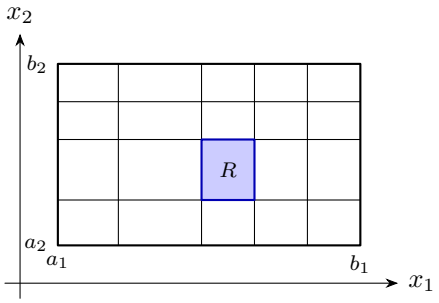


Figure. *A partition $P = (P_1, P_2)$ of $Q = [a_1, b_1] \times [a_2, b_2]$ divides it into subrectangles R. The lower and upper sums are $L(f, P) = \sum_R m_R(f)\, v(R)$ and $U(f, P) = \sum_R M_R(f)\, v(R)$, with m_R, M_R the infimum and supremum of f on R.*

Definition 10.1 (rectangle, volume)

Let $Q = [a_1, b_1] \times \cdots \times [a_n, b_n]$ be a rectangle in $\mathbb{R}^n$. The intervals $[a_i, b_i]$ are its **component intervals**; the maximum of $b_i - a_i$ is its **width**; the **volume** of Q is $\mathrm{v}(Q) = (b_1 - a_1)(b_2 - a_2) \cdots (b_n - a_n)$. For $n = 1$ volume and width coincide and are called the **length** of $[a, b]$.

Definition 10.2 (partition)

A **partition** of $[a, b]$ is a finite set P of points of $[a, b]$ containing a and b; we index it as $a = t_0 < t_1 < \cdots < t_k = b$, and each $[t_{i-1}, t_i]$ is a **subinterval determined by** P. A **partition** of a rectangle Q as above is an n-tuple $P = (P_1, \ldots, P_n)$ where P_j is a partition of $[a_j, b_j]$. If I_j is a subinterval determined by P_j, then $R = I_1 \times \cdots \times I_n$ is a **subrectangle determined by** P. The maximum width of these subrectangles is the **mesh** of P.

Definition 10.3 (upper and lower sums)

Let $f : Q \to \mathbb{R}$ be bounded and P a partition of Q. For each subrectangle R determined by P let $m_R(f) = \inf\{f(\mathbf{x}) : \mathbf{x} \in R\}$ and $M_R(f) = \sup\{f(\mathbf{x}) : \mathbf{x} \in R\}$. The **lower sum** and

upper sum of f determined by P are

$$L(f,P)=\sum_R m_R(f)\ \mathrm{v}(R), \qquad U(f,P)=\sum_R M_R(f)\ \mathrm{v}(R),$$

the sums extending over all subrectangles R determined by P.

If P'' is obtained from $P=(P_1,\dots,P_n)$ by adjoining points to some or all of the P_j, then P'' is a **refinement** of P. The partition $P''=(P_1\cup P_1',\dots,P_n\cup P_n')$ is a refinement of both P and P', their **common refinement**. Refining increases lower sums and decreases upper sums:

Lemma 10.1

Let P be a partition of the rectangle Q; let $f:Q\to\mathbb{R}$ be a bounded function. If P'' is a refinement of P, then

$$L(f,P)\le L(f,P'') \qquad \text{and} \qquad U(f,P'')\le U(f,P).$$

Proof. It suffices to treat P'' obtained by adjoining a single point q to one component partition, say $P_1=\{a_1=t_0<\cdots<t_k=b_1\}$, with q interior to $[t_{i-1},t_i]$. Most subrectangles of P are also subrectangles of P''. The exceptions are those of the form $R_S=[t_{i-1},t_i]\times S$, where S is a subrectangle of $[a_2,b_2]\times\cdots\times[a_n,b_n]$ determined by $(P_2,\dots,P_n)$; in the lower sum the term for R_S is replaced by the terms for $R_S'=[t_{i-1},q]\times S$ and $R_S''=[q,t_i]\times S$. Since $m_{R_S}(f)\le f(\mathbf{x})$ for $\mathbf{x}$ in R_S' and in R_S'', we get $m_{R_S}(f)\le m_{R_S'}(f)$ and $m_{R_S}(f)\le m_{R_S''}(f)$. Because $\mathrm{v}(R_S)=\mathrm{v}(R_S')+\mathrm{v}(R_S'')$,

$$m_{R_S}(f)\,\mathrm{v}(R_S)\le m_{R_S'}(f)\,\mathrm{v}(R_S')+m_{R_S''}(f)\,\mathrm{v}(R_S'').$$

Summing over S gives $L(f,P)\le L(f,P'')$. Similarly $U(f,P)\ge U(f,P'')$. ■

Lemma 10.2

Let Q be a rectangle; let $f:Q\to\mathbb{R}$ be a bounded function. If P and P' are any two partitions of Q, then $L(f,P)\le U(f,P')$.

Proof. If $P=P'$, this follows from $m_R(f)\le M_R(f)$. In general let P'' be the common refinement; by Lemma 10.1, $L(f,P)\le L(f,P'')\le U(f,P'')\le U(f,P')$. ■

Definition 10.4 (the integral)

Let $f:Q\to\mathbb{R}$ be bounded. As P ranges over all partitions of Q, define the **lower integral** and **upper integral**

$$\underline{\int_Q} f=\sup_P\{L(f,P)\}, \qquad \overline{\int_Q} f=\inf_P\{U(f,P)\}.$$

They exist since the $L(f,P)$ are bounded above by any $U(f,P')$ and the $U(f,P)$ are bounded below by any $L(f,P')$. If they are equal, f is **integrable** over Q, and the common value is the **integral** $\int_Q f$ or $\int_{\mathbf{x}\in Q} f(\mathbf{x})$.

Example 10.1

If $f:[a,b]\to\mathbb{R}$ is non-negative and bounded, $L(f,P)$ is the total area of rectangles inscribed in the region between the graph of f and the x-axis, and $U(f,P)$ the total area of rectangles

circumscribed about it. The lower integral is the "inner area" and the upper integral the "outer area" of the region; f is integrable when they agree. Similarly for $Q \subset \mathbb{R}^2$, with boxes in place of rectangles.

Example 10.2

Let $f : [0,1] \to \mathbb{R}$, $f(x) = 0$ if x is rational and $f(x) = 1$ if x is irrational. Every subinterval R contains rational and irrational numbers, so $m_R(f) = 0$, $M_R(f) = 1$, $L(f,P) = 0$ and $U(f,P) = \sum_R \mathrm{v}(R) = 1$ for every P. The lower integral is 0 and the upper integral 1: f is not integrable.

Theorem 10.3 (The Riemann condition)

Let Q be a rectangle; let $f : Q \to \mathbb{R}$ be a bounded function. Then

$$\underline{\int_Q} f \le \overline{\int_Q} f;$$

equality holds if and only if given $\epsilon > 0$ there exists a corresponding partition P of Q for which $U(f,P) - L(f,P) < \epsilon$.

Proof. Fix P'. Since $L(f,P) \le U(f,P')$ for all P, the lower integral is $\le U(f,P')$; as P' is arbitrary, the lower integral is $\le$ the upper integral.

Suppose they are equal. Choose P with $L(f,P)$ within $\epsilon/2$ of $\int_Q f$, and P' with $U(f,P')$ within $\epsilon/2$ of $\int_Q f$; let P'' be their common refinement. Since $L(f,P) \le L(f,P'') \le \int_Q f \le U(f,P'') \le U(f,P')$, the sums for P'' are within ϵ of each other. Conversely, if the upper and lower integrals differ, let ϵ be their difference; then for every P, $L(f,P) \le \underline{\int_Q} f < \overline{\int_Q} f \le U(f,P)$, so $U(f,P) - L(f,P) \ge \epsilon$. ■

Theorem 10.4

Every constant function $f(\mathbf{x}) = c$ is integrable. Indeed, if Q is a rectangle and if P is a partition of Q, then

$$\int_Q c = c \cdot \mathrm{v}(Q) = c \sum_R \mathrm{v}(R),$$

where the summation extends over all subrectangles determined by P.

Proof. $m_R(f) = c = M_R(f)$ for every R, so $L(f,P) = c\sum_R \mathrm{v}(R) = U(f,P)$ and the Riemann condition holds trivially; $\int_Q c$ lies between them, so equals $c\sum_R \mathrm{v}(R)$. This holds for every P; for the trivial partition (whose only subrectangle is Q) it gives $c \cdot \mathrm{v}(Q)$. ■

Corollary 10.5

Let Q be a rectangle in $\mathbb{R}^n$; let $\{Q_1, \ldots, Q_k\}$ be a finite collection of rectangles that covers Q. Then $\mathrm{v}(Q) \le \sum_{i=1}^k \mathrm{v}(Q_i)$.

Proof. Choose a rectangle Q' containing $Q, Q_1, \ldots, Q_k$, and use the end points of the component intervals of all these rectangles to define a partition P of Q'. Each of $Q, Q_1, \ldots, Q_k$ is a union of subrectangles determined by P. By Theorem 10.4 (applied to the partition of Q induced by P), $\mathrm{v}(Q) = \sum_{R \subset Q} \mathrm{v}(R)$. Each such R lies in at least one Q_i, so

$$\sum_{R \subset Q} \mathrm{v}(R) \le \sum_{i=1}^{k} \sum_{R \subset Q_i} \mathrm{v}(R) = \sum_{i=1}^{k} \mathrm{v}(Q_i).$$

■

Notation

For $n = 1$ we often write $\int_a^b f$ or $\int_{x=a}^{x=b} f(x)$ for $\int_{[a,b]} f$. The calculus notation $\int_a^b f(x)\,dx$, in which "dx" has no independent meaning, is avoided for now; later dx will be given a meaning. This definition is due to Darboux; an equivalent one due to Riemann uses sums $\sum_R f(\mathbf{x}_R)\,\mathrm{v}(R)$. Either way, it is called the **Riemann integral**.

§11. Existence of the Integral

Motivation. Which bounded functions are integrable? The answer is clean: exactly those whose discontinuities form a set of measure zero. This criterion is what lets us integrate over sets with complicated boundaries later.

Definition 11.1 (measure zero)

A subset A of $\mathbb{R}^n$ has **measure zero** in $\mathbb{R}^n$ if for every $\epsilon > 0$ there is a covering $Q_1, Q_2, \ldots$ of A by countably many rectangles such that $\sum_{i=1}^{\infty} \mathrm{v}(Q_i) < \epsilon$ ("the total volume of the Q_i is less than ϵ").

Theorem 11.1

(a) If $B \subset A$ and A has measure zero in $\mathbb{R}^n$, then so does B.

(b) Let A be the union of the countable collection of sets $A_1, A_2, \ldots$. If each A_i has measure zero in $\mathbb{R}^n$, so does A.

(c) A set A has measure zero in $\mathbb{R}^n$ if and only if for every $\epsilon > 0$ there is a countable covering of A by open rectangles $\operatorname{Int} Q_1, \operatorname{Int} Q_2, \ldots$ such that $\sum_{i=1}^{\infty} \mathrm{v}(Q_i) < \epsilon$.

(d) If Q is a rectangle in $\mathbb{R}^n$, then $\operatorname{Bd} Q$ has measure zero in $\mathbb{R}^n$ but Q does not.

Proof. (a) is immediate.

(b) Cover A_j by countably many rectangles $Q_{1j}, Q_{2j}, \ldots$ of total volume less than $\epsilon/2^j$. The collection $\{Q_{ij}\}$ is countable, covers A, and has total volume less than $\sum_j \epsilon/2^j = \epsilon$.

(c) If open rectangles $\operatorname{Int} Q_i$ cover A, so do the Q_i; this gives "if". Conversely, cover A by rectangles $Q_1', Q_2', \ldots$ of total volume less than $\epsilon/2$, and for each i choose a rectangle Q_i with $Q_i' \subset \operatorname{Int} Q_i$ and $\mathrm{v}(Q_i) \le 2\,\mathrm{v}(Q_i')$ (possible since v is a continuous function of the end points of the component intervals). Then the $\operatorname{Int} Q_i$ cover A and $\sum \mathrm{v}(Q_i) < \epsilon$.

(d) Let $Q = [a_1, b_1] \times \cdots \times [a_n, b_n]$. The subset of Q where $x_i = a_i$ is one of the ith **faces** of Q, the other where $x_i = b_i$. Each face has measure zero: e.g. the face $x_i = a_i$ is covered by the single rectangle $[a_1, b_1] \times \cdots \times [a_i, a_i + \delta] \times \cdots \times [a_n, b_n]$, whose volume is as small as desired. $\operatorname{Bd} Q$ is the union of the finitely many faces, so it has measure zero. Suppose Q had measure zero, and let $\epsilon = \mathrm{v}(Q)$. Cover Q by open rectangles $\operatorname{Int} Q_1, \operatorname{Int} Q_2, \ldots$ with $\sum \mathrm{v}(Q_i) < \epsilon$. Since Q is compact, finitely many cover Q, say $\operatorname{Int} Q_1, \ldots, \operatorname{Int} Q_k$; but $\sum_{i=1}^{k} \mathrm{v}(Q_i) < \epsilon = \mathrm{v}(Q)$ contradicts Corollary 10.5. ■

Example 11.1

Allowing *countably infinite* coverings is essential. The set A of rationals in $I = [0,1]$ is a countable union of one-point sets, hence has measure zero by (b). But A cannot be covered by finitely many intervals of total length less than $\epsilon < 1$: if $I_1, \dots, I_k$ cover A, their union B is closed (a finite union of closed sets) and contains all rationals of I, hence all their limit points, i.e. all of I; by Corollary 10.5, $\sum \mathrm{v}(I_i) \geq \mathrm{v}(I) = 1$.

Theorem 11.2

Let Q be a rectangle in $\mathbb{R}^n$; let $f : Q \to \mathbb{R}$ be a bounded function. Let D be the set of points of Q at which f fails to be continuous. Then $\int_Q f$ exists if and only if D has measure zero in $\mathbb{R}^n$.

Proof. Choose M with $|f(\mathbf{x})| \leq M$ for $\mathbf{x} \in Q$.

Step 1. ("If") Assume D has measure zero. Given $\epsilon > 0$, let $\epsilon' = \epsilon/(2M + 2\,\mathrm{v}(Q))$. Cover D by open rectangles $\operatorname{Int} Q_1, \operatorname{Int} Q_2, \dots$ with total volume less than ϵ' (Theorem 11.1(c)). For each $\mathbf{a} \in Q - D$ choose an open rectangle $\operatorname{Int} Q_{\mathbf{a}}$ containing $\mathbf{a}$ such that $|f(\mathbf{x}) - f(\mathbf{a})| < \epsilon'$ for $\mathbf{x} \in Q_{\mathbf{a}} \cap Q$ (continuity at $\mathbf{a}$). The open sets $\operatorname{Int} Q_i$ and $\operatorname{Int} Q_{\mathbf{a}}$ cover the compact set Q, so finitely many do: $\operatorname{Int} Q_1, \dots, \operatorname{Int} Q_k, \operatorname{Int} Q_{\mathbf{a}_1}, \dots, \operatorname{Int} Q_{\mathbf{a}_l}$. Write $Q'_j = Q_{\mathbf{a}_j}$. Then

$$\sum_{i=1}^{k} \mathrm{v}(Q_i) < \epsilon', \tag{1}$$

$$|f(\mathbf{x}) - f(\mathbf{y})| \leq 2\epsilon' \quad \text{for } \mathbf{x}, \mathbf{y} \in Q'_j \cap Q. \tag{2}$$

Replace each Q_i and Q'_j by its intersection with Q; (1), (2) persist. Use the end points of the component intervals of $Q_1, \dots, Q_k, Q'_1, \dots, Q'_l$ to define a partition P of Q; each Q_i, Q'_j is a union of subrectangles determined by P. Divide these subrectangles into two disjoint collections $\mathcal{R}$ and $\mathcal{R}'$, with each $R \in \mathcal{R}$ lying in some Q_i and each $R \in \mathcal{R}'$ lying in some Q'_j. Since $|f(\mathbf{x}) - f(\mathbf{y})| \leq 2M$ in general and $\leq 2\epsilon'$ on $R \in \mathcal{R}'$,

$$\sum_{R \in \mathcal{R}} (M_R(f) - m_R(f))\,\mathrm{v}(R) \leq 2M \sum_{R \in \mathcal{R}} \mathrm{v}(R), \qquad \sum_{R \in \mathcal{R}'} (M_R(f) - m_R(f))\,\mathrm{v}(R) \leq 2\epsilon' \sum_{R \in \mathcal{R}'} \mathrm{v}(R).$$

Now $\sum_{R \in \mathcal{R}} \mathrm{v}(R) \leq \sum_{i=1}^{k} \sum_{R \subset Q_i} \mathrm{v}(R) = \sum_i \mathrm{v}(Q_i) < \epsilon'$ and $\sum_{R \in \mathcal{R}'} \mathrm{v}(R) \leq \sum_{R \subset Q} \mathrm{v}(R) = \mathrm{v}(Q)$. Thus $U(f,P) - L(f,P) < 2M\epsilon' + 2\epsilon'\,\mathrm{v}(Q) = \epsilon$, and f is integrable by the Riemann condition.

Step 2. Oscillation. For $\mathbf{a} \in Q$ and $\delta > 0$ let $A_\delta = \{f(\mathbf{x}) : \mathbf{x} \in Q,\ |\mathbf{x} - \mathbf{a}| < \delta\}$, $M_\delta(f) = \sup A_\delta$, $m_\delta(f) = \inf A_\delta$. The **oscillation** of f at $\mathbf{a}$ is

$$\nu(f;\mathbf{a}) = \inf_{\delta > 0} [M_\delta(f) - m_\delta(f)] \ \geq 0.$$

f is continuous at $\mathbf{a}$ iff $\nu(f;\mathbf{a}) = 0$. If f is continuous at $\mathbf{a}$, given ϵ choose δ with $|f(\mathbf{x}) - f(\mathbf{a})| < \epsilon$ for $|\mathbf{x} - \mathbf{a}| < \delta$; then $M_\delta(f) \leq f(\mathbf{a}) + \epsilon$, $m_\delta(f) \geq f(\mathbf{a}) - \epsilon$, so $\nu(f;\mathbf{a}) \leq 2\epsilon$. Conversely if $\nu(f;\mathbf{a}) = 0$, given ϵ there is δ with $M_\delta(f) - m_\delta(f) < \epsilon$; for $|\mathbf{x} - \mathbf{a}| < \delta$ both $f(\mathbf{x})$ and $f(\mathbf{a})$ lie in $[m_\delta(f), M_\delta(f)]$, so $|f(\mathbf{x}) - f(\mathbf{a})| < \epsilon$.

Step 3. ("Only if") Assume f is integrable. Let $D_m = \{\mathbf{a} : \nu(f;\mathbf{a}) \geq 1/m\}$; by Step 2, $D = \bigcup_m D_m$, so it suffices to show each D_m has measure zero. Fix m and $\epsilon > 0$. Choose P with $U(f,P) - L(f,P) < \epsilon/2m$. Let D'_m be the points of D_m lying on $\operatorname{Bd} R$ for some subrectangle R determined by P, and D''_m the rest of D_m. D'_m lies in the finite union $\bigcup_R \operatorname{Bd} R$, which has measure zero, so it can be covered by countably many rectangles of total volume $< \epsilon/2$. Let $R_1, \dots, R_k$ be the subrectangles that contain points of D''_m. If $\mathbf{a} \in D''_m \cap R_i$, then $\mathbf{a} \notin \operatorname{Bd} R_i$, so some cubical neighborhood of radius δ about $\mathbf{a}$ lies in R_i, and

$$1/m \leq \nu(f;\mathbf{a}) \leq M_\delta(f) - m_\delta(f) \leq M_{R_i}(f) - m_{R_i}(f).$$

Multiplying by $\mathrm{v}(R_i)$ and summing: $\sum_{i=1}^{k}(1/m)\,\mathrm{v}(R_i) \le U(f,P) - L(f,P) < \epsilon/2m$, so $R_1,\dots,R_k$ cover D_m'' with total volume $< \epsilon/2$. ■

Theorem 11.3

Let Q be a rectangle in $\mathbb{R}^n$; let $f : Q \to \mathbb{R}$; assume f is integrable over Q.

(a) If f vanishes except on a set of measure zero, then $\int_Q f = 0$.

(b) If f is non-negative and if $\int_Q f = 0$, then f vanishes except on a set of measure zero.

Proof. (a) Suppose f vanishes except on a set E of measure zero. A subrectangle R determined by a partition P is not contained in E (Theorem 11.1(d),(a)), so f vanishes at some point of R, whence $m_R(f) \le 0 \le M_R(f)$. Hence $L(f,P) \le 0 \le U(f,P)$ for all P, so $\underline{\int}_Q f \le 0 \le \overline{\int}_Q f$; since $\int_Q f$ exists, it is 0.

(b) Suppose $f \ge 0$ and $\int_Q f = 0$. We show: if f is continuous at $\mathbf{a}$, then $f(\mathbf{a}) = 0$; then f vanishes except on its set of discontinuities, which has measure zero by Theorem 11.2. Suppose $f(\mathbf{a}) > 0$ and let $\epsilon = f(\mathbf{a})$; choose δ with $f(\mathbf{x}) > \epsilon/2$ for $|\mathbf{x}-\mathbf{a}| < \delta$, $\mathbf{x} \in Q$. Take a partition P of mesh less than δ; if R_0 is a subrectangle containing $\mathbf{a}$, then $m_{R_0}(f) \ge \epsilon/2$, and $m_R(f) \ge 0$ for all R. So $L(f,P) \ge (\epsilon/2)\,\mathrm{v}(R_0) > 0$, contradicting $L(f,P) \le \int_Q f = 0$. ■

Example 11.2

The assumption that $\int_Q f$ exists is necessary in (a): the function f on $I = [0,1]$ with $f(x) = 1$ for x rational and 0 otherwise vanishes except on a set of measure zero, but $\int_I f$ does not exist (cf. Example 10.2).

§12. Evaluation of the Integral

Motivation. Integrals over rectangles are computed by integrating one variable at a time. Fubini's theorem justifies this and reduces multiple integrals to single ones.

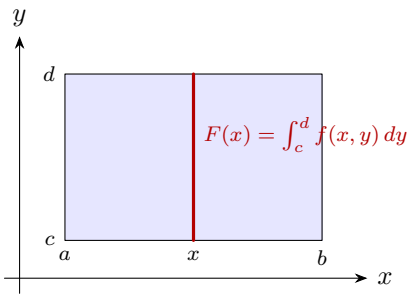


Figure. *Fubini's theorem: integrate f along each vertical segment to get $F(x)$, then integrate F over $[a,b]$:* $\int_Q f = \int_a^b \int_c^d f(x,y)\,dy\,dx$.

Theorem 12.1 (Fundamental theorem of calculus)

(a) If f is continuous on $[a,b]$, and if $F(x) = \int_a^x f$ for $x \in [a,b]$, then $F'(x)$ exists and equals $f(x)$.
(b) If f is continuous on $[a,b]$, and if g is a function such that $g'(x) = f(x)$ for $x \in [a,b]$, then $\int_a^b f = g(b) - g(a)$.

(At the end points, F' and g' mean one-sided derivatives.) In short: $D\int_a^x f = f(x)$ and $\int_a^x Dg = g(x) - g(a)$.

Proof [Supplement]. (a) By additivity (Theorem 13.3(d), or directly for intervals), $F(x+h) - F(x) = \int_x^{x+h} f$ for $h > 0$ (similarly for $h < 0$). By comparison, this lies between $h \min f$ and $h \max f$ over $[x, x+h]$, so $\frac{F(x+h)-F(x)}{h}$ lies between $f(s)$ and $f(t)$ for some $s, t \in [x, x+h]$; by continuity it tends to $f(x)$ as $h \to 0$.

(b) $F - g$ has derivative 0 on $[a, b]$, so it is constant by the mean-value theorem (Theorem 6.1); hence $\int_a^b f = F(b) - F(a) = g(b) - g(a)$. ∎

So we can compute the integral of a continuous f by finding an **antiderivative** g, $g' = f$; part (a) shows one always exists in theory.

For n-dimensional integrals we reduce to a sequence of lower-dimensional integrals. For continuous f on $Q = [a, b] \times [c, d]$ calculus uses

$$\int_Q f = \int_{x=a}^{x=b} \int_{y=c}^{y=d} f(x, y)$$

without proof. When f is merely integrable, the inner integral may fail to exist for some x (the function can behave badly along a single vertical line without affecting $\int_Q f$). We therefore replace the inner integral by the corresponding lower (or upper) integral, which always exists; this gives a correct general theorem.

Theorem 12.2 (Fubini's theorem)

Let $Q = A \times B$, where A is a rectangle in $\mathbb{R}^k$ and B is a rectangle in $\mathbb{R}^n$. Let $f : Q \to \mathbb{R}$ be a bounded function; write f in the form $f(\mathbf{x}, \mathbf{y})$ for $\mathbf{x} \in A$ and $\mathbf{y} \in B$. For each $\mathbf{x} \in A$, consider the lower and upper integrals

$$\underline{\int_{\mathbf{y}\in B}} f(\mathbf{x}, \mathbf{y}) \qquad \text{and} \qquad \overline{\int_{\mathbf{y}\in B}} f(\mathbf{x}, \mathbf{y}).$$

If f is integrable over Q, then these two functions of $\mathbf{x}$ are integrable over A, and

$$\int_Q f = \int_{\mathbf{x}\in A} \underline{\int_{\mathbf{y}\in B}} f(\mathbf{x}, \mathbf{y}) = \int_{\mathbf{x}\in A} \overline{\int_{\mathbf{y}\in B}} f(\mathbf{x}, \mathbf{y}).$$

Proof. Let $\underline{I}(\mathbf{x}) = \underline{\int}_{\mathbf{y}\in B} f(\mathbf{x}, \mathbf{y})$ and $\overline{I}(\mathbf{x}) = \overline{\int}_{\mathbf{y}\in B} f(\mathbf{x}, \mathbf{y})$. A partition P of Q consists of a partition P_A of A and P_B of B; if R_A, R_B are general subrectangles determined by P_A, P_B, then $R_A \times R_B$ is the general subrectangle determined by P.

Step 1. $L(f, P) \le L(\underline{I}, P_A)$. Fix $R_A \times R_B$ and $\mathbf{x}_0 \in R_A$. Then $m_{R_A\times R_B}(f) \le f(\mathbf{x}_0, \mathbf{y})$ for all $\mathbf{y} \in R_B$, so $m_{R_A\times R_B}(f) \le m_{R_B}(f(\mathbf{x}_0, \mathbf{y}))$. Multiply by $\mathrm{v}(R_B)$ and sum over R_B:

$$\sum_{R_B} m_{R_A\times R_B}(f)\, \mathrm{v}(R_B) \le L(f(\mathbf{x}_0, \cdot), P_B) \le \underline{I}(\mathbf{x}_0).$$

This holds for each $\mathbf{x}_0 \in R_A$, so $\sum_{R_B} m_{R_A\times R_B}(f)\, \mathrm{v}(R_B) \le m_{R_A}(\underline{I})$. Multiply by $\mathrm{v}(R_A)$ and sum over R_A; since $\mathrm{v}(R_A)\,\mathrm{v}(R_B) = \mathrm{v}(R_A \times R_B)$, $L(f, P) \le L(\underline{I}, P_A)$.

Step 2. $U(f, P) \ge U(\overline{I}, P_A)$, by the symmetric argument: for $\mathbf{x}_0 \in R_A$, $M_{R_A\times R_B}(f) \ge M_{R_B}(f(\mathbf{x}_0, \mathbf{y}))$, so $\sum_{R_B} M_{R_A\times R_B}(f)\, \mathrm{v}(R_B) \ge U(f(\mathbf{x}_0, \cdot), P_B) \ge \overline{I}(\mathbf{x}_0)$; hence the left side is $\ge M_{R_A}(\overline{I})$, and summing gives the claim.

Step 3. Hence the diagram

$$L(f, P) \le L(\underline{I}, P_A) \le \begin{Bmatrix} U(\underline{I}, P_A) \\ L(\overline{I}, P_A) \end{Bmatrix} \le U(\overline{I}, P_A) \le U(f, P),$$

in which the two middle terms are each between $L(\underline{I}, P_A)$ and $U(\overline{I}, P_A)$: the first and last inequalities are Steps 1 and 2, and the middle ones follow from $L(h, P_A) \le U(h, P_A)$ and $\underline{I}(\mathbf{x}) \le \overline{I}(\mathbf{x})$.

Step 4. Since f is integrable, given $\epsilon > 0$ choose P with $U(f,P) - L(f,P) < \epsilon$. Then the upper and lower sums for $\underline{I}$ are within ϵ of each other, and likewise for $\overline{I}$; so both are integrable over A. The numbers $\int_A \underline{I}$, $\int_A \overline{I}$, $\int_Q f$ all lie between the extreme ends of the diagram. As ϵ is arbitrary, they are equal. ∎

One may equally compute $\int_Q f$ by first taking the lower (or upper) integral with respect to $\mathbf{x}$ and then integrating in $\mathbf{y}$.

Corollary 12.3

Let $Q = A \times B$, where A is a rectangle in $\mathbb{R}^k$ and B is a rectangle in $\mathbb{R}^n$. Let $f : Q \to \mathbb{R}$ be a bounded function. If $\int_Q f$ exists, and if $\int_{\mathbf{y}\in B} f(\mathbf{x},\mathbf{y})$ exists for each $\mathbf{x} \in A$, then

$$\int_Q f = \int_{\mathbf{x}\in A} \int_{\mathbf{y}\in B} f(\mathbf{x},\mathbf{y}).$$

Proof [Supplement]. If the inner integral exists, it equals both $\underline{I}(\mathbf{x})$ and $\overline{I}(\mathbf{x})$; apply Fubini's theorem. ∎

Corollary 12.4

Let $Q = I_1 \times \cdots \times I_n$, where I_j is a closed interval in $\mathbb{R}$ for each j. If $f : Q \to \mathbb{R}$ is continuous, then

$$\int_Q f = \int_{x_1\in I_1} \cdots \int_{x_n \in I_n} f(x_1, \ldots, x_n).$$

Proof [Supplement]. Continuous functions on rectangles are integrable (Theorem 11.2). Write $Q = I_1 \times Q'$ with $Q' = I_2 \times \cdots \times I_n$. For fixed x_1, $f(x_1, \cdot)$ is continuous on Q', hence integrable, so Corollary 12.3 gives $\int_Q f = \int_{x_1\in I_1} \int_{Q'} f(x_1, \cdot)$. Apply induction on n to the inner integral. ∎

§13. The Integral over a Bounded Set

Motivation. Regions such as discs are not rectangles. We extend f by zero outside the set and integrate over a rectangle; integrability then depends on the boundary of the set, which must have measure zero.

To integrate over non-rectangular regions (mass of a circular plate, center of gravity of a spherical cap, ...) we generalize the definition.

Definition 13.1 (integral over a bounded set)

Let S be a bounded set in $\mathbb{R}^n$ and $f : S \to \mathbb{R}$ a bounded function. Define $f_S : \mathbb{R}^n \to \mathbb{R}$ by $f_S(\mathbf{x}) = f(\mathbf{x})$ for $\mathbf{x} \in S$ and $f_S(\mathbf{x}) = 0$ otherwise. Choose a rectangle Q containing S and define

$$\int_S f = \int_Q f_S,$$

provided the latter integral exists.

Lemma 13.1

Let Q and Q' be two rectangles in $\mathbb{R}^n$. If $f : \mathbb{R}^n \to \mathbb{R}$ is a bounded function that vanishes outside $Q \cap Q'$, then

$$\int_Q f = \int_{Q'} f;$$

one integral exists if and only if the other does.

Proof. First suppose $Q \subset Q'$. Let E be the set of points of $\operatorname{Int} Q$ where f is discontinuous. Both $f : Q \to \mathbb{R}$ and $f : Q' \to \mathbb{R}$ are continuous except at points of E and possibly points of $\operatorname{Bd} Q$, which has measure zero; so (Theorem 11.2) existence of either integral is equivalent to E having measure zero.

Suppose both exist. Let P be a partition of Q' and P'' its refinement obtained by adjoining the end points of the component intervals of Q. Then Q is a union of subrectangles R determined by P''. If R is not contained in Q, f vanishes at some point of R, so $m_R(f) \leq 0$. Hence

$$L(f, P) \leq L(f, P'') \leq \sum_{R \subset Q} m_R(f)\, \mathrm{v}(R) \leq \int_Q f.$$

Similarly $U(f, P) \geq \int_Q f$. As P is an arbitrary partition of Q', $\int_{Q'} f = \int_Q f$. For arbitrary Q, Q' choose Q'' containing both; then $\int_Q f = \int_{Q''} f = \int_{Q'} f$. ■

Lemma 13.2

Let S be a subset of $\mathbb{R}^n$; let $f, g : S \to \mathbb{R}$. Let $F, G : S \to \mathbb{R}$ be defined by

$$F(\mathbf{x}) = \max\{f(\mathbf{x}), g(\mathbf{x})\}, \qquad G(\mathbf{x}) = \min\{f(\mathbf{x}), g(\mathbf{x})\}.$$

(a) If f and g are continuous at $\mathbf{x}_0$, so are F and G.

(b) If f and g are integrable over S, so are F and G.

Proof. (a) If $f(\mathbf{x}_0) = g(\mathbf{x}_0) = r$, then given ϵ choose δ with $|f(\mathbf{x}) - r| < \epsilon$ and $|g(\mathbf{x}) - r| < \epsilon$ for $|\mathbf{x} - \mathbf{x}_0| < \delta$, $\mathbf{x} \in S$; then automatically $|F(\mathbf{x}) - F(\mathbf{x}_0)| < \epsilon$ and $|G(\mathbf{x}) - G(\mathbf{x}_0)| < \epsilon$. If $f(\mathbf{x}_0) > g(\mathbf{x}_0)$, then by continuity $f(\mathbf{x}) - g(\mathbf{x}) > 0$ on $U \cap S$ for a neighborhood U of $\mathbf{x}_0$, where $F = f$ and $G = g$; so F, G are continuous at $\mathbf{x}_0$. Similarly if $f(\mathbf{x}_0) < g(\mathbf{x}_0)$.

(b) Let $Q \supset S$. f_S, g_S are continuous on Q except on sets D, E of measure zero. Since $F_S = \max\{f_S, g_S\}$ and $G_S = \min\{f_S, g_S\}$, by (a) they are continuous except on $D \cup E$, which has measure zero; they are bounded, hence integrable over Q. ■

Theorem 13.3 (Properties of the integral)

Let S be a bounded set in $\mathbb{R}^n$; let $f, g : S \to \mathbb{R}$ be bounded functions.

(a) (Linearity) If f and g are integrable over S, so is $af + bg$, and $\int_S (af + bg) = a \int_S f + b \int_S g$.

(b) (Comparison) Suppose f and g are integrable over S. If $f(\mathbf{x}) \leq g(\mathbf{x})$ for $\mathbf{x} \in S$, then $\int_S f \leq \int_S g$. Furthermore, $|f|$ is integrable over S and $|\int_S f| \leq \int_S |f|$.

(c) (Monotonicity) Let $T \subset S$. If f is non-negative on S and integrable over T and S, then $\int_T f \leq \int_S f$.

(d) (Additivity) If $S = S_1 \cup S_2$ and f is integrable over S_1 and S_2, then f is integrable over S and $S_1 \cap S_2$; furthermore

$$\int_S f = \int_{S_1} f + \int_{S_2} f - \int_{S_1 \cap S_2} f.$$

Proof. (a) Since $(af+bg)_S = af_S + bg_S$, it suffices to treat integrals over a rectangle Q. If f, g are integrable over Q, they are continuous except on sets D, E of measure zero; $af+bg$ is continuous except on $D \cup E$, so it is integrable. Suppose first $a, b \geq 0$. For any partition P'' and subrectangle R, $am_R(f) + bm_R(g) \leq af(\mathbf{x}) + bg(\mathbf{x})$ for $\mathbf{x} \in R$, so $am_R(f) + bm_R(g) \leq m_R(af+bg)$, hence

$$aL(f, P'') + bL(g, P'') \leq L(af+bg, P'') \leq \int_Q (af+bg),$$

and similarly $aU(f, P'') + bU(g, P'') \geq \int_Q (af+bg)$. For arbitrary P, P' with common refinement P'', $aL(f,P) + bL(g,P') \leq \int_Q (af+bg) \leq aU(f,P) + bU(g,P')$. The number $a\int_Q f + b\int_Q g$ also lies between these ends for all P, P', so it equals $\int_Q (af+bg)$. It remains to show $\int_Q (-f) = -\int_Q f$: for $\mathbf{x} \in R$, $-M_R(f) \leq -f(\mathbf{x}) \leq -m_R(f)$, so $-M_R(f) \leq m_R(-f)$ and $M_R(-f) \leq -m_R(f)$; hence $-U(f,P) \leq L(-f,P) \leq \int_Q (-f) \leq U(-f,P) \leq -L(f,P)$, and $-\int_Q f$ also lies between the ends.

(b) It suffices to treat a rectangle Q. If $f \leq g$ on Q, then $m_R(f) \leq m_R(g)$, so $L(f,P) \leq L(g,P) \leq \int_Q g$ for every P, whence $\int_Q f \leq \int_Q g$. $|f| = \max\{f, -f\}$ is integrable by Lemma 13.2; apply comparison to $-|f| \leq f \leq |f|$.

(c) $f_T(\mathbf{x}) \leq f_S(\mathbf{x})$ for all $\mathbf{x}$; apply comparison.

(d) Let $T = S_1 \cap S_2$. First suppose $f \geq 0$ on S. With $Q \supset S$, f_{S_1}, f_{S_2} are integrable over Q, and $f_S = \max\{f_{S_1}, f_{S_2}\}$, $f_T = \min\{f_{S_1}, f_{S_2}\}$ are integrable by Lemma 13.2. In general put $f_+ = \max\{f, 0\}$, $f_- = \max\{-f, 0\}$; these are integrable over S_1 and S_2 (Lemma 13.2), hence over S and T by the special case, and so is $f = f_+ - f_-$ by linearity. The formula follows by linearity from $f_S(\mathbf{x}) = f_{S_1}(\mathbf{x}) + f_{S_2}(\mathbf{x}) - f_T(\mathbf{x})$. ■

Corollary 13.4

Let $S_1, \ldots, S_k$ be bounded sets in $\mathbb{R}^n$; assume $S_i \cap S_j$ has measure zero whenever $i \neq j$. Let $S = S_1 \cup \cdots \cup S_k$. If $f : S \to \mathbb{R}$ is integrable over each set S_i, then f is integrable over S and

$$\int_S f = \int_{S_1} f + \cdots + \int_{S_k} f.$$

Proof. For $k = 2$ this is additivity, since $\int_{S_1 \cap S_2} f$ exists and vanishes by Theorem 11.3. Induction gives the general case (note $(S_1 \cup \cdots \cup S_{k-1}) \cap S_k = \bigcup_{i<k} (S_i \cap S_k)$ has measure zero). ■

So far f was only required to be bounded; continuity could not be required, since f_S need not be continuous at points of $\operatorname{Bd} S$ even when f is continuous on S. Our main interest, however, is in continuous integrands.

Convention

Henceforth, we restrict ourselves in studying integration theory to the integration of continuous functions $f : S \to \mathbb{R}$.

Theorem 13.5

Let S be a bounded set in $\mathbb{R}^n$; let $f : S \to \mathbb{R}$ be a bounded continuous function. Let E be the set of points $\mathbf{x}_0$ of $\operatorname{Bd} S$ for which the condition

$$\lim_{\mathbf{x} \to \mathbf{x}_0} f(\mathbf{x}) = 0$$

fails to hold. If E has measure zero, then f is integrable over S.

Proof. We show f_S is continuous at every $\mathbf{x}_0 \notin E$; then Theorem 11.2 applies. If $\mathbf{x}_0 \in \operatorname{Int} S$, f and f_S agree near $\mathbf{x}_0$, and f is continuous there. If $\mathbf{x}_0 \in \operatorname{Ext} S$, f_S vanishes near $\mathbf{x}_0$. Suppose

$\mathbf{x}_0 \in \operatorname{Bd} S - E$ ($\mathbf{x}_0$ may or may not be in S). Then $f(\mathbf{x}) \to 0$ as $\mathbf{x} \to \mathbf{x}_0$ through S; since f is continuous, $f(\mathbf{x}_0) = 0$ if $\mathbf{x}_0 \in S$. As $f_S(\mathbf{x})$ is either $f(\mathbf{x})$ or 0, $f_S(\mathbf{x}) \to 0$ as $\mathbf{x} \to \mathbf{x}_0$. And $f_S(\mathbf{x}_0) = 0$: if $\mathbf{x}_0 \notin S$ by definition, if $\mathbf{x}_0 \in S$ because $f(\mathbf{x}_0) = 0$. ■

The converse also holds (we shall not need it).

Theorem 13.6

Let S be a bounded set in $\mathbb{R}^n$; let $f : S \to \mathbb{R}$ be a bounded continuous function; let $A = \operatorname{Int} S$. If f is integrable over S, then f is integrable over A, and $\int_S f = \int_A f$.

Proof.

Step 1. If f_S is continuous at $\mathbf{x}_0$, then f_A is continuous at $\mathbf{x}_0$ and agrees with f_S there. If $\mathbf{x}_0 \in \operatorname{Int} S$ or $\operatorname{Ext} S$, f_S and f_A agree near $\mathbf{x}_0$. Let $\mathbf{x}_0 \in \operatorname{Bd} S$. Arbitrarily near $\mathbf{x}_0$ are points $\mathbf{x} \notin S$, where $f_S(\mathbf{x}) = 0$; continuity of f_S at $\mathbf{x}_0$ forces $f_S(\mathbf{x}_0) = 0$. Since $f_A(\mathbf{x})$ is either $f_S(\mathbf{x})$ or 0, $f_A(\mathbf{x}) \to 0 = f_S(\mathbf{x}_0)$; and $f_A(\mathbf{x}_0) = 0$ because $\mathbf{x}_0 \notin A$.

Step 2. If f is integrable over S, f_S is continuous except on a set D of measure zero; by Step 1 so is f_A, so f is integrable over A. Since $f_S - f_A$ vanishes at points not in D, $\int_Q (f_S - f_A) = 0$ by Theorem 11.3, i.e. $\int_S f = \int_A f$. ■

§14. Rectifiable Sets

Motivation. A set has a volume when its characteristic function is integrable, i.e. when its boundary has measure zero. Rectifiable sets are the domains over which integration behaves as expected.

Definition 14.1 (rectifiable set, volume)

Let S be a bounded set in $\mathbb{R}^n$. If the constant function 1 is integrable over S, S is **rectifiable**, and its (n-dimensional) **volume** is

$$\mathrm{v}(S) = \int_S 1.$$

This agrees with the earlier definition when S is a rectangle.

Theorem 14.1

A subset S of $\mathbb{R}^n$ is rectifiable if and only if S is bounded and $\operatorname{Bd} S$ has measure zero.

Proof. The function 1_S (equal to 1 on S and 0 outside) is continuous on the open sets $\operatorname{Ext} S$ and $\operatorname{Int} S$ and fails to be continuous at each point of $\operatorname{Bd} S$. By Theorem 11.2, 1_S is integrable over a rectangle $Q \supset S$ iff $\operatorname{Bd} S$ has measure zero. ■

Theorem 14.2

(a) (Positivity) If S is rectifiable, $\mathrm{v}(S) \geq 0$.

(b) (Monotonicity) If S_1 and S_2 are rectifiable and if $S_1 \subset S_2$, then $\mathrm{v}(S_1) \leq \mathrm{v}(S_2)$.

(c) (Additivity) If S_1 and S_2 are rectifiable, so are $S_1 \cup S_2$ and $S_1 \cap S_2$, and $\mathrm{v}(S_1 \cup S_2) = \mathrm{v}(S_1) + \mathrm{v}(S_2) - \mathrm{v}(S_1 \cap S_2)$.

(d) Suppose S is rectifiable. Then $\mathrm{v}(S) = 0$ if and only if S has measure zero.

(e) If S is rectifiable, so is the set $A = \operatorname{Int} S$, and $\mathrm{v}(S) = \mathrm{v}(A)$.

(f) If S is rectifiable, and if $f : S \to \mathbb{R}$ is a bounded continuous function, then f is integrable over S.

Proof. (a), (b), (c) follow from Theorem 13.3.

(d) follows by applying Theorem 11.3 to the non-negative function 1_S.

(e) follows from Theorem 13.6, and (f) from Theorem 13.5 (here $E \subset \operatorname{Bd} S$, which has measure zero). ■

Terminology

Classically, this volume is called *content* (Jordan content), as opposed to *measure* (Lebesgue measure), the basis of the Lebesgue integral. Measure is defined for more sets, and the two agree when both are defined; a "set of measure zero" as defined here is exactly a set of Lebesgue measure zero, which need not be rectifiable. Munkres uses "rectifiable" for sets having volume (content).

Rectifiable sets are not easy to describe other than by Theorem 14.1. Bounded open or bounded closed sets need not be rectifiable:

Example 14.1

A bounded open set $A \subset \mathbb{R}$ such that $\operatorname{Bd} A$ does not have measure zero. Enumerate the rationals of $(0,1)$ as $q_1, q_2, \ldots$. Fix $0 < a < 1$ and for each i choose an open interval $(a_i, b_i) \subset (0,1)$ of length less than $a/2^i$ containing q_i. Let $A = \bigcup_i (a_i, b_i)$. Suppose $\operatorname{Bd} A$ has measure zero; let $\epsilon = 1 - a$ and cover $\operatorname{Bd} A$ by countably many open intervals of total length less than ϵ. Since $A \subset [0,1]$ contains every rational of $(0,1)$, $\bar{A} = [0,1]$; and $\bar{A} = A \cup \operatorname{Bd} A$. So these intervals, together with the intervals (a_i, b_i), form an open covering of $[0,1]$ of total length less than $\epsilon + \sum a/2^i = \epsilon + a = 1$. By compactness finitely many cover $[0,1]$, contradicting Corollary 10.5.

Definition 14.2 (simple region)

Let C be a compact rectifiable set in $\mathbb{R}^{n-1}$ and $\phi, \psi : C \to \mathbb{R}$ continuous with $\phi(\mathbf{x}) \le \psi(\mathbf{x})$ for $\mathbf{x} \in C$. The set

$$S = \{(\mathbf{x}, t) : \mathbf{x} \in C \text{ and } \phi(\mathbf{x}) \le t \le \psi(\mathbf{x})\}$$

is a **simple region** in $\mathbb{R}^n$. Nothing is special about the last coordinate: if $k + l = n - 1$ and $\mathbf{y}, \mathbf{z}$ denote general points of $\mathbb{R}^k, \mathbb{R}^l$, the set $\{(\mathbf{y}, t, \mathbf{z}) : (\mathbf{y}, \mathbf{z}) \in C,\ \phi(\mathbf{y}, \mathbf{z}) \le t \le \psi(\mathbf{y}, \mathbf{z})\}$ is also called a simple region.

Lemma 14.3

If S is a simple region in $\mathbb{R}^n$, then S is compact and rectifiable.

Proof.

Step 1. Let $G_\phi = \{(\mathbf{x}, t) : \mathbf{x} \in C,\ t = \phi(\mathbf{x})\}$ be the graph of ϕ, similarly G_ψ, and $D = \{(\mathbf{x}, t) : \mathbf{x} \in \operatorname{Bd} C,\ \phi(\mathbf{x}) \le t \le \psi(\mathbf{x})\}$. We show $\operatorname{Bd} S \subset G_\phi \cup G_\psi \cup D$. Each of these lies in S, so then $\operatorname{Bd} S \subset S$, S is closed, and being bounded, compact. Suppose $(\mathbf{x}_0, t_0)$ lies in none of the three sets. There are three cases:

(1) $\mathbf{x}_0 \notin C$;

(2) $\mathbf{x}_0 \in C$ and $t_0 < \phi(\mathbf{x}_0)$ or $t_0 > \psi(\mathbf{x}_0)$;

(3) $\mathbf{x}_0 \in \operatorname{Int} C$ and $\phi(\mathbf{x}_0) < t_0 < \psi(\mathbf{x}_0)$. In case (1) a neighborhood U of $\mathbf{x}_0$ is disjoint from C (C is closed), so $U \times \mathbb{R}$ is disjoint from S and $(\mathbf{x}_0, t_0) \in \operatorname{Ext} S$. In case (2), say $t_0 < \phi(\mathbf{x}_0)$: by continuity of ϕ there is a neighborhood W of $(\mathbf{x}_0, t_0)$ on which $\phi(\mathbf{x}) - t > 0$ for $\mathbf{x} \in C$; W is disjoint from S, so $(\mathbf{x}_0, t_0) \in \operatorname{Ext} S$; similarly if $t_0 > \psi(\mathbf{x}_0)$. In case (3) there is a neighborhood $U \times V$ of $(\mathbf{x}_0, t_0)$ with $U \subset C$ on which $t - \phi(\mathbf{x})$ and $\psi(\mathbf{x}) - t$ are positive; then $U \times V \subset S$ and $(\mathbf{x}_0, t_0) \in \operatorname{Int} S$.

Step 2. G_ϕ (and likewise G_ψ) has measure zero. Choose a rectangle $Q \supset C$ in $\mathbb{R}^{n-1}$. Given ϵ, let $\epsilon' = \epsilon / 2\,\mathrm{v}(Q)$. By uniform continuity of ϕ on the compact set C there is δ with $|\phi(\mathbf{x}) - \phi(\mathbf{y})| < \epsilon'$ for $\mathbf{x}, \mathbf{y} \in C$, $|\mathbf{x} - \mathbf{y}| < \delta$. Choose a partition P of Q of mesh less than δ. For each subrectangle R meeting C pick $\mathbf{x}_R \in R \cap C$ and let $I_R = [\phi(\mathbf{x}_R) - \epsilon', \phi(\mathbf{x}_R) + \epsilon']$. Then $R \times I_R$ contains every point $(\mathbf{x}, \phi(\mathbf{x}))$ with $\mathbf{x} \in C \cap R$, so these rectangles cover G_ϕ, with total volume $\sum_R \mathrm{v}(R)(2\epsilon') \le 2\epsilon'\,\mathrm{v}(Q) = \epsilon$.

Step 3. D has measure zero. Since ϕ, ψ are continuous on the compact set C, there is M with $-M \le \phi(\mathbf{x}) \le \psi(\mathbf{x}) \le M$. Given ϵ, cover $\operatorname{Bd} C$ (measure zero, as C is rectifiable) by rectangles $Q_1, Q_2, \ldots$ in $\mathbb{R}^{n-1}$ of total volume less than $\epsilon / 2M$; then the rectangles $Q_i \times [-M, M]$ cover D with total volume less than ϵ. ∎

Theorem 14.4 (Fubini's theorem for simple regions)

Let $S = \{(\mathbf{x}, t) : \mathbf{x} \in C \text{ and } \phi(\mathbf{x}) \le t \le \psi(\mathbf{x})\}$ be a simple region in $\mathbb{R}^n$. Let $f : S \to \mathbb{R}$ be a continuous function. Then f is integrable over S, and

$$\int_S f = \int_{\mathbf{x} \in C} \int_{t=\phi(\mathbf{x})}^{t=\psi(\mathbf{x})} f(\mathbf{x}, t).$$

Proof. Let $Q \times [-M, M]$ be a rectangle in $\mathbb{R}^n$ containing S. Since f is continuous and bounded on the compact rectifiable set S, it is integrable over S (Theorem 14.2(f)). For fixed $\mathbf{x}_0 \in Q$, the function $t \mapsto f_S(\mathbf{x}_0, t)$ is either identically zero (if $\mathbf{x}_0 \notin C$) or continuous at all but two points of $\mathbb{R}$; so it is integrable over $[-M, M]$, and Corollary 12.3 gives

$$\int_Q f_S = \int_{\mathbf{x} \in Q} \int_{t=-M}^{t=M} f_S(\mathbf{x}, t).$$

The inner integral vanishes if $\mathbf{x} \notin C$; and $f_S(\mathbf{x}, t)$ vanishes unless $\phi(\mathbf{x}) \le t \le \psi(\mathbf{x})$, in which case it equals $f(\mathbf{x}, t)$. So $\int_S f = \int_{\mathbf{x} \in C} \int_{t=\phi(\mathbf{x})}^{t=\psi(\mathbf{x})} f(\mathbf{x}, t)$. ∎

If S is not a simple region, one can often write it as a union of simple regions overlapping in sets of measure zero and use Corollary 13.4.

Example 14.2

A set $S \subset \mathbb{R}^2$ that is not a simple region (e.g. a "notched" polygon) can often be broken by a vertical line into simple regions overlapping in a set of measure zero; one then integrates over each piece and adds.

Example 14.3

The annulus $S = \{(x, y) : 1 \le x^2 + y^2 \le 4\}$ is not a simple region, but it is the union of two simple regions (its upper and lower halves) overlapping in a set of measure zero. The limits of integration are unpleasant; in practice one passes to polar coordinates—a special case of the change of variables theorem (Chapter 4).

Volume and rigid motions

Our discussion lacks one thing: why is volume independent of position? If $h : \mathbb{R}^n \to \mathbb{R}^n$ satisfies $\|h(\mathbf{x}) - h(\mathbf{y})\| = \|\mathbf{x} - \mathbf{y}\|$ (an **isometry**) and S is rectifiable, is $T = h(S)$ rectifiable with $\mathrm{v}(T) = \mathrm{v}(S)$? For instance, the square S with vertices $(0,0), (5,0), (5,5), (0,5)$ rotated through $\theta = \arctan \frac{3}{4}$ is the square T with vertices $(0,0), (4,3), (1,7), (-3,4)$. T is a simple region, but showing $\mathrm{v}(T) = 25$ directly (integrating $\psi - \phi$ over $[-3,4]$, or estimating $L(1_T, P) \le 25 \le U(1_T, P)$) is unappealing. The general result, proved in Chapter 4 (Theorem 20.7), is that isometries preserve volume.

§15. Improper Integrals

Motivation. Many natural integrals live on unbounded sets or have unbounded integrands, for example $\int_{\mathbb{R}^2} e^{-(x^2+y^2)}$. The extended integral over open sets handles them and is the version used in the change of variables theorem.

We now define $\int_S f$ when S or f is unbounded (an *improper integral*), but only for S open in $\mathbb{R}^n$.

Definition 15.1 (extended integral)

Let A be open in $\mathbb{R}^n$ and $f : A \to \mathbb{R}$ continuous. If $f \ge 0$ on A, the (**extended**) **integral** $\int_A f$ is the supremum of the numbers $\int_D f$ as D ranges over all compact rectifiable subsets of A, provided this supremum exists; then f is **integrable over A (in the extended sense)**. For general continuous f let

$$f_+(\mathbf{x}) = \max\{f(\mathbf{x}), 0\}, \qquad f_-(\mathbf{x}) = \max\{-f(\mathbf{x}), 0\};$$

f is integrable over A if both f_+ and f_- are, and then $\int_A f = \int_A f_+ - \int_A f_-$.

If A and f are bounded, $\int_A f$ could mean the extended or the ordinary integral. If the ordinary integral exists, so does the extended one and they agree (Theorem 15.4), but the extended integral may exist when the ordinary one does not.

Convention

If A is an open set in $\mathbb{R}^n$, then $\int_A f$ will denote the extended integral unless specifically stated otherwise.

Lemma 15.1

Let A be an open set in $\mathbb{R}^n$. Then there exists a sequence $C_1, C_2, \ldots$ of compact rectifiable subsets of A whose union is A, such that $C_N \subset \operatorname{Int} C_{N+1}$ for each N.

Proof. Let d be the sup metric and $B = \mathbb{R}^n - A$. For each positive integer N let

$$D_N = \{\mathbf{x} : d(\mathbf{x}, B) \ge 1/N \text{ and } d(\mathbf{x}, \mathbf{0}) \le N\}.$$

Since $d(\mathbf{x}, B)$ and $d(\mathbf{x}, \mathbf{0})$ are continuous (proof of Theorem 4.6), D_N is closed; it lies in the cube of radius N, so it is compact. $D_N \subset A$, since $d(\mathbf{x}, B) \ge 1/N$ excludes $\mathbf{x} \in B$. The D_N cover A: for $\mathbf{x} \in A$, $d(\mathbf{x}, B) > 0$ (as A is open), so $d(\mathbf{x}, B) \ge 1/N$ and $d(\mathbf{x}, \mathbf{0}) \le N$ for large N. The set

$$A_{N+1} = \{\mathbf{x} : d(\mathbf{x}, B) > 1/(N+1) \text{ and } d(\mathbf{x}, \mathbf{0}) < N+1\}$$

is open, contained in D_{N+1}, and contains D_N; hence $D_N \subset \operatorname{Int} D_{N+1}$.

The D_N may not be rectifiable, so we modify them. For each $\mathbf{x} \in D_N$ choose a closed cube centered at $\mathbf{x}$ and contained in $\operatorname{Int} D_{N+1}$. The interiors of these cubes cover D_N; choose finitely many whose interiors cover D_N and let C_N be their union. As a finite union of rectangles, C_N is compact and rectifiable, and $D_N \subset \operatorname{Int} C_N \subset C_N \subset \operatorname{Int} D_{N+1}$. Hence $\bigcup C_N = A$ and $C_N \subset \operatorname{Int} C_{N+1}$. ■

Theorem 15.2

Let A be open in $\mathbb{R}^n$; let $f : A \to \mathbb{R}$ be continuous. Choose a sequence C_N of compact rectifiable subsets of A whose union is A such that $C_N \subset \operatorname{Int} C_{N+1}$ for each N. Then f is integrable over A if and only if the sequence $\int_{C_N} |f|$ is bounded. In this case,

$$\int_A f = \lim_{N\to\infty} \int_{C_N} f.$$

In particular, f is integrable over A iff $|f|$ is.

Proof.

Step 1. Let $f \geq 0$, so $f = |f|$. The sequence $\int_{C_N} f$ is increasing (monotonicity), so it converges iff it is bounded. If f is integrable over A, then since each C_N is a compact rectifiable subset of A, $\int_{C_N} f \leq \int_A f$; so the sequence is bounded and $\lim \int_{C_N} f \leq \int_A f$. Conversely, suppose the sequence is bounded, and let D be any compact rectifiable subset of A. D is covered by the open sets $\operatorname{Int} C_1 \subset \operatorname{Int} C_2 \subset \cdots$, hence by finitely many, hence by one, $\operatorname{Int} C_M$. Then $\int_D f \leq \int_{C_M} f \leq \lim_N \int_{C_N} f$. So f is integrable over A and $\int_A f \leq \lim \int_{C_N} f$.

Step 2. For arbitrary continuous f: f is integrable over A iff f_+ and f_- are, iff the sequences $\int_{C_N} f_+$, $\int_{C_N} f_-$ are bounded (Step 1). Since $0 \leq f_\pm \leq |f|$ and $|f| = f_+ + f_-$, this holds iff $\int_{C_N} |f|$ is bounded. Then $\int_{C_N} f_\pm \to \int_A f_\pm$, and $\int_{C_N} f = \int_{C_N} f_+ - \int_{C_N} f_-$ converges to $\int_A f_+ - \int_A f_- = \int_A f$. ■

Theorem 15.3

Let A be an open set in $\mathbb{R}^n$. Let $f, g : A \to \mathbb{R}$ be continuous functions.

(a) (Linearity) If f and g are integrable over A, so is $af+bg$; and $\int_A (af+bg) = a\int_A f + b\int_A g$.

(b) (Comparison) Let f and g be integrable over A. If $f(\mathbf{x}) \leq g(\mathbf{x})$ for $\mathbf{x} \in A$, then $\int_A f \leq \int_A g$. In particular, $|\int_A f| \leq \int_A |f|$.

(c) (Monotonicity) Assume B is open and $B \subset A$. If f is non-negative on A and integrable over A, then f is integrable over B and $\int_B f \leq \int_A f$.

(d) (Additivity) Suppose A and B are open in $\mathbb{R}^n$ and f is continuous on $A \cup B$. If f is integrable on A and B, then f is integrable on $A \cup B$ and $A \cap B$, and $\int_{A\cup B} f = \int_A f + \int_B f - \int_{A\cap B} f$.

Here all integrals are extended integrals.

Proof. Let C_N be compact rectifiable sets with union A and $C_N \subset \operatorname{Int} C_{N+1}$ (Lemma 15.1).

(a) By the properties of the ordinary integral, $\int_{C_N} |af+bg| \leq |a| \int_{C_N} |f| + |b| \int_{C_N} |g|$, which is bounded; so $af + bg$ is integrable (Theorem 15.2). Linearity follows by taking limits in $\int_{C_N} (af + bg) = a\int_{C_N} f + b\int_{C_N} g$.

(b) Take limits in $\int_{C_N} f \leq \int_{C_N} g$.

(c) A compact rectifiable subset D of B is one of A, so $\int_D f \leq \int_A f$; hence f is integrable over B with $\int_B f \leq \int_A f$.

(d) Let D_N be a corresponding sequence for B, and $E_N = C_N \cup D_N$, $F_N = C_N \cap D_N$. These are compact rectifiable with unions $A \cup B$ and $A \cap B$. Also $E_N \subset \operatorname{Int} E_{N+1}$: if $\mathbf{x} \in C_N$, a neighborhood of

$\mathbf{x}$ lies in $C_{N+1} \subset E_{N+1}$; similarly for D_N. And $F_N \subset \operatorname{Int} F_{N+1}$: for $\mathbf{x} \in F_N$ there are neighborhoods $U \subset C_{N+1}$, $V \subset D_{N+1}$, so $U \cap V \subset F_{N+1}$. By additivity of the ordinary integral,

$$\int_{E_N} f = \int_{C_N} f + \int_{D_N} f - \int_{F_N} f. \qquad (*)$$

Applying $(*)$ to $|f|$, $\int_{E_N} |f|$ and $\int_{F_N} |f|$ are bounded by $\int_{C_N} |f| + \int_{D_N} |f|$. So f is integrable over $A \cup B$ and $A \cap B$, and the formula follows by taking limits in $(*)$. ■

Theorem 15.4

Let A be a bounded open set in $\mathbb{R}^n$; let $f : A \to \mathbb{R}$ be a bounded continuous function. Then the extended integral $\int_A f$ exists. If the ordinary integral $\int_A f$ also exists, then these two integrals are equal.

Proof. Let Q be a rectangle containing A.

Step 1. Choose M with $|f| \leq M$ on A. For any compact rectifiable $D \subset A$, $\int_D |f| \leq \int_D M \leq M\, \mathrm{v}(Q)$. So f is integrable over A in the extended sense.

Step 2. Let $f \geq 0$ and suppose the ordinary integral exists; it equals $\int_Q f_A$. For compact rectifiable $D \subset A$, $\int_D f = \int_D f_A \leq \int_Q f_A$ by monotonicity, so (extended) $\int_A f \leq$ (ordinary) $\int_A f$. On the other hand, let P be a partition of Q, let $R_1, \ldots, R_k$ be the subrectangles determined by P that lie in A, and $D = R_1 \cup \cdots \cup R_k$. Since $m_R(f_A) = m_R(f)$ for $R \subset A$ and $m_R(f_A) \leq 0$ otherwise (as f_A vanishes at some point of R),

$$L(f_A, P) \leq \sum_{i=1}^{k} m_{R_i}(f)\, \mathrm{v}(R_i) \leq \sum_{i=1}^{k} \int_{R_i} f = \int_D f \leq \text{(extended)} \int_A f,$$

using comparison and additivity (Corollary 13.4). As P is arbitrary, (ordinary) $\int_A f \leq$ (extended) $\int_A f$.

Step 3. In general, if the ordinary integral exists, so do the ordinary integrals of f_+, f_- (Lemma 13.2), and (ordinary) $\int_A f =$ (ordinary) $\int_A f_+ -$ (ordinary) $\int_A f_- =$ (extended) $\int_A f_+ -$ (extended) $\int_A f_- =$ (extended) $\int_A f$. ■

Example 15.1

If A is a bounded open set and f a bounded continuous function on A, the extended integral exists but the ordinary one may not. With A the open subset of $\mathbb{R}$ of Example 14.1, $\operatorname{Bd} A$ does not have measure zero, so the ordinary integral $\int_A 1$ does not exist (Theorem 14.1), although the extended integral $\int_A 1$ does.

Corollary 15.5

Let S be a bounded set in $\mathbb{R}^n$; let $f : S \to \mathbb{R}$ be a bounded continuous function. If f is integrable over S in the ordinary sense, then

$$\text{(ordinary)} \int_S f = \text{(extended)} \int_{\operatorname{Int} S} f.$$

Proof. Apply Theorems 13.6 and 15.4. ■

So theorems about extended integrals have consequences for ordinary integrals; the change of variables theorem (Chapter 4) is an important example. A third formulation of the extended integral is given in Chapter 4 (via partitions of unity). The following one is often convenient in practice:

Theorem 15.6

Let A be open in $\mathbb{R}^n$; let $f : A \to \mathbb{R}$ be continuous. Let $U_1 \subset U_2 \subset \cdots$ be a sequence of open sets whose union is A. Then $\int_A f$ exists if and only if the sequence $\int_{U_N} |f|$ exists and is bounded; in this case,

$$\int_A f = \lim_{N\to\infty} \int_{U_N} f.$$

Proof. As usual it suffices to consider $f \geq 0$. If $\int_A f$ exists, monotonicity (Theorem 15.3(c)) shows f is integrable over each U_N with $\int_{U_N} f \leq \int_A f$; so the increasing sequence $\int_{U_N} f$ converges, and $\lim \int_{U_N} f \leq \int_A f$. Conversely, suppose $\int_{U_N} f$ exists and is bounded. A compact rectifiable $D \subset A$ is covered by $U_1 \subset U_2 \subset \cdots$, hence by one U_M; by definition $\int_D f \leq \int_{U_M} f \leq \lim_N \int_{U_N} f$. As D is arbitrary, $\int_A f \leq \lim_N \int_{U_N} f$. ■

Usually one chooses U_N rectifiable with f bounded on U_N; then $\int_{U_N} f$ exists as an ordinary integral (hence as an extended one) and can be computed by familiar techniques.

Example 15.2

Let $A = \{(x, y) : x > 1,\ y > 1\}$ and $f(x, y) = 1/x^2y^2$; f is bounded on A but A is unbounded. Let $U_N = (1, N)^2$; U_N is rectifiable, f is bounded and continuous on $\bar{U}_N$, so $\int_{U_N} f$ exists as an ordinary integral and Fubini's theorem applies:

$$\int_{U_N} f = \int_{x=1}^{x=N} \int_{y=1}^{y=N} 1/x^2y^2 = ((N-1)/N)^2.$$

Hence $\int_A f = 1$. (One could also use Theorem 15.2 with $C_N = [(N+1)/N, N]^2$.)

Example 15.3

Let $B = (0, 1)^2$ and $f(x, y) = 1/x^2y^2$. Now B is bounded but f is unbounded near each point of the x- and y-axes. With $U_N = (1/N, 1)^2$, f is bounded on U_N and

$$\int_{U_N} f = (-1 + N)^2,$$

which is unbounded. Hence $\int_B f$ does not exist.

Chapter 4. Changes of Variables

Overview

The substitution rule of calculus, e.g. $\int_0^1 (2x^2+1)^{10}(4x)\,dx = \int_1^3 y^{10}\,dy$ with $y = 2x^2+1$, is generalized in two ways:

(1) to n-dimensional integrals, and (2) to the *extended* integral over open sets (no serious restriction, by Corollary 15.5). The result is the *change of variables theorem* (§17), proved in §19 using partitions of unity (§16) and the local factorization of diffeomorphisms into primitive ones (§18). §20 gives the geometric meaning of the determinant and the invariance of volume under isometries.

§16. Partitions of Unity

Instead of breaking the *set* A into compact rectifiable pieces C_N, we now break the *function* f into functions f_N each vanishing outside a compact set, and sum the integrals $\int_A f_N$. This approach recurs throughout the book.

Lemma 16.1

Let Q be a rectangle in $\mathbb{R}^n$. There is a C^∞ function $\phi : \mathbb{R}^n \to \mathbb{R}$ such that $\phi(\mathbf{x}) > 0$ for $\mathbf{x} \in \operatorname{Int} Q$ and $\phi(\mathbf{x}) = 0$ otherwise.

Proof. Let $f(x) = e^{-1/x}$ for $x > 0$ and $f(x) = 0$ otherwise. Then $f(x) > 0$ for $x > 0$, and f is of class C^∞ (see the remark below). Let $g(x) = f(x) \cdot f(1-x)$; g is C^∞, positive for $0 < x < 1$ and zero otherwise. For $Q = [a_1, b_1] \times \cdots \times [a_n, b_n]$ define

$$\phi(\mathbf{x}) = g\Big(\frac{x_1 - a_1}{b_1 - a_1}\Big) \cdot g\Big(\frac{x_2 - a_2}{b_2 - a_2}\Big) \cdots g\Big(\frac{x_n - a_n}{b_n - a_n}\Big).$$

■

Why f is C^∞ [Supplement]

For $x > 0$, induction shows $f^{(k)}(x) = p_k(1/x)e^{-1/x}$ with p_k a polynomial. Since $t^m e^{-t} \to 0$ as $t \to \infty$ for every m, $f^{(k)}(x) \to 0$ as $x \to 0^+$; and $f^{(k)}(0) = 0$ follows inductively from $\lim_{x\to 0^+} f^{(k)}(x)/x = \lim_{t\to\infty} t\,p_k(t)e^{-t} = 0$.

Lemma 16.2

Let $\mathcal{A}$ be a collection of open sets in $\mathbb{R}^n$; let A be their union. Then there exists a countable collection $Q_1, Q_2, \ldots$ of rectangles contained in A such that:

(1) The sets $\operatorname{Int} Q_i$ cover A.

(2) Each Q_i is contained in an element of $\mathcal{A}$.

(3) Each point of A has a neighborhood that intersects only finitely many of the sets Q_i.

(1) and (2) are easy; (3), the *local finiteness condition*, is the point.

Proof.

Step 1. Let $D_1, D_2, \ldots$ be compact subsets of A with union A and $D_i \subset \operatorname{Int} D_{i+1}$ (Lemma 15.1); put $D_i = \emptyset$ for $i \le 0$. Let $B_i = D_i - \operatorname{Int} D_{i-1}$. B_i is bounded and closed (the intersection of the

closed sets D_i and $\mathbb{R}^n - \operatorname{Int} D_{i-1}$), hence compact, and it is disjoint from the closed set D_{i-2} since $D_{i-2} \subset \operatorname{Int} D_{i-1}$. For each $\mathbf{x} \in B_i$ choose a closed cube $C_\mathbf{x}$ centered at $\mathbf{x}$, contained in A, disjoint from D_{i-2}, and small enough to lie in an element of $\mathcal{A}$. The interiors of the $C_\mathbf{x}$ cover B_i; choose finitely many whose interiors cover B_i, and call this finite collection $\mathcal{C}_i$.

Step 2. Let $\mathcal{C} = \mathcal{C}_1 \cup \mathcal{C}_2 \cup \cdots$, a countable collection of cubes, each contained in an element of $\mathcal{A}$. Given $\mathbf{x} \in A$, let i be the smallest integer with $\mathbf{x} \in \operatorname{Int} D_i$; then $\mathbf{x} \in B_i = D_i - \operatorname{Int} D_{i-1}$, so $\mathbf{x}$ lies interior to a cube of $\mathcal{C}_i$. Finally, every cube of $\mathcal{C}_{i+2}, \mathcal{C}_{i+3}, \ldots$ is disjoint from D_i by construction, so the open set $\operatorname{Int} D_i \ni \mathbf{x}$ meets only cubes of $\mathcal{C}_1, \ldots, \mathcal{C}_{i+1}$, finitely many. ■

Local finiteness holds at each point of A, but *not* at points of $\operatorname{Bd} A$: every neighborhood of such a point meets infinitely many cubes of $\mathcal{C}$.

Definition 16.1 (support)

If $\phi : \mathbb{R}^n \to \mathbb{R}$, the **support** of ϕ is the closure of $\{\mathbf{x} : \phi(\mathbf{x}) \neq 0\}$. Equivalently, if $\mathbf{x} \notin \operatorname{Support} \phi$, there is a neighborhood of $\mathbf{x}$ on which ϕ vanishes identically.

Theorem 16.3 (Existence of a partition of unity)

Let $\mathcal{A}$ be a collection of open sets in $\mathbb{R}^n$; let A be their union. There exists a sequence $\phi_1, \phi_2, \ldots$ of continuous functions $\phi_i : \mathbb{R}^n \to \mathbb{R}$ such that:

(1) $\phi_i(\mathbf{x}) \geq 0$ for all $\mathbf{x}$.

(2) The set $S_i = \operatorname{Support} \phi_i$ is contained in A.

(3) Each point of A has a neighborhood that intersects only finitely many of the sets S_i.

(4) $\sum_{i=1}^{\infty} \phi_i(\mathbf{x}) = 1$ for each $\mathbf{x} \in A$.

(5) The functions ϕ_i are of class C^∞.

(6) The sets S_i are compact.

(7) For each i, the set S_i is contained in an element of $\mathcal{A}$.

Definition 16.2 (partition of unity)

A collection $\{\phi_i\}$ satisfying (1)–(4) is a **partition of unity** on A. If it satisfies (5) it is **of class** C^∞; if (6), it has **compact supports**; if (7), it is **dominated by** the collection $\mathcal{A}$.

Proof. Let $Q_1, Q_2, \ldots$ be rectangles as in Lemma 16.2, and for each i let $\psi_i : \mathbb{R}^n \to \mathbb{R}$ be a C^∞ function positive on $\operatorname{Int} Q_i$ and zero elsewhere (Lemma 16.1). Then $\psi_i \geq 0$, $\operatorname{Support} \psi_i = Q_i$ is a compact subset of A contained in an element of $\mathcal{A}$, and each point of A has a neighborhood meeting only finitely many Q_i. So $\{\psi_i\}$ satisfies everything except (4). By (3), for $\mathbf{x} \in A$ only finitely many $\psi_i(\mathbf{x})$ are nonzero, so $\lambda(\mathbf{x}) = \sum_i \psi_i(\mathbf{x})$ converges trivially; near each point of A it is a finite sum of C^∞ functions, so λ is C^∞ on A. Also $\lambda(\mathbf{x}) > 0$ for $\mathbf{x} \in A$, since $\mathbf{x} \in \operatorname{Int} Q_i$ for some i. Define $\phi_i(\mathbf{x}) = \psi_i(\mathbf{x})/\lambda(\mathbf{x})$ (and 0 outside A; this is C^∞ on $\mathbb{R}^n$ since ψ_i vanishes outside the compact set $Q_i \subset A$). ■

By (1) and (4), the numbers $\phi_i(\mathbf{x})$ "partition unity" into non-negative pieces. Local finiteness implies that for any compact $C \subset A$, there is an open set about C on which all but finitely many ϕ_i vanish identically (cover C by finitely many such neighborhoods).

Example 16.1

Let $f(x) = (1+\cos x)/2$ for $-\pi \le x \le \pi$ and $f(x) = 0$ otherwise; f is of class C^1. For $m \ge 0$ let $\phi_{2m+1}(x) = f(x - m\pi)$, and for $m \ge 1$ let $\phi_{2m}(x) = f(x + m\pi)$. Then $\{\phi_i\}$ is a partition of unity on $\mathbb{R}$: each support is a compact interval $[k\pi, (k+2)\pi]$, each point of $\mathbb{R}$ has a neighborhood meeting at most three supports, and $\sum \phi_i(x) = 1$ (since $f(x) + f(x-\pi) = 1$ on $[0,\pi]$, as $\cos(x-\pi) = -\cos x$).

Lemma 16.4

Let A be open in $\mathbb{R}^n$; let $f : A \to \mathbb{R}$ be continuous. If f vanishes outside the compact subset C of A, then the integrals $\int_A f$ and $\int_C f$ exist and are equal.

Proof. $\int_C f$ exists because C is bounded and f_C, which equals f on A and vanishes outside C, is continuous and bounded on $\mathbb{R}^n$. Let C_N be compact rectifiable sets with union A and $C_N \subset \operatorname{Int} C_{N+1}$. C is covered by finitely many $\operatorname{Int} C_N$, hence by one, $\operatorname{Int} C_M$. Since f vanishes outside C, $\int_C f = \int_{C_N} f$ for $N \ge M$. Applying this to $|f|$, $\lim \int_{C_N} |f|$ exists, so f is integrable over A (Theorem 15.2), and $\int_A f = \lim \int_{C_N} f = \int_C f$. ∎

Theorem 16.5

Let A be open in $\mathbb{R}^n$; let $f : A \to \mathbb{R}$ be continuous. Let $\{\phi_i\}$ be a partition of unity on A having compact supports. The integral $\int_A f$ exists if and only if the series

$$\sum_{i=1}^{\infty} \Big[\int_A \phi_i |f| \Big]$$

converges; in this case,

$$\int_A f = \sum_{i=1}^{\infty} \Big[\int_A \phi_i f \Big].$$

Note that $\int_A \phi_i f$ exists and equals the ordinary integral $\int_{S_i} \phi_i f$, by Lemma 16.4.

Proof. First let $f \ge 0$.

Step 1. Suppose $\sum_i [\int_A \phi_i f]$ converges. Let D be a compact rectifiable subset of A. There is M such that ϕ_i vanishes identically on D for $i > M$, so $f(\mathbf{x}) = \sum_{i=1}^{M} \phi_i(\mathbf{x}) f(\mathbf{x})$ for $\mathbf{x} \in D$. Hence

$$\int_D f = \sum_{i=1}^{M} \Big[\int_D \phi_i f \Big] \le \sum_{i=1}^{M} \Big[\int_{D \cup S_i} \phi_i f \Big] = \sum_{i=1}^{M} \Big[\int_A \phi_i f \Big] \le \sum_{i=1}^{\infty} \Big[\int_A \phi_i f \Big],$$

by linearity, monotonicity and Lemma 16.4. So f is integrable over A and $\int_A f \le \sum_i [\int_A \phi_i f]$.

Step 2. Suppose f is integrable over A. Given N, $D = S_1 \cup \cdots \cup S_N$ is compact, and for $i \le N$, $\phi_i f$ vanishes outside D, so $\int_A \phi_i f = \int_D \phi_i f$. Thus

$$\sum_{i=1}^{N} \Big[\int_A \phi_i f \Big] = \int_D \Big[\sum_{i=1}^{N} \phi_i f \Big] \le \int_D f \le \int_A f,$$

using $\sum_{i \le N} \phi_i \le 1$. The partial sums are bounded, so the series converges with sum $\le \int_A f$.

Step 3. For arbitrary continuous f: by Theorem 15.2, $\int_A f$ exists iff $\int_A |f|$ exists, iff (Steps 1, 2) $\sum [\int_A \phi_i |f|]$ converges. If $\int_A f$ exists,

$$\int_A f = \int_A f_+ - \int_A f_- = \sum_{i=1}^{\infty} \Big[\int_A \phi_i f_+ \Big] - \sum_{i=1}^{\infty} \Big[\int_A \phi_i f_- \Big] = \sum_{i=1}^{\infty} \Big[\int_A \phi_i f \Big],$$

since convergent series can be added term by term. ∎

§17. The Change of Variables Theorem

Motivation. In one variable, $\int f(g(u))g'(u)du = \int f(x)dx$. In $\mathbb{R}^n$ the factor $g'(u)$ becomes $|\det Dg(u)|$, the factor by which g scales small volumes. This chapter states and proves this formula, which underlies polar and spherical coordinates and later the integration of forms.

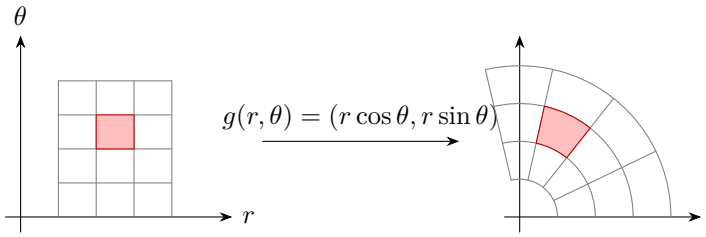


Figure. *A small rectangle $\Delta r\,\Delta\theta$ is mapped by polar coordinates to an approximate rectangle of area $r\,\Delta r\,\Delta\theta$; the factor $r = |\det Dg|$ is the local volume-scaling factor in the change of variables formula.*

Recall the convention $\int_b^a f = -\int_a^b f$.

Theorem 17.1 (Substitution rule)

Let $I = [a, b]$. Let $g : I \to \mathbb{R}$ be a function of class C^1, with $g'(x) \neq 0$ for $x \in (a, b)$. Then the set $g(I)$ is a closed interval J with end points $g(a)$ and $g(b)$. If $f : J \to \mathbb{R}$ is continuous, then

$$\int_{g(a)}^{g(b)} f = \int_a^b (f \circ g)g', \qquad \text{or equivalently,} \qquad \int_J f = \int_I (f \circ g)|g'|.$$

Proof. By continuity of g' and the intermediate-value theorem, $g' > 0$ on all of (a, b) or $g' < 0$ on all of (a, b); by the mean-value theorem g is strictly increasing or strictly decreasing, hence one-to-one, and $g(a) < g(b)$ or $g(a) > g(b)$ accordingly. Let $J = [c, d]$ be the interval with end points $g(a), g(b)$; by the intermediate-value theorem g carries I onto J. Let $F(y) = \int_c^y f$; then $F' = f$ by the fundamental theorem of calculus, and $h = F \circ g$ has $h'(x) = f(g(x))g'(x)$, which is continuous. Hence

$$\int_a^b (f \circ g)g' = h(b) - h(a) = F(g(b)) - F(g(a)) = \int_c^{g(b)} f - \int_c^{g(a)} f = \int_{g(a)}^{g(b)} f. \qquad (*)$$

If $g' > 0$, then $J = [g(a), g(b)]$ and $|g'| = g'$, so $(*)$ reads

$$\int_I (f \circ g)|g'| = \int_J f. \qquad (**)$$

If $g' < 0$, then $J = [g(b), g(a)]$ and $|g'| = -g'$, and $(*)$ again becomes $(**)$. ∎

Example 17.1

For $\int_{x=0}^{x=1}(2x^2 + 1)^{10}(4x)$, set $f(y) = y^{10}$, $g(x) = 2x^2 + 1$; $g'(x) = 4x > 0$ for $0 < x < 1$, so $\int_0^1 (2x^2 + 1)^{10}(4x) = \int_0^1 f(g(x))g'(x) = \int_1^3 y^{10}$.

Example 17.2

For $\int_{-1}^1 (1 - y^2)^{-1/2}$, calculus sets $y = g(x) = \sin x$, $-\pi/2 \leq x \leq \pi/2$; $g'(x) = \cos x > 0$ on

$(-\pi/2, \pi/2)$, and formally

$$\int_{-1}^{1} \frac{1}{(1-y^2)^{1/2}} = \int_{-\pi/2}^{\pi/2} \frac{\cos x}{(1-\sin^2 x)^{1/2}} = \int_{-\pi/2}^{\pi/2} 1 = \pi.$$

But the substitution rule does *not* apply: $f(y) = (1-y^2)^{-1/2}$ is not continuous (not even bounded) on $[-1,1]$; the integral is improper. This is one reason to prove the theorem for the extended integral.

Definition 17.1 (change of variables, diffeomorphism)

Let A be open in $\mathbb{R}^n$. A one-to-one function $g : A \to \mathbb{R}^n$ of class C^r with $\det Dg(\mathbf{x}) \neq 0$ for $\mathbf{x} \in A$ is a **change of variables** in $\mathbb{R}^n$. If A and B are open in $\mathbb{R}^n$ and $g : A \to B$ is a one-to-one map of A onto B such that g and g^{-1} are of class C^r, then g is a **diffeomorphism** (of class C^r).

These notions coincide. If g is a diffeomorphism, the chain rule shows Dg is non-singular, so g is a change of variables. Conversely, if $g : A \to \mathbb{R}^n$ is a change of variables, Theorem 8.2 shows $B = g(A)$ is open and $g^{-1} : B \to A$ is of class C^r.

Theorem 17.2 (Change of variables theorem)

Let $g : A \to B$ be a diffeomorphism of open sets in $\mathbb{R}^n$. Let $f : B \to \mathbb{R}$ be a continuous function. Then f is integrable over B if and only if the function $(f \circ g)|\det Dg|$ is integrable over A; in this case,

$$\int_B f = \int_A (f \circ g)|\det Dg|.$$

For $n = 1$, Dg is the 1 by 1 matrix $[g']$, so this contains the substitution rule; since the integrals are extended, it also justifies Example 17.2. The proof is in §19.

Example 17.3 (polar coordinates)

Let $B = \{(x,y) : x > 0,\ y > 0,\ x^2 + y^2 < a^2\}$. The **polar coordinate transformation** $g : \mathbb{R}^2 \to \mathbb{R}^2$, $g(r,\theta) = (r\cos\theta, r\sin\theta)$, has $\det Dg(r,\theta) = r$ and carries the open rectangle $A = \{(r,\theta) : 0 < r < a,\ 0 < \theta < \pi/2\}$ one-to-one onto B. Since $\det Dg = r > 0$ on A, $g : A \to B$ is a diffeomorphism, and

$$\int_B x^2y^2 = \int_A (r\cos\theta)^2 (r\sin\theta)^2 r;$$

the latter exists as an ordinary integral and is evaluated by Fubini's theorem.

Example 17.4

To integrate x^2y^2 over $W = \{(x,y) : x^2 + y^2 < a^2\}$, note that g is not a diffeomorphism of an open set onto W. It is a diffeomorphism of $U = (0,a) \times (0, 2\pi)$ onto $V = \{(x,y) : x^2 + y^2 < a^2$, and $x < 0$ if $y = 0\}$, which is W with the non-negative x-axis deleted. Since that axis has measure zero, $\int_W x^2y^2 = \int_V x^2y^2$, which is expressed as an integral over U by polar coordinates.

Example 17.5 (spherical coordinates)

Let $B = \{(x,y,z) : x > 0,\ y > 0,\ x^2+y^2+z^2 < a^2\}$. The **spherical coordinate transformation** is

$$g(\rho,\phi,\theta) = (\rho\sin\phi\cos\theta,\ \rho\sin\phi\sin\theta,\ \rho\cos\phi), \qquad \det Dg = \rho^2\sin\phi,$$

positive for $0<\phi<\pi$, $\rho\neq 0$. g carries $A = \{(\rho,\phi,\theta) : 0<\rho<a,\ 0<\phi<\pi,\ 0<\theta<\pi/2\}$ one-to-one onto B, so

$$\int_B x^2 z = \int_A (\rho\sin\phi\cos\theta)^2(\rho\cos\phi)\rho^2\sin\phi,$$

evaluated by Fubini's theorem.

§18. Diffeomorphisms in $\mathbb{R}^n$

Motivation. To prove change of variables we break a general diffeomorphism into simple pieces, each changing only one coordinate. This section develops the local decomposition needed for that reduction.

Two basic facts: the image of a compact rectifiable set under a diffeomorphism is compact and rectifiable; and any diffeomorphism factors locally into "primitive" diffeomorphisms.

Lemma 18.1

Let A be open in $\mathbb{R}^n$; let $g : A \to \mathbb{R}^n$ be a function of class C^1. If the subset E of A has measure zero in $\mathbb{R}^n$, then the set $g(E)$ also has measure zero in $\mathbb{R}^n$.

Proof.

Step 1. Let $\epsilon,\delta > 0$. If S has measure zero, S can be covered by countably many closed cubes, each of width less than δ, of total volume less than ϵ. It suffices to show that a rectangle $Q = [a_1,b_1]\times\cdots\times[a_n,b_n]$ can be covered by finitely many cubes of width $< \delta$ with total volume $< 2\,\mathrm{v}(Q)$. Choose $\lambda > 0$ with $Q_\lambda = [a_1-\lambda,b_1+\lambda]\times\cdots\times[a_n-\lambda,b_n+\lambda]$ of volume $< 2\,\mathrm{v}(Q)$. Choose N with $1/N$ less than both δ and λ. Among the numbers m/N ($m\in\mathbb{Z}$), let c_i be the largest with $c_i \le a_i$ and d_i the smallest with $d_i \ge b_i$; then $[a_i,b_i]\subset[c_i,d_i]\subset[a_i-\lambda,b_i+\lambda]$. The rectangle $Q' = [c_1,d_1]\times\cdots\times[c_n,d_n]$ contains Q, has $\mathrm{v}(Q') < 2\,\mathrm{v}(Q)$, and is partitioned by the points m/N into cubes of width $1/N<\delta$, whose total volume is $\mathrm{v}(Q')$ (Theorem 10.4).

Step 2. Let C be a closed cube in A with $|Dg(\mathbf{x})|\le M$ on C. If C has width w, then $g(C)$ lies in a closed cube of width $(nM)w$. Indeed, let $\mathbf{a}$ be the center of C. For $\mathbf{x}\in C$ the mean-value theorem gives $\mathbf{c}_j$ on the segment from $\mathbf{a}$ to $\mathbf{x}$ with $g_j(\mathbf{x})-g_j(\mathbf{a}) = Dg_j(\mathbf{c}_j)\cdot(\mathbf{x}-\mathbf{a})$, so

$$|g_j(\mathbf{x})-g_j(\mathbf{a})| \le n|Dg_j(\mathbf{c}_j)|\cdot|\mathbf{x}-\mathbf{a}| \le nM(w/2).$$

So $g(\mathbf{x})$ lies in the cube $\{\mathbf{y} : |\mathbf{y}-g(\mathbf{a})|\le nM(w/2)\}$ of width $(nM)w$.

Step 3. Let $E\subset A$ have measure zero. Let C_i be compact sets with union A and $C_i\subset \operatorname{Int} C_{i+1}$; with $E_k = C_k\cap E$ it suffices to show each $g(E_k)$ has measure zero. Given $\epsilon>0$, choose $\delta>0$ so that the δ-neighborhood of C_k (sup metric) lies in $\operatorname{Int} C_{k+1}$ (Theorem 4.6), and M with $|Dg(\mathbf{x})|\le M$ for $\mathbf{x}\in C_{k+1}$. By Step 1 cover E_k by countably many cubes of width $<\delta$ with total volume less than $\epsilon' = \epsilon/(nM)^n$; let $D_1,D_2,\ldots$ be those that actually meet E_k. Each D_i has width $<\delta$, so it lies in C_{k+1}, where $|Dg|\le M$; by Step 2, $g(D_i)$ lies in a cube D_i' of width nM(width D_i), so $\mathrm{v}(D_i') = (nM)^n\,\mathrm{v}(D_i)$. The cubes D_i' cover $g(E_k)$ with total volume less than $(nM)^n\epsilon' = \epsilon$. ∎

Example 18.1

Differentiability is needed: if g is merely continuous, the image of a set of measure zero need not have measure zero. There is a continuous map $f : [0,1] \to [0,1]^2$ whose image is the entire square (the *Peano space-filling curve*).

Theorem 18.2

Let $g : A \to B$ be a diffeomorphism of class C^r, where A and B are open sets in $\mathbb{R}^n$. Let D be a compact subset of A, and let $E = g(D)$.

(a) We have $g(\operatorname{Int} D) = \operatorname{Int} E$ and $g(\operatorname{Bd} D) = \operatorname{Bd} E$.

(b) If D is rectifiable, so is E.

Proof. (a) Since g^{-1} is continuous, g maps open sets of A to open sets of B (hence of $\mathbb{R}^n$). So $g(\operatorname{Int} D)$ is open and contained in $g(D) = E$:

$$g(\operatorname{Int} D) \subset \operatorname{Int} E. \tag{1}$$

Similarly g carries the open set $(\operatorname{Ext} D) \cap A$ onto an open set, disjoint from $g(D) = E$ since g is one-to-one:

$$g((\operatorname{Ext} D) \cap A) \subset \operatorname{Ext} E. \tag{2}$$

Hence

$$g(\operatorname{Bd} D) \supset \operatorname{Bd} E. \tag{3}$$

For let $\mathbf{y} \in \operatorname{Bd} E$. E is compact (continuous image of compact D), hence closed, so $\mathbf{y} \in E \subset B$; let $\mathbf{x} \in A$ with $g(\mathbf{x}) = \mathbf{y}$. By (1) $\mathbf{x} \notin \operatorname{Int} D$, and by (2) $\mathbf{x} \notin \operatorname{Ext} D$; so $\mathbf{x} \in \operatorname{Bd} D$. By symmetry the same holds for $g^{-1} : B \to A$: (1′) $g^{-1}(\operatorname{Int} E) \subset \operatorname{Int} D$, (3′) $g^{-1}(\operatorname{Bd} E) \supset \operatorname{Bd} D$. Combining (1),(1′) gives $g(\operatorname{Int} D) = \operatorname{Int} E$, and (3),(3′) give $g(\operatorname{Bd} D) = \operatorname{Bd} E$.

(b) If D is rectifiable, $\operatorname{Bd} D$ has measure zero; by Lemma 18.1 so does $g(\operatorname{Bd} D) = \operatorname{Bd} E$, and E is rectifiable. ■

Definition 18.1 (primitive diffeomorphism)

Let $h : A \to B$ be a diffeomorphism of open sets in $\mathbb{R}^n$ $(n \geq 2)$, $h(\mathbf{x}) = (h_1(\mathbf{x}), \ldots, h_n(\mathbf{x}))$. h **preserves the ith coordinate** if $h_i(\mathbf{x}) = x_i$ for all $\mathbf{x} \in A$. If h preserves the ith coordinate for some i, h is a **primitive diffeomorphism**.

Theorem 18.3

Let $g : A \to B$ be a diffeomorphism of open sets in $\mathbb{R}^n$, where $n \geq 2$. Given $\mathbf{a} \in A$, there is a neighborhood U_0 of $\mathbf{a}$ contained in A, and a sequence of diffeomorphisms of open sets in $\mathbb{R}^n$,

$$U_0 \xrightarrow{h_1} U_1 \xrightarrow{h_2} U_2 \to \cdots \xrightarrow{h_k} U_k,$$

such that the composite $h_k \circ \cdots \circ h_2 \circ h_1$ equals $g|U_0$, and such that each h_i is a primitive diffeomorphism.

Proof.

Step 1. Linear case: $T(\mathbf{x}) = C \cdot \mathbf{x}$ with C non-singular factors into primitive non-singular linear maps. C is a product of elementary matrices (Theorem 2.4), whose transformations either (1) switch two coordinates, (2) replace the ith coordinate by itself plus a multiple of another, or (3) multiply the ith coordinate by a nonzero scalar. Types (2), (3) leave all but the ith coordinate fixed, so are

primitive (for $n \geq 2$). Type (1) is a composite of types (2) and (3): starting from rows i, j equal to $\mathbf{a}, \mathbf{b}$, replace row i by row i − row j (giving $\mathbf{a}-\mathbf{b}, \mathbf{b}$), then row j by row j + row i (giving $\mathbf{a}-\mathbf{b}, \mathbf{a}$), then row i by row i − row j (giving $-\mathbf{b}, \mathbf{a}$), then multiply row i by -1 (giving $\mathbf{b}, \mathbf{a}$).

Step 2. A translation $t(\mathbf{x}) = \mathbf{x} + \mathbf{c}$ is the composite of $t_1(\mathbf{x}) = \mathbf{x} + (0, c_2, \ldots, c_n)$ and $t_2(\mathbf{x}) = \mathbf{x} + (c_1, 0, \ldots, 0)$, both primitive.

Step 3. Special case $\mathbf{a} = \mathbf{0}$, $g(\mathbf{0}) = \mathbf{0}$, $Dg(\mathbf{0}) = I_n$: g factors locally into two primitive diffeomorphisms. Define $h : A \to \mathbb{R}^n$ by $h(\mathbf{x}) = (g_1(\mathbf{x}), \ldots, g_{n-1}(\mathbf{x}), x_n)$. Then $h(\mathbf{0}) = \mathbf{0}$ and

$$Dh(\mathbf{x}) = \begin{bmatrix} \partial(g_1, \ldots, g_{n-1})/\partial \mathbf{x} \\ 0 \ \cdots \ 0 \ \ 1 \end{bmatrix};$$

since the first $n-1$ rows are those of Dg, $Dh(\mathbf{0}) = I_n$. By the inverse function theorem h is a diffeomorphism of a neighborhood V_0 of $\mathbf{0}$ with an open set V_1. Define $k : V_1 \to \mathbb{R}^n$ by $k(\mathbf{y}) = (y_1, \ldots, y_{n-1}, g_n(h^{-1}(\mathbf{y})))$. Then $k(\mathbf{0}) = \mathbf{0}$, and

$$Dk(\mathbf{y}) = \begin{bmatrix} I_{n-1} & 0 \\ D(g_n \circ h^{-1})(\mathbf{y}) \end{bmatrix}, \qquad D(g_n \circ h^{-1})(\mathbf{0}) = Dg_n(\mathbf{0}) \cdot [Dh(\mathbf{0})]^{-1} = [0 \ \cdots \ 0 \ 1],$$

so $Dk(\mathbf{0}) = I_n$ and k is a diffeomorphism of a neighborhood W_1 of $\mathbf{0}$ with an open set W_2. Let $W_0 = h^{-1}(W_1)$. The diffeomorphisms $W_0 \xrightarrow{h} W_1 \xrightarrow{k} W_2$ are primitive, and $k \circ h = g|W_0$: for $\mathbf{x} \in W_0$ and $\mathbf{y} = h(\mathbf{x}) = (g_1(\mathbf{x}), \ldots, g_{n-1}(\mathbf{x}), x_n)$,

$$k(\mathbf{y}) = (y_1, \ldots, y_{n-1}, g_n(h^{-1}(\mathbf{y}))) = (g_1(\mathbf{x}), \ldots, g_{n-1}(\mathbf{x}), g_n(\mathbf{x})) = g(\mathbf{x}).$$

Step 4. General case. Given $g : A \to B$ and $\mathbf{a} \in A$, let $C = Dg(\mathbf{a})$ and define $t_1(\mathbf{x}) = \mathbf{x} + \mathbf{a}$, $t_2(\mathbf{x}) = \mathbf{x} - g(\mathbf{a})$, $T(\mathbf{x}) = C^{-1} \cdot \mathbf{x}$. Let $\tilde{g} = T \circ t_2 \circ g \circ t_1$, a diffeomorphism of the open set $t_1^{-1}(A)$ with $T(t_2(B))$, satisfying $\tilde{g}(\mathbf{0}) = \mathbf{0}$ and $D\tilde{g}(\mathbf{0}) = C^{-1} \cdot Dg(\mathbf{a}) \cdot I_n = I_n$. By Step 3 there is an open set $W_0 \ni \mathbf{0}$ in $t_1^{-1}(A)$ on which $\tilde{g}$ factors into two primitive diffeomorphisms. Let $W_2 = \tilde{g}(W_0)$, $A_0 = t_1(W_0)$, $B_0 = t_2^{-1}T^{-1}(W_2)$. Then g carries A_0 onto B_0, and $g|A_0$ is the composite

$$A_0 \xrightarrow{t_1^{-1}} W_0 \xrightarrow{\tilde{g}} W_2 \xrightarrow{T^{-1}} T^{-1}(W_2) \xrightarrow{t_2^{-1}} B_0.$$

By Steps 1 and 2, t_1^{-1}, t_2^{-1} and T^{-1} factor into primitive transformations. ■

§19. Proof of the Change of Variables Theorem

Motivation. The proof reduces to $n = 1$ through Fubini's theorem and the decomposition of diffeomorphisms into primitive ones, using partitions of unity to patch local results together.

We first prove the "only if" part.

Lemma 19.1

Let $g : A \to B$ be a diffeomorphism of open sets in $\mathbb{R}^n$. Then for every continuous function $f : B \to \mathbb{R}$ that is integrable over B, the function $(f \circ g)|\det Dg|$ is integrable over A, and

$$\int_B f = \int_A (f \circ g)|\det Dg|.$$

Proof.

Step 1. If the lemma holds for diffeomorphisms $g : U \to V$ and $h : V \to W$, it holds for $h \circ g$. Let $f : W \to \mathbb{R}$ be continuous and integrable over W. Then

$$\int_W f = \int_V (f \circ h)|\det Dh| = \int_U (f \circ h \circ g)\,|(\det Dh) \circ g|\,|\det Dg|;$$

the second integral exists and equals the first because the lemma holds for h, the third because it holds for g. It remains to note that by the chain rule $D(h \circ g)(\mathbf{x}) = Dh(g(\mathbf{x})) \cdot Dg(\mathbf{x})$, so $\det D(h \circ g) = [(\det Dh) \circ g] \cdot [\det Dg]$.

Step 2. Suppose each $\mathbf{x} \in A$ has a neighborhood $U \subset A$ such that the lemma holds for $g : U \to V$ ($V = g(U)$) and all continuous $f : V \to \mathbb{R}$ whose supports are compact subsets of V. Then the lemma holds for g. This is where partitions of unity are used. Write A as a union of such open sets U_α, with $V_\alpha = g(U_\alpha)$; their union is B. Choose a partition of unity $\{\phi_i\}$ on B with compact supports, dominated by $\{V_\alpha\}$. Then $\{\phi_i \circ g\}$ is a partition of unity on A with compact supports: $\phi_i(g(\mathbf{x})) \geq 0$; with $T_i = \operatorname{Support} \phi_i$, $g^{-1}(T_i)$ is compact and $\phi_i \circ g$ vanishes outside it, so $S_i = \operatorname{Support}(\phi_i \circ g) \subset g^{-1}(T_i)$ is compact; if W is a neighborhood of $g(\mathbf{x})$ meeting only finitely many T_i, then $g^{-1}(W)$ is a neighborhood of $\mathbf{x}$ meeting S_i only for these i; and $\sum_i \phi_i(g(\mathbf{x})) = 1$.

Let $f : B \to \mathbb{R}$ be continuous and integrable over B. By Theorem 16.5, $\int_B f = \sum_i [\int_B \phi_i f]$. Given i, choose α with $T_i \subset V_\alpha$. Since $\phi_i f$ is continuous on B and vanishes outside the compact set T_i, Lemma 16.4 gives $\int_B \phi_i f = \int_{T_i} \phi_i f = \int_{V_\alpha} \phi_i f$. By hypothesis for $g : U_\alpha \to V_\alpha$,

$$\int_{V_\alpha} \phi_i f = \int_{U_\alpha} (\phi_i \circ g)(f \circ g)|\det Dg|,$$

and since this integrand vanishes outside the compact set S_i, Lemma 16.4 gives $\int_B \phi_i f = \int_A (\phi_i \circ g)(f \circ g)|\det Dg|$. Summing,

$$\int_B f = \sum_{i=1}^{\infty} \Big[\int_A (\phi_i \circ g)(f \circ g)|\det Dg| \Big]. \qquad (*)$$

Since $|f|$ is integrable over B, $(*)$ holds with $|f|$ in place of f; as $\{\phi_i \circ g\}$ is a partition of unity on A, Theorem 16.5 shows $(f \circ g)|\det Dg|$ is integrable over A, and then $(*)$ and Theorem 16.5 give $\int_B f = \int_A (f \circ g)|\det Dg|$.

Step 3. The lemma holds for $n = 1$. Let $g : A \to B$ be a diffeomorphism of open sets in $\mathbb{R}$. Given $x \in A$, let I be a closed interval in A whose interior contains x, and $J = g(I)$, an interval with $g(\operatorname{Int} I) = \operatorname{Int} J$ (Theorems 17.1, 18.2). By Step 2 it suffices to verify, for continuous $f : \operatorname{Int} J \to \mathbb{R}$ with compact support in $\operatorname{Int} J$,

$$\int_{\operatorname{Int} J} f = \int_{\operatorname{Int} I} (f \circ g)|g'|. \qquad (**)$$

Extending f by 0 on $\operatorname{Bd} J$, $(**)$ is equivalent (Theorem 15.4 and Lemma 16.4) to $\int_J f = \int_I (f \circ g)|g'|$ for ordinary integrals, which is Theorem 17.1.

Step 4. Let $n > 1$. To prove the lemma for an arbitrary diffeomorphism $g : A \to B$ in $\mathbb{R}^n$, it suffices to prove it for primitive diffeomorphisms $h : U \to V$ of open sets in $\mathbb{R}^n$. Suppose so. Given $\mathbf{x} \in A$, Theorem 18.3 gives a neighborhood U_0 of $\mathbf{x}$ and primitive diffeomorphisms $h_1, \ldots, h_k$ whose composite is $g|U_0$. By Step 1 the lemma holds for $g|U_0$; as $\mathbf{x}$ is arbitrary, Step 2 gives it for g.

Step 5. If the lemma holds in dimension $n-1$, it holds in dimension n. By Step 4 it suffices to treat a primitive diffeomorphism $h : U \to V$; say h preserves the last coordinate. Let $\mathbf{p} \in U$, $\mathbf{q} = h(\mathbf{p})$. Choose a rectangle $Q \subset V$ whose interior contains $\mathbf{q}$ and let $S = h^{-1}(Q)$; by Theorem 18.2, h is a diffeomorphism of $\operatorname{Int} S$ onto $\operatorname{Int} Q$. By Step 2 it suffices to prove the lemma for $h : \operatorname{Int} S \to \operatorname{Int} Q$ and continuous $f : \operatorname{Int} Q \to \mathbb{R}$ whose support is a compact subset of $\operatorname{Int} Q$. Then $(f \circ h)|\det Dh|$ vanishes outside a compact subset of $\operatorname{Int} S$, so is integrable over $\operatorname{Int} S$ (Lemma 16.4); we must show

$$\int_{\operatorname{Int} Q} f = \int_{\operatorname{Int} S} (f \circ h)|\det Dh|.$$

These integrals exist as ordinary integrals, so by Theorem 15.4 this is an equation in ordinary integrals. Extend f to $\mathbb{R}^n$ by 0 outside $\operatorname{Int} Q$, and define $F : \mathbb{R}^n \to \mathbb{R}$ by $F = (f \circ h)|\det Dh|$ on $\operatorname{Int} S$ and 0 elsewhere; both are continuous, and we must show $\int_Q f = \int_S F$.

Write $Q = D \times I$ with D a rectangle in $\mathbb{R}^{n-1}$ and I a closed interval. S is compact, so its projection to $\mathbb{R}^{n-1} \times 0$ lies in a set $E \times 0$ with E a rectangle in $\mathbb{R}^{n-1}$; since h preserves the last coordinate, $S \subset E \times I$. As F vanishes outside S, we must show $\int_Q f = \int_{E\times I} F$, which by Fubini's theorem is equivalent to

$$\int_{t\in I}\int_{\mathbf{y}\in D} f(\mathbf{y},t) = \int_{t\in I}\int_{\mathbf{x}\in E} F(\mathbf{x},t).$$

We show the inner integrals are equal. The slices of U, V at height t have the form $U_t \times t$, $V_t \times t$ with U_t, V_t open in $\mathbb{R}^{n-1}$; the slice of S is $S_t \times t$ with S_t compact. As $F(\cdot,t)$ vanishes outside S_t, and $f(\cdot,t)$ outside a compact subset of V_t, the needed equality is equivalent (Lemma 16.4) to

$$\int_{\mathbf{y}\in V_t} f(\mathbf{y},t) = \int_{\mathbf{x}\in U_t} F(\mathbf{x},t),$$

an equation in $(n-1)$-dimensional integrals. Now $h(\mathbf{x},t) = (k(\mathbf{x},t),t)$ for a C^1 function $k : U \to \mathbb{R}^{n-1}$, and

$$Dh = \begin{bmatrix} \partial k/\partial \mathbf{x} & \partial k/\partial t \\ 0 \;\cdots\; 0 & 1 \end{bmatrix}, \qquad \det Dh = \det \partial k/\partial \mathbf{x}.$$

For fixed t, $\mathbf{x} \mapsto k(\mathbf{x},t)$ is a C^1 one-to-one map of U_t onto V_t with $\det \partial k/\partial \mathbf{x} \neq 0$, hence a diffeomorphism of open sets in $\mathbb{R}^{n-1}$. By the induction hypothesis,

$$\int_{\mathbf{y}\in V_t} f(\mathbf{y},t) = \int_{\mathbf{x}\in U_t} f(k(\mathbf{x},t),t)|\det \partial k/\partial \mathbf{x}|,$$

and for $\mathbf{x} \in U_t$ the integrand on the right is $f(h(\mathbf{x},t))|\det Dh| = F(\mathbf{x},t)$. ■

Now the "if" part.

Lemma 19.2

Let $g : A \to B$ be a diffeomorphism of open sets in $\mathbb{R}^n$; let $f : B \to \mathbb{R}$ be continuous. If $(f \circ g)|\det Dg|$ is integrable over A, then f is integrable over B.

Proof. Apply Lemma 19.1 to the diffeomorphism $g^{-1} : B \to A$ and the function $F = (f \circ g)|\det Dg|$, continuous and integrable over A: the function $(F \circ g^{-1})|\det Dg^{-1}|$ is integrable over B. It equals f: if $g(\mathbf{x}) = \mathbf{y}$, then $D(g^{-1})(\mathbf{y}) = [Dg(\mathbf{x})]^{-1}$ (Theorem 7.4), so

$$(F \circ g^{-1})(\mathbf{y}) \cdot |\det D(g^{-1})(\mathbf{y})| = F(\mathbf{x}) \cdot |1/\det Dg(\mathbf{x})| = f(\mathbf{y}).$$

■

Lemmas 19.1 and 19.2 together prove Theorem 17.2.

Example 19.1

If both integrals in the change of variables theorem exist as ordinary integrals, the theorem says they are equal. But it can happen that only one exists as an ordinary integral. In Example 17.2, the diffeomorphism $g : (-\pi/2, \pi/2) \to (-1,1)$, $g(x) = \sin x$, gives

$$\int_{(-1,1)} 1/(1-y^2)^{1/2} = \int_{(-\pi/2,\pi/2)} 1;$$

the right side exists as an ordinary integral, but the left does not.

§20. Applications of Change of Variables

Motivation. With the theorem proved, we explain why $|\det A|$ is the volume of the parallelepiped spanned by the columns of A and why volume is invariant under isometries, the geometric content of the determinant used in Chapter 5.

20.1 The meaning of the determinant

Theorem 20.1

Let A be an n by n matrix. Let $h : \mathbb{R}^n \to \mathbb{R}^n$ be the linear transformation $h(\mathbf{x}) = A \cdot \mathbf{x}$. Let S be a rectifiable set in $\mathbb{R}^n$, and let $T = h(S)$. Then

$$\mathrm{v}(T) = |\det A| \cdot \mathrm{v}(S).$$

Proof. If A is non-singular, h is a diffeomorphism of $\mathbb{R}^n$ with itself carrying $\operatorname{Int} S$ onto $\operatorname{Int} T$, and T is rectifiable (Theorem 18.2; for non-compact S apply it to $\bar{S}$). By the change of variables theorem (and Theorem 14.2(e)),

$$\mathrm{v}(T) = \mathrm{v}(\operatorname{Int} T) = \int_{\operatorname{Int} T} 1 = \int_{\operatorname{Int} S} |\det Dh| = \int_{\operatorname{Int} S} |\det A| = |\det A| \cdot \mathrm{v}(S).$$

If A is singular, $\det A = 0$ and we show $\mathrm{v}(T) = 0$. T is bounded, and h carries $\mathbb{R}^n$ onto a linear subspace V of dimension $p < n$, which has measure zero in $\mathbb{R}^n$. Then $\bar{T}$ is closed, bounded and of measure zero; 1_T is continuous except on $\bar{T}$ and vanishes outside $\bar{T}$, so $\int_T 1$ exists and equals 0. ■

So $|\det A|$ is the factor by which $h(\mathbf{x}) = A \cdot \mathbf{x}$ multiplies volumes.

Definition 20.1 (parallelopiped)

Let $\mathbf{a}_1, \ldots, \mathbf{a}_k$ be independent vectors in $\mathbb{R}^n$. The **k-dimensional parallelopiped** $\mathcal{P} = \mathcal{P}(\mathbf{a}_1, \ldots, \mathbf{a}_k)$ is the set of all $\mathbf{x} = c_1\mathbf{a}_1 + \cdots + c_k\mathbf{a}_k$ with $0 \le c_i \le 1$. The vectors $\mathbf{a}_i$ are the **edges** of $\mathcal{P}$.

A 2-dimensional parallelopiped is a parallelogram; a 3-dimensional one is the usual parallelopiped. For $k = n$ we already have a notion of volume:

Theorem 20.2

Let $\mathbf{a}_1, \ldots, \mathbf{a}_n$ be n independent vectors in $\mathbb{R}^n$. Let $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n]$ be the n by n matrix with columns $\mathbf{a}_1, \ldots, \mathbf{a}_n$. Then

$$\mathrm{v}(\mathcal{P}(\mathbf{a}_1, \ldots, \mathbf{a}_n)) = |\det A|.$$

Proof. $h(\mathbf{x}) = A \cdot \mathbf{x}$ carries $\mathbf{e}_j$ to $\mathbf{a}_j$ and the unit cube $I^n = [0,1]^n$ onto $\mathcal{P}(\mathbf{a}_1, \ldots, \mathbf{a}_n)$. By Theorem 20.1, $\mathrm{v}(\mathcal{P}) = |\det A| \cdot \mathrm{v}(I^n) = |\det A|$. ■

Example 20.1

In calculus, the volume of the parallelopiped with edges $\mathbf{a}, \mathbf{b}, \mathbf{c}$ in $\mathbb{R}^3$ is, up to sign, the triple scalar product $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \det[\mathbf{a}\ \mathbf{b}\ \mathbf{c}]$, and its sign depends on whether $\mathbf{a}, \mathbf{b}, \mathbf{c}$ is "right-handed" or "left-handed". We generalize this second notion.

Definition 20.2 (frame, orientation)

Let V be an n-dimensional vector space. An n-tuple $(\mathbf{a}_1, \ldots, \mathbf{a}_n)$ of independent vectors in V is an **n-frame** in V. In $\mathbb{R}^n$, a frame is **right-handed** if $\det[\mathbf{a}_1 \ \cdots \ \mathbf{a}_n] > 0$ and **left-handed** otherwise. The collection of all right-handed frames in $\mathbb{R}^n$ is an **orientation** of $\mathbb{R}^n$, and so is the collection of all left-handed frames. In general, choose a linear isomorphism $T : \mathbb{R}^n \to V$; one orientation of V consists of all frames $(T(\mathbf{a}_1), \ldots, T(\mathbf{a}_n))$ with $(\mathbf{a}_1, \ldots, \mathbf{a}_n)$ right-handed in $\mathbb{R}^n$, the other of those with $(\mathbf{a}_1, \ldots, \mathbf{a}_n)$ left-handed. Each is the **reverse** (or **opposite**) of the other.

This is independent of the choice of T (by Theorem 20.3 below applied to $T'^{-1} \circ T$). In an arbitrary n-dimensional vector space "right-handed" is not well-defined, but "orientation" is.

Example 20.2

In $\mathbb{R}^1$ a frame is a single nonzero number; it is right-handed iff positive. In $\mathbb{R}^2$, $(\mathbf{a}_1, \mathbf{a}_2)$ is right-handed iff one must rotate $\mathbf{a}_1$ counterclockwise through an angle less than π to point it in the direction of $\mathbf{a}_2$. In $\mathbb{R}^3$, $(\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3)$ is right-handed iff curling the fingers of one's right hand from $\mathbf{a}_1$ to $\mathbf{a}_2$ makes the thumb point in the direction of $\mathbf{a}_3$. One justification: if a frame $(\mathbf{a}_1(t), \mathbf{a}_2(t), \mathbf{a}_3(t))$ varies continuously for $0 \le t \le 1$ and is right-handed at $t = 0$, it stays right-handed, since $\det[\mathbf{a}_1 \ \mathbf{a}_2 \ \mathbf{a}_3]$ cannot change sign (intermediate-value theorem); and $(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$ satisfies both conditions.

Theorem 20.3

Let C be a non-singular n by n matrix. Let $h : \mathbb{R}^n \to \mathbb{R}^n$ be the linear transformation $h(\mathbf{x}) = C \cdot \mathbf{x}$. Let $(\mathbf{a}_1, \ldots, \mathbf{a}_n)$ be a frame in $\mathbb{R}^n$. If $\det C > 0$, the frames $(\mathbf{a}_1, \ldots, \mathbf{a}_n)$ and $(h(\mathbf{a}_1), \ldots, h(\mathbf{a}_n))$ belong to the same orientation of $\mathbb{R}^n$; if $\det C < 0$, they belong to opposite orientations of $\mathbb{R}^n$.

If $\det C > 0$, h is **orientation-preserving**; if $\det C < 0$, **orientation-reversing**.

Proof. Let $\mathbf{b}_i = h(\mathbf{a}_i)$. Then $C \cdot [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n] = [\mathbf{b}_1 \ \cdots \ \mathbf{b}_n]$, so $(\det C) \det[\mathbf{a}_1 \cdots \mathbf{a}_n] = \det[\mathbf{b}_1 \cdots \mathbf{b}_n]$. ■

20.2 Invariance of volume under isometries

Definition 20.3 (orthogonal and orthonormal sets)

Vectors $\mathbf{a}_1, \ldots, \mathbf{a}_k$ of $\mathbb{R}^n$ form an **orthogonal set** if $\langle \mathbf{a}_i, \mathbf{a}_j \rangle = 0$ for $i \ne j$, and an **orthonormal set** if moreover $\langle \mathbf{a}_i, \mathbf{a}_i \rangle = 1$ for all i. If they are orthogonal and nonzero, the vectors $\mathbf{a}_i / \|\mathbf{a}_i\|$ are orthonormal.

An orthogonal set of nonzero vectors is independent: taking the dot product of $d_1\mathbf{a}_1 + \cdots + d_k\mathbf{a}_k = \mathbf{0}$ with $\mathbf{a}_i$ gives $d_i \langle \mathbf{a}_i, \mathbf{a}_i \rangle = 0$, so $d_i = 0$. Hence an orthogonal set of n nonzero vectors in $\mathbb{R}^n$ is a basis; $\mathbf{e}_1, \ldots, \mathbf{e}_n$ is one, but there are many others.

Definition 20.4 (orthogonal matrix)

An n by n matrix A is **orthogonal** if its columns form an orthonormal set; equivalently, $A^{\mathrm{tr}} \cdot A = I_n$.

If A is orthogonal, A^{tr} is a left inverse, hence (Theorem 2.5) a right inverse: A is orthogonal iff A is non-singular and $A^{\mathrm{tr}} = A^{-1}$. Also $\det A = \pm 1$, since $(\det A)^2 = (\det A^{\mathrm{tr}})(\det A) = \det(A^{\mathrm{tr}} \cdot A) =$

$\det I_n = 1$. The orthogonal matrices form a *group*:

Theorem 20.4

Let A, B, C be orthogonal n by n matrices. Then:
(a) $A \cdot B$ is orthogonal.
(b) $A \cdot (B \cdot C) = (A \cdot B) \cdot C$.
(c) There is an orthogonal matrix I_n such that $A \cdot I_n = I_n \cdot A = A$ for all orthogonal A.
(d) Given A, there is an orthogonal matrix A^{-1} such that $A \cdot A^{-1} = A^{-1} \cdot A = I_n$.

Proof. (a) $(A \cdot B)^{\text{tr}} \cdot (A \cdot B) = B^{\text{tr}} \cdot A^{\text{tr}} \cdot A \cdot B = B^{\text{tr}} \cdot B = I_n$.

(b) is immediate and (c) holds since I_n is orthogonal.

(d) Since $A^{\text{tr}} = A^{-1}$, $I_n = A \cdot A^{\text{tr}} = (A^{\text{tr}})^{\text{tr}} \cdot A^{\text{tr}} = (A^{-1})^{\text{tr}} \cdot A^{-1}$, so A^{-1} is orthogonal. ■

Definition 20.5 (orthogonal transformation, isometry)

$h(\mathbf{x}) = A \cdot \mathbf{x}$ is an **orthogonal transformation** if A is an orthogonal matrix; equivalently, h carries the basis $\mathbf{e}_1, \ldots, \mathbf{e}_n$ to an orthonormal basis. A map $h : \mathbb{R}^n \to \mathbb{R}^n$ is a (euclidean) **isometry** if $\|h(\mathbf{x}) - h(\mathbf{y})\| = \|\mathbf{x} - \mathbf{y}\|$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$.

Theorem 20.5

Let $h : \mathbb{R}^n \to \mathbb{R}^n$ be a map such that $h(\mathbf{0}) = \mathbf{0}$.
(a) The map h is an isometry if and only if it preserves dot products.
(b) The map h is an isometry if and only if it is an orthogonal transformation.

Proof. (a) We have

$$\|h(\mathbf{x}) - h(\mathbf{y})\|^2 = \langle h(\mathbf{x}), h(\mathbf{x})\rangle - 2\langle h(\mathbf{x}), h(\mathbf{y})\rangle + \langle h(\mathbf{y}), h(\mathbf{y})\rangle, \tag{1}$$

$$\|\mathbf{x} - \mathbf{y}\|^2 = \langle \mathbf{x}, \mathbf{x}\rangle - 2\langle \mathbf{x}, \mathbf{y}\rangle + \langle \mathbf{y}, \mathbf{y}\rangle. \tag{2}$$

If h preserves dot products, the right sides are equal, so h is an isometry. Conversely, if h is an isometry, then $\|h(\mathbf{x}) - h(\mathbf{0})\| = \|\mathbf{x} - \mathbf{0}\|$ gives $\|h(\mathbf{x})\| = \|\mathbf{x}\|$; so the first and last terms of (1) and (2) agree, as do the left sides, whence $\langle h(\mathbf{x}), h(\mathbf{y})\rangle = \langle \mathbf{x}, \mathbf{y}\rangle$.

(b) If $h(\mathbf{x}) = A \cdot \mathbf{x}$ with A orthogonal, then $\langle h(\mathbf{x}), h(\mathbf{y})\rangle = h(\mathbf{x})^{\text{tr}} \cdot h(\mathbf{y}) = \mathbf{x}^{\text{tr}} \cdot A^{\text{tr}} \cdot A \cdot \mathbf{y} = \mathbf{x}^{\text{tr}} \cdot \mathbf{y}$, so h preserves dot products and is an isometry by (a). Conversely, let h be an isometry with $h(\mathbf{0}) = \mathbf{0}$, let $\mathbf{a}_i = h(\mathbf{e}_i)$ and $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n]$. By (a) the $\mathbf{a}_i$ are orthonormal, so A is orthogonal. Since the $\mathbf{a}_i$ form a basis, $h(\mathbf{x}) = \sum_i \alpha_i(\mathbf{x})\mathbf{a}_i$ for real functions α_i; orthonormality gives $\langle h(\mathbf{x}), \mathbf{a}_j\rangle = \alpha_j(\mathbf{x})$, while $\langle h(\mathbf{x}), \mathbf{a}_j\rangle = \langle h(\mathbf{x}), h(\mathbf{e}_j)\rangle = \langle \mathbf{x}, \mathbf{e}_j\rangle = x_j$. So $\alpha_j(\mathbf{x}) = x_j$ and $h(\mathbf{x}) = \sum_i x_i \mathbf{a}_i = A \cdot \mathbf{x}$. ■

Theorem 20.6

Let $h : \mathbb{R}^n \to \mathbb{R}^n$. Then h is an isometry if and only if it equals an orthogonal transformation followed by a translation, that is, if and only if h has the form $h(\mathbf{x}) = A \cdot \mathbf{x} + \mathbf{p}$, where A is an orthogonal matrix.

Proof. Let $\mathbf{p} = h(\mathbf{0})$ and $k(\mathbf{x}) = h(\mathbf{x}) - \mathbf{p}$. Then $\|k(\mathbf{x}) - k(\mathbf{y})\| = \|h(\mathbf{x}) - h(\mathbf{y})\|$, so k is an isometry iff h is. Since $k(\mathbf{0}) = \mathbf{0}$, k is an isometry iff $k(\mathbf{x}) = A \cdot \mathbf{x}$ with A orthogonal (Theorem 20.5), i.e. iff $h(\mathbf{x}) = A \cdot \mathbf{x} + \mathbf{p}$. ■

Theorem 20.7

Let $h : \mathbb{R}^n \to \mathbb{R}^n$ be an isometry. If S is a rectifiable set in $\mathbb{R}^n$, then the set $T = h(S)$ is rectifiable, and $\mathrm{v}(T) = \mathrm{v}(S)$.

Proof. $h(\mathbf{x}) = A \cdot \mathbf{x} + \mathbf{p}$ with A orthogonal, so $Dh(\mathbf{x}) = A$ and $|\det A| = 1$; by the change of variables theorem (as in Theorem 20.1), $\mathrm{v}(T) = |\det A| \cdot \mathrm{v}(S) = \mathrm{v}(S)$. ■

Chapter 5. Manifolds

Overview

For a bounded rectifiable set $A \subset \mathbb{R}^k$, $\mathrm{v}(A) = \int_A 1$ (length for $k = 1$, area for $k = 2$). Calculus also measures the length of curves and the area of surfaces in $\mathbb{R}^3$. This chapter introduces their k-dimensional analogues, *k-manifolds in $\mathbb{R}^n$*, defines k-dimensional volume for them, and defines the integral of a scalar function over a k-manifold with respect to volume.

§21. The Volume of a Parallelopiped

Let $\mathcal{P}$ be a k-dimensional parallelopiped in $\mathbb{R}^n$, $k < n$. Its n-dimensional volume is zero, since it lies in a k-dimensional subspace, which has measure zero in $\mathbb{R}^n$. A reasonable k-dimensional volume should (1) be preserved by orthogonal transformations of $\mathbb{R}^n$, and (2) agree with the usual volume when $\mathcal{P}$ lies in $\mathbb{R}^k \times 0$. These conditions determine it.

Lemma 21.1

Let W be a linear subspace of $\mathbb{R}^n$ of dimension k. Then there is an orthonormal basis for $\mathbb{R}^n$ whose first k elements form a basis for W.

Proof. By Theorem 1.2 there is a basis $\mathbf{a}_1, \ldots, \mathbf{a}_n$ of $\mathbb{R}^n$ whose first k elements form a basis of W. The *Gram–Schmidt process* produces an orthogonal set $\mathbf{b}_1, \ldots, \mathbf{b}_n$ such that for each i, $\mathbf{b}_1, \ldots, \mathbf{b}_i$ span the same space as $\mathbf{a}_1, \ldots, \mathbf{a}_i$: set $\mathbf{b}_1 = \mathbf{a}_1$ and in general

$$\mathbf{b}_i = \mathbf{a}_i - \lambda_{i1}\mathbf{b}_1 - \lambda_{i2}\mathbf{b}_2 - \cdots - \lambda_{i,i-1}\mathbf{b}_{i-1}.$$

Whatever the scalars λ_{ij}, each $\mathbf{a}_j$ is a combination of $\mathbf{b}_1, \ldots, \mathbf{b}_j$ and each $\mathbf{b}_j$ is a combination of $\mathbf{a}_1, \ldots, \mathbf{a}_j$ (induction), so $\mathbf{a}_1, \ldots, \mathbf{a}_i$ and $\mathbf{b}_1, \ldots, \mathbf{b}_i$ span the same subspace; in particular the n vectors $\mathbf{b}_i$ span $\mathbb{R}^n$, so they are independent and nonzero. If $\mathbf{b}_1, \ldots, \mathbf{b}_{i-1}$ are mutually orthogonal, taking the dot product of the equation for $\mathbf{b}_i$ with $\mathbf{b}_j$ $(j < i)$ gives $\langle \mathbf{b}_i, \mathbf{b}_j \rangle = \langle \mathbf{a}_i, \mathbf{b}_j \rangle - \lambda_{ij}\langle \mathbf{b}_j, \mathbf{b}_j \rangle$; since $\mathbf{b}_j \neq \mathbf{0}$, a unique λ_{ij} makes this vanish. With these choices the $\mathbf{b}_i$ are mutually orthogonal. Dividing each by its norm gives the desired orthonormal basis. ■

Theorem 21.2

Let W be a k-dimensional linear subspace of $\mathbb{R}^n$. There is an orthogonal transformation $h : \mathbb{R}^n \to \mathbb{R}^n$ that carries W onto the subspace $\mathbb{R}^k \times 0$ of $\mathbb{R}^n$.

Proof. Choose an orthonormal basis $\mathbf{b}_1, \ldots, \mathbf{b}_n$ of $\mathbb{R}^n$ with $\mathbf{b}_1, \ldots, \mathbf{b}_k$ a basis of W. Let $g(\mathbf{x}) = B \cdot \mathbf{x}$, where B has columns $\mathbf{b}_1, \ldots, \mathbf{b}_n$; g is orthogonal and $g(\mathbf{e}_i) = \mathbf{b}_i$, so g carries $\mathbb{R}^k \times 0$ (basis $\mathbf{e}_1, \ldots, \mathbf{e}_k$) onto W. Its inverse is the desired map. ■

Theorem 21.3

There is a unique function V that assigns, to each k-tuple $(\mathbf{x}_1, \ldots, \mathbf{x}_k)$ of elements of $\mathbb{R}^n$, a non-negative number such that:

(1) If $h : \mathbb{R}^n \to \mathbb{R}^n$ is an orthogonal transformation, then $V(h(\mathbf{x}_1), \ldots, h(\mathbf{x}_k)) = V(\mathbf{x}_1, \ldots, \mathbf{x}_k)$.

(2) If $\mathbf{y}_1, \ldots, \mathbf{y}_k$ belong to the subspace $\mathbb{R}^k \times 0$ of $\mathbb{R}^n$, so that $\mathbf{y}_i = \begin{bmatrix} \mathbf{z}_i \\ 0 \end{bmatrix}$ for $\mathbf{z}_i \in \mathbb{R}^k$, then

$$V(\mathbf{y}_1,\dots,\mathbf{y}_k) = |\det[\mathbf{z}_1 \ \cdots \ \mathbf{z}_k]|.$$

The function V vanishes if and only if the vectors $\mathbf{x}_1,\dots,\mathbf{x}_k$ are dependent. It satisfies the equation

$$V(\mathbf{x}_1,\dots,\mathbf{x}_k) = [\det(X^{\mathrm{tr}}\cdot X)]^{1/2},$$

where X is the n by k matrix $X = [\mathbf{x}_1 \ \cdots \ \mathbf{x}_k]$.

We often write $V(\mathbf{x}_1,\dots,\mathbf{x}_k)$ simply as $V(X)$.

Proof. Let $F(X) = \det(X^{\mathrm{tr}}\cdot X)$.

Step 1. If $h(\mathbf{x}) = A\cdot\mathbf{x}$ with A orthogonal, $F(A\cdot X) = \det(X^{\mathrm{tr}}A^{\mathrm{tr}}AX) = \det(X^{\mathrm{tr}}X) = F(X)$. If Z is k by k and $Y = \begin{bmatrix} Z \\ 0 \end{bmatrix}$, then $F(Y) = \det([Z^{\mathrm{tr}}\ 0]\cdot\begin{bmatrix} Z \\ 0 \end{bmatrix}) = \det(Z^{\mathrm{tr}}Z) = \det^2 Z$.

Step 2. F is non-negative: given $\mathbf{x}_1,\dots,\mathbf{x}_k$, let W be a k-dimensional subspace containing them, and $h(\mathbf{x}) = A\cdot\mathbf{x}$ an orthogonal transformation carrying W onto $\mathbb{R}^k\times 0$ (Theorem 21.2). Then $A\cdot X = \begin{bmatrix} Z \\ 0 \end{bmatrix}$ and $F(X) = F(A\cdot X) = \det^2 Z \geq 0$. Also $F(X) = 0$ iff the columns of Z are dependent, iff $\mathbf{x}_1,\dots,\mathbf{x}_k$ are dependent.

Step 3. Define $V(X) = (F(X))^{1/2}$. By Step 1, V satisfies (1) and (2), and by Step 2's computation V is uniquely determined by them. ■

Definition 21.1 (k-volume of a parallelopiped)

If $\mathbf{x}_1,\dots,\mathbf{x}_k$ are independent vectors in $\mathbb{R}^n$, the **k-dimensional volume** of $\mathcal{P} = \mathcal{P}(\mathbf{x}_1,\dots,\mathbf{x}_k)$ is the positive number $V(\mathbf{x}_1,\dots,\mathbf{x}_k)$.

Example 21.1

Let $\mathbf{a},\mathbf{b}$ be independent in $\mathbb{R}^3$, $X = [\mathbf{a}\ \mathbf{b}]$, and θ the angle between them, $\langle\mathbf{a},\mathbf{b}\rangle = \|\mathbf{a}\|\|\mathbf{b}\|\cos\theta$. Then

$$(V(X))^2 = \det\begin{bmatrix} \|\mathbf{a}\|^2 & \langle\mathbf{a},\mathbf{b}\rangle \\ \langle\mathbf{b},\mathbf{a}\rangle & \|\mathbf{b}\|^2 \end{bmatrix} = \|\mathbf{a}\|^2\|\mathbf{b}\|^2(1-\cos^2\theta) = \|\mathbf{a}\|^2\|\mathbf{b}\|^2\sin^2\theta,$$

the square of the area (base $\|\mathbf{a}\|$ times height $\|\mathbf{b}\|\sin\theta$) of the parallelogram with edges $\mathbf{a},\mathbf{b}$.

Calculus also gives the area of $\mathcal{P}(\mathbf{a},\mathbf{b})$ as $\|\mathbf{a}\times\mathbf{b}\|$, where

$$\mathbf{a}\times\mathbf{b} = \det\begin{bmatrix} a_2 & b_2 \\ a_3 & b_3 \end{bmatrix}\mathbf{e}_1 - \det\begin{bmatrix} a_1 & b_1 \\ a_3 & b_3 \end{bmatrix}\mathbf{e}_2 + \det\begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix}\mathbf{e}_3,$$

justified by verifying $\|\mathbf{a}\|^2\|\mathbf{b}\|^2 - \langle\mathbf{a},\mathbf{b}\rangle^2 = \|\mathbf{a}\times\mathbf{b}\|^2$. Theorem 21.3 is convenient in theory; the following generalization of the cross-product formula is often more convenient in practice.

Definition 21.2 (ascending k-tuple)

Let $X = [\mathbf{x}_1 \ \cdots \ \mathbf{x}_k]$ be an n by k matrix, $k\leq n$. If $I = (i_1,\dots,i_k)$ with $1\leq i_1 < i_2 < \cdots < i_k \leq n$, I is an **ascending k-tuple** from $\{1,\dots,n\}$, and X_I or $X(i_1,\dots,i_k)$ denotes the k by k submatrix of X consisting of rows $i_1,\dots,i_k$. More generally, for any k-tuple I from $\{1,\dots,n\}$ (not necessarily distinct or ordered), X_I denotes the k by k matrix whose successive rows are rows $i_1,\dots,i_k$ of X.

Theorem 21.4

Let X be an n by k matrix with $k \le n$. Then

$$V(X) = \Big[\sum_{[I]} \det{}^2 X_I\Big]^{1/2},$$

where the symbol $[I]$ indicates that the summation extends over all ascending k-tuples from the set $\{1, \dots, n\}$.

This is a "Pythagorean theorem for k-volume": the square of the volume of $\mathcal{P}$ is the sum of the squares of the volumes of its projections onto the coordinate k-planes.

Proof. Let $F(X) = \det(X^{\mathrm{tr}}X)$ and $G(X) = \sum_{[I]} \det^2 X_I$; we show $F = G$.

Step 1. If $k = 1$, X is a column with entries $\lambda_1, \dots, \lambda_n$ and $F(X) = \sum \lambda_i^2 = G(X)$. If $k = n$, G has one term and $F(X) = \det^2 X = G(X)$.

Step 2. If the columns $\mathbf{x}_i$ are orthogonal, $X^{\mathrm{tr}}X$ is diagonal with entries $\|\mathbf{x}_i\|^2$, so $F(X) = \|\mathbf{x}_1\|^2 \cdots \|\mathbf{x}_k\|^2$.

Step 3. Neither (1) exchanging columns $j \neq l$ nor (2) replacing column j by itself plus c times column l changes F or G. Such a column operation transforms X into $X \cdot E^{\mathrm{tr}}$, where E is the corresponding elementary matrix, with $\det E = \pm 1$. So $F(XE^{\mathrm{tr}}) = \det(EX^{\mathrm{tr}}XE^{\mathrm{tr}}) = F(X)$. Deleting rows commutes with column operations, so $(XE^{\mathrm{tr}})_I = X_I E^{\mathrm{tr}}$, and $G(XE^{\mathrm{tr}}) = \sum_{[I]} \det^2(X_I) \det^2(E^{\mathrm{tr}}) = G(X)$.

Step 4. It suffices to prove the theorem when all entries of the bottom row of X are zero except possibly the last, and the columns are orthogonal. Given X, if the last row has a nonzero entry, column operations of the above types bring X to the form whose last row is $(0, \dots, 0, \lambda)$, $\lambda \neq 0$; if the last row is zero this holds with $\lambda = 0$. Then apply Gram–Schmidt to the columns: the jth column is replaced by itself minus multiples of earlier columns—operations of type (2)—which keep the zeros in the last row. Afterwards the columns are orthogonal and the matrix still has this form.

Step 5. Induction on n. For $n = 1$, $k = 1$ (Step 1); for $n = 2$, $k = 1$ or 2 (Step 1). Assume the result for fewer than n rows and let $1 < k < n$. By Step 4 we may assume

$$X = \begin{bmatrix} \mathbf{b}_1 & \cdots & \mathbf{b}_{k-1} & \mathbf{b}_k \\ 0 & \cdots & 0 & \lambda \end{bmatrix}$$

with orthogonal columns; the $\mathbf{b}_i \in \mathbb{R}^{n-1}$ are then orthogonal. Let $B = [\mathbf{b}_1 \cdots \mathbf{b}_k]$, $C = [\mathbf{b}_1 \cdots \mathbf{b}_{k-1}]$. By Step 2,

$$F(X) = \|\mathbf{b}_1\|^2 \cdots \|\mathbf{b}_{k-1}\|^2(\|\mathbf{b}_k\|^2 + \lambda^2) = F(B) + \lambda^2 F(C).$$

Split $G(X) = \sum_{i_k<n} \det^2 X_I + \sum_{i_k=n} \det^2 X_I$. If $i_k < n$, $X_I = B_I$, so the first sum is $G(B)$. If $i_k = n$, expanding along the last row, $\det X(i_1, \dots, i_{k-1}, n) = \pm\lambda \det C(i_1, \dots, i_{k-1})$, so the second sum is $\lambda^2 G(C)$. By the induction hypothesis $F(B) = G(B)$ and $F(C) = G(C)$, hence $F(X) = G(X)$. ∎

§22. The Volume of a Parametrized-Manifold

Motivation. How do we measure the area of a surface in $\mathbb{R}^3$, or the k-dimensional volume of a k-dimensional object in $\mathbb{R}^n$? Linear algebra gives the volume of a parallelepiped as $\sqrt{\det(X^T X)}$; integrating this over a parametrization defines volume, independently of the parametrization chosen.

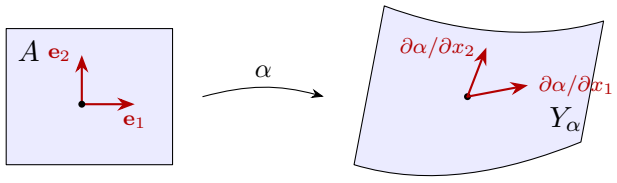


Figure. *A parametrized surface $\alpha : A \to \mathbb{R}^3$. A small square in A is mapped approximately onto the parallelogram spanned by the columns of $D\alpha$, whose area is $V(D\alpha) = \sqrt{\det(D\alpha^{\mathsf{T}} D\alpha)}$; integrating this factor over A gives the area of Y_α.*

Definition 22.1 (parametrized-manifold, volume)

Let $k \le n$, A open in $\mathbb{R}^k$, and $\alpha : A \to \mathbb{R}^n$ of class C^r ($r \ge 1$). The set $Y = \alpha(A)$ together with the map α is a **parametrized-manifold** of dimension k, denoted Y_α. Its (k-dimensional) **volume** is

$$\mathrm{v}(Y_\alpha) = \int_A V(D\alpha),$$

provided the integral exists.

Plausibility. Suppose $A = \operatorname{Int} Q$ and α extends to be C^r near Q. For a partition P of Q and a subrectangle $R = [a_1, a_1 + h_1] \times \cdots \times [a_k, a_k + h_k]$, α maps R onto a "curved rectangle" $\alpha(R) \subset Y$. The edge from $\mathbf{a}$ to $\mathbf{a} + h_i \mathbf{e}_i$ goes to a curve whose chord $\alpha(\mathbf{a} + h_i \mathbf{e}_i) - \alpha(\mathbf{a})$ is approximated to first order by $\mathbf{v}_i = D\alpha(\mathbf{a}) \cdot h_i \mathbf{e}_i = (\partial\alpha/\partial x_i) h_i$. The parallelopiped with edges $\mathbf{v}_i$ has k-volume

$$V(\mathbf{v}_1, \ldots, \mathbf{v}_k) = V(\partial\alpha/\partial x_1, \ldots, \partial\alpha/\partial x_k) \cdot (h_1 \cdots h_k) = V(D\alpha(\mathbf{a})) \cdot \mathrm{v}(R).$$

Summing over R gives a number between the lower and upper sums of $V(D\alpha)$ for P, which approximates $\int_A V(D\alpha)$ as closely as we wish.

Definition 22.2 (integral with respect to volume)

Let A be open in $\mathbb{R}^k$, $\alpha : A \to \mathbb{R}^n$ of class C^r, $Y = \alpha(A)$, and f a real-valued function continuous at each point of Y. The **integral of f over Y_α with respect to volume** is

$$\int_{Y_\alpha} f \, dV = \int_A (f \circ \alpha) V(D\alpha),$$

provided this integral exists. In particular $\mathrm{v}(Y_\alpha) = \int_{Y_\alpha} dV$.

Here dV is a meaningless symbol meaning "integral with respect to volume"; it is written dV to avoid confusion with the integral $\int_A \omega$ of a differential form (Chapter 7). The literature often uses the same symbol d for both.

Theorem 22.1

Let $g : A \to B$ be a diffeomorphism of open sets in $\mathbb{R}^k$. Let $\beta : B \to \mathbb{R}^n$ be a map of class C^r; let $Y = \beta(B)$. Let $\alpha = \beta \circ g$; then $\alpha : A \to \mathbb{R}^n$ and $Y = \alpha(A)$. If $f : Y \to \mathbb{R}$ is a continuous function, then f is integrable over Y_β if and only if it is integrable over Y_α; in this case

$$\int_{Y_\alpha} f \, dV = \int_{Y_\beta} f \, dV.$$

In particular, $\mathrm{v}(Y_\alpha) = \mathrm{v}(Y_\beta)$.

Proof. We must show $\int_B (f \circ \beta) V(D\beta) = \int_A (f \circ \alpha) V(D\alpha)$, one existing iff the other does. By the

change of variables theorem,

$$\int_B (f\circ\beta)V(D\beta) = \int_A ((f\circ\beta)\circ g)\,(V(D\beta)\circ g)\,|\det Dg|,$$

so it suffices to show $(V(D\beta)\circ g)|\det Dg| = V(D\alpha)$. With $\mathbf{y} = g(\mathbf{x})$, $D\alpha(\mathbf{x}) = D\beta(\mathbf{y})\cdot Dg(\mathbf{x})$, so

$$[V(D\alpha(\mathbf{x}))]^2 = \det(Dg(\mathbf{x})^{\mathrm{tr}}\, D\beta(\mathbf{y})^{\mathrm{tr}}\, D\beta(\mathbf{y})\, Dg(\mathbf{x})) = (\det Dg(\mathbf{x}))^2[V(D\beta(\mathbf{y}))]^2.$$

■

Example 22.1 (length of a curve)

Let A be an open interval in $\mathbb{R}^1$ and $\alpha : A \to \mathbb{R}^n$ of class C^r; Y_α is a **parametrized-curve**, and its 1-dimensional volume is its **length**:

$$\mathrm{v}(Y_\alpha) = \int_A V(D\alpha) = \int_A \Big[\Big(\frac{d\alpha_1}{dt}\Big)^2 + \cdots + \Big(\frac{d\alpha_n}{dt}\Big)^2\Big]^{1/2},$$

since $D\alpha$ is the column matrix of the $d\alpha_i/dt$—the familiar arc-length formula.

Example 22.2

For $\alpha(t) = (a\cos t, a\sin t)$, $0 < t < 3\pi$, the formula gives $\int_0^{3\pi}[a^2\sin^2 t + a^2\cos^2 t]^{1/2} = 3\pi a$. Since α is not one-to-one, this is not the length of the image (the circle of radius a) but the distance travelled by a particle moving by $\mathbf{x} = \alpha(t)$. Later we restrict to one-to-one parametrizations.

Example 22.3 (area of a surface)

If A is open in $\mathbb{R}^2$, Y_α is a **parametrized-surface**, and its 2-dimensional volume is its **area**. For $n = 3$, with (x, y) the general point of $\mathbb{R}^2$, $D\alpha = [\partial\alpha/\partial x \;\; \partial\alpha/\partial y]$ and (Example 21.1)

$$\mathrm{v}(Y_\alpha) = \int_A \Big\|\frac{\partial\alpha}{\partial x}\times\frac{\partial\alpha}{\partial y}\Big\|.$$

If $\alpha(x, y) = (x, y, f(x, y))$ with $f : A \to \mathbb{R}$ of class C^r, Y is the graph of f, and since $D\alpha$ has rows $(1, 0)$, $(0, 1)$, $(\partial f/\partial x, \partial f/\partial y)$, Theorem 21.4 gives

$$\mathrm{v}(Y_\alpha) = \int_A [1 + (\partial f/\partial x)^2 + (\partial f/\partial y)^2]^{1/2},$$

the surface-area formula of calculus.

Example 22.4 (hemisphere)

Let A be the open disc $x^2+y^2 < a^2$ and $f(x, y) = [a^2-x^2-y^2]^{1/2}$; the graph is a **hemisphere** of radius a. With $\alpha(x, y) = (x, y, f(x, y))$ one checks $V(D\alpha) = a/(a^2-x^2-y^2)^{1/2}$, so in polar coordinates

$$\mathrm{v}(Y_\alpha) = \int_B ar/(a^2 - r^2)^{1/2}, \qquad B = (0, a)\times(0, 2\pi).$$

This is improper, so Fubini does not apply directly. Integrate over $(0, a_n)\times(0, 2\pi)$ with $0 < a_n < a$ and let $a_n \to a$: $\mathrm{v}(Y_\alpha) = \lim_{n\to\infty}(-2\pi a)[(a^2 - a_n^2)^{1/2} - a] = 2\pi a^2$. (A method avoiding improper integrals is in §25.)

§23. Manifolds in $\mathbb{R}^n$

Motivation. Curves and surfaces with corners or self-intersections should be excluded, and global objects like spheres need several parametrizations. A manifold is a set covered by compatible local parametrizations, and the transition maps are diffeomorphisms by the inverse function theorem.

We restrict ourselves to submanifolds of $\mathbb{R}^n$; abstract manifolds appear in Chapter 9.

Definition 23.1 (manifold without boundary)

Let $k > 0$. Suppose M is a subspace of $\mathbb{R}^n$ such that for each $\mathbf{p} \in M$ there are a set V containing $\mathbf{p}$ and open in M, a set U open in $\mathbb{R}^k$, and a continuous map $\alpha : U \to V$ carrying U onto V in a one-to-one fashion, such that:
(1) α is of class C^r;
(2) $\alpha^{-1} : V \to U$ is continuous;
(3) $D\alpha(\mathbf{x})$ has rank k for each $\mathbf{x} \in U$. Then M is a **k-manifold without boundary** in $\mathbb{R}^n$, of class C^r, and α is a **coordinate patch** on M about $\mathbf{p}$.

Example 23.1

For $k = 1$, $D\alpha$ has rank 1 iff $D\alpha \neq \mathbf{0}$; this rules out "cusps" and "corners". If $\alpha(t) = (t^3, t^2)$, the image M has a cusp at the origin; α is C^∞ and α^{-1} is continuous, but $D\alpha(0)$ has rank 0. If $\beta(t) = (t^3, |t^3|)$, the image has a corner at the origin; β is C^2 and β^{-1} continuous, but $D\beta(0)$ has rank 0.

Example 23.2

For $k = 2$, rank 2 means the columns $\partial\alpha/\partial x_1$, $\partial\alpha/\partial x_2$ are independent at $\mathbf{a}$. Since $\partial\alpha/\partial x_j$ is the velocity of the curve $t \mapsto \alpha(\mathbf{a} + t\mathbf{e}_j)$ in M, these span a 2-dimensional "tangent plane". When the condition fails, e.g. $\alpha(x, y) = (x(x^2 + y^2),\ y(x^2 + y^2),\ x^2 + y^2)$, the image has no tangent plane at the origin (a cone-like point); α is C^∞ and α^{-1} continuous, but $D\alpha(\mathbf{0})$ has rank 0.

Example 23.3

Continuity of α^{-1} rules out other pathologies. Let $\alpha(t) = (\sin 2t)(|\cos t|, \sin t)$ for $0 < t < \pi$; its image M is a "figure eight". α is C^1 with $D\alpha$ of rank 1 and maps $(0, \pi)$ one-to-one onto M, but α^{-1} is not continuous: the image of a small interval U_0 about $\pi/2$ is not open in M (points of M near $\mathbf{0}$ need not come from points near $\pi/2$).

Example 23.4

If $\alpha : A \to \mathbb{R}^n$ is C^r on A open in $\mathbb{R}^k$ and $Y = \alpha(A)$, then Y_α is a parametrized-manifold, but Y need not be a manifold. If α is one-to-one, α^{-1} is continuous and $D\alpha$ has rank k, then Y is a manifold without boundary, covered by the single coordinate patch α.

Definition 23.2 (C^r on an arbitrary set)

Let $S \subset \mathbb{R}^k$ and $f : S \to \mathbb{R}^n$. f is **of class C^r on** S if f extends to a function $g : U \to \mathbb{R}^n$ of class C^r on an open set $U \supset S$.

Composites of C^r functions are C^r: if $f_1 : S \to \mathbb{R}^n$ and $f_2 : T \to \mathbb{R}^p$ are C^r with $f_1(S) \subset T$, with C^r extensions g_1 on U and g_2 on V, then $g_2 \circ g_1$ is a C^r extension of $f_2 \circ f_1$ on $g_1^{-1}(V) \supset S$.

Lemma 23.1

Let S be a subset of $\mathbb{R}^k$; let $f : S \to \mathbb{R}^n$. If for each $\mathbf{x} \in S$, there is a neighborhood $U_\mathbf{x}$ of $\mathbf{x}$ and a function $g_\mathbf{x} : U_\mathbf{x} \to \mathbb{R}^n$ of class C^r that agrees with f on $U_\mathbf{x} \cap S$, then f is of class C^r on S.

Proof. Cover S by the $U_\mathbf{x}$; let A be their union, and $\{\phi_i\}$ a C^∞ partition of unity on A dominated by $\{U_\mathbf{x}\}$. For each i choose $U_\mathbf{x}$ containing $\operatorname{Support} \phi_i$ and let $g_i = g_\mathbf{x}$. The C^r function $\phi_i g_i : U_\mathbf{x} \to \mathbb{R}^n$ vanishes outside a closed subset of $U_\mathbf{x}$; extend it by 0 to a C^r function h_i on A. Let $g(\mathbf{x}) = \sum_i h_i(\mathbf{x})$ for $\mathbf{x} \in A$. Near each point of A, g is a finite sum of h_i's, so g is C^r on A. For $\mathbf{x} \in S$, $h_i(\mathbf{x}) = \phi_i(\mathbf{x})g_i(\mathbf{x}) = \phi_i(\mathbf{x})f(\mathbf{x})$ whenever $\phi_i(\mathbf{x}) \neq 0$, so $g(\mathbf{x}) = \sum_i \phi_i(\mathbf{x})f(\mathbf{x}) = f(\mathbf{x})$. ■

Definition 23.3 (half-space)

$\mathbb{H}^k$, **upper half-space** in $\mathbb{R}^k$, consists of the $\mathbf{x} \in \mathbb{R}^k$ with $x_k \geq 0$; $\mathbb{H}^k_+$, the **open upper half-space**, of those with $x_k > 0$.

Lemma 23.2

Let U be open in $\mathbb{H}^k$ but not in $\mathbb{R}^k$; let $\alpha : U \to \mathbb{R}^n$ be of class C^r. Let $\beta : U' \to \mathbb{R}^n$ be a C^r extension of α defined on an open set U' of $\mathbb{R}^k$. Then for $\mathbf{x} \in U$, the derivative $D\beta(\mathbf{x})$ depends only on the function α and is independent of the extension β. It follows that we may denote this derivative by $D\alpha(\mathbf{x})$ without ambiguity.

Proof. $\partial\beta_i/\partial x_j$ at $\mathbf{x}$ is the limit of $[\beta(\mathbf{x} + h\mathbf{e}_j) - \beta(\mathbf{x})]/h$ as $h \to 0$; it may be computed letting $h \to 0$ through positive values. If $\mathbf{x} \in \mathbb{H}^k$, so is $\mathbf{x} + h\mathbf{e}_j$ for $h > 0$, and there β agrees with α; so $D\beta(\mathbf{x})$ depends only on α. ■

Definition 23.4 (k-manifold)

Let $k > 0$. A **k-manifold in $\mathbb{R}^n$ of class C^r** is a subspace M of $\mathbb{R}^n$ such that for each $\mathbf{p} \in M$ there are an open set V of M containing $\mathbf{p}$, a set U open in either $\mathbb{R}^k$ or $\mathbb{H}^k$, and a continuous map $\alpha : U \to V$ carrying U onto V in a one-to-one fashion, such that:
(1) α is of class C^r;
(2) $\alpha^{-1} : V \to U$ is continuous;
(3) $D\alpha(\mathbf{x})$ has rank k for each $\mathbf{x} \in U$. The map α is a **coordinate patch** on M about $\mathbf{p}$. A discrete collection of points in $\mathbb{R}^n$ is a **0-manifold**.

A manifold without boundary is the special case where all coordinate patches have domains open in $\mathbb{R}^k$. A k-manifold has two kinds of points: those with neighborhoods like open k-balls, and those with neighborhoods only like open half-balls; the latter form the *boundary* (§24).

Lemma 23.3

Let M be a manifold in $\mathbb{R}^n$, and let $\alpha : U \to V$ be a coordinate patch on M. If U_0 is a subset of U that is open in U, then the restriction of α to U_0 is also a coordinate patch on M.

Proof. Since U_0 is open in U and α^{-1} is continuous, $V_0 = \alpha(U_0)$ is open in V, hence in M; and U_0 is open in $\mathbb{R}^k$ or $\mathbb{H}^k$ according as U is. $\alpha|U_0$ maps U_0 one-to-one onto V_0, is C^r, its inverse is a restriction of α^{-1}, hence continuous, and its derivative has rank k. ■

This fails without continuity of α^{-1}: in Example 23.3, $\alpha|U_0$ is not a coordinate patch since its image is not open in M.

§24. The Boundary of a Manifold

Motivation. Stokes' theorem relates an integral over a manifold to one over its boundary, so we need the boundary itself to be a manifold of one lower dimension. Using half-spaces as local models makes this automatic.

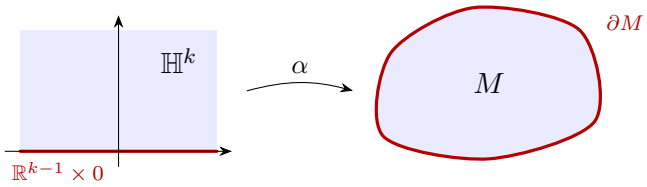


Figure. *Points of a k-manifold M with boundary have neighbourhoods modelled either on $\mathbb{R}^k$ (interior points) or on the half-space $\mathbb{H}^k$ (boundary points); the boundary ∂M is a $(k-1)$-manifold without boundary.*

Theorem 24.1

Let M be a k-manifold in $\mathbb{R}^n$, of class C^r. Let $\alpha_0 : U_0 \to V_0$ and $\alpha_1 : U_1 \to V_1$ be coordinate patches on M, with $W = V_0 \cap V_1$ non-empty. Let $W_i = \alpha_i^{-1}(W)$. Then the map

$$\alpha_1^{-1} \circ \alpha_0 : W_0 \to W_1$$

is of class C^r, and its derivative is non-singular.

We call $\alpha_1^{-1} \circ \alpha_0$ the **transition function** between α_0 and α_1.

Proof. It suffices to show: if $\alpha : U \to V$ is a coordinate patch on M, then $\alpha^{-1} : V \to \mathbb{R}^k$ is of class C^r as a map of the subset V of $\mathbb{R}^n$. Then $\alpha_1^{-1} \circ \alpha_0$ and $\alpha_0^{-1} \circ \alpha_1$ are C^r, inverse to each other, and the chain rule shows their derivatives are non-singular.

By Lemma 23.1 it suffices to show α^{-1} is locally C^r. Let $\mathbf{p}_0 \in V$, $\mathbf{x}_0 = \alpha^{-1}(\mathbf{p}_0)$. Suppose U is open in $\mathbb{H}^k$ but not $\mathbb{R}^k$; extend α to a C^r map β on an open set U' of $\mathbb{R}^k$. $D\alpha(\mathbf{x}_0)$ has rank k, so some k rows are independent; say the first k. Let $\pi : \mathbb{R}^n \to \mathbb{R}^k$ project onto the first k coordinates. Then $g = \pi \circ \beta$ maps U' into $\mathbb{R}^k$ with $Dg(\mathbf{x}_0)$ non-singular, so by the inverse function theorem g is a C^r diffeomorphism of an open set W of $\mathbb{R}^k$ about $\mathbf{x}_0$ with an open set of $\mathbb{R}^k$. We claim $h = g^{-1} \circ \pi$, which is C^r, extends α^{-1} near $\mathbf{p}_0$. $U_0 = W \cap U$ is open in U, so $V_0 = \alpha(U_0)$ is open in V, hence there is an open $A \subset \mathbb{R}^n$ with $A \cap V = V_0$; shrink A into the domain of h (intersect with $\pi^{-1}(g(W))$). For $\mathbf{p} \in A \cap V = V_0$, with $\mathbf{x} = \alpha^{-1}(\mathbf{p})$,

$$h(\mathbf{p}) = g^{-1}(\pi(\alpha(\mathbf{x}))) = g^{-1}(g(\mathbf{x})) = \mathbf{x} = \alpha^{-1}(\mathbf{p}).$$

If U is open in $\mathbb{R}^k$, take $U' = U$, $\beta = \alpha$; the argument is unchanged. ∎

Definition 24.1 (interior and boundary points)

Let M be a k-manifold in $\mathbb{R}^n$ and $\mathbf{p} \in M$. If there is a coordinate patch $\alpha : U \to V$ on M about $\mathbf{p}$ with U open in $\mathbb{R}^k$, $\mathbf{p}$ is an **interior point** of M; otherwise $\mathbf{p}$ is a **boundary point**. The set of boundary points is the **boundary** ∂M of M.

These terms are unrelated to the topological notions $\operatorname{Int} S$, $\operatorname{Bd} S$, $\operatorname{Ext} S$ of a subset $S \subset \mathbb{R}^n$. For a manifold we write ∂M for its boundary and $M - \partial M$ for its interior.

Lemma 24.2

Let M be a k-manifold in $\mathbb{R}^n$; let $\alpha : U \to V$ be a coordinate patch about the point $\mathbf{p}$ of M.

(a) If U is open in $\mathbb{R}^k$, then $\mathbf{p}$ is an interior point of M.

(b) If U is open in $\mathbb{H}^k$ and if $\mathbf{p} = \alpha(\mathbf{x}_0)$ for $\mathbf{x}_0 \in \mathbb{H}^k_+$, then $\mathbf{p}$ is an interior point of M.

(c) If U is open in $\mathbb{H}^k$ and $\mathbf{p} = \alpha(\mathbf{x}_0)$ for $\mathbf{x}_0 \in \mathbb{R}^{k-1} \times 0$, then $\mathbf{p}$ is a boundary point of M.

Proof. (a) is the definition.

(b) Let $U_0 = U \cap \mathbb{H}^k_+$, open in $\mathbb{R}^k$, and $V_0 = \alpha(U_0)$; then $\alpha|U_0$ is a coordinate patch about $\mathbf{p}$ (Lemma 23.3) with domain open in $\mathbb{R}^k$.

(c) Let $\alpha_0 : U_0 \to V_0$ be a patch about $\mathbf{p}$ with U_0 open in $\mathbb{H}^k$ and $\mathbf{p} = \alpha_0(\mathbf{x}_0)$, $\mathbf{x}_0 \in \mathbb{R}^{k-1} \times 0$. Suppose there were a patch $\alpha_1 : U_1 \to V_1$ about $\mathbf{p}$ with U_1 open in $\mathbb{R}^k$. $W = V_0 \cap V_1$ is open in M; $W_0 = \alpha_0^{-1}(W)$ is open in $\mathbb{H}^k$ and contains $\mathbf{x}_0$, and $W_1 = \alpha_1^{-1}(W)$ is open in $\mathbb{R}^k$. By Theorem 24.1, $\alpha_0^{-1} \circ \alpha_1 : W_1 \to W_0$ is C^r, one-to-one onto W_0, with non-singular derivative. By Theorem 8.2 its image W_0 is open in $\mathbb{R}^k$. But $W_0 \subset \mathbb{H}^k$ contains the point $\mathbf{x}_0$ of $\mathbb{R}^{k-1} \times 0$, so it is not open in $\mathbb{R}^k$—a contradiction. ■

$\mathbb{H}^k$ is itself a k-manifold in $\mathbb{R}^k$, and by this lemma $\partial\mathbb{H}^k = \mathbb{R}^{k-1} \times 0$.

Theorem 24.3

Let M be a k-manifold in $\mathbb{R}^n$, of class C^r. If ∂M is non-empty, then ∂M is a $k-1$ manifold without boundary in $\mathbb{R}^n$ of class C^r.

Proof. Let $\mathbf{p} \in \partial M$ and $\alpha : U \to V$ a patch about $\mathbf{p}$. Then U is open in $\mathbb{H}^k$ and $\mathbf{p} = \alpha(\mathbf{x}_0)$ with $\mathbf{x}_0 \in \partial\mathbb{H}^k$. By Lemma 24.2, α maps points of $U \cap \mathbb{H}^k_+$ to interior points of M and points of $U \cap \partial\mathbb{H}^k$ to points of ∂M. So α restricted to $U \cap \partial\mathbb{H}^k$ maps it one-to-one onto the open set $V_0 = V \cap \partial M$ of ∂M. Let U_0 be the open set of $\mathbb{R}^{k-1}$ with $U_0 \times 0 = U \cap \partial\mathbb{H}^k$, and define $\alpha_0(\mathbf{x}) = \alpha(\mathbf{x}, 0)$ for $\mathbf{x} \in U_0$. Then $\alpha_0 : U_0 \to V_0$ is a coordinate patch on ∂M: it is C^r because α is; $D\alpha_0(\mathbf{x})$ consists of the first $k-1$ columns of $D\alpha(\mathbf{x}, 0)$, so it has rank $k-1$; and α_0^{-1} is the restriction to V_0 of α^{-1} followed by projection of $\mathbb{R}^k$ onto its first $k-1$ coordinates, hence continuous. ■

The patch α_0 on ∂M is said to be obtained by **restricting** the patch α on M.

Theorem 24.4

Let $\mathcal{O}$ be open in $\mathbb{R}^n$; let $f : \mathcal{O} \to \mathbb{R}$ be of class C^r. Let M be the set of points $\mathbf{x}$ for which $f(\mathbf{x}) = 0$; let N be the set of points for which $f(\mathbf{x}) \geq 0$. Suppose M is non-empty and $Df(\mathbf{x})$ has rank 1 at each point of M. Then N is an n-manifold in $\mathbb{R}^n$ and $\partial N = M$.

Proof. If $\mathbf{p} \in N$ with $f(\mathbf{p}) > 0$, let $U = \{\mathbf{x} : f(\mathbf{x}) > 0\}$, open in $\mathbb{R}^n$; the identity $\alpha : U \to U$ is a coordinate patch on N about $\mathbf{p}$ with domain open in $\mathbb{R}^n$. If $f(\mathbf{p}) = 0$, then $Df(\mathbf{p}) \neq 0$, so some $D_i f(\mathbf{p}) \neq 0$; say $D_n f(\mathbf{p}) \neq 0$. Define $F : \mathcal{O} \to \mathbb{R}^n$ by $F(\mathbf{x}) = (x_1, \ldots, x_{n-1}, f(\mathbf{x}))$. Then

$$DF = \begin{bmatrix} I_{n-1} & 0 \\ * & D_n f \end{bmatrix}$$

is non-singular at $\mathbf{p}$, so F is a diffeomorphism of a neighborhood A of $\mathbf{p}$ with an open set B of $\mathbb{R}^n$. F carries $A \cap N$ onto $B \cap \mathbb{H}^n$ (as $\mathbf{x} \in N$ iff $f(\mathbf{x}) \geq 0$) and $A \cap M$ onto $B \cap \partial\mathbb{H}^n$ (as $\mathbf{x} \in M$ iff $f(\mathbf{x}) = 0$). So $F^{-1} : B \cap \mathbb{H}^n \to A \cap N$ is a coordinate patch on N about $\mathbf{p}$ with domain open in $\mathbb{H}^n$, and $\mathbf{p} = F^{-1}(F(\mathbf{p}))$ with $F(\mathbf{p}) \in \partial\mathbb{H}^n$. By Lemma 24.2, $\partial N = M$. ■

Definition 24.2 (ball and sphere)

$B^n(a) = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x}\| \le a\}$ is the n-**ball** and $S^{n-1}(a) = \{\mathbf{x} : \|\mathbf{x}\| = a\}$ the $(n-1)$-**sphere** of radius a.

Corollary 24.5

The n-ball $B^n(a)$ is an n-manifold in $\mathbb{R}^n$ of class C^∞, and $S^{n-1}(a) = \partial B^n(a)$.

Proof. Apply Theorem 24.4 to $f(\mathbf{x}) = a^2 - \|\mathbf{x}\|^2$; $Df(\mathbf{x}) = [-2x_1 \ \cdots \ -2x_n]$ is nonzero at each point of $S^{n-1}(a)$. ■

§25. Integrating a Scalar Function over a Manifold

Motivation. To integrate over a manifold covered by several patches we cut the function into pieces supported in single patches, using partitions of unity, and add the results. The result does not depend on the choices made.

For simplicity we restrict to compact manifolds M; the general case can be handled as in §16.

Definition 25.1 (integral, support in one patch)

Let M be a compact k-manifold in $\mathbb{R}^n$ of class C^r, and $f : M \to \mathbb{R}$ continuous with $C = \operatorname{Support} f$ (compact). Suppose there is a coordinate patch $\alpha : U \to V$ on M with $C \subset V$; since $\alpha^{-1}(C)$ is compact we may assume U bounded. Define

$$\int_M f\,dV = \int_{\operatorname{Int} U} (f \circ \alpha) V(D\alpha),$$

where $\operatorname{Int} U = U$ if U is open in $\mathbb{R}^k$, and $\operatorname{Int} U = U \cap \mathbb{H}^k_+$ if U is open in $\mathbb{H}^k$ but not in $\mathbb{R}^k$.

This exists as an ordinary integral (hence as an extended one): $F = (f \circ \alpha)V(D\alpha)$ is continuous on U and vanishes outside the compact set $\alpha^{-1}(C)$, so it is bounded. If U is open in $\mathbb{R}^k$, F vanishes near each point of $\operatorname{Bd} U$; otherwise F vanishes near each point of $\operatorname{Bd} U$ not in $\partial\mathbb{H}^k$, a set of measure zero. Either way F is integrable over U and over $\operatorname{Int} U$.

Lemma 25.1

If the support of f can be covered by a single coordinate patch, the integral $\int_M f\,dV$ is well-defined, independent of the choice of coordinate patch.

Proof. First, if $\alpha : U \to V$ contains $\operatorname{Support} f$ and $W \subset U$ is open with $\alpha(W) \supset \operatorname{Support} f$, then $\int_{\operatorname{Int} W}(f \circ \alpha)V(D\alpha) = \int_{\operatorname{Int} U}(f \circ \alpha)V(D\alpha)$: the integrand vanishes outside W, then apply Theorem 13.6. Now let $\alpha_i : U_i \to V_i$ $(i = 0, 1)$ be patches with $V_0, V_1 \supset \operatorname{Support} f$. Let $W = V_0 \cap V_1$, $W_i = \alpha_i^{-1}(W)$. By the preceding remark it suffices to show $\int_{\operatorname{Int} W_0}(f \circ \alpha_0)V(D\alpha_0) = \int_{\operatorname{Int} W_1}(f \circ \alpha_1)V(D\alpha_1)$. Since $\alpha_1^{-1} \circ \alpha_0 : \operatorname{Int} W_0 \to \operatorname{Int} W_1$ is a diffeomorphism (Theorem 24.1, and Lemma 24.2 for interiors), this follows from Theorem 22.1. ■

Lemma 25.2

Let M be a compact k-manifold in $\mathbb{R}^n$, of class C^r. Given a covering of M by coordinate patches, there exists a finite collection of C^∞ functions $\phi_1, \dots, \phi_l$ mapping $\mathbb{R}^n$ into $\mathbb{R}$ such that:

(1) $\phi_i(\mathbf{x}) \geq 0$ for all $\mathbf{x}$.
(2) Given i, the support of ϕ_i is compact and there is a coordinate patch $\alpha_i : U_i \to V_i$ belonging to the given covering such that $(\text{Support}\,\phi_i) \cap M \subset V_i$.
(3) $\sum \phi_i(\mathbf{x}) = 1$ for $\mathbf{x} \in M$.

We call $\{\phi_1, \ldots, \phi_l\}$ a **partition of unity on** M dominated by the given collection of coordinate patches.

Proof. For each patch $\alpha : U \to V$ in the covering choose an open set A_V of $\mathbb{R}^n$ with $A_V \cap M = V$. Let A be the union of the A_V, and choose a C^∞ partition of unity on A with compact supports dominated by $\{A_V\}$ (Theorem 16.3). By local finiteness and compactness of M, all but finitely many of these functions vanish identically on M; let $\phi_1, \ldots, \phi_l$ be those that do not. ■

Definition 25.2 (integral over a compact manifold; volume)

Let M be a compact k-manifold in $\mathbb{R}^n$ of class C^r and $f : M \to \mathbb{R}$ continuous. Choose a partition of unity $\phi_1, \ldots, \phi_l$ on M dominated by the collection of all coordinate patches on M. Define

$$\int_M f\,dV = \sum_{i=1}^{l} \Big[\int_M (\phi_i f)\,dV \Big],$$

and the (k-dimensional) **volume** of M by $\mathrm{v}(M) = \int_M 1\,dV$.

If Support f lies in a single patch $\alpha : U \to V$, this agrees with Definition 25.1: with $A = \operatorname{Int} U$,

$$\sum_{i=1}^{l} \Big[\int_M (\phi_i f)\,dV \Big] = \sum_{i=1}^{l} \Big[\int_A (\phi_i \circ \alpha)(f \circ \alpha) V(D\alpha) \Big] = \int_A \Big[\sum_i (\phi_i \circ \alpha) \Big] (f \circ \alpha) V(D\alpha) = \int_A (f \circ \alpha) V(D\alpha),$$

since $\sum_i (\phi_i \circ \alpha) = 1$ on A. It is independent of the partition of unity: if $\psi_1, \ldots, \psi_m$ is another, then since Support $\psi_j f$ lies in a single patch, the computation just given (with $\psi_j f$ in place of f) gives $\sum_i \int_M \phi_i \psi_j f\,dV = \int_M \psi_j f\,dV$. Summing over j, $\sum_j \sum_i \int_M \phi_i \psi_j f\,dV = \sum_j \int_M \psi_j f\,dV$, and by symmetry the double sum also equals $\sum_i \int_M \phi_i f\,dV$.

Theorem 25.3

Let M be a compact k-manifold in $\mathbb{R}^n$, of class C^r. Let $f, g : M \to \mathbb{R}$ be continuous. Then

$$\int_M (af + bg)\,dV = a \int_M f\,dV + b \int_M g\,dV.$$

Proof [Supplement]. For f, g supported in one patch this is linearity of the integral over $\operatorname{Int} U$; in general apply it to each $\phi_i f, \phi_i g$ and sum. ■

This definition is good for theory but not for computation: to integrate over S^{n-1}, one breaks it into pieces, integrates over each, and adds. The next theorem justifies this.

Definition 25.3 (measure zero in M)

Let M be a compact k-manifold in $\mathbb{R}^n$ of class C^r. A subset D of M has **measure zero in** M if it can be covered by countably many coordinate patches $\alpha_i : U_i \to V_i$ such that each $D_i = \alpha_i^{-1}(D \cap V_i)$ has measure zero in $\mathbb{R}^k$.

Equivalently, $\alpha^{-1}(D \cap V)$ has measure zero in $\mathbb{R}^k$ for *every* patch $\alpha : U \to V$: it is the union of the sets $\alpha^{-1}(D \cap V \cap V_i)$, and each is the image of a subset of D_i under the C^r map $\alpha^{-1} \circ \alpha_i$, hence has measure zero (Lemma 18.1).

Theorem 25.4

Let M be a compact k-manifold in $\mathbb{R}^n$, of class C^r. Let $f : M \to \mathbb{R}$ be a continuous function. Suppose that $\alpha_i : A_i \to M_i$, for $i = 1, \dots, N$, is a coordinate patch on M, such that A_i is open in $\mathbb{R}^k$ and M is the disjoint union of the open sets $M_1, \dots, M_N$ of M and a set K of measure zero in M. Then

$$\int_M f\,dV = \sum_{i=1}^{N} \Big[\int_{A_i} (f \circ \alpha_i) V(D\alpha_i) \Big]. \tag{$*$}$$

So $\int_M f\,dV$ can be computed by breaking M into pieces that are parametrized-manifolds and integrating over each.

Proof. Both sides of $(*)$ are linear in f, so we may assume $C = \operatorname{Support} f$ lies in a single patch $\alpha : U \to V$, U bounded. Then $\int_M f\,dV = \int_{\operatorname{Int} U} (f \circ \alpha) V(D\alpha)$.

Step 1. Let $W_i = \alpha^{-1}(M_i \cap V)$ and $L = \alpha^{-1}(K \cap V)$. Then W_i is open in $\mathbb{R}^k$, L has measure zero in $\mathbb{R}^k$, and U is the disjoint union of L and the W_i. We show $\int_M f\,dV = \sum_i \int_{W_i} F$, where $F = (f \circ \alpha) V(D\alpha)$. These are ordinary integrals: F is bounded and vanishes near each point of $\operatorname{Bd} W_i$ not in L. Then

$$\sum_i \Big[\int_{W_i} F \Big] = \int_{(\operatorname{Int} U) - L} F = \int_{\operatorname{Int} U} F = \int_M f\,dV,$$

by additivity, since L has measure zero, and by definition.

Step 2. We show $\int_{W_i} F = \int_{A_i} F_i$, with $F_i = (f \circ \alpha_i) V(D\alpha_i)$. The map $\alpha_i^{-1} \circ \alpha$ is a diffeomorphism of W_i onto the open set $B_i = \alpha_i^{-1}(M_i \cap V)$ of $\mathbb{R}^k$, so by the change of variables theorem, as in Theorem 22.1, $\int_{W_i} F = \int_{B_i} F_i$. These may not be ordinary integrals, so care is needed. Since C is closed in M, $\alpha_i^{-1}(C)$ is closed in A_i, so $D_i = A_i - \alpha_i^{-1}(C)$ is open in $\mathbb{R}^k$, and F_i vanishes on D_i. Additivity of the extended integral gives

$$\int_{A_i} F_i = \int_{B_i} F_i + \int_{D_i} F_i - \int_{B_i \cap D_i} F_i;$$

the last two integrals vanish ($A_i = B_i \cup D_i$ since $\alpha_i^{-1}(C) \subset B_i$). ∎

Example 25.1

The open upper hemisphere of $S^2(a)$ has area $2\pi a^2$ (Example 22.4). The reflection $(x, y, z) \mapsto (x, y, -z)$ is an isometry of $\mathbb{R}^3$, so the open lower hemisphere also has area $2\pi a^2$. The two hemispheres make up all of the sphere except a set of measure zero (the equator), so by Theorem 25.4, $S^2(a)$ has area $4\pi a^2$.

Example 25.2

A method with no improper integrals. For $|z_0| < a$, the plane $z = z_0$ meets $S^2(a)$ in the circle $x^2 + y^2 = a^2 - z_0^2$. This suggests parametrizing $S^2(a)$ by $\alpha : A \to \mathbb{R}^3$,

$$\alpha(t, z) = ((a^2 - z^2)^{1/2} \cos t,\ (a^2 - z^2)^{1/2} \sin t,\ z), \qquad A = \{(t, z) : 0 < t < 2\pi,\ |z| < a\}.$$

α is a coordinate patch covering all of $S^2(a)$ except a great-circle arc, which has measure zero. Then

$$D\alpha = \begin{bmatrix} -(a^2 - z^2)^{1/2} \sin t & (-z \cos t)/(a^2 - z^2)^{1/2} \\ (a^2 - z^2)^{1/2} \cos t & (-z \sin t)/(a^2 - z^2)^{1/2} \\ 0 & 1 \end{bmatrix},$$

and one checks $V(D\alpha) = a$. Hence $\mathrm{v}(S^2(a)) = \int_A a = 4\pi a^2$.

Chapter 6. Differential Forms

Overview

Differentiation and integration have been treated in generality; the third topic is "vector integral calculus", whose major theorems carry the names of Green, Gauss and Stokes. For k-manifolds in $\mathbb{R}^n$ the concepts of linear algebra and vector calculus are no longer adequate; one needs *multilinear algebra*. Sections 26–28 develop it using only the linear algebra of Chapter 1. The rest of the chapter combines it with differentiation to define *differential forms* in $\mathbb{R}^n$ and their operators, which replace the vector and scalar fields and the operators grad, curl and div of $\mathbb{R}^3$. Integration and Stokes' theorem follow in Chapter 7.

§26. Multilinear Algebra

Motivation. The integrand over a k-dimensional surface must assign a number to each infinitesimal parallelepiped, that is, to k tangent vectors, linearly in each. This is a multilinear function, a tensor; this section develops their algebra.

26.1 Tensors

Definition 26.1 (k-tensor)

Let V be a vector space and $V^k = V \times \cdots \times V$ the set of k-tuples $(\mathbf{v}_1, \ldots, \mathbf{v}_k)$ of vectors of V. A function $f : V^k \to \mathbb{R}$ is **linear in the ith variable** if, for fixed $\mathbf{v}_j$ $(j \neq i)$, the function $T(\mathbf{v}) = f(\mathbf{v}_1, \ldots, \mathbf{v}_{i-1}, \mathbf{v}, \mathbf{v}_{i+1}, \ldots, \mathbf{v}_k)$ is linear. f is **multilinear** if it is linear in the ith variable for each i; it is then called a **k-tensor**, or a **tensor of order k**, on V. The set of k-tensors on V is $\mathcal{L}^k(V)$. $\mathcal{L}^1(V)$ is the set of linear maps $V \to \mathbb{R}$, the **dual space** V^* of V.

Theorem 26.1

The set of all k-tensors on V constitutes a vector space if we define

$$(f+g)(\mathbf{v}_1, \ldots, \mathbf{v}_k) = f(\mathbf{v}_1, \ldots, \mathbf{v}_k) + g(\mathbf{v}_1, \ldots, \mathbf{v}_k), \qquad (cf)(\mathbf{v}_1, \ldots, \mathbf{v}_k) = c(f(\mathbf{v}_1, \ldots, \mathbf{v}_k)).$$

Proof. Sums and scalar multiples of functions linear in each variable are linear in each variable; the vector space axioms hold because they hold for real numbers pointwise. The zero tensor is the function whose value on every k-tuple is zero. ∎

Lemma 26.2

Let $\mathbf{a}_1, \ldots, \mathbf{a}_n$ be a basis for V. If $f, g : V^k \to \mathbb{R}$ are k-tensors on V, and if $f(\mathbf{a}_{i_1}, \ldots, \mathbf{a}_{i_k}) = g(\mathbf{a}_{i_1}, \ldots, \mathbf{a}_{i_k})$ for every k-tuple $I = (i_1, \ldots, i_k)$ of integers from the set $\{1, \ldots, n\}$, then $f = g$.

(The integers $i_1, \ldots, i_k$ need not be distinct or in any order.)

Proof. Write $\mathbf{v}_i = \sum_{j=1}^n c_{ij}\mathbf{a}_j$. Expanding by multilinearity, one variable at a time,

$$f(\mathbf{v}_1, \ldots, \mathbf{v}_k) = \sum_{1 \le j_1, \ldots, j_k \le n} c_{1j_1} c_{2j_2} \cdots c_{kj_k}\, f(\mathbf{a}_{j_1}, \ldots, \mathbf{a}_{j_k}),$$

and likewise for g. ∎

Theorem 26.3

Let V be a vector space with basis $\mathbf{a}_1, \dots, \mathbf{a}_n$. Let $I = (i_1, \dots, i_k)$ be a k-tuple of integers from the set $\{1, \dots, n\}$. There is a unique k-tensor ϕ_I on V such that, for every k-tuple $J = (j_1, \dots, j_k)$ from the set $\{1, \dots, n\}$,

$$\phi_I(\mathbf{a}_{j_1}, \dots, \mathbf{a}_{j_k}) = \begin{cases} 0 & \text{if } I \neq J, \\ 1 & \text{if } I = J. \end{cases} \qquad (*)$$

The tensors ϕ_I form a basis for $\mathcal{L}^k(V)$.

The ϕ_I are the **elementary k-tensors** on V corresponding to the basis $\mathbf{a}_1, \dots, \mathbf{a}_n$. There are n^k of them, so $\dim \mathcal{L}^k(V) = n^k$. For $k = 1$, the basis $\phi_1, \dots, \phi_n$ of V^* is the basis **dual** to $\mathbf{a}_1, \dots, \mathbf{a}_n$.

Proof. Uniqueness follows from Lemma 26.2.

Existence: for $k = 1$, define the linear map ϕ_i by $\phi_i(\mathbf{a}_j) = 0$ $(i \neq j)$, 1 $(i = j)$ (Theorem 1.4). For $k > 1$ set

$$\phi_I(\mathbf{v}_1, \dots, \mathbf{v}_k) = [\phi_{i_1}(\mathbf{v}_1)] \cdot [\phi_{i_2}(\mathbf{v}_2)] \cdots [\phi_{i_k}(\mathbf{v}_k)];$$

it is multilinear and has the values $(*)$. Given a k-tensor f, let $d_I = f(\mathbf{a}_{i_1}, \dots, \mathbf{a}_{i_k})$ and $g = \sum_J d_J \phi_J$ (sum over all k-tuples J). Then $g(\mathbf{a}_{i_1}, \dots, \mathbf{a}_{i_k}) = d_I$, so $f = g$ by Lemma 26.2; the representation is unique by the same lemma. ■

Thus a k-tensor may be defined by specifying its values arbitrarily on k-tuples of basis elements.

Example 26.1

Let $V = \mathbb{R}^n$ with the usual basis and dual basis $\phi_1, \dots, \phi_n$. Then $\phi_i(\mathbf{x}) = \phi_i(x_1\mathbf{e}_1 + \cdots + x_n\mathbf{e}_n) = x_i$: ϕ_i is projection onto the ith coordinate. More generally $\phi_I(\mathbf{x}_1, \dots, \mathbf{x}_k) = \phi_{i_1}(\mathbf{x}_1) \cdots \phi_{i_k}(\mathbf{x}_k)$; writing $X = [\mathbf{x}_1 \cdots \mathbf{x}_k]$ with entries x_{ij},

$$\phi_I(\mathbf{x}_1, \dots, \mathbf{x}_k) = x_{i_1 1} x_{i_2 2} \cdots x_{i_k k},$$

a *monomial* in the components of the $\mathbf{x}_j$. The general k-tensor on $\mathbb{R}^n$ is a linear combination of such monomials: the general 1-tensor is $f(\mathbf{x}) = d_1 x_1 + \cdots + d_n x_n$, the general 2-tensor is $g(\mathbf{x}, \mathbf{y}) = \sum_{i,j=1}^n d_{ij} x_i y_j$, and so on.

26.2 The tensor product

Definition 26.2 (tensor product)

Let f be a k-tensor and g an l-tensor on V. The $(k + l)$-tensor $f \otimes g$ is

$$(f \otimes g)(\mathbf{v}_1, \dots, \mathbf{v}_{k+l}) = f(\mathbf{v}_1, \dots, \mathbf{v}_k) \cdot g(\mathbf{v}_{k+1}, \dots, \mathbf{v}_{k+l}).$$

It is multilinear, and is the **tensor product** of f and g.

Theorem 26.4

Let f, g, h be tensors on V. Then the following properties hold:

(1) (Associativity) $f \otimes (g \otimes h) = (f \otimes g) \otimes h$.

(2) (Homogeneity) $(cf) \otimes g = c(f \otimes g) = f \otimes (cg)$.

(3) (Distributivity) Suppose f and g have the same order. Then $(f + g) \otimes h = f \otimes h + g \otimes h$

and $h \otimes (f+g) = h \otimes f + h \otimes g$.

(4) Given a basis $\mathbf{a}_1, \ldots, \mathbf{a}_n$ for V, the corresponding elementary tensors ϕ_I satisfy $\phi_I = \phi_{i_1} \otimes \phi_{i_2} \otimes \cdots \otimes \phi_{i_k}$, where $I = (i_1, \ldots, i_k)$.

No parentheses are needed in (4) by associativity; nothing is said about commutativity, which almost never holds.

Proof. Straightforward. For instance, if f, g, h have orders k, l, m, both $(f \otimes (g \otimes h))(\mathbf{v}_1, \ldots, \mathbf{v}_{k+l+m})$ and $((f \otimes g) \otimes h)(\mathbf{v}_1, \ldots, \mathbf{v}_{k+l+m})$ equal $f(\mathbf{v}_1, \ldots, \mathbf{v}_k) \cdot g(\mathbf{v}_{k+1}, \ldots, \mathbf{v}_{k+l}) \cdot h(\mathbf{v}_{k+l+1}, \ldots, \mathbf{v}_{k+l+m})$. Property (4) is the formula in the proof of Theorem 26.3. ■

26.3 The action of a linear transformation

Definition 26.3 (dual transformation)

Let $T : V \to W$ be linear. The **dual transformation** $T^* : \mathcal{L}^k(W) \to \mathcal{L}^k(V)$ (in the opposite direction) is

$$(T^* f)(\mathbf{v}_1, \ldots, \mathbf{v}_k) = f(T(\mathbf{v}_1), \ldots, T(\mathbf{v}_k)),$$

i.e. $T^* f$ is the composite of $T \times \cdots \times T : V^k \to W^k$ with $f : W^k \to \mathbb{R}$.

$T^* f$ is multilinear since T is linear and f multilinear.

Theorem 26.5

Let $T : V \to W$ be a linear transformation; let $T^* : \mathcal{L}^k(W) \to \mathcal{L}^k(V)$ be the dual transformation. Then:
(1) T^* is linear.
(2) $T^*(f \otimes g) = T^* f \otimes T^* g$.
(3) If $S : W \to X$ is a linear transformation, then $(S \circ T)^* f = T^*(S^* f)$.

Proof. Straightforward; e.g. for (1),

$$(T^*(af + bg))(\mathbf{v}_1, \ldots, \mathbf{v}_k) = (af + bg)(T(\mathbf{v}_1), \ldots, T(\mathbf{v}_k)) = aT^* f(\mathbf{v}_1, \ldots, \mathbf{v}_k) + bT^* g(\mathbf{v}_1, \ldots, \mathbf{v}_k).$$

For (2), both sides evaluated at $(\mathbf{v}_1, \ldots, \mathbf{v}_{k+l})$ give $f(T\mathbf{v}_1, \ldots, T\mathbf{v}_k)\, g(T\mathbf{v}_{k+1}, \ldots, T\mathbf{v}_{k+l})$. For (3), both sides give $f(S(T(\mathbf{v}_1)), \ldots, S(T(\mathbf{v}_k)))$. ■

Property (3) says the diagram $V \xrightarrow{T} W \xrightarrow{S} X$ induces $\mathcal{L}^k(X) \xrightarrow{S^*} \mathcal{L}^k(W) \xrightarrow{T^*} \mathcal{L}^k(V)$ with $(S \circ T)^* = T^* \circ S^*$.

§27. Alternating Tensors

Motivation. Orientation matters in integration: reversing the order of two vectors should reverse the sign, and degenerate parallelepipeds should get zero. Alternating tensors have exactly these properties, and the determinant is the model example.

27.1 Permutations

Definition 27.1 (permutation, symmetric group)

Let $k \geq 2$. A **permutation** of $\{1, \ldots, k\}$ is a one-to-one map σ of this set onto itself; the set of them is S_k. With composition and inverses, S_k is a group, the **symmetric group** (**permutation group**); it has $k!$ elements.

Definition 27.2 (elementary permutation)

For $1 \leq i < k$, let $e_i \in S_k$ be defined by $e_i(j) = j$ for $j \neq i, i+1$, $e_i(i) = i+1$, $e_i(i+1) = i$. e_i is an **elementary permutation**; $e_i \circ e_i$ is the identity.

Lemma 27.1

If $\sigma \in S_k$, then σ equals a composite of elementary permutations.

Proof. Say σ *fixes the first* i *integers* if $\sigma(j) = j$ for $1 \leq j \leq i$ $(0 \leq i \leq k)$. If $i = k$, σ is the identity, which equals $e_j \circ e_j$. We show: if σ fixes the first $i-1$ integers $(0 < i \leq k)$, then $\sigma = \pi \circ \sigma'$ with π a composite of elementary permutations and σ' fixing the first i integers; induction completes the proof. Since σ fixes $1, \ldots, i-1$ and is one-to-one, $\sigma(i) = l \geq i$. If $l = i$, take $\sigma' = \sigma$, $\pi =$ identity. If $l > i$, let $\sigma' = e_i \circ \cdots \circ e_{l-1} \circ \sigma$. Then σ' fixes $1, \ldots, i-1$ (as σ and $e_i, \ldots, e_{l-1}$ do), and $\sigma'(i) = e_i(\cdots(e_{l-1}(l))\cdots) = i$. And $\sigma = e_{l-1} \circ \cdots \circ e_i \circ \sigma'$. ■

Definition 27.3 (sign of a permutation)

Let $\sigma \in S_k$. A pair $i < j$ with $\sigma(i) > \sigma(j)$ is an **inversion** in σ. The **sign** $\operatorname{sgn}\sigma$ is -1 if the number of inversions is odd and $+1$ if it is even; σ is **odd** or **even** accordingly.

Lemma 27.2

Let $\sigma, \tau \in S_k$.

(a) If σ equals a composite of m elementary permutations, then $\operatorname{sgn}\sigma = (-1)^m$.

(b) $\operatorname{sgn}(\sigma \circ \tau) = (\operatorname{sgn}\sigma) \cdot (\operatorname{sgn}\tau)$.

(c) $\operatorname{sgn}\sigma^{-1} = \operatorname{sgn}\sigma$.

(d) If $p \neq q$, and if τ is the permutation that exchanges p and q and leaves all other integers fixed, then $\operatorname{sgn}\tau = -1$.

Proof.

Step 1. For any σ, $\operatorname{sgn}(\sigma \circ e_l) = -\operatorname{sgn}\sigma$. Write the values of σ in order: $(\sigma(1), \ldots, \sigma(l), \sigma(l+1), \ldots, \sigma(k))$ $(*)$; for $\tau = \sigma \circ e_l$ the sequence is $(\sigma(1), \ldots, \sigma(l+1), \sigma(l), \ldots, \sigma(k))$ $(**)$. Inversions are pairs of entries appearing in reverse natural order. For $p \neq q$: if neither is l or $l+1$, $\sigma(p), \sigma(q)$ occupy the same slots in both sequences; if one is l or $l+1$ and the other is not, they are in the same relative order in both. Only the pair $\sigma(l), \sigma(l+1)$ changes: it is an inversion in exactly one of the two sequences. So $(**)$ has one more or one fewer inversion than $(*)$.

Step 2. The identity has sign $+1$, and composing with m elementary permutations changes the sign m times: (a). Writing σ, τ as composites of m, n elementary permutations, (b) follows from $(-1)^{m+n} = (-1)^m(-1)^n$.

(c) follows from (b) since $\sigma^{-1} \circ \sigma$ is the identity.

(d) Count inversions: for $p < q$, $q = p + l$, the values of τ in order are $(1, \dots, p-1, q, p+1, \dots, p+l-1, p, p+l+1, \dots, k)$. The pairs $\{q, p+1\}, \dots, \{q, p+l-1\}$ and $\{p+1, p\}, \dots, \{p+l-1, p\}$ are inversions, and so is $\{q, p\}$: $2l-1$ in all, an odd number. ■

27.2 Alternating tensors

Definition 27.4 (symmetric and alternating tensors)

Let f be a k-tensor on V and $\sigma \in S_k$. Define the k-tensor f^σ by

$$f^\sigma(\mathbf{v}_1, \dots, \mathbf{v}_k) = f(\mathbf{v}_{\sigma(1)}, \dots, \mathbf{v}_{\sigma(k)}).$$

f is **symmetric** if $f^e = f$ for each elementary permutation e, and **alternating** if $f^e = -f$ for every elementary permutation e; i.e. f is alternating if

$$f(\mathbf{v}_1, \dots, \mathbf{v}_{i+1}, \mathbf{v}_i, \dots, \mathbf{v}_k) = -f(\mathbf{v}_1, \dots, \mathbf{v}_i, \mathbf{v}_{i+1}, \dots, \mathbf{v}_k) \quad \text{for all } i.$$

The set of alternating k-tensors on V is $\mathcal{A}^k(V)$, a linear subspace of $\mathcal{L}^k(V)$. By convention, $\mathcal{A}^1(V) = \mathcal{L}^1(V)$.

We are primarily interested in alternating tensors.

Example 27.1

Elementary tensors of order $k > 1$ are not alternating, but certain combinations are. With $V = \mathbb{R}^n$ and the dual basis ϕ_i, $f = \phi_{i,j} - \phi_{j,i}$ satisfies

$$f(\mathbf{x}, \mathbf{y}) = x_i y_j - x_j y_i = \det \begin{bmatrix} x_i & y_i \\ x_j & y_j \end{bmatrix},$$

so $f(\mathbf{y}, \mathbf{x}) = -f(\mathbf{x}, \mathbf{y})$. Similarly $g(\mathbf{x}, \mathbf{y}, \mathbf{z}) = \det \begin{bmatrix} x_i & y_i & z_i \\ x_j & y_j & z_j \\ x_k & y_k & z_k \end{bmatrix}$ is an alternating 3-tensor on $\mathbb{R}^n$; $g = \phi_{i,j,k} + \phi_{j,k,i} + \phi_{k,i,j} - \phi_{j,i,k} - \phi_{i,k,j} - \phi_{k,j,i}$. Alternating tensors and determinants are intimately related.

Lemma 27.3

Let f be a k-tensor on V; let $\sigma, \tau \in S_k$.

(a) The transformation $f \to f^\sigma$ is a linear transformation of $\mathcal{L}^k(V)$ to $\mathcal{L}^k(V)$. It has the property that for all σ, τ, $(f^\sigma)^\tau = f^{\tau \circ \sigma}$.

(b) The tensor f is alternating if and only if $f^\sigma = (\operatorname{sgn} \sigma) f$ for all σ. If f is alternating and if $\mathbf{v}_p = \mathbf{v}_q$ with $p \neq q$, then $f(\mathbf{v}_1, \dots, \mathbf{v}_k) = 0$.

Proof. (a) Linearity says $(af + bg)^\sigma = af^\sigma + bg^\sigma$. With $\mathbf{w}_i = \mathbf{v}_{\tau(i)}$,

$$(f^\sigma)^\tau(\mathbf{v}_1, \dots, \mathbf{v}_k) = f^\sigma(\mathbf{w}_1, \dots, \mathbf{w}_k) = f(\mathbf{w}_{\sigma(1)}, \dots, \mathbf{w}_{\sigma(k)}) = f(\mathbf{v}_{\tau(\sigma(1))}, \dots, \mathbf{v}_{\tau(\sigma(k))}) = f^{\tau\circ\sigma}(\mathbf{v}_1, \dots, \mathbf{v}_k).$$

(b) If $f^\sigma = (\operatorname{sgn} \sigma) f$ for all σ, then $f^e = -f$ for elementary e. Conversely, write $\sigma = \sigma_1 \circ \cdots \circ \sigma_m$ with σ_i elementary; by (a), $f^\sigma = ((\cdots (f^{\sigma_m}) \cdots)^{\sigma_2})^{\sigma_1} = (-1)^m f = (\operatorname{sgn} \sigma) f$. If $\mathbf{v}_p = \mathbf{v}_q$ $(p \neq q)$ and τ exchanges p and q, then $f^\tau(\mathbf{v}_1, \dots, \mathbf{v}_k) = f(\mathbf{v}_1, \dots, \mathbf{v}_k)$, while $f^\tau = -f$ since $\operatorname{sgn} \tau = -1$; so $f(\mathbf{v}_1, \dots, \mathbf{v}_k) = 0$. ■

For $k = 1$ nothing is to be done. For $k > n$, $\mathcal{A}^k(V)$ is trivial: some basis element appears twice in each k-tuple of basis elements, so an alternating f vanishes on all of them (Lemma 26.2). For $1 < k \leq n$, an alternating tensor is determined by its values on k-tuples of basis elements with indices in ascending order, which may be specified arbitrarily.

Lemma 27.4

Let $\mathbf{a}_1, \ldots, \mathbf{a}_n$ be a basis for V. If f, g are alternating k-tensors on V, and if $f(\mathbf{a}_{i_1}, \ldots, \mathbf{a}_{i_k}) = g(\mathbf{a}_{i_1}, \ldots, \mathbf{a}_{i_k})$ for every ascending k-tuple of integers $I = (i_1, \ldots, i_k)$ from the set $\{1, \ldots, n\}$, then $f = g$.

Proof. By Lemma 26.2 it suffices to show f, g agree on an arbitrary $(\mathbf{a}_{j_1}, \ldots, \mathbf{a}_{j_k})$. If two indices j_p, j_q coincide, both values are 0 (Lemma 27.3). Otherwise let σ be the permutation such that $I = (j_{\sigma(1)}, \ldots, j_{\sigma(k)})$ is ascending. Then $f(\mathbf{a}_{i_1}, \ldots, \mathbf{a}_{i_k}) = f^\sigma(\mathbf{a}_{j_1}, \ldots, \mathbf{a}_{j_k}) = (\operatorname{sgn}\sigma) f(\mathbf{a}_{j_1}, \ldots, \mathbf{a}_{j_k})$, and similarly for g. Since f, g agree on the ascending tuple, they agree on $(\mathbf{a}_{j_1}, \ldots, \mathbf{a}_{j_k})$. ■

Theorem 27.5

Let V be a vector space with basis $\mathbf{a}_1, \ldots, \mathbf{a}_n$. Let $I = (i_1, \ldots, i_k)$ be an ascending k-tuple from the set $\{1, \ldots, n\}$. There is a unique alternating k-tensor ψ_I on V such that for every ascending k-tuple $J = (j_1, \ldots, j_k)$ from the set $\{1, \ldots, n\}$,

$$\psi_I(\mathbf{a}_{j_1}, \ldots, \mathbf{a}_{j_k}) = \begin{cases} 0 & \text{if } I \neq J, \\ 1 & \text{if } I = J. \end{cases}$$

The tensors ψ_I form a basis for $\mathcal{A}^k(V)$. The tensor ψ_I in fact satisfies the formula

$$\psi_I = \sum_\sigma (\operatorname{sgn}\sigma)(\phi_I)^\sigma,$$

where the summation extends over all $\sigma \in S_k$.

The ψ_I are the **elementary alternating k-tensors** on V corresponding to the basis $\mathbf{a}_1, \ldots, \mathbf{a}_n$.

Proof. Uniqueness follows from Lemma 27.4. Define ψ_I by the formula. It is alternating: for $\tau \in S_k$,

$$(\psi_I)^\tau = \sum_\sigma (\operatorname{sgn}\sigma)((\phi_I)^\sigma)^\tau = \sum_\sigma (\operatorname{sgn}\sigma)(\phi_I)^{\tau\circ\sigma} = (\operatorname{sgn}\tau)\sum_\sigma (\operatorname{sgn}(\tau\circ\sigma))(\phi_I)^{\tau\circ\sigma} = (\operatorname{sgn}\tau)\psi_I,$$

since $\tau \circ \sigma$ ranges over S_k as σ does. Its values: $\psi_I(\mathbf{a}_{j_1}, \ldots, \mathbf{a}_{j_k}) = \sum_\sigma (\operatorname{sgn}\sigma)\phi_I(\mathbf{a}_{j_{\sigma(1)}}, \ldots, \mathbf{a}_{j_{\sigma(k)}})$. At most one term is nonzero, the one with $I = (j_{\sigma(1)}, \ldots, j_{\sigma(k)})$; as I and J are both ascending this happens only if $I = J$ and σ is the identity, giving value 1.

The ψ_I form a basis: given alternating f, let $d_I = f(\mathbf{a}_{i_1}, \ldots, \mathbf{a}_{i_k})$ for ascending I, and $g = \sum_{[J]} d_J \psi_J$ (sum over ascending J). Then f and g agree on ascending k-tuples of basis elements, so $f = g$ by Lemma 27.4; uniqueness of the representation follows from the same lemma. ■

Thus, once a basis is chosen, each alternating k-tensor can be written uniquely as $f = \sum_{[J]} d_J \psi_J$; the d_J are the **components** of f relative to $\{\psi_J\}$. Since each k-element subset of $\{1, \ldots, n\}$ gives exactly one ascending k-tuple,

$$\dim \mathcal{A}^k(V) = \binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Theorem 27.6

Let $T : V \to W$ be a linear transformation. If f is an alternating tensor on W, then T^*f is an alternating tensor on V.

Proof [Supplement]. $(T^*f)(\dots, \mathbf{v}_{i+1}, \mathbf{v}_i, \dots) = f(\dots, T\mathbf{v}_{i+1}, T\mathbf{v}_i, \dots) = -f(\dots, T\mathbf{v}_i, T\mathbf{v}_{i+1}, \dots) = -(T^*f)(\dots, \mathbf{v}_i, \mathbf{v}_{i+1}, \dots)$. ■

27.3 Determinants

We now (at long last) construct the determinant for matrices larger than 3 by 3 (cf. §2).

Definition 27.5 (determinant)

Let $\phi_1, \dots, \phi_n$ be the basis of $\mathcal{L}^1(\mathbb{R}^n)$ dual to the usual basis. $\mathcal{A}^n(\mathbb{R}^n)$ has dimension 1; its unique elementary alternating n-tensor is $\psi_{1,\dots,n}$. For an n by n matrix $X = [\mathbf{x}_1 \ \cdots \ \mathbf{x}_n]$ define

$$\det X = \psi_{1,\dots,n}(\mathbf{x}_1, \dots, \mathbf{x}_n).$$

This satisfies the axioms of §2. Let $g(X) = \psi_I(\mathbf{x}_1, \dots, \mathbf{x}_n)$ with $I = (1, \dots, n)$; g is multilinear and alternating in the *columns* of X. So $f(A) = g(A^{\text{tr}})$ is multilinear and alternating in the rows of A, and $f(I_n) = g(I_n) = \psi_I(\mathbf{e}_1, \dots, \mathbf{e}_n) = 1$; hence f satisfies the axioms. By Theorem 2.11, $f(A) = f(A^{\text{tr}})$, so $f(A) = g(A)$ and g satisfies the axioms too. Theorem 27.5 gives the formula

$$\det X = \sum_\sigma (\operatorname{sgn}\sigma)\phi_I(\mathbf{x}_{\sigma(1)}, \dots, \mathbf{x}_{\sigma(n)}) = \sum_\sigma (\operatorname{sgn}\sigma) x_{1,\sigma(1)} \cdot x_{2,\sigma(2)} \cdots x_{n,\sigma(n)},$$

sometimes used as the definition of the determinant.

Theorem 27.7

Let ψ_I be an elementary alternating tensor on $\mathbb{R}^n$ corresponding to the usual basis for $\mathbb{R}^n$, where $I = (i_1, \dots, i_k)$. Given vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$ of $\mathbb{R}^n$, let X be the matrix $X = [\mathbf{x}_1 \ \cdots \ \mathbf{x}_k]$. Then

$$\psi_I(\mathbf{x}_1, \dots, \mathbf{x}_k) = \det X_I,$$

where X_I denotes the matrix whose successive rows are rows $i_1, \dots, i_k$ of X.

Proof. $\psi_I(\mathbf{x}_1, \dots, \mathbf{x}_k) = \sum_\sigma (\operatorname{sgn}\sigma)\phi_I(\mathbf{x}_{\sigma(1)}, \dots, \mathbf{x}_{\sigma(k)}) = \sum_\sigma (\operatorname{sgn}\sigma) x_{i_1,\sigma(1)} \cdot x_{i_2,\sigma(2)} \cdots x_{i_k,\sigma(k)}$, which is the formula for $\det X_I$. ■

Example 27.2

The elementary alternating 3-tensors on $\mathbb{R}^4$ are

$$\psi_{i,j,k}(\mathbf{x}, \mathbf{y}, \mathbf{z}) = \det \begin{bmatrix} x_i & y_i & z_i \\ x_j & y_j & z_j \\ x_k & y_k & z_k \end{bmatrix},$$

for $(i, j, k) = (1, 2, 3), (1, 2, 4), (1, 3, 4), (2, 3, 4)$; the general element of $\mathcal{A}^3(\mathbb{R}^4)$ is a linear combination of these four.

Notation

Multilinear algebra has the *exterior product* construction $\Lambda^k(W)$, a quotient of the k-fold tensor product of W. For finite-dimensional V, $\Lambda^k(V^*)$ is naturally isomorphic to the space of alternating k-tensors on V, which some authors therefore denote $\Lambda^k(V^*)$, others $\Lambda^k(V)$. To avoid this confusion we use the neutral notation $\mathcal{A}^k(V)$.

§28. The Wedge Product

Motivation. We need to multiply alternating tensors so that $dx \wedge dy$ represents oriented area. The wedge product is the alternating version of the tensor product and makes the alternating tensors into an algebra with basis dx_I.

$f \otimes g$ is almost never alternating, even if f and g are; something else is needed. The actual definition matters less than its properties:

Theorem 28.1

Let V be a vector space. There is an operation that assigns, to each $f \in \mathcal{A}^k(V)$ and each $g \in \mathcal{A}^l(V)$, an element $f \wedge g \in \mathcal{A}^{k+l}(V)$, such that the following properties hold:

(1) (Associativity) $f \wedge (g \wedge h) = (f \wedge g) \wedge h$.

(2) (Homogeneity) $(cf) \wedge g = c(f \wedge g) = f \wedge (cg)$.

(3) (Distributivity) If f and g have the same order, $(f + g) \wedge h = f \wedge h + g \wedge h$ and $h \wedge (f + g) = h \wedge f + h \wedge g$.

(4) (Anticommutativity) If f and g have orders k and l, respectively, then $g \wedge f = (-1)^{kl} f \wedge g$.

(5) Given a basis $\mathbf{a}_1, \ldots, \mathbf{a}_n$ for V, let ϕ_i denote the dual basis for V^*, and let ψ_I denote the corresponding elementary alternating tensors. If $I = (i_1, \ldots, i_k)$ is an ascending k-tuple of integers from the set $\{1, \ldots, n\}$, then $\psi_I = \phi_{i_1} \wedge \phi_{i_2} \wedge \cdots \wedge \phi_{i_k}$.

These five properties characterize the product $\wedge$ uniquely for finite-dimensional spaces V. Furthermore, it has the following additional property:

(6) If $T : V \to W$ is a linear transformation, and if f and g are alternating tensors on W, then $T^*(f \wedge g) = T^* f \wedge T^* g$.

$f \wedge g$ is the **wedge product** of f and g. By (4), $f \wedge f = 0$ for an alternating tensor f of odd order.

Proof.

Step 1. For a k-tensor F on V (not necessarily alternating) define $A : \mathcal{L}^k(V) \to \mathcal{L}^k(V)$ by

$$AF = \sum_\sigma (\operatorname{sgn}\sigma) F^\sigma,$$

summed over $\sigma \in S_k$. (Some authors include a factor $1/k!$.) Thus $\psi_I = A\phi_I$. A has the properties:

(i) A is linear;

(ii) AF is alternating;

(iii) if F is already alternating, $AF = (k!)F$. Indeed (i) holds since $F \mapsto F^\sigma$ is linear;

(ii) by the computation $(AF)^\tau = \sum_\sigma (\operatorname{sgn}\sigma) F^{\tau\circ\sigma} = (\operatorname{sgn}\tau) \sum_\sigma (\operatorname{sgn}\tau\circ\sigma) F^{\tau\circ\sigma} = (\operatorname{sgn}\tau) AF$;

(iii) since $F^\sigma = (\operatorname{sgn}\sigma) F$ gives $AF = \sum_\sigma (\operatorname{sgn}\sigma)^2 F = (k!)F$.

Step 2. For $f \in \mathcal{A}^k(V)$, $g \in \mathcal{A}^l(V)$ define

$$f \wedge g = \frac{1}{k!\,l!} A(f \otimes g),$$

an alternating tensor of order $k+l$. *Motivation for* $1/k!l!$: in $(f\wedge g)(\mathbf{v}_1, \dots, \mathbf{v}_{k+l}) = \frac{1}{k!l!}\sum_\sigma (\operatorname{sgn}\sigma) f(\mathbf{v}_{\sigma(1)}, \dots, \mathbf{v}_{\sigma(k)})$ $g(\mathbf{v}_{\sigma(k+1)}, \dots, \mathbf{v}_{\sigma(k+l)})$, permuting $\mathbf{v}_{\sigma(1)}, \dots, \mathbf{v}_{\sigma(k)}$ among themselves and $\mathbf{v}_{\sigma(k+1)}, \dots, \mathbf{v}_{\sigma(k+l)}$ among themselves gives $k!l!$ terms with the same value (since f, g are alternating, the sign changes cancel); dividing by $k!l!$ removes this redundancy.

Step 3. Homogeneity: $(cf) \wedge g = A((cf)\otimes g)/k!l! = A(c(f\otimes g))/k!l! = cA(f\otimes g)/k!l! = c(f\wedge g)$; similarly for the other part. Distributivity follows from distributivity of $\otimes$ and linearity of A.

Step 4. More generally, for tensors F, G of orders k, l (not necessarily alternating), $A(F\otimes G) = (-1)^{kl} A(G\otimes F)$. Let $\pi \in S_{k+l}$ with $(\pi(1), \dots, \pi(k+l)) = (k+1, k+2, \dots, k+l, 1, 2, \dots, k)$; counting inversions, $\operatorname{sgn}\pi = (-1)^{kl}$. Since $(G\otimes F)^\pi(\mathbf{v}_1, \dots, \mathbf{v}_{k+l}) = G(\mathbf{v}_{k+1}, \dots, \mathbf{v}_{k+l}) \cdot F(\mathbf{v}_1, \dots, \mathbf{v}_k)$, we have $(G\otimes F)^\pi = F \otimes G$. Hence

$$A(F\otimes G) = \sum_\sigma (\operatorname{sgn}\sigma)((G\otimes F)^\pi)^\sigma = (\operatorname{sgn}\pi)\sum_\sigma (\operatorname{sgn}\sigma\circ\pi)(G\otimes F)^{\sigma\circ\pi} = (\operatorname{sgn}\pi) A(G\otimes F),$$

since $\sigma\circ\pi$ runs over S_{k+l}. Taking $F = f$, $G = g$ gives anticommutativity.

Step 5. If F, G have orders k, l and $AF = 0$, then $A(F\otimes G) = 0$. In $A(F\otimes G)$, group together the terms with the same last factor $G(\mathbf{v}_{\sigma(k+1)}, \dots, \mathbf{v}_{\sigma(k+l)})$. They are

$$(\operatorname{sgn}\sigma)\Big[\sum_\tau (\operatorname{sgn}\tau) F(\mathbf{v}_{\sigma(\tau(1))}, \dots, \mathbf{v}_{\sigma(\tau(k))})\Big] \cdot G(\mathbf{v}_{\sigma(k+1)}, \dots, \mathbf{v}_{\sigma(k+l)}),$$

τ ranging over S_k, and the bracket is $AF(\mathbf{v}_{\sigma(1)}, \dots, \mathbf{v}_{\sigma(k)}) = 0$. Each group cancels.

Step 6. If F is any tensor and h alternating of order m, then $(AF)\wedge h = \frac{1}{m!} A(F\otimes h)$. With F of order k this reads $\frac{1}{k!m!}A((AF)\otimes h) = \frac{1}{m!}A(F\otimes h)$, i.e. (by linearity and distributivity) $A\{[AF - (k!)F]\otimes h\} = 0$. By Step 5 it suffices that $A[AF - (k!)F] = 0$, which follows from property (iii) of A, since AF is alternating: $A(AF) = (k!)AF$.

Step 7. For alternating f, g, h of orders k, l, m: $(f\wedge g)\wedge h = \frac{1}{k!l!m!}A((f\otimes g)\otimes h)$. Indeed with $F = f\otimes g$, $f\wedge g = \frac{1}{k!l!}AF$, so by Step 6, $(f\wedge g)\wedge h = \frac{1}{k!l!}(AF)\wedge h = \frac{1}{k!l!m!}A(F\otimes h)$.

Step 8. Associativity:

$$\begin{aligned}
(k!l!m!)(f\wedge g)\wedge h &= A((f\otimes g)\otimes h) && \text{by Step 7,}\\
&= A(f\otimes(g\otimes h)) && \text{by associativity of } \otimes,\\
&= (-1)^{k(l+m)} A((g\otimes h)\otimes f) && \text{by Step 4,}\\
&= (-1)^{k(l+m)}(l!m!k!)(g\wedge h)\wedge f && \text{by Step 7,}\\
&= (k!l!m!) f\wedge(g\wedge h) && \text{by anticommutativity.}
\end{aligned}$$

Step 9. Property (5). More generally, for 1-tensors $f_1, \dots, f_k$,

$$A(f_1\otimes\cdots\otimes f_k) = f_1\wedge\cdots\wedge f_k. \qquad (*)$$

Then (5) follows since $\psi_I = A\phi_I = A(\phi_{i_1}\otimes\cdots\otimes\phi_{i_k})$. $(*)$ is trivial for $k = 1$; if true for $k-1$, let $F = f_1\otimes\cdots\otimes f_{k-1}$; by Step 6 with $m = 1$, $A(F\otimes f_k) = (AF)\wedge f_k = (f_1\wedge\cdots\wedge f_{k-1})\wedge f_k$.

Step 10. Uniqueness: with only (1)–(5) one can compute wedge products in finite dimensions. Write $f = \sum_{[I]} b_I\psi_I$, $g = \sum_{[J]} c_J\psi_J$; distributivity and homogeneity give $f\wedge g = \sum_{[I]}\sum_{[J]} b_I c_J \psi_I\wedge\psi_J$, and $\psi_I\wedge\psi_J = (\phi_{i_1}\wedge\cdots\wedge\phi_{i_k})\wedge(\phi_{j_1}\wedge\cdots\wedge\phi_{j_l})$. By associativity and the rules $\phi_i\wedge\phi_j = -\phi_j\wedge\phi_i$, $\phi_i\wedge\phi_i = 0$ (from anticommutativity), $\psi_I\wedge\psi_J = 0$ if two indices coincide, and otherwise equals $\operatorname{sgn}\pi$ times ψ_K, where K is the ascending rearrangement of (I, J) and π the permutation effecting it.

Step 11. Property (6). For any tensor F on W, $T^*(F^\sigma) = (T^*F)^\sigma$; since T^* is linear, $T^*(AF) = A(T^*F)$. For alternating f, g of orders k, l on W,

$$T^*(f \wedge g) = \frac{1}{k!l!} T^*(A(f \otimes g)) = \frac{1}{k!l!} A(T^*(f \otimes g)) = \frac{1}{k!l!} A((T^*f) \otimes (T^*g)) = (T^*f) \wedge (T^*g),$$

using Theorem 26.5. ∎

Remark

Some texts define the wedge product with coefficient $1/(k+l)!$ instead of $1/k!l!$. This also gives an associative operation, and all properties of Theorem 28.1 persist except (5), which acquires a factor $k!$ on the right side of the formula for ψ_I. We need only alternating tensors and the properties above.

§29. Tangent Vectors and Differential Forms

Calculus studies vector algebra in $\mathbb{R}^3$, scalar and vector fields, and the operators $\operatorname{grad} f = \nabla f$, $\operatorname{curl} \vec{F} = \nabla \times \vec{F}$, $\operatorname{div} \vec{G} = \nabla \cdot \vec{G}$. Analogously, having studied tensor algebra in $\mathbb{R}^n$, we now introduce tensor fields, especially alternating tensor fields (*differential forms*), and in §30 the *differential operator* d, the analogue of grad, curl and div.

29.1 Tangent vectors and vector fields

Definition 29.1 (tangent vector, tangent space)

Given $\mathbf{x} \in \mathbb{R}^n$, a **tangent vector** to $\mathbb{R}^n$ at $\mathbf{x}$ is a pair $(\mathbf{x}; \mathbf{v})$ with $\mathbf{v} \in \mathbb{R}^n$. The tangent vectors at $\mathbf{x}$ form a vector space, the **tangent space** $\mathcal{T}_{\mathbf{x}}(\mathbb{R}^n)$, under

$$(\mathbf{x}; \mathbf{v}) + (\mathbf{x}; \mathbf{w}) = (\mathbf{x}; \mathbf{v} + \mathbf{w}), \qquad c(\mathbf{x}; \mathbf{v}) = (\mathbf{x}; c\mathbf{v}).$$

We picture $\mathbf{x}$ as a point and $\mathbf{v}$ as an arrow with initial point $\mathbf{x}$; $\mathcal{T}_{\mathbf{x}}(\mathbb{R}^n) = \mathbf{x} \times \mathbb{R}^n$. We do not add $(\mathbf{x}; \mathbf{v}) + (\mathbf{y}; \mathbf{w})$ if $\mathbf{x} \neq \mathbf{y}$.

Definition 29.2 (velocity vector)

Let (a, b) be an open interval and $\gamma : (a, b) \to \mathbb{R}^n$ of class C^r. The **velocity vector** of γ at parameter value t is $(\gamma(t); D\gamma(t))$.

For $\gamma(t) = x(t)\mathbf{e}_1 + y(t)\mathbf{e}_2 + z(t)\mathbf{e}_3$ in $\mathbb{R}^3$, calculus defines the velocity as $D\gamma(t) = \frac{dx}{dt}\mathbf{e}_1 + \frac{dy}{dt}\mathbf{e}_2 + \frac{dz}{dt}\mathbf{e}_3$.

Definition 29.3 (induced transformation α_*)

Let A be open in $\mathbb{R}^k$ or $\mathbb{H}^k$, $\alpha : A \to \mathbb{R}^n$ of class C^r, $\mathbf{x} \in A$, $\mathbf{p} = \alpha(\mathbf{x})$. The linear transformation $\alpha_* : \mathcal{T}_{\mathbf{x}}(\mathbb{R}^k) \to \mathcal{T}_{\mathbf{p}}(\mathbb{R}^n)$ **induced** by α is

$$\alpha_*(\mathbf{x}; \mathbf{v}) = (\mathbf{p}; D\alpha(\mathbf{x}) \cdot \mathbf{v}).$$

By the chain rule, $\alpha_*(\mathbf{x}; \mathbf{v})$ is the velocity vector of the curve $\gamma(t) = \alpha(\mathbf{x} + t\mathbf{v})$ at $t = 0$.

Lemma 29.1

Let A be open in $\mathbb{R}^k$ or $\mathbb{H}^k$; let $\alpha : A \to \mathbb{R}^m$ be of class C^r. Let B be an open set of $\mathbb{R}^m$ or $\mathbb{H}^m$ containing $\alpha(A)$; let $\beta : B \to \mathbb{R}^n$ be of class C^r. Then $(\beta \circ \alpha)_* = \beta_* \circ \alpha_*$.

Proof. With $\mathbf{y} = \alpha(\mathbf{x})$, $\mathbf{z} = \beta(\mathbf{y})$, the chain rule gives $(\beta \circ \alpha)_*(\mathbf{x}; \mathbf{v}) = (\mathbf{z}; D\beta(\mathbf{y}) \cdot D\alpha(\mathbf{x}) \cdot \mathbf{v}) = \beta_*(\mathbf{y}; D\alpha(\mathbf{x}) \cdot \mathbf{v}) = \beta_*(\alpha_*(\mathbf{x}; \mathbf{v}))$. ■

Definition 29.4 (tangent vector field)

If A is open in $\mathbb{R}^n$, a **tangent vector field** in A is a continuous $F : A \to \mathbb{R}^n \times \mathbb{R}^n$ with $F(\mathbf{x}) \in \mathcal{T}_{\mathbf{x}}(\mathbb{R}^n)$ for each $\mathbf{x}$; so $F(\mathbf{x}) = (\mathbf{x}; f(\mathbf{x}))$ for some $f : A \to \mathbb{R}^n$. If f is of class C^r, F is of class C^r.

Definition 29.5 (tangent space to a manifold)

Let M be a k-manifold of class C^r in $\mathbb{R}^n$, $\mathbf{p} \in M$, and $\alpha : U \to V$ a coordinate patch about $\mathbf{p}$ with $\alpha(\mathbf{x}) = \mathbf{p}$. The **tangent space to M at $\mathbf{p}$** is $\mathcal{T}_{\mathbf{p}}(M) = \alpha_*(\mathcal{T}_{\mathbf{x}}(\mathbb{R}^k))$, the set of vectors $\alpha_*(\mathbf{x}; \mathbf{v})$, $\mathbf{v} \in \mathbb{R}^k$.

$\mathcal{T}_{\mathbf{p}}(M)$ is a linear subspace of $\mathcal{T}_{\mathbf{p}}(\mathbb{R}^n)$, independent of the choice of α. It is spanned by the vectors $(\mathbf{p}; D\alpha(\mathbf{x}) \cdot \mathbf{e}_j) = (\mathbf{p}; \partial\alpha/\partial x_j)$, $j = 1, \ldots, k$, which are independent since $D\alpha$ has rank k; so they form a basis. The union of the $\mathcal{T}_{\mathbf{p}}(M)$ is the **tangent bundle** $\mathcal{T}(M)$; a **tangent vector field to** M is a continuous $F : M \to \mathcal{T}(M)$ with $F(\mathbf{p}) \in \mathcal{T}_{\mathbf{p}}(M)$.

29.2 Tensor fields

Definition 29.6 (tensor field, differential form)

Let A be open in $\mathbb{R}^n$. A **k-tensor field** in A is a function ω assigning to each $\mathbf{x} \in A$ a k-tensor $\omega(\mathbf{x}) \in \mathcal{L}^k(\mathcal{T}_{\mathbf{x}}(\mathbb{R}^n))$, such that $\omega(\mathbf{x})((\mathbf{x}; \mathbf{v}_1), \ldots, (\mathbf{x}; \mathbf{v}_k))$ is continuous as a function of $(\mathbf{x}, \mathbf{v}_1, \ldots, \mathbf{v}_k)$; if it is of class C^r, ω is a tensor field of class C^r. If $\omega(\mathbf{x})$ is alternating for each $\mathbf{x}$, ω is a **differential form** (or simply a **form**) of order k on A. More generally, a k-tensor field on an m-manifold M assigns to each $\mathbf{p} \in M$ an element of $\mathcal{L}^k(\mathcal{T}_{\mathbf{p}}(M))$; if alternating, it is a differential form on M.

A tensor field on an open set of $\mathbb{R}^n$ containing M restricts to one on M; conversely any tensor field on M extends to an open set containing M, but the proof is non-trivial. We restrict ourselves to tensor fields defined on open sets of $\mathbb{R}^n$.

Definition 29.7 (elementary forms)

$(\mathbf{x}; \mathbf{e}_1), \ldots, (\mathbf{x}; \mathbf{e}_n)$ is the **usual basis** of $\mathcal{T}_{\mathbf{x}}(\mathbb{R}^n)$. The **elementary 1-forms** $\tilde{\phi}_i$ on $\mathbb{R}^n$ are defined by $\tilde{\phi}_i(\mathbf{x})(\mathbf{x}; \mathbf{e}_j) = 0$ if $i \neq j$ and 1 if $i = j$. For an ascending k-tuple $I = (i_1, \ldots, i_k)$, the **elementary k-form** is $\tilde{\psi}_I(\mathbf{x}) = \tilde{\phi}_{i_1}(\mathbf{x}) \wedge \cdots \wedge \tilde{\phi}_{i_k}(\mathbf{x})$.

For each $\mathbf{x}$, $\tilde{\phi}_1(\mathbf{x}), \ldots, \tilde{\phi}_n(\mathbf{x})$ is the basis of $\mathcal{L}^1(\mathcal{T}_{\mathbf{x}}(\mathbb{R}^n))$ dual to the usual basis, and $\tilde{\psi}_I(\mathbf{x})$ is the corresponding elementary alternating tensor. They are of class C^∞, since

$$\tilde{\phi}_i(\mathbf{x})(\mathbf{x}; \mathbf{v}) = v_i, \qquad \tilde{\psi}_I(\mathbf{x})((\mathbf{x}; \mathbf{v}_1), \ldots, (\mathbf{x}; \mathbf{v}_k)) = \det X_I, \quad X = [\mathbf{v}_1 \ \cdots \ \mathbf{v}_k].$$

A k-form on A can be written uniquely as $\omega(\mathbf{x}) = \sum_{[I]} b_I(\mathbf{x})\tilde{\psi}_I(\mathbf{x})$; the scalar functions b_I are the **components** of ω relative to the standard elementary forms.

Lemma 29.2

Let ω be a k-form on the open set A of $\mathbb{R}^n$. Then ω is of class C^r if and only if its component functions b_I are of class C^r on A.

Proof. Write $\omega = \sum_{[I]} b_I \tilde{\psi}_I$. The $\tilde{\psi}_I$ are C^∞, so if the b_I are C^r, so is ω. Conversely, if ω is C^r as a function of $(\mathbf{x}, \mathbf{v}_1, \dots, \mathbf{v}_k)$, then for ascending J the function $\omega(\mathbf{x})((\mathbf{x};\mathbf{e}_{j_1}), \dots, (\mathbf{x};\mathbf{e}_{j_k}))$ is C^r in $\mathbf{x}$; it equals $b_J(\mathbf{x})$. ■

Lemma 29.3

Let ω and η be k-forms, and let θ be an l-form, on the open set A of $\mathbb{R}^n$. If ω and η and θ are of class C^r, so are $a\omega + b\eta$ and $\omega \wedge \theta$.

Proof. $a\omega + b\eta$ is a linear combination of C^r functions. Write $\omega = \sum_{[I]} b_I \tilde{\psi}_I$, $\theta = \sum_{[J]} c_J \tilde{\psi}_J$; then $\omega \wedge \theta = \sum_{[I]} \sum_{[J]} b_I c_J \tilde{\psi}_I \wedge \tilde{\psi}_J$. To express this in elementary forms, drop terms with repeated indices, rewrite the rest with indices in ascending order, and collect terms: each component of $\omega \wedge \theta$ is a sum, with signs ± 1, of functions $b_I c_J$, hence of class C^r (Lemma 29.2). ■

29.3 Differential forms of order zero

Definition 29.8 (scalar field; 0-form)

If A is open in $\mathbb{R}^n$ and $f : A \to \mathbb{R}$ is of class C^r, f is a **scalar field** in A, also called a **differential form of order** 0. The wedge product of 0-forms is $f \wedge g = f \cdot g$, and of a 0-form with a k-form is $(\omega \wedge f)(\mathbf{x}) = (f \wedge \omega)(\mathbf{x}) = f(\mathbf{x}) \cdot \omega(\mathbf{x})$.

All formal algebraic properties of the wedge product hold; anticommutativity holds since 0-forms have order 0: $f \wedge g = (-1)^0 g \wedge f$, $f \wedge \omega = (-1)^0 \omega \wedge f$.

Convention

Henceforth, Roman letters such as f, g, h denote 0-forms, and Greek letters such as ω, η, θ denote k-forms for $k > 0$.

§30. The Differential Operator

Motivation. The exterior derivative d unifies gradient, curl and divergence in one operator, and $d^2 = 0$ contains the identities $\operatorname{curl}\operatorname{grad} = 0$ and $\operatorname{div}\operatorname{curl} = 0$. It is what appears on the left side of Stokes' theorem.

The operator d carries k-forms to $(k+1)$-forms. We first define it on 0-forms.

30.1 The differential of a 0-form

Definition 30.1 (differential of a function)

Let A be open in $\mathbb{R}^n$ and $f : A \to \mathbb{R}$ of class C^r. The 1-form df on A is

$$df(\mathbf{x})(\mathbf{x};\mathbf{v}) = f'(\mathbf{x};\mathbf{v}) = Df(\mathbf{x}) \cdot \mathbf{v}.$$

It is the **differential** of f; it is of class C^{r-1} as a function of $\mathbf{x}$ and $\mathbf{v}$.

Theorem 30.1

The operator d is linear on 0-forms.

Proof. If $h = af+bg$, then $Dh(\mathbf{x}) = aDf(\mathbf{x})+bDg(\mathbf{x})$, so $dh(\mathbf{x})(\mathbf{x};\mathbf{v}) = a\,df(\mathbf{x})(\mathbf{x};\mathbf{v})+b\,dg(\mathbf{x})(\mathbf{x};\mathbf{v})$. ■

Lemma 30.2

Let $\tilde{\phi}_1,\dots,\tilde{\phi}_n$ be the elementary 1-forms in $\mathbb{R}^n$. Let $\pi_i : \mathbb{R}^n \to \mathbb{R}$ be the ith projection function, defined by the equation $\pi_i(x_1,\dots,x_n) = x_i$. Then $d\pi_i = \tilde{\phi}_i$.

Proof. $d\pi_i$ is a C^∞ 1-form, and $d\pi_i(\mathbf{x})(\mathbf{x};\mathbf{v}) = D\pi_i(\mathbf{x})\cdot\mathbf{v} = [0\cdots0\ 1\ 0\cdots0]\cdot\mathbf{v} = v_i$. ■

Convention

If $\mathbf{x}$ denotes the general point of $\mathbb{R}^n$, we denote the ith projection function $\mathbb{R}^n \to \mathbb{R}$ by x_i. Then dx_i equals the elementary 1-form $\tilde{\phi}_i$ in $\mathbb{R}^n$. If $I = (i_1,\dots,i_k)$ is an ascending k-tuple, we write $dx_I = dx_{i_1}\wedge\cdots\wedge dx_{i_k}$ for the elementary k-form $\tilde{\psi}_I$. The general k-form is uniquely $\omega = \sum_{[I]} b_I\,dx_I$. For an arbitrary k-tuple J we also set $dx_J = dx_{j_1}\wedge\cdots\wedge dx_{j_k}$; note dx_J is not the differential of a form but a wedge product of elementary 1-forms.

These forms are characterized by

$$dx_i(\mathbf{x})(\mathbf{x};\mathbf{v}) = v_i, \qquad dx_I(\mathbf{x})((\mathbf{x};\mathbf{v}_1),\dots,(\mathbf{x};\mathbf{v}_k)) = \det X_I, \quad X = [\mathbf{v}_1\cdots\mathbf{v}_k].$$

(Using x_i for the function π_i confuses a function with its value—an abuse as common as speaking of "the function x^3+2x+1".)

Theorem 30.3

Let A be open in $\mathbb{R}^n$; let $f : A \to \mathbb{R}$ be of class C^r. Then

$$df = (D_1 f)\,dx_1 + \cdots + (D_n f)\,dx_n.$$

In particular, $df = 0$ if f is a constant function.

In Leibniz notation, $df = \frac{\partial f}{\partial x_1}dx_1 + \cdots + \frac{\partial f}{\partial x_n}dx_n$—a formula in calculus books whose meaning is not explained there.

Proof. Evaluate both sides on $(\mathbf{x};\mathbf{v})$: $df(\mathbf{x})(\mathbf{x};\mathbf{v}) = Df(\mathbf{x})\cdot\mathbf{v}$, while $\sum_i D_i f(\mathbf{x})\,dx_i(\mathbf{x})(\mathbf{x};\mathbf{v}) = \sum_i D_i f(\mathbf{x})v_i$. ■

Convention

Since df is only C^{r-1} when f is C^r, to avoid counting degrees of differentiability we henceforth restrict ourselves to manifolds, maps, vector fields, and forms that are of class C^∞.

30.2 The differential of a k-form

Definition 30.2 ($\Omega^k(A)$)

For A open in $\mathbb{R}^n$, $\Omega^k(A)$ is the vector space of all (C^∞) k-forms on A, the **linear space of k-forms** on A.

Theorem 30.4

Let A be an open set in $\mathbb{R}^n$. There exists a unique linear transformation

$$d : \Omega^k(A) \to \Omega^{k+1}(A),$$

defined for $k \geq 0$, such that:

(1) If f is a 0-form, then df is the 1-form $df(\mathbf{x})(\mathbf{x};\mathbf{v}) = Df(\mathbf{x})\cdot\mathbf{v}$.

(2) If ω and η are forms of orders k and l, respectively, then $d(\omega\wedge\eta) = d\omega\wedge\eta + (-1)^k\omega\wedge d\eta$.

(3) For every form ω, $d(d\omega) = 0$.

d is the **differential operator**, and $d\omega$ the **differential** of ω.

Proof.

Step 1. Uniqueness. First, (2) and (3) imply $d(d\omega_1\wedge\cdots\wedge d\omega_k) = 0$ for any forms $\omega_1,\dots,\omega_k$: for $k=1$ this is (3); assuming it for $k-1$ and setting $\eta = d\omega_2\wedge\cdots\wedge d\omega_k$, (2) gives $d(d\omega_1\wedge\eta) = d(d\omega_1)\wedge\eta \pm d\omega_1\wedge d\eta$, both terms vanishing. Since d is linear, it suffices to compute d on $\omega = f\,dx_I$:

$$d\omega = d(f\wedge dx_I) = df\wedge dx_I + f\wedge d(dx_I) = df\wedge dx_I,$$

since $dx_I = dx_{i_1}\wedge\cdots\wedge dx_{i_k}$ with $dx_{i_j} = d(x_{i_j})$. So $d\omega$ is determined by d on 0-forms, which is (1).

Step 2. Definition. For $\omega = \sum_{[I]} f_I\,dx_I$ define $d\omega = \sum_{[I]} df_I\wedge dx_I$. It is C^∞: $d\omega = \sum_{[I]}[\sum_{j=1}^n (D_jf_I)dx_j]\wedge dx_I$; deleting terms with j among the indices of I, rearranging the rest into ascending order and collecting terms, each component of $d\omega$ is a linear combination of the D_jf_I. (If ω were only C^r, $d\omega$ would be C^{r-1}.) d is linear: for $\omega = \sum f_I dx_I$, $\eta = \sum g_I dx_I$, $d(a\omega+b\eta) = \sum d(af_I+bg_I)\wedge dx_I = \sum(a\,df_I + b\,dg_I)\wedge dx_I = a\,d\omega + b\,d\eta$.

Step 3. For an *arbitrary* k-tuple J, $d(f\,dx_J) = df\wedge dx_J$. If two indices of J coincide, $dx_J = 0$. Otherwise let I be the ascending rearrangement of J by the permutation π; by anticommutativity $dx_I = (\operatorname{sgn}\pi)dx_J$, and since d is linear and $\wedge$ homogeneous, $d(f\,dx_I) = df\wedge dx_I$ gives $(\operatorname{sgn}\pi)d(f\,dx_J) = (\operatorname{sgn}\pi)df\wedge dx_J$.

Step 4. Property (2) for $k=l=0$: $d(f\wedge g) = \sum_j D_j(f\cdot g)dx_j = \sum_j (D_jf)g\,dx_j + \sum_j f(D_jg)dx_j = df\wedge g + f\wedge dg$.

Step 5. Property (2) in general. For $k,l>0$, by linearity take $\omega = f\,dx_I$, $\eta = g\,dx_J$:

$$\begin{aligned} d(\omega\wedge\eta) &= d(fg\,dx_I\wedge dx_J) = d(fg)\wedge dx_I\wedge dx_J && \text{by Step 3,}\\ &= (df\,g + f\,dg)\wedge dx_I\wedge dx_J && \text{by Step 4,}\\ &= (df\wedge dx_I)\wedge(g\,dx_J) + (-1)^k(f\,dx_I)\wedge(dg\wedge dx_J) = d\omega\wedge\eta + (-1)^k\omega\wedge d\eta, \end{aligned}$$

the sign arising because dx_I is a k-form and dg a 1-form. If k or l is 0 the same argument works with dx_I or dx_J absent.

Step 6. If f is a 0-form, $d(df) = 0$:

$$d(df) = d\Big(\sum_j D_jf\,dx_j\Big) = \sum_j d(D_jf)\wedge dx_j = \sum_j\sum_i D_iD_jf\,dx_i\wedge dx_j = \sum_{i<j}(D_iD_jf - D_jD_if)\,dx_i\wedge dx_j,$$

which vanishes by equality of mixed partials (Theorem 6.3).

Step 7. If ω is a k-form, $k > 0$, $d(d\omega) = 0$. By linearity take $\omega = f\,dx_I$: $d(d\omega) = d(df \wedge dx_I) = d(df) \wedge dx_I - df \wedge d(dx_I)$ by (2). $d(df) = 0$ by Step 6, and $d(dx_I) = d(1) \wedge dx_I = 0$ by definition. ■

Definition 30.3 (closed and exact forms)

Let A be open in $\mathbb{R}^n$. A 0-form f on A is **exact** on A if it is constant on A; a k-form ω with $k > 0$ is **exact** on A if $\omega = d\theta$ for some $(k-1)$-form θ on A. A k-form ω $(k \geq 0)$ is **closed** on A if $d\omega = 0$.

Every exact form is closed: $df = 0$ for constant f, and $d(d\theta) = 0$. Conversely every closed form on A is exact if $A = \mathbb{R}^n$, or more generally if A is "star-convex" (Chapter 8), but not in general. If every closed k-form on A is exact, A is **homologically trivial in dimension** k.

Example 30.1

Let $A = \{(x, y) \in \mathbb{R}^2 : x \neq 0\}$ and $f(x, y) = x/|x|$. f is C^∞ on A and $df = 0$, but f is not exact on A since it is not constant.

Example 30.2

In differential equations, $P(x, y)\,dx + Q(x, y)\,dy = 0$ is called exact if there is f with $P = \partial f/\partial x$, $Q = \partial f/\partial y$; in our terms the 1-form $P\,dx + Q\,dy$ is the differential of f, hence exact. In $\mathbb{R}^3$, a vector field $\vec{F} = P\vec{i} + Q\vec{j} + R\vec{k}$ is *conservative* if it is the gradient of a scalar field f; this is the same as saying the form $P\,dx + Q\,dy + R\,dz$ is the differential of the 0-form f.

§31. Application to Vector and Scalar Fields

We show that the tensor-field formalism generalizes vector analysis in $\mathbb{R}^3$; these results are used in §38. $\Omega^k(A)$ and the set of C^∞ vector fields on A are linear spaces; we define linear maps translating between them.

Definition 31.1 (gradient, divergence)

Let A be open in $\mathbb{R}^n$, $f : A \to \mathbb{R}$ a scalar field. The **gradient** of f is the vector field

$$(\operatorname{grad} f)(\mathbf{x}) = (\mathbf{x}; D_1 f(\mathbf{x})\mathbf{e}_1 + \cdots + D_n f(\mathbf{x})\mathbf{e}_n).$$

If $G(\mathbf{x}) = (\mathbf{x}; g(\mathbf{x}))$ is a vector field with $g(\mathbf{x}) = g_1(\mathbf{x})\mathbf{e}_1 + \cdots + g_n(\mathbf{x})\mathbf{e}_n$, the **divergence** of G is the scalar field

$$(\operatorname{div} G)(\mathbf{x}) = D_1 g_1(\mathbf{x}) + \cdots + D_n g_n(\mathbf{x}).$$

Theorem 31.1

Let A be an open set in $\mathbb{R}^n$. There exist vector space isomorphisms α_i and β_j as in the following diagram:

$$\begin{array}{ccc} \text{Scalar fields in } A & \xrightarrow{\alpha_0} & \Omega^0(A) \\ {\scriptstyle \operatorname{grad}}\downarrow & & \downarrow{\scriptstyle d} \\ \text{Vector fields in } A & \xrightarrow{\alpha_1} & \Omega^1(A) \end{array} \qquad \begin{array}{ccc} \text{Vector fields in } A & \xrightarrow{\beta_{n-1}} & \Omega^{n-1}(A) \\ {\scriptstyle \operatorname{div}}\downarrow & & \downarrow{\scriptstyle d} \\ \text{Scalar fields in } A & \xrightarrow{\beta_n} & \Omega^n(A) \end{array}$$

such that $d \circ \alpha_0 = \alpha_1 \circ \operatorname{grad}$ and $d \circ \beta_{n-1} = \beta_n \circ \operatorname{div}$.

Proof. For scalar fields f, h and vector fields $F(\mathbf{x}) = (\mathbf{x}; \sum f_i(\mathbf{x})\mathbf{e}_i)$, $G(\mathbf{x}) = (\mathbf{x}; \sum g_i(\mathbf{x})\mathbf{e}_i)$ define

$$\alpha_0 f = f, \quad \alpha_1 F = \sum_{i=1}^{n} f_i\, dx_i, \quad \beta_{n-1} G = \sum_{i=1}^{n} (-1)^{i-1} g_i\, dx_1 \wedge \cdots \wedge \widehat{dx_i} \wedge \cdots \wedge dx_n, \quad \beta_n h = h\, dx_1 \wedge \cdots \wedge dx_n,$$

where $\widehat{a}$ means a is omitted. [Supplement]Each is linear, and bijective because $\{dx_i\}$, $\{dx_1 \wedge \cdots \wedge \widehat{dx_i} \wedge \cdots \wedge dx_n\}$ and $\{dx_1 \wedge \cdots \wedge dx_n\}$ are the elementary bases of Ω^1, Ω^{n-1}, Ω^n (Lemma 29.2). By Theorem 30.3, $d(\alpha_0 f) = df = \sum_i D_i f\, dx_i = \alpha_1(\operatorname{grad} f)$. And

$$d(\beta_{n-1} G) = \sum_i (-1)^{i-1} \sum_j D_j g_i\, dx_j \wedge dx_1 \wedge \cdots \wedge \widehat{dx_i} \wedge \cdots \wedge dx_n = \sum_i (-1)^{i-1} D_i g_i (-1)^{i-1} dx_1 \wedge \cdots \wedge dx_n = \beta_n(\operatorname{div} G),$$

since only $j = i$ survives and moving dx_i past $i-1$ factors gives the sign $(-1)^{i-1}$. ∎

For $\mathbb{R}^3$ there is also a *curl* operator.

Definition 31.2 (curl)

Let A be open in $\mathbb{R}^3$ and $F(\mathbf{x}) = (\mathbf{x}; \sum f_i(\mathbf{x})\mathbf{e}_i)$ a vector field. The **curl** of F is

$$(\operatorname{curl} F)(\mathbf{x}) = (\mathbf{x}; (D_2 f_3 - D_3 f_2)\mathbf{e}_1 + (D_3 f_1 - D_1 f_3)\mathbf{e}_2 + (D_1 f_2 - D_2 f_1)\mathbf{e}_3),$$

remembered as the symbolic determinant $\det \begin{bmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 \\ D_1 & D_2 & D_3 \\ f_1 & f_2 & f_3 \end{bmatrix}$.

Theorem 31.2

Let A be an open set in $\mathbb{R}^3$. There exist vector space isomorphisms α_i and β_j as in the following diagram:

$$\begin{array}{ccc} \text{Scalar fields in } A & \xrightarrow{\alpha_0} & \Omega^0(A) \\ \downarrow{\scriptstyle \operatorname{grad}} & & \downarrow{\scriptstyle d} \\ \text{Vector fields in } A & \xrightarrow{\alpha_1} & \Omega^1(A) \\ \downarrow{\scriptstyle \operatorname{curl}} & & \downarrow{\scriptstyle d} \\ \text{Vector fields in } A & \xrightarrow{\beta_2} & \Omega^2(A) \\ \downarrow{\scriptstyle \operatorname{div}} & & \downarrow{\scriptstyle d} \\ \text{Scalar fields in } A & \xrightarrow{\beta_3} & \Omega^3(A) \end{array}$$

such that $d \circ \alpha_0 = \alpha_1 \circ \operatorname{grad}$, $d \circ \alpha_1 = \beta_2 \circ \operatorname{curl}$, and $d \circ \beta_2 = \beta_3 \circ \operatorname{div}$.

Proof. The maps are those of Theorem 31.1 with $n = 3$: $\beta_2 G = g_1\, dx_2 \wedge dx_3 - g_2\, dx_1 \wedge dx_3 + g_3\, dx_1 \wedge dx_2$. Only the middle equation needs checking [Supplement]:

$$\begin{aligned} d(\alpha_1 F) &= d(f_1 dx_1 + f_2 dx_2 + f_3 dx_3) = \sum_{i,j} D_j f_i\, dx_j \wedge dx_i \\ &= (D_2 f_3 - D_3 f_2)\, dx_2 \wedge dx_3 + (D_3 f_1 - D_1 f_3)\, dx_3 \wedge dx_1 + (D_1 f_2 - D_2 f_1)\, dx_1 \wedge dx_2, \end{aligned}$$

and since $dx_3 \wedge dx_1 = -dx_1 \wedge dx_3$, this is $\beta_2(\operatorname{curl} F)$. ∎

Consequently, $d(d\omega) = 0$ translates into $\operatorname{curl} \operatorname{grad} f = 0$ and $\operatorname{div} \operatorname{curl} F = 0$ in $\mathbb{R}^3$.

§32. The Action of a Differentiable Map

A C^∞ map $\alpha : A \to \mathbb{R}^n$, A open in $\mathbb{R}^k$, induces α_* on tangent spaces, and linear maps T induce dual maps T^* on alternating tensors. Combining these, α induces a dual transformation α^* of forms, preserving the vector space structure, the wedge product and d.

Definition 32.1 (pullback α^*)

Let A be open in $\mathbb{R}^k$, $\alpha : A \to \mathbb{R}^m$ of class C^∞, and B open in $\mathbb{R}^m$ containing $\alpha(A)$. The **dual transformation of forms** $\alpha^* : \Omega^l(B) \to \Omega^l(A)$ is defined as follows: for a 0-form f on B, $(\alpha^* f)(\mathbf{x}) = f(\alpha(\mathbf{x}))$; for an l-form ω on B, $l > 0$,

$$(\alpha^*\omega)(\mathbf{x})((\mathbf{x};\mathbf{v}_1),\dots,(\mathbf{x};\mathbf{v}_l)) = \omega(\alpha(\mathbf{x}))(\alpha_*(\mathbf{x};\mathbf{v}_1),\dots,\alpha_*(\mathbf{x};\mathbf{v}_l)).$$

Since $f, \omega, \alpha, D\alpha$ are C^∞, so are $\alpha^* f$ and $\alpha^*\omega$. If α is constant, $\alpha^* f$ is constant and $\alpha^*\omega$ is the zero tensor. If $\alpha(\mathbf{x}) = \mathbf{y}$ and $T = \alpha_* : \mathcal{T}_{\mathbf{x}}(\mathbb{R}^k) \to \mathcal{T}_{\mathbf{y}}(\mathbb{R}^m)$, then $T^*(\omega(\mathbf{y})) = (\alpha^*\omega)(\mathbf{x})$, since

$$T^*(\omega(\mathbf{y}))((\mathbf{x};\mathbf{v}_1),\dots,(\mathbf{x};\mathbf{v}_l)) = \omega(\alpha(\mathbf{x}))(\alpha_*(\mathbf{x};\mathbf{v}_1),\dots,\alpha_*(\mathbf{x};\mathbf{v}_l)).$$

So earlier results on dual transformations become results about forms:

Theorem 32.1

Let A be open in $\mathbb{R}^k$; let $\alpha : A \to \mathbb{R}^m$ be a C^∞ map. Let B be open in $\mathbb{R}^m$ and contain $\alpha(A)$; let $\beta : B \to \mathbb{R}^n$ be a C^∞ map. Let ω, η, θ be forms defined in an open set C of $\mathbb{R}^n$ containing $\beta(B)$; assume ω and η have the same order. The transformations α^* and β^* have the following properties:
(1) $\beta^*(a\omega + b\eta) = a(\beta^*\omega) + b(\beta^*\eta)$.
(2) $\beta^*(\omega \wedge \theta) = \beta^*\omega \wedge \beta^*\theta$.
(3) $(\beta \circ \alpha)^*\omega = \alpha^*(\beta^*\omega)$.

Proof. For forms of positive order, (1) and (3) restate Theorem 26.5 (using Lemma 29.1 for (3)), and (2) restates property (6) of Theorem 28.1. [Supplement](Order zero.) If ω, η are 0-forms, $\beta^*(af+bg) = (af+bg)\circ\beta = a(f\circ\beta)+b(g\circ\beta)$. If $\theta = g$ is a 0-form, $\beta^*(\omega\wedge g)(\mathbf{y}) = g(\beta(\mathbf{y}))\cdot\beta^*\omega(\mathbf{y}) = (\beta^*\omega \wedge \beta^* g)(\mathbf{y})$, and similarly if ω is a 0-form. For 0-forms, (3) reads $f \circ (\beta \circ \alpha) = (f \circ \beta) \circ \alpha$. ■

We now compute $\alpha^*\omega$ when ω is a 1-form and when ω is a k-form on $\mathbb{R}^n$ with $\alpha : A \to \mathbb{R}^n$, A open in $\mathbb{R}^k$; since α^* is linear, preserves wedge products and $\alpha^* f = f \circ \alpha$, it suffices to treat elementary forms.

Theorem 32.2

Let A be open in $\mathbb{R}^k$; let $\alpha : A \to \mathbb{R}^n$ be a C^∞ map. Let $\mathbf{x}$ denote the general point of $\mathbb{R}^k$; let $\mathbf{y}$ denote the general point of $\mathbb{R}^n$. Then dx_i and dy_i denote the elementary 1-forms in $\mathbb{R}^k$ and $\mathbb{R}^n$, respectively.
(a) $\alpha^*(dy_i) = d\alpha_i$.
(b) If $I = (i_1,\dots,i_k)$ is an ascending k-tuple from the set $\{1,\dots,n\}$, then

$$\alpha^*(dy_I) = \left(\det \frac{\partial \alpha_I}{\partial \mathbf{x}}\right) dx_1 \wedge \cdots \wedge dx_k, \qquad \text{where} \quad \frac{\partial \alpha_I}{\partial \mathbf{x}} = \frac{\partial(\alpha_{i_1},\dots,\alpha_{i_k})}{\partial(x_1,\dots,x_k)}.$$

Proof. (a) With $\mathbf{y} = \alpha(\mathbf{x})$,

$$(\alpha^*(dy_i))(\mathbf{x})(\mathbf{x};\mathbf{v}) = dy_i(\mathbf{y})(\alpha_*(\mathbf{x};\mathbf{v})) = i\text{th component of } D\alpha(\mathbf{x})\cdot\mathbf{v} = \sum_{j=1}^k D_j\alpha_i(\mathbf{x})v_j = \sum_{j=1}^k \frac{\partial \alpha_i}{\partial x_j}(\mathbf{x})\, dx_j(\mathbf{x})(\mathbf{x};\mathbf{v}).$$

So $\alpha^*(dy_i) = \sum_j \frac{\partial \alpha_i}{\partial x_j} dx_j$, which equals $d\alpha_i$ by Theorem 30.3.

(b) $\alpha^*(dy_I)$ is a k-form on an open set of $\mathbb{R}^k$, so $\alpha^*(dy_I) = h\, dx_1 \wedge \cdots \wedge dx_k$ for a scalar function h, found by evaluating on $((\mathbf{x};\mathbf{e}_1),\dots,(\mathbf{x};\mathbf{e}_k))$:

$$h(\mathbf{x}) = dy_I(\mathbf{y})(\alpha_*(\mathbf{x};\mathbf{e}_1),\dots,\alpha_*(\mathbf{x};\mathbf{e}_k)) = dy_I(\mathbf{y})((\mathbf{y};\partial\alpha/\partial x_1),\dots,(\mathbf{y};\partial\alpha/\partial x_k)) = \det[D\alpha(\mathbf{x})]_I = \det\frac{\partial \alpha_I}{\partial \mathbf{x}}.$$

■

Formula (a) is easy to remember: to compute $\alpha^*(dy_i)$, take dy_i and substitute $y_i = \alpha_i(\mathbf{x})$. One could compute $\alpha^*(dy_I) = d\alpha_{i_1} \wedge \cdots \wedge d\alpha_{i_k}$, but the wedge product is laborious if $k > 2$.

Theorem 32.3

Let A be open in $\mathbb{R}^k$; let $\alpha : A \to \mathbb{R}^n$ be of class C^∞. If ω is an l-form defined in an open set of $\mathbb{R}^n$ containing $\alpha(A)$, then

$$\alpha^*(d\omega) = d(\alpha^*\omega).$$

Proof. Let $\mathbf{x}$, $\mathbf{y}$ be the general points of $\mathbb{R}^k$, $\mathbb{R}^n$.

Step 1. For a 0-form f:

$$\alpha^*(df) = \alpha^*\Big(\sum_{i=1}^n (D_i f)\, dy_i\Big) = \sum_{i=1}^n ((D_i f) \circ \alpha)\, d\alpha_i. \tag{$*$}$$

On the other hand

$$d(\alpha^* f) = d(f \circ \alpha) = \sum_{j=1}^k D_j(f \circ \alpha)\, dx_j. \tag{$**$}$$

By the chain rule, $D(f \circ \alpha)(\mathbf{x}) = Df(\mathbf{y}) \cdot D\alpha(\mathbf{x})$, so $D_j(f \circ \alpha) = \sum_i ((D_i f) \circ \alpha) \cdot D_j \alpha_i$. Substituting in $(**)$,

$$d(\alpha^* f) = \sum_{j=1}^k \sum_{i=1}^n ((D_i f) \circ \alpha) D_j \alpha_i\, dx_j = \sum_{i=1}^n ((D_i f) \circ \alpha)\, d\alpha_i, \tag{$***$}$$

which agrees with $(*)$.

Step 2. For forms of positive order, by linearity it suffices to take $\omega = f\, dy_I$ with I ascending. Then

$$\alpha^*(d\omega) = \alpha^*(df \wedge dy_I) = \alpha^*(df) \wedge \alpha^*(dy_I), \tag{$\dagger$}$$

$$d(\alpha^*\omega) = d[(\alpha^* f) \wedge \alpha^*(dy_I)] = d(\alpha^* f) \wedge \alpha^*(dy_I) + (\alpha^* f) \wedge 0, \tag{$\dagger\dagger$}$$

since $d(\alpha^*(dy_I)) = d(d\alpha_{i_1} \wedge \cdots \wedge d\alpha_{i_l}) = 0$. Compare $(\dagger)$ and $(\dagger\dagger)$ using Step 1. ■

With this, the algebra of differential forms and d, as summarized in §30 and this section, is all we need.

Forms versus vector fields

Here forms appear more natural than vector fields. A C^∞ map $\alpha : A \to \mathbb{R}^n$ induces α_* on tangent vectors, but there is no way to carry a vector *field* on A to one on $\alpha(A)$: if $\alpha(\mathbf{x}_1) = \alpha(\mathbf{x}_2) = \mathbf{y}$ with $\mathbf{x}_1 \neq \mathbf{x}_2$, a field $F(\mathbf{x}) = (\mathbf{x}; f(\mathbf{x}))$ gives two possibly different vectors $\alpha_*(\mathbf{x}_1; f(\mathbf{x}_1))$, $\alpha_*(\mathbf{x}_2; f(\mathbf{x}_2))$ at $\mathbf{y}$. For a diffeomorphism $\alpha : A \to B$ one can define $\tilde{\alpha}_* F = G$ by $G(\mathbf{y}) = \alpha_*(F(\alpha^{-1}(\mathbf{y})))$, and $\tilde{\alpha}_* h = h \circ \alpha^{-1}$ for scalar fields; but this does not in general commute with grad, curl, div, nor with the translation operators α_i, β_j of §31. Forms pull back under *any* C^∞ map, compatibly with $\wedge$ and d.

Chapter 7. Stokes' Theorem

Overview

Chapter 6 showed how k-forms generalize scalar and vector fields in $\mathbb{R}^3$, and how d generalizes grad, curl and div. We now define the integral of a k-form over a k-manifold, generalizing line and surface integrals in $\mathbb{R}^3$; these appear in the classical Stokes' theorem and divergence theorem, and integrals of forms appear in the generalized Stokes' theorem (§37). Recall the convention: all manifolds, forms, vector fields and scalar fields are of class C^∞.

§33. Integrating Forms over Parametrized-Manifolds

Motivation. A k-form can be integrated over a parametrized k-manifold by pulling it back to $\mathbb{R}^k$. The change of variables theorem shows that the result depends only on the orientation of the parametrization.

Definition 33.1 (integral of a k-form over an open set of $\mathbb{R}^k$)

Let A be open in $\mathbb{R}^k$ and η a k-form defined in A; η can be written uniquely as $\eta = f\,dx_1 \wedge \cdots \wedge dx_k$. The **integral of η over A** is

$$\int_A \eta = \int_A f,$$

provided the latter integral exists.

This seems coordinate-dependent, since dx_i depends on the standard basis of $\mathbb{R}^k$. But if $\mathbf{a}_1, \ldots, \mathbf{a}_k$ is *any* right-handed orthonormal basis of $\mathbb{R}^k$, one checks $\int_A \eta = \int_{\mathbf{x}\in A} \eta(\mathbf{x})((\mathbf{x};\mathbf{a}_1), \ldots, (\mathbf{x};\mathbf{a}_k))$; so the integral depends on the orientation of $\mathbb{R}^k$ but not on the choice of basis.

Definition 33.2 (integral over a parametrized-manifold)

Let A be open in $\mathbb{R}^k$, $\alpha : A \to \mathbb{R}^n$ of class C^∞, and $Y = \alpha(A)$, so that Y_α is a parametrized-manifold. If ω is a k-form defined in an open set of $\mathbb{R}^n$ containing Y, the **integral of ω over Y_α** is

$$\int_{Y_\alpha} \omega = \int_A \alpha^*\omega,$$

provided the latter integral exists. It is linear in ω.

The integral is invariant under reparametrization *up to sign*:

Theorem 33.1

Let $g : A \to B$ be a diffeomorphism of open sets in $\mathbb{R}^k$. Assume $\det Dg$ does not change sign on A. Let $\beta : B \to \mathbb{R}^n$ be a map of class C^∞; let $Y = \beta(B)$. Let $\alpha = \beta \circ g$; then $\alpha : A \to \mathbb{R}^n$ and $Y = \alpha(A)$. If ω is a k-form defined in an open set of $\mathbb{R}^n$ containing Y, then ω is integrable over Y_β if and only if it is integrable over Y_α; in this case,

$$\int_{Y_\alpha} \omega = \pm \int_{Y_\beta} \omega,$$

where the sign agrees with the sign of $\det Dg$.

Proof. Let $\mathbf{x}$, $\mathbf{y}$ denote the general points of A, B. We must show $\int_A \alpha^*\omega = \epsilon \int_B \beta^*\omega$, with $\epsilon = \pm 1$ the sign of $\det Dg$. Put $\eta = \beta^*\omega$; since $\alpha^*\omega = g^*(\beta^*\omega)$, we need $\int_A g^*\eta = \epsilon \int_B \eta$. Write $\eta = f\, dy_1 \wedge \cdots \wedge dy_k$. By Theorem 32.2 (with $k = n$),

$$g^*\eta = (f \circ g) g^*(dy_1 \wedge \cdots \wedge dy_k) = (f \circ g) \det(Dg)\, dx_1 \wedge \cdots \wedge dx_k,$$

so we need $\int_A (f \circ g) \det Dg = \epsilon \int_B f$. This is the change of variables theorem, since $\det Dg = \epsilon |\det Dg|$. ■

If A is connected, the hypothesis that $\det Dg$ does not change sign is automatic: $\{\det Dg > 0\}$ and $\{\det Dg < 0\}$ are disjoint open sets with union A.

Theorem 33.2

Let A be open in $\mathbb{R}^k$; let $\alpha : A \to \mathbb{R}^n$ be of class C^∞; let $Y = \alpha(A)$. Let $\mathbf{x}$ denote the general point of A; and let $\mathbf{z}$ denote the general point of $\mathbb{R}^n$. If $\omega = f\, dz_I$ is a k-form defined in an open set of $\mathbb{R}^n$ containing Y, then

$$\int_{Y_\alpha} \omega = \int_A (f \circ \alpha) \det(\partial \alpha_I / \partial \mathbf{x}).$$

Proof. By Theorem 32.2, $\alpha^*\omega = (f \circ \alpha) \det(\partial \alpha_I / \partial \mathbf{x})\, dx_1 \wedge \cdots \wedge dx_k$. ■

The notion of a k-form, and even more of its integral over a parametrized-manifold, is abstract; §36 gives a geometric interpretation.

The "dx" notation of calculus

If $\eta = f\, dx$ is a 1-form on $A = (a, b)$, then $\int_A \eta = \int_A f$, i.e. $\int_A f\, dx = \int_a^b f$: the left side is the integral of a *form*, the right of a *function*, and they are equal by definition. Line integrals also make sense: for a 1-form $P\, dx + Q\, dy + R\, dz$ on an open set of $\mathbb{R}^3$ and a parametrized-curve $\gamma : (a, b) \to \mathbb{R}^3$ with image C, Theorem 33.2 gives

$$\int_{C_\gamma} P\, dx + Q\, dy + R\, dz = \int_{(a,b)} \Big[P(\gamma(t)) \frac{d\gamma_1}{dt} + Q(\gamma(t)) \frac{d\gamma_2}{dt} + R(\gamma(t)) \frac{d\gamma_3}{dt} \Big],$$

the calculus formula. The "$dx\, dy$" notation for double integrals is harder: $\iint_A f(x, y)\, dx\, dy$ resembles both the iterated integral, $\int_c^d [\int_a^b f(x, y)\, dx]\, dy = \int_A f$, and the integral of a 2-form, $\int_A f\, dx \wedge dy = \int_A f$. Reversing x and y, the iterated integral still gives $\int_A f$ but $\int_A f\, dy \wedge dx = -\int_A f$. To avoid confusion we do not use "$dx\, dy$". The "dV" notation of Chapter 5 is unambiguous: if A is open in $\mathbb{R}^k$, parametrized by the identity, then $\int_{A_\alpha} f\, dV = \int_A (f \circ \alpha) V(D\alpha) = \int_A f$; the d there has nothing to do with the operator d.

§34. Orientable Manifolds

Motivation. To integrate forms over a whole manifold we need all patches to have compatible orientations. Orientability is the condition that this is possible; the Möbius band is the standard example where it fails.

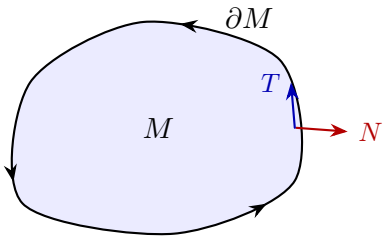


Figure. *The induced orientation on ∂M for a region M in $\mathbb{R}^2$: with N the outward normal, the tangent T is chosen so that (N,T) is positively oriented, i.e. the boundary is traversed with M on the left. This is the convention under which Green's and Stokes' theorems hold without sign changes.*

For ω supported in a single patch $\alpha : U \to V$ on M we want $\int_M \omega = \int_{\operatorname{Int} U} \alpha^*\omega$. Since this is invariant under reparametrization only up to sign, we need an extra condition on M: *orientability*.

Definition 34.1 (orientation-preserving diffeomorphism)

A diffeomorphism $g : A \to B$ of open sets in $\mathbb{R}^k$ is **orientation-preserving** if $\det Dg > 0$ on A, and **orientation-reversing** if $\det Dg < 0$ on A.

This generalizes §20: g is orientation-preserving iff for each $\mathbf{x}$ the linear transformation of $\mathbb{R}^k$ with matrix $Dg(\mathbf{x})$, i.e. $g_* : \mathcal{T}_{\mathbf{x}}(\mathbb{R}^k) \to \mathcal{T}_{g(\mathbf{x})}(\mathbb{R}^k)$, is orientation-preserving.

Definition 34.2 (orientable manifold, orientation)

Let M be a k-manifold in $\mathbb{R}^n$. Coordinate patches $\alpha_i : U_i \to V_i$ $(i = 0, 1)$ on M **overlap positively** if the transition function $\alpha_1^{-1} \circ \alpha_0$ is orientation-preserving (when $V_0 \cap V_1 \neq \emptyset$). If M can be covered by coordinate patches each pair of which overlap positively (if they overlap at all), M is **orientable**; otherwise **non-orientable**. Given such a covering, adjoin all other patches overlapping these positively; the expanded collection still overlaps positively and is an **orientation** on M. A manifold with an orientation is an **oriented manifold**.

This makes no sense for 0-manifolds; see §37. For a k-dimensional vector space V we now have two notions of orientation; they correspond: given an orientation of V in the sense of §20, for each frame $(\mathbf{v}_1, \ldots, \mathbf{v}_k)$ in it the linear isomorphism $\alpha : \mathbb{R}^k \to V$ with $\alpha(\mathbf{e}_i) = \mathbf{v}_i$ is a coordinate patch on V; any two such overlap positively, and they specify an orientation in the present sense.

34.1 Oriented manifolds in $\mathbb{R}^n$ of dimensions 1, $n-1$ and n

Definition 34.3 (unit tangent field)

Let M be an oriented 1-manifold in $\mathbb{R}^n$. For $\mathbf{p} \in M$ choose a patch $\alpha : U \to V$ about $\mathbf{p}$ in the orientation, with $\alpha(t_0) = \mathbf{p}$, and define

$$T(\mathbf{p}) = (\mathbf{p}; D\alpha(t_0)/\|D\alpha(t_0)\|).$$

T is the **unit tangent field corresponding to the orientation** of M.

$(\mathbf{p}; D\alpha(t_0))$ is the velocity of the curve α at t_0, and $T(\mathbf{p})$ is this vector normalized. T is well-defined: if β is another patch in the orientation with $\mathbf{p} = \beta(t_1)$ and $g = \beta^{-1} \circ \alpha$, then $D\alpha(t_0) = D\beta(t_1) \cdot Dg(t_0)$ with $Dg(t_0) > 0$ (orientation-preserving), so $D\alpha(t_0)/\|D\alpha(t_0)\| = D\beta(t_1)/\|D\beta(t_1)\|$. T is C^∞ since $t_0 = \alpha^{-1}(\mathbf{p})$ is a C^∞ function of $\mathbf{p}$.

Example 34.1

An oriented 1-manifold M with unit tangent field T is pictured by drawing arrows along M in the direction of T; this is a *directed curve* in calculus.

If M has boundary points $\mathbf{p}, \mathbf{q}$ (e.g. an arc), a patch $\alpha : U \to V$ about the boundary point $\mathbf{p}$ has U open in $\mathbb{H}^1$, so $T(\mathbf{p})$ must point *into* M from $\mathbf{p}$; likewise at $\mathbf{q}$. Then no unit tangent field points into M at both ends, and M would seem non-orientable—an anomaly. It disappears if we also allow patches with domains open in the left half-line $\mathbb{L}^1 = \{x : x \leq 0\}$.

Convention

In the case of a 1-manifold M, we allow the domains of the coordinate patches on M to be open sets in $\mathbb{R}^1$ or in $\mathbb{H}^1$ or in $\mathbb{L}^1$.

With this, every 1-manifold is orientable (we shall not prove this).

Definition 34.4 (unit normal field)

Let M be an $(n-1)$-manifold in $\mathbb{R}^n$ and $\mathbf{p} \in M$. Let $(\mathbf{p}; \mathbf{n})$ be a unit vector in $\mathcal{T}_\mathbf{p}(\mathbb{R}^n)$ orthogonal to the $(n-1)$-dimensional subspace $\mathcal{T}_\mathbf{p}(M)$; $\mathbf{n}$ is determined up to sign. Given an orientation of M, choose a patch $\alpha : U \to V$ in it with $\alpha(\mathbf{x}) = \mathbf{p}$; the columns $\partial\alpha/\partial x_i$ give a basis of $\mathcal{T}_\mathbf{p}(M)$. Fix the sign of $\mathbf{n}$ by requiring the frame $(\mathbf{n}, \partial\alpha/\partial x_1, \ldots, \partial\alpha/\partial x_{n-1})$ to be right-handed, i.e. $\det[\mathbf{n}\ D\alpha(\mathbf{x})] > 0$. The field $N(\mathbf{p}) = (\mathbf{p}; \mathbf{n}(\mathbf{p}))$ is the **unit normal field to M corresponding to the orientation** of M. (It is well-defined and C^∞; see §38.)

Example 34.2 (Möbius band)

The Möbius band in $\mathbb{R}^3$ has no continuous unit normal vector field; it is not orientable.

Example 34.3 (Klein bottle)

The Klein bottle K can be pictured in $\mathbb{R}^3$ as a self-intersecting surface swept out by a moving circle $C_0, C_1, C_2, \ldots$. It can be represented as a 2-manifold *without* self-intersections in $\mathbb{R}^4$: keep the circle in $\mathbb{R}^3 \times 0$, but as it approaches the spot where it would cross part of the surface already generated, let it move "up" into $\mathbb{R}^3 \times \mathbb{H}^1_+$ until past that spot, then come back down. K contains a copy of the Möbius band M; if K were orientable, so would be M (take all patches on M that overlap positively the patches of the orientation of K). So K is not orientable.

Definition 34.5 (natural orientation)

Let M be an n-manifold in $\mathbb{R}^n$. For a patch $\alpha : U \to V$ on M, $D\alpha$ is n by n. The **natural orientation** of M consists of all coordinate patches α on M with $\det D\alpha > 0$; any two overlap positively.

M can be covered by such patches: given $\mathbf{p}$ and a patch $\alpha : U \to V$ about $\mathbf{p}$ with U open in $\mathbb{R}^n$ or $\mathbb{H}^n$, shrink U to an open ϵ-ball or its intersection with $\mathbb{H}^n$; then U is connected and $\det D\alpha$ has constant sign on U. If negative, replace α by $\alpha \circ r$ with $r(x_1, \ldots, x_n) = (-x_1, x_2, \ldots, x_n)$.

34.2 Reversing the orientation of a manifold

Definition 34.6 (reverse orientation)

Let $r : \mathbb{R}^k \to \mathbb{R}^k$ be the reflection $r(x_1, \dots, x_k) = (-x_1, x_2, \dots, x_k)$; it is its own inverse, carries $\mathbb{H}^k$ to itself if $k > 1$, and $\mathbb{H}^1$ to $\mathbb{L}^1$ if $k = 1$. If M is an oriented k-manifold and $\alpha_i : U_i \to V_i$ are the patches of its orientation, let $\beta_i = \alpha_i \circ r : r(U_i) \to V_i$. Then β_i overlaps α_i negatively, while the β_i overlap each other positively; they form the **reverse** (or **opposite**) orientation.

So every orientable k-manifold has at least two orientations; a connected one has exactly two. (A 1-manifold with four components, for example, has four orientations.) If M is an oriented 1-manifold with tangent field T, reversing the orientation replaces T by $-T$, since $(\alpha \circ r)(t) = \alpha(-t)$. If M is an oriented $(n-1)$-manifold with normal field N, reversing replaces N by $-N$: $\partial(\alpha \circ r)/\partial x_1 = -\partial\alpha/\partial x_1$, $\partial(\alpha \circ r)/\partial x_i = \partial\alpha/\partial x_i$ for $i > 1$, and $(\mathbf{n}, \partial\alpha/\partial x_1, \dots)$ is right-handed iff $(-\mathbf{n}, -\partial\alpha/\partial x_1, \partial\alpha/\partial x_2, \dots)$ is.

34.3 The induced orientation of ∂M

Theorem 34.1

Let $k > 1$. If M is an orientable k-manifold with non-empty boundary, then ∂M is orientable.

Proof. Let $\mathbf{p} \in \partial M$ and $\alpha : U \to V$ a patch about $\mathbf{p}$. The restricted patch on ∂M (§24) is $\alpha_0 = \alpha \circ b$, where $b : \mathbb{R}^{k-1} \to \mathbb{R}^k$, $b(x_1, \dots, x_{k-1}) = (x_1, \dots, x_{k-1}, 0)$. We show: if α, β are patches about $\mathbf{p}$ overlapping positively, then so do α_0, β_0. Let $g = \beta^{-1} \circ \alpha : W_0 \to W_1$, with W_0, W_1 open in $\mathbb{H}^k$; then $\det Dg > 0$. If $\mathbf{x} \in \partial\mathbb{H}^k$, the last row of $Dg(\mathbf{x})$ is $Dg_k = [0 \ \cdots \ 0 \ \partial g_k/\partial x_k]$ with $\partial g_k/\partial x_k \geq 0$: incrementing one of $x_1, \dots, x_{k-1}$ from a point of $\partial\mathbb{H}^k$ does not change g_k (which stays 0, as g maps $\partial\mathbb{H}^k$ into $\partial\mathbb{H}^k$), while incrementing x_k positively increases g_k (as g maps into $\mathbb{H}^k$). Since $\det Dg \neq 0$, $\partial g_k/\partial x_k > 0$, and since $\det Dg > 0$,

$$\det \frac{\partial(g_1, \dots, g_{k-1})}{\partial(x_1, \dots, x_{k-1})} > 0.$$

This matrix is the derivative of the transition function $\beta_0^{-1} \circ \alpha_0$. ■

So an orientation of M gives one of ∂M by restricting the patches of the orientation; but this is not always the one we prefer:

Definition 34.7 (induced orientation of ∂M)

Let M be an orientable k-manifold with non-empty boundary and fix an orientation of M. The **induced orientation** of ∂M is: if k is even, the orientation obtained by restricting the patches of the orientation of M; if k is odd, the opposite of that orientation.

Example 34.4

S^2 and the torus T are orientable 2-manifolds, being boundaries of orientable 3-manifolds in $\mathbb{R}^3$. If M is a 3-manifold in $\mathbb{R}^3$ oriented naturally, the induced orientation of ∂M corresponds to the unit normal field pointing *outwards* from M (proved in §38). Informally: let α be a patch about $\mathbf{p} \in \partial M$ in the natural orientation; $(\alpha \circ b)(\mathbf{x}) = \alpha(x_1, x_2, 0)$ is the restricted patch. Since $\dim M = 3$ is odd, the induced orientation of ∂M is opposite to that of the restrictions, so the normal $N = (\mathbf{p}; \mathbf{n})$ for the induced orientation satisfies: $(-\mathbf{n}, \partial\alpha/\partial x_1, \partial\alpha/\partial x_2)$ is right-

handed. Since $\det D\alpha > 0$, $(\partial\alpha/\partial x_3, \partial\alpha/\partial x_1, \partial\alpha/\partial x_2)$ is right-handed. So $-\mathbf{n}$ and $\partial\alpha/\partial x_3$ lie on the same side of the tangent plane. Since $\partial\alpha/\partial x_3$ points *into* M, $\mathbf{n}$ points outwards.

Example 34.5

Let M be an oriented 2-manifold in $\mathbb{R}^3$ with non-empty boundary, N the unit normal field for the orientation of M, and T the unit tangent field for the induced orientation of ∂M. For $\mathbf{p} \in \partial M$ let $W(\mathbf{p})$ be the unit vector perpendicular to $N(\mathbf{p})$ and $T(\mathbf{p})$ with (N, T, W) right-handed. Then $W(\mathbf{p})$ is tangent to M and points into M. (This makes precise the calculus description: walking around ∂M in the direction of T with head pointing along N, M is on your left.) Indeed, let $\alpha : U \to V$ be a patch about $\mathbf{p}$ in the orientation of M; since $\dim M = 2$ is even, $\alpha \circ b$ belongs to the induced orientation of ∂M, so $\partial\alpha/\partial x_1$ points in the direction of T. $\partial\alpha/\partial x_2$ is the velocity of a curve moving into M, so it points into M; choose λ with $\mathbf{w} = \partial\alpha/\partial x_2 + \lambda\,\partial\alpha/\partial x_1$ orthogonal to $\partial\alpha/\partial x_1$; then $\mathbf{w}$ points into M, and $W(\mathbf{p}) = (\mathbf{p}; \mathbf{w}/\|\mathbf{w}\|)$. By definition $(\mathbf{n}, \partial\alpha/\partial x_1, \partial\alpha/\partial x_2)$ is right-handed, and $\det[\mathbf{n}\ \partial\alpha/\partial x_1\ \partial\alpha/\partial x_2] = \det[\mathbf{n}\ \partial\alpha/\partial x_1\ \mathbf{w}]$; so (N, T, W) is right-handed.

§35. Integrating Forms over Oriented Manifolds

Motivation. With orientations fixed, the patch-by-patch definition with partitions of unity gives a well-defined integral of a k-form over an oriented compact k-manifold.

The procedure parallels §25, so we abbreviate the details.

Definition 35.1 (integral, support in one patch)

Let M be a compact oriented k-manifold in $\mathbb{R}^n$ and ω a k-form defined in an open set of $\mathbb{R}^n$ containing M. Let $C = M \cap (\text{Support}\,\omega)$, a compact set. Suppose there is a coordinate patch $\alpha : U \to V$ on M belonging to the orientation of M with $C \subset V$; we may assume U bounded. Define

$$\int_M \omega = \int_{\text{Int}\,U} \alpha^*\omega,$$

where $\text{Int}\,U = U$ if U is open in $\mathbb{R}^k$ and $\text{Int}\,U = U \cap \mathbb{H}^k_+$ if U is open in $\mathbb{H}^k$ but not in $\mathbb{R}^k$.

Existence: α extends to a C^∞ map on an open set U' of $\mathbb{R}^k$, and $\alpha^*\omega$ extends to a C^∞ form $h\,dx_1 \wedge \cdots \wedge dx_k$ on U'; then $\int_{\text{Int}\,U} \alpha^*\omega = \int_{\text{Int}\,U} h$. h is continuous on U and vanishes outside the compact set $\alpha^{-1}(C)$, so it is bounded; it vanishes near each point of $\text{Bd}\,U$ (if U is open in $\mathbb{R}^k$), or near each point of $\text{Bd}\,U$ not in $\partial\mathbb{H}^k$, a set of measure zero. So h is integrable over U and over $\text{Int}\,U$. *Well-definedness*: as in Lemma 25.1, using now that transition functions between patches of the orientation are orientation-preserving, so the sign in Theorem 33.1 is $+$. *Linearity*: if ω and η have supports whose intersections with M lie in a single patch of the orientation, $\int_M a\omega + b\eta = a\int_M \omega + b\int_M \eta$, since α^* and $\int_{\text{Int}\,U}$ are linear. *Orientation*: if $-M$ denotes M with the opposite orientation, $\int_{-M} \omega = -\int_M \omega$, by Theorem 33.1.

Definition 35.2 (integral of a form over a compact oriented manifold)

Let M be a compact oriented k-manifold in $\mathbb{R}^n$ and ω a k-form defined in an open set of $\mathbb{R}^n$ containing M. Cover M by coordinate patches belonging to its orientation, and choose a

partition of unity $\phi_1, \dots, \phi_l$ on M dominated by this collection (Lemma 25.2). Define

$$\int_M \omega = \sum_{i=1}^{l} \Big[\int_M \phi_i \omega \Big].$$

This agrees with Definition 35.1 when $M \cap \operatorname{Support} \omega$ lies in one patch, by linearity and $\omega(\mathbf{x}) = \sum_i \phi_i(\mathbf{x})\omega(\mathbf{x})$ for $\mathbf{x} \in M$; it is independent of the partition of unity by the argument used for $\int_M f\, dV$.

Theorem 35.1

Let M be a compact oriented k-manifold in $\mathbb{R}^n$. Let ω, η be k-forms defined in an open set of $\mathbb{R}^n$ containing M. Then

$$\int_M (a\omega + b\eta) = a \int_M \omega + b \int_M \eta.$$

If $-M$ denotes M with the opposite orientation, then $\int_{-M} \omega = -\int_M \omega$.

Proof [Supplement]. Apply the single-patch facts above to each $\phi_i\omega$, $\phi_i\eta$ and sum; for $-M$, the patches $\alpha_i \circ r$ of the reverse orientation and the same partition of unity may be used. ■

As with $\int_M f\, dV$, in practice one evaluates $\int_M \omega$ by breaking M into pieces:

Theorem 35.2

Let M be a compact oriented k-manifold in $\mathbb{R}^n$. Let ω be a k-form defined in an open set of $\mathbb{R}^n$ containing M. Suppose that $\alpha_i : A_i \to M_i$, for $i = 1, \dots, N$, is a coordinate patch on M belonging to the orientation of M, such that A_i is open in $\mathbb{R}^k$ and M is the disjoint union of the open sets $M_1, \dots, M_N$ of M and a set K of measure zero in M. Then

$$\int_M \omega = \sum_{i=1}^{N} \Big[\int_{A_i} \alpha_i^* \omega \Big].$$

Proof [Supplement]. Copy the proof of Theorem 25.4: by linearity assume $M \cap \operatorname{Support} \omega$ lies in one patch $\alpha : U \to V$ of the orientation, and write $\alpha^*\omega = F\, dx_1 \wedge \cdots \wedge dx_k$, $\alpha_i^*\omega = F_i\, dx_1 \wedge \cdots \wedge dx_k$. With $W_i = \alpha^{-1}(M_i \cap V)$ and $L = \alpha^{-1}(K \cap V)$, $\int_M \omega = \int_{\operatorname{Int} U} F = \sum_i \int_{W_i} F$ as in Step 1 there. The transition map $\alpha_i^{-1} \circ \alpha : W_i \to B_i = \alpha_i^{-1}(M_i \cap V)$ is an orientation-preserving diffeomorphism, so by Theorem 33.1 (sign $+$) $\int_{W_i} F = \int_{B_i} F_i$, and $\int_{B_i} F_i = \int_{A_i} F_i$ by additivity as in Step 2 there. (Alternatively combine Theorems 25.4 and 36.2.) ■

§36. A Geometric Interpretation of Forms and Integrals

Motivation. Integrals of forms are related to the scalar integrals of Chapter 5 via unit normal and tangent fields. This explains how line integrals, flux integrals and volume integrals of vector calculus are all integrals of forms.

The integral of a k-form over an oriented k-manifold is abstract; we relate it to the integral of a scalar function with respect to volume. First, alternating tensors versus volume:

Theorem 36.1

Let W be a k-dimensional linear subspace of $\mathbb{R}^n$; let $(\mathbf{a}_1, \dots, \mathbf{a}_k)$ be an orthonormal k-frame in W, and let f be an alternating k-tensor on W. If $(\mathbf{x}_1, \dots, \mathbf{x}_k)$ is an arbitrary k-tuple in W, then

$$f(\mathbf{x}_1, \dots, \mathbf{x}_k) = \epsilon\, V(\mathbf{x}_1, \dots, \mathbf{x}_k)\, f(\mathbf{a}_1, \dots, \mathbf{a}_k),$$

where $\epsilon = \pm 1$. If the $\mathbf{x}_i$ are independent, then $\epsilon = +1$ if the frames $(\mathbf{x}_1, \dots, \mathbf{x}_k)$ and $(\mathbf{a}_1, \dots, \mathbf{a}_k)$ belong to the same orientation of W and $\epsilon = -1$ otherwise.

(If the $\mathbf{x}_i$ are dependent, $V(\mathbf{x}_1, \dots, \mathbf{x}_k) = 0$ by Theorem 21.3 and ϵ does not matter.)

Proof.

Step 1. $W = \mathbb{R}^k$. Then f is a multiple of the determinant: $f(\mathbf{x}_1, \dots, \mathbf{x}_k) = c \det[\mathbf{x}_1 \cdots \mathbf{x}_k]$. If the $\mathbf{x}_i$ are dependent, f vanishes. Otherwise $f(\mathbf{x}_1, \dots, \mathbf{x}_k) = c\,\epsilon_1 V(\mathbf{x}_1, \dots, \mathbf{x}_k)$ with $\epsilon_1 = +1$ iff $(\mathbf{x}_1, \dots, \mathbf{x}_k)$ is right-handed, and $f(\mathbf{a}_1, \dots, \mathbf{a}_k) = c\,\epsilon_2 V(\mathbf{a}_1, \dots, \mathbf{a}_k) = c\,\epsilon_2$ similarly. So $f(\mathbf{x}_1, \dots, \mathbf{x}_k) = (\epsilon_1/\epsilon_2) V(\mathbf{x}_1, \dots, \mathbf{x}_k) f(\mathbf{a}_1, \dots, \mathbf{a}_k)$, and $\epsilon_1/\epsilon_2 = +1$ iff the two frames belong to the same orientation.

Step 2. In general choose an orthogonal transformation h of $\mathbb{R}^n$ carrying W onto $\mathbb{R}^k \times 0$, with inverse $k : \mathbb{R}^k \times 0 \to W$. Then $k^* f$ is an alternating tensor on $\mathbb{R}^k \times 0$; $(h(\mathbf{x}_i))$ is a k-tuple and $(h(\mathbf{a}_i))$ an orthonormal frame in $\mathbb{R}^k \times 0$. By Step 1, $(k^* f)(h(\mathbf{x}_1), \dots, h(\mathbf{x}_k)) = \epsilon V(h(\mathbf{x}_1), \dots, h(\mathbf{x}_k))(k^* f)(h(\mathbf{a}_1), \dots, h(\mathbf{a}_k))$. Since V is unchanged by orthogonal transformations, this is the desired equation. If the $\mathbf{x}_i$ are independent, $\epsilon = +1$ iff $(h(\mathbf{x}_i))$, $(h(\mathbf{a}_i))$ belong to the same orientation of $\mathbb{R}^k \times 0$, iff $(\mathbf{x}_i)$, $(\mathbf{a}_i)$ belong to the same orientation of W. ■

So for two orthonormal frames $(\mathbf{a}_i)$, $(\mathbf{b}_i)$ in W, $f(\mathbf{a}_1, \dots, \mathbf{a}_k) = \pm f(\mathbf{b}_1, \dots, \mathbf{b}_k)$, the sign depending on whether they lie in the same orientation of W.

Definition 36.1 (natural orientation of $\mathcal{T}_{\mathbf{p}}(M)$)

Let M be an oriented k-manifold in $\mathbb{R}^n$ and $\mathbf{p} \in M$. Choose a patch $\alpha : U \to V$ in the orientation of M about $\mathbf{p}$, with $\alpha(\mathbf{x}) = \mathbf{p}$. The collection of k-frames in $\mathcal{T}_{\mathbf{p}}(M)$ of the form $(\alpha_*(\mathbf{x}; \mathbf{a}_1), \dots, \alpha_*(\mathbf{x}; \mathbf{a}_k))$, with $(\mathbf{a}_1, \dots, \mathbf{a}_k)$ right-handed in $\mathbb{R}^k$, is the **natural orientation** of $\mathcal{T}_{\mathbf{p}}(M)$ induced by the orientation of M; it is independent of α.

Theorem 36.2

Let M be a compact oriented k-manifold in $\mathbb{R}^n$; let ω be a k-form defined in an open set of $\mathbb{R}^n$ containing M. Let λ be the scalar function on M defined by the equation

$$\lambda(\mathbf{p}) = \omega(\mathbf{p})((\mathbf{p}; \mathbf{a}_1), \dots, (\mathbf{p}; \mathbf{a}_k)),$$

where $((\mathbf{p}; \mathbf{a}_1), \dots, (\mathbf{p}; \mathbf{a}_k))$ is any orthonormal frame in the linear space $\mathcal{T}_{\mathbf{p}}(M)$ belonging to its natural orientation. Then λ is continuous, and

$$\int_M \omega = \int_M \lambda\, dV.$$

Proof. By linearity assume $M \cap \operatorname{Support} \omega$ lies in one patch $\alpha : U \to V$ of the orientation. Write $\alpha^* \omega = h\, dx_1 \wedge \cdots \wedge dx_k$ and let $\alpha(\mathbf{x}) = \mathbf{p}$. Then

$$h(\mathbf{x}) = (\alpha^* \omega)(\mathbf{x})((\mathbf{x}; \mathbf{e}_1), \dots, (\mathbf{x}; \mathbf{e}_k)) = \omega(\mathbf{p})((\mathbf{p}; \partial\alpha/\partial x_1), \dots, (\mathbf{p}; \partial\alpha/\partial x_k)) = \pm V(D\alpha(\mathbf{x}))\lambda(\mathbf{p})$$

by Theorem 36.1; the sign is + because $((\mathbf{p}; \partial\alpha/\partial x_1), \dots, (\mathbf{p}; \partial\alpha/\partial x_k))$ belongs to the natural orientation of $\mathcal{T}_{\mathbf{p}}(M)$ by definition. Since $V(D\alpha) \neq 0$ ($D\alpha$ has rank k) and $\mathbf{x} = \alpha^{-1}(\mathbf{p})$ is continuous

in $\mathbf{p}$, $\lambda(\mathbf{p}) = h(\mathbf{x})/V(D\alpha(\mathbf{x}))$ is continuous. Hence

$$\int_M \lambda\, dV = \int_{\operatorname{Int} U} (\lambda \circ \alpha) V(D\alpha) = \int_{\operatorname{Int} U} h = \int_{\operatorname{Int} U} \alpha^*\omega = \int_M \omega.$$

■

So for each k-form ω there is a scalar function λ (in fact C^∞) with $\int_M \omega = \int_M \lambda\, dV$. The converse is true but harder: there is a k-form ω_V, defined on an open set about M, whose value on any orthonormal basis of $\mathcal{T}_{\mathbf{p}}(M)$ in its natural orientation is 1. Then for any C^∞ function λ on M, $\int_M \lambda\, dV = \int_M \lambda\omega_V$, so the integral of λ over M is the integral of the form $\lambda\omega_V$. ω_V is a **volume form** for M, since $\int_M \omega_V = \int_M dV = \mathrm{v}(M)$. This applies only to orientable M: on a non-orientable M the integral of a scalar function is defined but that of a form is not.

Notation

Some mathematicians denote the volume form ω_V by dV (or dV). This makes the above equations tautologies, but it can confuse: V is *not* a form, and d does *not* denote the differential operator in this context.

§37. The Generalized Stokes' Theorem

This is the culmination: a theorem about integrals of forms containing the three basic theorems of vector integral calculus—Green's theorem, Stokes' theorem and the divergence theorem. Let $I^k = [0,1]^k$ be the unit k-cube; $\operatorname{Int} I^k = (0,1)^k$ and $\operatorname{Bd} I^k = I^k - \operatorname{Int} I^k$.

Lemma 37.1

Let $k > 1$. Let η be a $k-1$ form defined in an open set U of $\mathbb{R}^k$ containing the unit k-cube I^k. Assume η vanishes at all points of $\operatorname{Bd} I^k$ except possibly at points of the subset $(\operatorname{Int} I^{k-1}) \times 0$. Then

$$\int_{\operatorname{Int} I^k} d\eta = (-1)^k \int_{\operatorname{Int} I^{k-1}} b^*\eta,$$

where $b : I^{k-1} \to I^k$ is the map $b(u_1, \dots, u_{k-1}) = (u_1, \dots, u_{k-1}, 0)$.

Proof. Let $\mathbf{x}$ and $\mathbf{u}$ be the general points of $\mathbb{R}^k$ and $\mathbb{R}^{k-1}$. For $1 \le j \le k$ let $I_j = (1, \dots, \hat{j}, \dots, k)$; the elementary $(k-1)$-forms in $\mathbb{R}^k$ are $dx_{I_j} = dx_1 \wedge \cdots \wedge \widehat{dx_j} \wedge \cdots \wedge dx_k$. By linearity of the integrals, d and b^*, it suffices to take $\eta = f\, dx_{I_j}$.

Step 1. We have

$$d\eta = df \wedge dx_{I_j} = \Big(\sum_{i=1}^k D_i f\, dx_i\Big) \wedge dx_{I_j} = (-1)^{j-1}(D_j f)\, dx_1 \wedge \cdots \wedge dx_k.$$

So, using the Fubini theorem with $\mathbf{v} = (x_1, \dots, \hat{x}_j, \dots, x_k)$,

$$\int_{\operatorname{Int} I^k} d\eta = (-1)^{j-1} \int_{I^k} D_j f = (-1)^{j-1} \int_{\mathbf{v} \in I^{k-1}} \int_{x_j \in I} D_j f(x_1, \dots, x_k).$$

By the fundamental theorem of calculus the inner integral is $f(x_1, \dots, 1, \dots, x_k) - f(x_1, \dots, 0, \dots, x_k)$, with 1 and 0 in the jth place. η, hence f, vanishes on $\operatorname{Bd} I^k$ except possibly on the open bottom face $(\operatorname{Int} I^{k-1}) \times 0$. If $j < k$ both terms vanish; if $j = k$ the result is $-f(x_1, \dots, x_{k-1}, 0)$. Hence

$$\int_{\operatorname{Int} I^k} d\eta = \begin{cases} 0 & j < k, \\ (-1)^k \int_{I^{k-1}} (f \circ b) & j = k. \end{cases}$$

Step 2. $Db = \begin{bmatrix} I_{k-1} \\ 0 \end{bmatrix}$, so by Theorem 32.2,

$$b^*(dx_{I_j}) = [\det Db(1,\dots,\hat{j},\dots,k)]\, du_1 \wedge \cdots \wedge du_{k-1} = \begin{cases} 0 & j < k, \\ du_1 \wedge \cdots \wedge du_{k-1} & j = k. \end{cases}$$

Hence $\int_{\operatorname{Int} I^{k-1}} b^*\eta$ is 0 if $j < k$ and $\int_{\operatorname{Int} I^{k-1}} (f \circ b)$ if $j = k$. Compare with Step 1. ■

Theorem 37.2 (Stokes' theorem)

Let $k > 1$. Let M be a compact oriented k-manifold in $\mathbb{R}^n$; give ∂M the induced orientation if ∂M is not empty. Let ω be a $k-1$ form defined in an open set of $\mathbb{R}^n$ containing M. Then

$$\int_M d\omega = \int_{\partial M} \omega$$

if ∂M is not empty; and $\int_M d\omega = 0$ if ∂M is empty.

Proof.

Step 1. A special covering of M by coordinate patches. If $\mathbf{p} \in M - \partial M$, choose a patch $\alpha : U \to V$ in the orientation of M with U open in $\mathbb{R}^k$, $U \supset I^k$, and α carrying a point of $\operatorname{Int} I^k$ to $\mathbf{p}$ (precede an arbitrary patch by a translation and a stretching of $\mathbb{R}^k$). Let $W = \operatorname{Int} I^k$ and $Y = \alpha(W)$; then $\alpha : W \to Y$ is a coordinate patch in the orientation about $\mathbf{p}$, with W open in $\mathbb{R}^k$. If $\mathbf{p} \in \partial M$, choose $\alpha : U \to V$ in the orientation with U open in $\mathbb{H}^k$, $U \supset I^k$, carrying a point of $(\operatorname{Int} I^{k-1}) \times 0$ to $\mathbf{p}$. Let $W = (\operatorname{Int} I^k) \cup ((\operatorname{Int} I^{k-1}) \times 0)$ and $Y = \alpha(W)$; then $\alpha : W \to Y$ is a patch in the orientation about $\mathbf{p}$ with W open in $\mathbb{H}^k$ but not in $\mathbb{R}^k$. In each case α can be extended to a C^∞ map on an open set of $\mathbb{R}^k$ containing I^k.

Step 2. Since d and the integrals are linear, and $\omega = \sum_i \phi_i \omega$ for a partition of unity $\{\phi_i\}$ on M dominated by these patches, it suffices to prove the theorem when $C = M \cap \operatorname{Support} \omega$ is covered by a single patch $\alpha : W \to Y$ of Step 1. Then $M \cap \operatorname{Support} d\omega \subset C$ is covered too. Let $\eta = \alpha^*\omega$, extended (without change of notation) to a C^∞ form on an open set of $\mathbb{R}^k$ containing I^k. η vanishes at all points of $\operatorname{Bd} I^k$ except possibly points of $(\operatorname{Int} I^{k-1}) \times 0$, so Lemma 37.1 applies.

Step 3. Case $W = \operatorname{Int} I^k$; Y is disjoint from ∂M. Since $\alpha^* d\omega = d\alpha^*\omega = d\eta$,

$$\int_M d\omega = \int_{\operatorname{Int} I^k} \alpha^* d\omega = \int_{\operatorname{Int} I^k} d\eta = (-1)^k \int_{\operatorname{Int} I^{k-1}} b^*\eta.$$

Here η vanishes outside $\operatorname{Int} I^k$, in particular on $I^{k-1} \times 0$, so $b^*\eta = 0$ and $\int_M d\omega = 0$. If ∂M is empty, this is what we wanted. If not, $\int_{\partial M} \omega = 0$ too, since $\operatorname{Support} \omega$ is disjoint from ∂M.

Step 4. Case W open in $\mathbb{H}^k$, $Y \cap \partial M \neq \emptyset$. As before, $\int_M d\omega = \int_{\operatorname{Int} I^k} d\eta = (-1)^k \int_{\operatorname{Int} I^{k-1}} b^*\eta$. Now $\partial M \cap \operatorname{Support} \omega$ is covered by the patch $\beta = \alpha \circ b : \operatorname{Int} I^{k-1} \to Y \cap \partial M$ on ∂M, obtained by restricting α. β belongs to the induced orientation of ∂M if k is even, and to the opposite orientation if k is odd; so computing $\int_{\partial M} \omega$ with β we must reverse the sign when k is odd:

$$\int_{\partial M} \omega = (-1)^k \int_{\operatorname{Int} I^{k-1}} \beta^*\omega.$$

Since $\beta^*\omega = b^*(\alpha^*\omega) = b^*\eta$, the theorem follows. ■

What if $k = 1$? If ∂M is empty, one proves readily that $\int_M d\omega = 0$. If ∂M is non-empty we must say what an "orientation" of a 0-manifold is and how to "integrate" a 0-form over it. First a special case.

Definition 37.1 (arc)

Let M be a 1-manifold in $\mathbb{R}^n$. If there is a one-to-one C^∞ map $\alpha : [a,b] \to M$ carrying $[a,b]$ onto M with $D\alpha(t) \neq 0$ for all t, M is a (smooth) **arc** in $\mathbb{R}^n$. If M is oriented so that $\alpha|(a,b)$ belongs to the orientation, $\mathbf{p} = \alpha(a)$ is the **initial point** and $\mathbf{q} = \alpha(b)$ the **final point** of M.

Theorem 37.3

Let M be a 1-manifold in $\mathbb{R}^n$; let f be a 0-form defined in an open set about M. If M is an arc with initial point $\mathbf{p}$ and final point $\mathbf{q}$, then

$$\int_M df = f(\mathbf{q}) - f(\mathbf{p}).$$

Proof. $\alpha : (a,b) \to M - \mathbf{p} - \mathbf{q}$ is a patch covering all of M except a set of measure zero, so by Theorem 35.2, $\int_M df = \int_{(a,b)} \alpha^*(df)$. With t the general point of $\mathbb{R}$, $\alpha^*(df) = d(f \circ \alpha) = D(f \circ \alpha)\, dt$, so $\int_{(a,b)} \alpha^*(df) = \int_{(a,b)} D(f \circ \alpha) = f(\alpha(b)) - f(\alpha(a))$ by the fundamental theorem of calculus. ■

Definition 37.2 (orientation and integral for 0-manifolds)

A compact 0-manifold N in $\mathbb{R}^n$ is a finite set $\{\mathbf{x}_1, \ldots, \mathbf{x}_m\}$. An **orientation** of N is a function $\epsilon : N \to \{-1, 1\}$. If f is a 0-form defined on an open set containing N, the **integral of f over the oriented manifold N** is

$$\int_N f = \sum_{i=1}^{m} \epsilon(\mathbf{x}_i) f(\mathbf{x}_i).$$

If M is an oriented 1-manifold with non-empty boundary, the **induced orientation** of ∂M is defined by $\epsilon(\mathbf{p}) = -1$ if there is a patch $\alpha : U \to V$ about $\mathbf{p}$ in the orientation of M with U open in $\mathbb{H}^1$, and $\epsilon(\mathbf{p}) = +1$ otherwise.

Theorem 37.4 (Stokes' theorem in dimension 1)

Let M be a compact oriented 1-manifold in $\mathbb{R}^n$; give ∂M the induced orientation if ∂M is not empty. Let f be a 0-form defined in an open set of $\mathbb{R}^n$ containing M. Then

$$\int_M df = \int_{\partial M} f$$

if ∂M is not empty; and $\int_M df = 0$ if ∂M is empty.

Proof [Supplement]. Cover M by patches $\alpha : W \to Y$ of its orientation, where W is one of the intervals $(0,1)$, $[0,1)$, $(-1,0]$, and use a partition of unity dominated by them; by linearity it suffices to treat f with $M \cap \operatorname{Support} f$ inside one such Y. Let $g = f \circ \alpha$, extended to a C^∞ function near $[-1,1]$; then $\int_M df = \int_{\operatorname{Int} W} \alpha^* df = \int_{\operatorname{Int} W} g'$. If $W = (0,1)$, g vanishes near 0 and 1, so $\int_0^1 g' = 0$, and $\operatorname{Support} f$ misses ∂M. If $W = [0,1)$, $\alpha(0) = \mathbf{p} \in \partial M$ has $\epsilon(\mathbf{p}) = -1$ and $g(1) = 0$: $\int_0^1 g' = -g(0) = -f(\mathbf{p}) = \int_{\partial M} f$. If $W = (-1,0]$, $\alpha(0) = \mathbf{p}$ has $\epsilon(\mathbf{p}) = +1$ (no patch of the orientation about $\mathbf{p}$ has domain open in $\mathbb{H}^1$, since such a patch would overlap α negatively), and $\int_{-1}^0 g' = g(0) = f(\mathbf{p}) = \int_{\partial M} f$. ■

§38. Applications to Vector Analysis

By §31, forms of order $k = 0, 1, n-1, n$ in $\mathbb{R}^n$ can be interpreted as scalar or vector fields. We show integrals of forms can be similarly interpreted, so that Stokes' theorem yields the classical theorems of vector integral calculus.

38.1 The gradient theorem for 1-manifolds in $\mathbb{R}^n$

Lemma 38.1

Let M be a compact oriented 1-manifold in $\mathbb{R}^n$; let T be the unit tangent vector to M corresponding to the orientation. Let $F(\mathbf{x}) = (\mathbf{x}; f(\mathbf{x})) = (\mathbf{x}; \sum f_i(\mathbf{x})\mathbf{e}_i)$ be a vector field defined in an open set of $\mathbb{R}^n$ containing M; it corresponds to the 1-form $\omega = \sum f_i\, dx_i$. Then

$$\int_M \omega = \int_M \langle F, T\rangle\, ds.$$

Here the classical ds (arc length) replaces dV. Replacing M by $-M$ changes the sign of $\int_M \omega$ and replaces T by $-T$, so the right side changes sign too.

Proof. *First proof.* By Theorem 36.2, $\int_M \omega = \int_M \lambda\, ds$, where $\lambda(\mathbf{p})$ is the value of $\omega(\mathbf{p})$ on an orthonormal basis of $\mathcal{T}_{\mathbf{p}}(M)$ in its natural orientation; here $T(\mathbf{p}) = (\mathbf{p}; \mathbf{t})$ is such a basis, and $\omega(\mathbf{p})(\mathbf{p}; \mathbf{t}) = \sum f_i(\mathbf{p})t_i$. So $\lambda(\mathbf{p}) = \langle F(\mathbf{p}), T(\mathbf{p})\rangle$.

Second proof. By linearity assume $C = M \cap \operatorname{Support}\omega$ lies in a single patch $\alpha : U \to V$ of the orientation. With t the general point of $\mathbb{R}$, $\alpha^*\omega = \sum_i (f_i \circ \alpha)\, d\alpha_i = \sum_i (f_i \circ \alpha)(D\alpha_i)\, dt = \langle f \circ \alpha, D\alpha\rangle\, dt$, so $\int_M \omega = \int_{\operatorname{Int} U} \langle f \circ \alpha, D\alpha\rangle$. On the other hand

$$\int_M \langle F, T\rangle\, ds = \int_{\operatorname{Int} U} \langle F \circ \alpha, T \circ \alpha\rangle V(D\alpha) = \int_{\operatorname{Int} U} \langle f \circ \alpha, D\alpha/\|D\alpha\|\rangle V(D\alpha) = \int_{\operatorname{Int} U} \langle f \circ \alpha, D\alpha\rangle,$$

since $V(D\alpha) = [\det(D\alpha^{\mathrm{tr}} D\alpha)]^{1/2} = \|D\alpha\|$. ■

Theorem 38.2 (The gradient theorem)

Let M be a compact 1-manifold in $\mathbb{R}^n$; let T be a unit tangent vector field to M. Let f be a C^∞ function defined in an open set about M. If ∂M is empty, then $\int_M \langle \operatorname{grad} f, T\rangle\, ds = 0$. If ∂M consists of the points $\mathbf{x}_1, \ldots, \mathbf{x}_m$, let $\epsilon_i = -1$ if T points into M at $\mathbf{x}_i$ and $\epsilon_i = +1$ otherwise. Then

$$\int_M \langle \operatorname{grad} f, T\rangle\, ds = \sum_{i=1}^m \epsilon_i f(\mathbf{x}_i).$$

Proof. The 1-form df corresponds to $\operatorname{grad} f$ (Theorem 31.1), so $\int_M df = \int_M \langle \operatorname{grad} f, T\rangle\, ds$ by the lemma. Apply Stokes' theorem in dimension 1. ■

38.2 The divergence theorem for $n-1$ manifolds in $\mathbb{R}^n$

Recall from §34 the unit normal field $N(\mathbf{p}) = (\mathbf{p}; \mathbf{n}(\mathbf{p}))$ of an oriented $(n-1)$-manifold M, defined by: $\mathbf{n}$ is a unit vector orthogonal to $\mathcal{T}_{\mathbf{p}}(M)$ and $(\mathbf{n}, \partial\alpha/\partial x_1, \ldots, \partial\alpha/\partial x_{n-1})$ is right-handed for a patch α in the orientation with $\alpha(\mathbf{x}) = \mathbf{p}$. *$N$ is well-defined*: if β is another such patch, $g = \beta^{-1} \circ \alpha$ and $g(\mathbf{x}) = \mathbf{y}$, then $D\alpha(\mathbf{x}) = D\beta(\mathbf{y})Dg(\mathbf{x})$, so for any $\mathbf{v} \in \mathbb{R}^n$,

$$[\mathbf{v}\ D\alpha(\mathbf{x})] = [\mathbf{v}\ D\beta(\mathbf{y})] \cdot \begin{bmatrix} 1 & 0 \\ 0 & Dg(\mathbf{x}) \end{bmatrix}, \qquad \det[\mathbf{v}\ D\alpha(\mathbf{x})] = \det[\mathbf{v}\ D\beta(\mathbf{y})] \cdot \det Dg(\mathbf{x}).$$

Since $\det Dg > 0$, $[\mathbf{v}\ D\alpha(\mathbf{x})]$ has positive determinant iff $[\mathbf{v}\ D\beta(\mathbf{y})]$ does. To see N is C^∞ we find a formula for it.

Example 38.1

For $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$, calculus teaches that $\mathbf{c} = \mathbf{a} \times \mathbf{b}$ is perpendicular to both, $(\mathbf{c}, \mathbf{a}, \mathbf{b})$ is right-handed, and $\|\mathbf{c}\| = V(\mathbf{a}, \mathbf{b})$; here $c_1 = \det \begin{bmatrix} a_2 & b_2 \\ a_3 & b_3 \end{bmatrix}$, $c_2 = -\det \begin{bmatrix} a_1 & b_1 \\ a_3 & b_3 \end{bmatrix}$, $c_3 = \det \begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix}$. So for an oriented 2-manifold in $\mathbb{R}^3$ and a patch α in its orientation, $\mathbf{n} = \mathbf{c}/\|\mathbf{c}\|$ with $\mathbf{c} = \partial\alpha/\partial x_1 \times \partial\alpha/\partial x_2$ is the unit normal.

Lemma 38.3

Given independent vectors $\mathbf{x}_1, \ldots, \mathbf{x}_{n-1}$ in $\mathbb{R}^n$, let X be the n by $n-1$ matrix $X = [\mathbf{x}_1\ \cdots\ \mathbf{x}_{n-1}]$, and let $\mathbf{c}$ be the vector $\mathbf{c} = \sum c_i \mathbf{e}_i$, where

$$c_i = (-1)^{i-1} \det X(1, \ldots, \hat{\imath}, \ldots, n).$$

The vector $\mathbf{c}$ has the following properties:
(1) $\mathbf{c}$ is non-zero and orthogonal to each $\mathbf{x}_i$.
(2) The frame $(\mathbf{c}, \mathbf{x}_1, \ldots, \mathbf{x}_{n-1})$ is right-handed.
(3) $\|\mathbf{c}\| = V(X)$.

Proof. For $\mathbf{a} \in \mathbb{R}^n$, expanding by cofactors of the first column,

$$\det[\mathbf{a}\ \mathbf{x}_1\ \cdots\ \mathbf{x}_{n-1}] = \sum_{i=1}^{n} a_i (-1)^{i-1} \det X(1, \ldots, \hat{\imath}, \ldots, n) = \langle \mathbf{a}, \mathbf{c} \rangle.$$

(1) With $\mathbf{a} = \mathbf{x}_i$ the matrix has two equal columns, so $\langle \mathbf{x}_i, \mathbf{c} \rangle = 0$. Since the columns of X span an $(n-1)$-dimensional space, so do its rows; some $n-1$ rows are independent, say all but the ith; then $c_i \neq 0$.

(2) With $\mathbf{a} = \mathbf{c}$, $\det[\mathbf{c}\ \mathbf{x}_1 \cdots \mathbf{x}_{n-1}] = \|\mathbf{c}\|^2 > 0$.

(3) This follows from Theorem 21.4; alternatively,

$$[\mathbf{c}\ \mathbf{x}_1 \cdots \mathbf{x}_{n-1}]^{\mathrm{tr}} \cdot [\mathbf{c}\ \mathbf{x}_1 \cdots \mathbf{x}_{n-1}] = \begin{bmatrix} \|\mathbf{c}\|^2 & 0 \\ 0 & X^{\mathrm{tr}} X \end{bmatrix},$$

and taking determinants with (2), $\|\mathbf{c}\|^4 = \|\mathbf{c}\|^2 V(X)^2$. ■

Corollary 38.4

If M is an oriented $n-1$ manifold in $\mathbb{R}^n$, then the unit normal vector $N(\mathbf{p})$ corresponding to the orientation of M is a C^∞ function of $\mathbf{p}$.

Proof. For a patch $\alpha : U \to V$ in the orientation let $c_i(\mathbf{x}) = (-1)^{i-1} \det D\alpha(1, \ldots, \hat{\imath}, \ldots, n)(\mathbf{x})$ and $\mathbf{c}(\mathbf{x}) = \sum c_i(\mathbf{x}) \mathbf{e}_i$. Then $N(\mathbf{p}) = (\mathbf{p}; \mathbf{c}(\mathbf{x})/\|\mathbf{c}(\mathbf{x})\|)$ with $\mathbf{x} = \alpha^{-1}(\mathbf{p})$, a C^∞ function of $\mathbf{p}$. ■

Lemma 38.5

Let M be a compact oriented $n-1$ manifold in $\mathbb{R}^n$; let N be the corresponding unit normal vector field. Let G be a vector field defined in an open set U of $\mathbb{R}^n$ containing M. If we denote the general point of $\mathbb{R}^n$ by $\mathbf{y}$, this vector field has the form $G(\mathbf{y}) = (\mathbf{y}; g(\mathbf{y})) = (\mathbf{y}; \sum g_i(\mathbf{y}) \mathbf{e}_i)$;

it corresponds to the $n-1$ form

$$\omega = \sum_{i=1}^{n} (-1)^{i-1} g_i \, dy_1 \wedge \cdots \wedge \widehat{dy_i} \wedge \cdots \wedge dy_n.$$

Then $\int_M \omega = \int_M \langle G, N \rangle \, dV$.

Replacing M by $-M$ changes the sign of $\int_M \omega$ and replaces N by $-N$.

Proof. *First proof.* By Theorem 36.2, $\int_M \omega = \int_M \lambda \, dV$, where $\lambda(\mathbf{p})$ is the value of $\omega(\mathbf{p})$ on an orthonormal basis $((\mathbf{p}; \mathbf{a}_1), \ldots, (\mathbf{p}; \mathbf{a}_{n-1}))$ of $\mathcal{T}_\mathbf{p}(M)$ in its natural orientation. Let $A = [\mathbf{a}_1 \cdots \mathbf{a}_{n-1}]$ and $\mathbf{c} = \sum c_i \mathbf{e}_i$ with $c_i = (-1)^{i-1} \det A(1, \ldots, \hat{\imath}, \ldots, n)$. By Lemma 38.3, $\mathbf{c}$ is orthogonal to each $\mathbf{a}_i$, $(\mathbf{c}, \mathbf{a}_1, \ldots, \mathbf{a}_{n-1})$ is right-handed, and $\|\mathbf{c}\| = V(A) = [\det(A^{\mathrm{tr}} A)]^{1/2} = [\det I_{n-1}]^{1/2} = 1$. So $N = (\mathbf{p}; \mathbf{c})$. By Theorem 27.7, $dy_1 \wedge \cdots \wedge \widehat{dy_i} \wedge \cdots \wedge dy_n((\mathbf{p}; \mathbf{a}_1), \ldots, (\mathbf{p}; \mathbf{a}_{n-1})) = \det A(1, \ldots, \hat{\imath}, \ldots, n)$, hence $\lambda(\mathbf{p}) = \sum_i (-1)^{i-1} g_i(\mathbf{p}) \det A(1, \ldots, \hat{\imath}, \ldots, n) = \sum_i g_i(\mathbf{p}) c_i = \langle G, N \rangle$.

Second proof. By linearity assume $C = M \cap \operatorname{Support} \omega$ lies in a single patch $\alpha : U \to V$ of the orientation. By Theorem 32.2,

$$\int_M \omega = \int_{\operatorname{Int} U} \alpha^* \omega = \int_{\operatorname{Int} U} \Big[\sum_{i=1}^{n} (-1)^{i-1} (g_i \circ \alpha) \det D\alpha(1, \ldots, \hat{\imath}, \ldots, n) \Big].$$

With $c_i = (-1)^{i-1} \det D\alpha(1, \ldots, \hat{\imath}, \ldots, n)$, $N(\alpha(\mathbf{x})) = (\alpha(\mathbf{x}); \mathbf{c}(\mathbf{x})/\|\mathbf{c}(\mathbf{x})\|)$ as in Corollary 38.4, and

$$\int_M \langle G, N \rangle \, dV = \int_{\operatorname{Int} U} \langle G \circ \alpha, N \circ \alpha \rangle V(D\alpha) = \int_{\operatorname{Int} U} \langle g \circ \alpha, \mathbf{c} \rangle = \int_{\operatorname{Int} U} \Big[\sum_i (g_i \circ \alpha)(-1)^{i-1} \det D\alpha(1, \ldots, \hat{\imath}, \ldots, n) \Big],$$

since $\|\mathbf{c}\| = V(D\alpha)$. ■

Lemma 38.6

Let M be a compact n-manifold in $\mathbb{R}^n$, oriented naturally. Let $\omega = h \, dx_1 \wedge \cdots \wedge dx_n$ be an n-form defined in an open set of $\mathbb{R}^n$ containing M. Then h is the corresponding scalar field, and

$$\int_M \omega = \int_M h \, dV.$$

Proof. *First proof.* $\int_M \omega = \int_M \lambda \, dV$ with λ the value of ω on an orthonormal basis of $\mathcal{T}_\mathbf{p}(M) = \mathcal{T}_\mathbf{p}(\mathbb{R}^n)$ in its natural orientation. A patch α of the natural orientation has $\det D\alpha > 0$, so the natural orientation of $\mathcal{T}_\mathbf{p}(M)$ consists of the right-handed frames; the usual basis is one, and the value of ω on it is h. *Second proof.* Assume $M \cap \operatorname{Support} \omega$ lies in one patch α of the orientation. By definition $\int_M \omega = \int_{\operatorname{Int} U} \alpha^* \omega = \int_{\operatorname{Int} U} (h \circ \alpha) \det D\alpha$ and $\int_M h \, dV = \int_{\operatorname{Int} U} (h \circ \alpha) V(D\alpha)$; and $V(D\alpha) = |\det D\alpha| = \det D\alpha$ since α belongs to the natural orientation. ■

$\int_M h \, dV$ equals the ordinary integral of h over the bounded set M: $A = M - \partial M$ is open in $\mathbb{R}^n$ and the identity $i : A \to A$ is a patch of the natural orientation covering M except a set of measure zero, so by Theorem 25.4, $\int_M h \, dV = \int_A (h \circ i) V(Di) = \int_A h$, which equals $\int_M h$ since ∂M has measure zero in $\mathbb{R}^n$.

For a naturally oriented n-manifold M in $\mathbb{R}^n$, the induced orientation of ∂M behaves as in Example 34.4 for $n = 3$. The **inward normal** to ∂M at $\mathbf{p}$ is the velocity vector of a curve that begins at $\mathbf{p}$ and moves into M as the parameter increases; the **outward normal** is its negative.

Lemma 38.7

Let M be an n-manifold in $\mathbb{R}^n$. If M is oriented naturally, then the induced orientation of ∂M corresponds to the unit normal field N to ∂M that points outwards from M at each point of ∂M.

Proof. Let $\alpha : U \to V$ be a patch about $\mathbf{p} \in \partial M$ in the orientation of M, so $\det D\alpha > 0$, and let $b(x_1, \ldots, x_{n-1}) = (x_1, \ldots, x_{n-1}, 0)$. Then $\alpha_0 = \alpha \circ b$ is a patch on ∂M about $\mathbf{p}$; it belongs to the induced orientation of ∂M if n is even and to its opposite if n is odd. Let $N(\mathbf{p}) = (\mathbf{p}; \mathbf{n})$ be the normal for the induced orientation. Then $\det[(-1)^n \mathbf{n} \;\; D\alpha_0] > 0$, which implies

$$\det[D\alpha_0 \;\; \mathbf{n}] = \det\left[\frac{\partial \alpha}{\partial x_1} \;\; \cdots \;\; \frac{\partial \alpha}{\partial x_{n-1}} \;\; \mathbf{n}\right] < 0,$$

(moving the first column to the last position changes the sign by $(-1)^{n-1}$). On the other hand $\det D\alpha = \det[\frac{\partial \alpha}{\partial x_1} \cdots \frac{\partial \alpha}{\partial x_{n-1}} \;\; \frac{\partial \alpha}{\partial x_n}] > 0$. So $\mathbf{n}$ and $\partial\alpha/\partial x_n$ lie on opposite sides of the tangent plane $\mathcal{T}_{\mathbf{p}}(\partial M)$. $\partial\alpha/\partial x_n$ is the velocity vector of a curve beginning at a point of ∂M and moving into M as the parameter increases; thus $\mathbf{n}$ is the outward normal. ■

Theorem 38.8 (The divergence theorem)

Let M be a compact n-manifold in $\mathbb{R}^n$. Let N be the unit normal vector field to ∂M that points outwards from M. If G is a vector field defined in an open set of $\mathbb{R}^n$ containing M, then

$$\int_M (\operatorname{div} G)\, dV = \int_{\partial M} \langle G, N \rangle \, dV.$$

The left side is an integral with respect to n-volume, the right with respect to $(n-1)$-volume.

Proof. Let $\omega = \beta_{n-1} G$ be the corresponding $(n-1)$-form. Orient M naturally and give ∂M the induced orientation; by Lemma 38.7 the normal field corresponding to this orientation is N, so $\int_{\partial M} \omega = \int_{\partial M} \langle G, N \rangle \, dV$ by Lemma 38.5. By Theorem 31.1, $d\omega = (\operatorname{div} G)\, dx_1 \wedge \cdots \wedge dx_n$, and Lemma 38.6 gives $\int_M d\omega = \int_M (\operatorname{div} G)\, dV$. Apply Stokes' theorem. ■

In $\mathbb{R}^3$ the divergence theorem is sometimes called *Gauss' theorem*.

38.3 Stokes' theorem for 2-manifolds in $\mathbb{R}^3$

Theorem 38.9 (Stokes' theorem—classical version)

Let M be a compact orientable 2-manifold in $\mathbb{R}^3$. Let N be a unit normal field to M. Let F be a C^∞ vector field defined in an open set about M. If ∂M is empty, then $\int_M \langle \operatorname{curl} F, N \rangle \, dV = 0$. If ∂M is non-empty, let T be the unit tangent vector field to ∂M chosen so that the vector $W(\mathbf{p}) = N(\mathbf{p}) \times T(\mathbf{p})$ points into M from ∂M. Then

$$\int_M \langle \operatorname{curl} F, N \rangle \, dV = \int_{\partial M} \langle F, T \rangle \, ds.$$

Proof. Let $\omega = \alpha_1 F$ be the corresponding 1-form; by Theorem 31.2, $\operatorname{curl} F$ corresponds to the 2-form $d\omega$. Orient M so that N is its unit normal field; by Lemma 38.5, $\int_M d\omega = \int_M \langle \operatorname{curl} F, N \rangle \, dV$. If ∂M is non-empty, its induced orientation corresponds to the unit tangent field T (Example 34.5: (N, T, W) is right-handed, i.e. $W = N \times T$ points into M), so Lemma 38.1 gives $\int_{\partial M} \omega = \int_{\partial M} \langle F, T \rangle \, ds$. Apply Stokes' theorem. ■

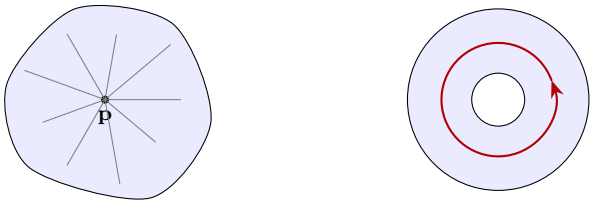
Chapter 8. Closed Forms and Exact Forms

Overview

In physics one often asks whether a vector field F in $\mathbb{R}^3$ is the gradient of a scalar field f (F is **conservative**, and f, or its negative, is a **potential function** for F): in the language of forms, whether a 1-form is the differential of a 0-form, i.e. exact. Similarly one asks whether G is the curl of a field F: whether a 2-form is exact. In $\mathbb{R}^n$: if ω is a k-form on an open set A, a necessary condition for exactness is $d\omega = 0$. It is not in general sufficient; this chapter explores what additional conditions on A, or on A and ω, ensure that ω is exact.

§39. The Poincaré Lemma

Motivation. Every closed form is locally exact, and on star-shaped domains it is globally exact. This settles when a vector field has a potential in a convex region, and it is the local fact whose global failure is measured by de Rham cohomology.



star-convex: closed ⇒ exact punctured: $d\theta$ closed, not exact

Figure. *On a star-convex set every segment from* $\mathbf{p}$ *stays in the set, which is what the Poincaré construction integrates along; there every closed form is exact. Around a hole this fails:* $d\theta$ *is closed on* $\mathbb{R}^2 - \mathbf{0}$ *but its integral around the loop is* $2\pi \neq 0$.

If A is *star-convex*, every closed form on A is exact (the Poincaré lemma). We begin with a preliminary result.

Theorem 39.1 (Leibniz's rule)

Let Q be a rectangle in $\mathbb{R}^n$; let $f : Q \times [a,b] \to \mathbb{R}$ be a continuous function. Denote f by $f(\mathbf{x},t)$ for $\mathbf{x} \in Q$ and $t \in [a,b]$. Then the function

$$F(\mathbf{x}) = \int_{t=a}^{t=b} f(\mathbf{x},t)$$

is continuous on Q. Furthermore, if $\partial f/\partial x_j$ is continuous on $Q \times [a,b]$, then

$$\frac{\partial F}{\partial x_j}(\mathbf{x}) = \int_{t=a}^{t=b} \frac{\partial f}{\partial x_j}(\mathbf{x},t).$$

This is *Leibniz's rule for differentiating under the integral sign.*

Proof.

Step 1. $Q \times [a,b]$ is compact, so f is uniformly continuous: given $\epsilon > 0$ there is $\delta > 0$ with $|f(\mathbf{x}_1,t_1) - f(\mathbf{x}_0,t_0)| < \epsilon$ whenever $|(\mathbf{x}_1,t_1) - (\mathbf{x}_0,t_0)| < \delta$. So for $|\mathbf{x}_1 - \mathbf{x}_0| < \delta$,

$$|F(\mathbf{x}_1) - F(\mathbf{x}_0)| \leq \int_{t=a}^{t=b} |f(\mathbf{x}_1,t) - f(\mathbf{x}_0,t)| \leq \epsilon(b-a).$$

Step 2. Only x_j and t are involved, so we may assume $n = 1$ and $Q = [c, d]$. Let $G(x) = \int_{t=a}^{t=b} D_1 f(x, t)$; we show $F'(x) = G(x)$. Since $D_1 f$ is continuous on $[c, d] \times [a, b]$, Fubini's theorem and the fundamental theorem of calculus give, for $x_0 \in [c, d]$,

$$\int_{x=c}^{x=x_0} G(x) = \int_{x=c}^{x=x_0} \int_{t=a}^{t=b} D_1 f(x,t) = \int_{t=a}^{t=b} \int_{x=c}^{x=x_0} D_1 f(x,t) = \int_{t=a}^{t=b} [f(x_0,t) - f(c,t)] = F(x_0) - F(c).$$

G is continuous by Step 1, so the fundamental theorem of calculus gives $G(x) = F'(x)$. ■

Definition 39.1 (differentiable homotopy)

Let A, B be open in $\mathbb{R}^n$, $\mathbb{R}^m$, and $g, h : A \to B$ of class C^∞. g and h are **differentiably homotopic** if there is a C^∞ map $H : A \times I \to B$ ($I = [0, 1]$) with

$$H(\mathbf{x}, 0) = g(\mathbf{x}) \qquad \text{and} \qquad H(\mathbf{x}, 1) = h(\mathbf{x}) \qquad (\mathbf{x} \in A).$$

H is a **differentiable homotopy** between g and h.

For each t, $\mathbf{x} \mapsto H(\mathbf{x}, t)$ is a C^∞ map $A \to B$; thinking of t as time, H "deforms" g into h as t goes from 0 to 1.

Theorem 39.2

Let A and B be open sets in $\mathbb{R}^n$ and $\mathbb{R}^m$, respectively. Let $g, h : A \to B$ be C^∞ maps that are differentiably homotopic. Then there is a linear transformation

$$\mathcal{P} : \Omega^{k+1}(B) \to \Omega^k(A),$$

defined for $k \geq 0$, such that for any form η of order $k > 0$,

$$d\mathcal{P}\eta + \mathcal{P}d\eta = h^*\eta - g^*\eta,$$

while for a form f of order 0,

$$\mathcal{P}df = h^*f - g^*f.$$

Hence if η is a closed form of positive order, $h^*\eta$ and $g^*\eta$ differ by an exact form ($= d\mathcal{P}\eta$); if f is a closed 0-form, $h^*f = g^*f$. (d raises order by 1 and $\mathcal{P}$ lowers it by 1; $\mathcal{P}f$ is undefined for a 0-form f.)

Proof.

Step 1. A special case. Given A open in $\mathbb{R}^n$, let U be a neighborhood of $A \times I$ in $\mathbb{R}^{n+1}$ and $\alpha, \beta : A \to U$, $\alpha(\mathbf{x}) = (\mathbf{x}, 0)$, $\beta(\mathbf{x}) = (\mathbf{x}, 1)$ (differentiably homotopic). We define for each $(k+1)$-form η on U a k-form $P\eta$ on A with

$$dP\eta + Pd\eta = \beta^*\eta - \alpha^*\eta \ \ (\text{order } \eta > 0), \qquad Pdf = \beta^*f - \alpha^*f \ \ (\text{order } f = 0). \qquad (*)$$

Let $\mathbf{x}$ be the general point of $\mathbb{R}^n$ and t that of $\mathbb{R}$; $dx_1, \ldots, dx_n, dt$ are the elementary 1-forms of $\mathbb{R}^{n+1}$. For a continuous scalar function g on $A \times I$ define $(\mathcal{I}g)(\mathbf{x}) = \int_{t=0}^{t=1} g(\mathbf{x}, t)$. For $k \geq 0$ write a $(k+1)$-form uniquely as

$$\eta = \sum_{[I]} f_I \, dx_I + \sum_{[J]} g_J \, dx_J \wedge dt,$$

I ascending $(k+1)$-tuples and J ascending k-tuples from $\{1, \ldots, n\}$, and define

$$P\eta = \sum_{[I]} P(f_I \, dx_I) + \sum_{[J]} P(g_J \, dx_J \wedge dt), \qquad P(f_I \, dx_I) = 0, \quad P(g_J \, dx_J \wedge dt) = (-1)^k (\mathcal{I}g_J) \, dx_J.$$

$P\eta$ is a k-form on A; P is linear (uniqueness of the representation and linearity of $\mathcal{I}$), and $\mathcal{I}g$ is C^∞ by Leibniz's rule, so $P\eta$ is C^∞. For $k=0$, $\eta = \sum f_i\,dx_i + g\,dt$, J is empty, and $P\eta = \mathcal{I}g$. (P "integrates η in the direction of the last coordinate".)

Step 2. The formulas $P(f\,dx_I) = 0$ and $P(g\,dx_J \wedge dt) = (-1)^k(\mathcal{I}g)\,dx_J$ hold even for arbitrary (non-ascending) I, J: trivially if indices repeat (then $dx_I = 0$, $dx_J = 0$); by definition if ascending; and in general since rearranging indices changes dx_I, dx_J only by a sign.

Step 3. $(*)$ for $k=0$:

$$P(df) = P\Big(\sum_j \frac{\partial f}{\partial x_j}dx_j\Big) + P\Big(\frac{\partial f}{\partial t}dt\Big) = 0 + (-1)^0\mathcal{I}\Big(\frac{\partial f}{\partial t}\Big) = f\circ\beta - f\circ\alpha = \beta^*f - \alpha^*f,$$

by the fundamental theorem of calculus.

Step 4. $(*)$ for $k>0$. Since $\alpha(\mathbf{x}) = (\mathbf{x},0)$, $\alpha^*(dx_i) = dx_i$ and $\alpha^*(dt) = d(0) = 0$; likewise for β. By linearity it suffices to treat $\eta = f\,dx_I$ and $\eta = g\,dx_J\wedge dt$. For $\eta = f\,dx_I$: the left side is

$$dP\eta + Pd\eta = d(0) + P\Big(\sum_j \frac{\partial f}{\partial x_j}dx_j\wedge dx_I\Big) + P\Big(\frac{\partial f}{\partial t}dt\wedge dx_I\Big) = 0 + (-1)^{k+1}P\Big(\frac{\partial f}{\partial t}dx_I\wedge dt\Big) = \mathcal{I}\Big(\frac{\partial f}{\partial t}\Big)dx_I = [f\circ\beta - f\circ\alpha$$

using Step 2 ($dt\wedge dx_I = (-1)^{k+1}dx_I\wedge dt$ as dx_I has order $k+1$), and the right side is $\beta^*\eta - \alpha^*\eta = (f\circ\beta)\beta^*(dx_I) - (f\circ\alpha)\alpha^*(dx_I) = [f\circ\beta - f\circ\alpha]\,dx_I$. For $\eta = g\,dx_J\wedge dt$:

$$d(P\eta) = d[(-1)^k(\mathcal{I}g)\,dx_J] = (-1)^k\sum_{j=1}^n D_j(\mathcal{I}g)\,dx_j\wedge dx_J. \qquad (**)$$

On the other hand $d\eta = \sum_j (D_jg)\,dx_j\wedge dx_J\wedge dt + (D_{n+1}g)\,dt\wedge dx_J\wedge dt$, and the last term vanishes, so by Step 2

$$P(d\eta) = (-1)^{k+1}\sum_{j=1}^n \mathcal{I}(D_jg)\,dx_j\wedge dx_J. \qquad (***)$$

By Leibniz's rule $D_j(\mathcal{I}g) = \mathcal{I}(D_jg)$, so $(**)+(***) = 0$. The right side is $\beta^*(g\,dx_J\wedge dt) - \alpha^*(g\,dx_J\wedge dt) = 0$, since $\beta^*(dt) = \alpha^*(dt) = 0$.

Step 5. General case. Let $H : A\times I \to B$ be a differentiable homotopy between g and h, $\alpha,\beta : A\to A\times I$ the maps of Step 1, and P the corresponding transformation. Define $\mathcal{P}\eta = P(H^*\eta)$; since $H^*\eta$ is a $(k+1)$-form on a neighborhood of $A\times I$, $\mathcal{P}\eta$ is a k-form on A. As $H\circ\alpha = g$ and $H\circ\beta = h$, for $k>0$

$$d\mathcal{P}\eta + \mathcal{P}d\eta = dP(H^*\eta) + P(H^*d\eta) = dP(H^*\eta) + P(dH^*\eta) = \beta^*(H^*\eta) - \alpha^*(H^*\eta) = h^*\eta - g^*\eta,$$

by Step 1; the case $k=0$ is similar. ■

Definition 39.2 (star-convex)

An open set A in $\mathbb{R}^n$ is **star-convex** with respect to the point $\mathbf{p}$ of A if for each $\mathbf{x}\in A$ the line segment joining $\mathbf{x}$ and $\mathbf{p}$ lies in A.

Example 39.1

A star-shaped region may be star-convex with respect to one of its points $\mathbf{p}$ but not with respect to another point $\mathbf{q}$. A convex set is star-convex with respect to each of its points. A "C"-shaped region is not star-convex with respect to any of its points.

Theorem 39.3 (The Poincaré lemma)

Let A be a star-convex open set in $\mathbb{R}^n$. If ω is a closed k-form on A, then ω is exact on A.

Proof. Let A be star-convex with respect to $\mathbf{p}$. Let $h : A \to A$ be the identity and $g : A \to A$ the constant map to $\mathbf{p}$. They are differentiably homotopic via the **straight-line homotopy**

$$H(\mathbf{x}, t) = t\, h(\mathbf{x}) + (1-t)\, g(\mathbf{x}),$$

which maps $A \times I$ into A, since $H(\mathbf{x}, t)$ lies on the segment from $h(\mathbf{x}) = \mathbf{x}$ to $g(\mathbf{x}) = \mathbf{p}$. Let $\mathcal{P}$ be as in Theorem 39.2. If f is a 0-form on A, $\mathcal{P}(df) = h^*f - g^*f = f \circ h - f \circ g$; if $df = 0$ then $0 = f(\mathbf{x}) - f(\mathbf{p})$ for all $\mathbf{x}$, so f is constant on A. If ω has order $k > 0$, $d\mathcal{P}\omega + \mathcal{P}d\omega = h^*\omega - g^*\omega$, where $h^*\omega = \omega$ (h is the identity) and $g^*\omega = 0$ (g is constant). If $d\omega = 0$, then $d\mathcal{P}\omega = \omega$, so ω is exact. ■

Theorem 39.4

Let A be a star-convex open set in $\mathbb{R}^n$. Let ω be a closed k-form on A. If $k > 1$, and if η and η_0 are two $k-1$ forms on A with $d\eta = \omega = d\eta_0$, then $\eta = \eta_0 + d\theta$ for some $k-2$ form θ on A. If $k = 1$, and if f and f_0 are two 0-forms on A with $df = \omega = df_0$, then $f = f_0 + c$ for some constant c.

Proof. $d(\eta - \eta_0) = 0$, so $\eta - \eta_0$ is a closed form on A, exact by the Poincaré lemma. Similarly $f - f_0$ is a closed 0-form, hence constant. ■

§40. The deRham Groups of Punctured Euclidean Space

Star-convex open sets are homologically trivial in all dimensions. The simplest set that is not is the punctured euclidean space $\mathbb{R}^n - \mathbf{0}$: earlier we found a closed $(n-1)$-form on $\mathbb{R}^n - \mathbf{0}$ that is not exact. We now give a definitive criterion for when a closed form on $\mathbb{R}^n - \mathbf{0}$ is exact, using vector spaces $H^k(A)$ called the *deRham groups* of A: A is homologically trivial in dimension k iff $H^k(A)$ is trivial.

Definition 40.1 (quotient space)

Let W be a linear subspace of a vector space V. V/W is the set of subsets of V of the form $\mathbf{v} + W = \{\mathbf{v} + \mathbf{w} : \mathbf{w} \in W\}$, called **cosets** of V determined by W. If $\mathbf{v}_1 - \mathbf{v}_2 \in W$ then $\mathbf{v}_1 + W = \mathbf{v}_2 + W$, and if $\mathbf{v}_1 - \mathbf{v}_2 \notin W$ they are disjoint; so V/W is a **partition** of V. With

$$(\mathbf{v}_1 + W) + (\mathbf{v}_2 + W) = (\mathbf{v}_1 + \mathbf{v}_2) + W, \qquad c(\mathbf{v} + W) = (c\mathbf{v}) + W,$$

V/W is a vector space, the **quotient space** of V by W.

The operations are well-defined: if $\mathbf{v}_1 + W = \mathbf{v}_1' + W$ and $\mathbf{v}_2 + W = \mathbf{v}_2' + W$, then $\mathbf{v}_1 - \mathbf{v}_1'$ and $\mathbf{v}_2 - \mathbf{v}_2'$ lie in W, hence so does $(\mathbf{v}_1 + \mathbf{v}_2) - (\mathbf{v}_1' + \mathbf{v}_2')$; similarly for scalar multiplication. If V is finite-dimensional so is V/W; but V/W may be finite-dimensional even when V and W are not.

Definition 40.2 (induced map of quotients)

If $W \subset V$ and $W' \subset V'$ are linear subspaces and $T : V \to V'$ is linear with $T(W) \subset W'$, then $\tilde{T} : V/W \to V'/W'$, $\tilde{T}(\mathbf{v} + W) = T(\mathbf{v}) + W'$, is a well-defined linear transformation, **induced** by T.

Definition 40.3 (deRham group)

Let A be open in $\mathbb{R}^n$. The closed k-forms $\mathcal{C}^k(A)$ and the exact k-forms $\mathcal{E}^k(A)$ are linear subspaces of $\Omega^k(A)$, and $\mathcal{E}^k(A) \subset \mathcal{C}^k(A)$. The **deRham group** of A in dimension k is the quotient vector space

$$H^k(A) = \mathcal{C}^k(A)/\mathcal{E}^k(A).$$

For a closed k-form ω we denote its coset $\omega + \mathcal{E}^k(A)$ by $\{\omega\}$.

$H^k(A)$ is the trivial (zero) vector space iff A is homologically trivial in dimension k. If $g : A \to B$ is a C^∞ map of open sets in $\mathbb{R}^n, \mathbb{R}^m$, then $g^* : \Omega^k(B) \to \Omega^k(A)$ commutes with d, so it carries closed forms to closed forms and exact forms to exact forms, and induces a linear map $g^* : H^k(B) \to H^k(A)$ (we write g^* instead of $\tilde{g}^*$). Two tools from algebraic topology compute these groups: *homotopy equivalence*, and a special case of the *Mayer–Vietoris theorem*.

Theorem 40.1 (Homotopy equivalence theorem)

Let A and B be open sets in $\mathbb{R}^n$ and $\mathbb{R}^m$, respectively. Let $g : A \to B$ and $h : B \to A$ be C^∞ maps. If $g \circ h : B \to B$ is differentiably homotopic to the identity map i_B of B, and if $h \circ g : A \to A$ is differentiably homotopic to the identity map i_A of A, then g^* and h^* are linear isomorphisms of the deRham groups.

If $g \circ h = i_B$ and $h \circ g = i_A$, g and h are diffeomorphisms; under the weaker hypotheses of the theorem they are called (differentiable) **homotopy equivalences**.

Proof. If η is a closed k-form on A ($k \geq 0$), Theorem 39.2 implies $(h \circ g)^*\eta - (i_A)^*\eta$ is exact (for $k = 0$ it is zero). So the induced maps satisfy $g^*(h^*(\{\eta\})) = \{\eta\}$: $g^* \circ h^*$ is the identity of $H^k(A)$. Similarly $h^* \circ g^*$ is the identity of $H^k(B)$. The first fact shows g^* maps $H^k(B)$ onto $H^k(A)$ (since $\{\eta\} = g^*(h^*\{\eta\})$); the second shows g^* is one-to-one (if $g^*\{\omega\} = 0$ then $\{\omega\} = h^*(g^*\{\omega\}) = 0$). By symmetry h^* is also a linear isomorphism. ■

Lemma 40.2

Let U and V be open sets in $\mathbb{R}^n$; let $X = U \cup V$; and suppose $A = U \cap V$ is non-empty. Then there exists a C^∞ function $\phi : X \to [0, 1]$ such that ϕ is identically 0 in a neighborhood of $U - A$ and ϕ is identically 1 in a neighborhood of $V - A$.

Proof. Let $\{\phi_i\}$ be a partition of unity on X dominated by $\{U, V\}$, $S_i = \text{Support}\,\phi_i$. Divide the indices into disjoint sets J and K such that $S_i \subset U$ for $i \in J$ and $S_i \subset V$ for $i \in K$ (e.g. $J = \{i : S_i \subset U\}$, K the rest). Let $\phi(\mathbf{x}) = \sum_{i \in K} \phi_i(\mathbf{x})$; by local finiteness ϕ is C^∞ on X. Let $\mathbf{a} \in U - A$. Choose a neighborhood W of $\mathbf{a}$ meeting only finitely many S_i; let D be the union of those among them with $i \in K$. D is closed and does not contain $\mathbf{a}$ (as $S_i \subset V$ for $i \in K$ and $\mathbf{a} \notin V$), so $W - D$ is a neighborhood of $\mathbf{a}$ on which each ϕ_i, $i \in K$, vanishes; thus $\phi = 0$ on $W - D$. Since $1 - \phi(\mathbf{x}) = \sum_{i \in J} \phi_i(\mathbf{x})$, symmetry shows $1 - \phi$ vanishes in a neighborhood of each point of $V - A$. ■

Theorem 40.3 (Mayer–Vietoris—special case)

Let U and V be open sets in $\mathbb{R}^n$ with U and V homologically trivial in all dimensions. Let $X = U \cup V$; suppose $A = U \cap V$ is non-empty. Then $H^0(X)$ is trivial, and for $k \geq 0$, the space $H^{k+1}(X)$ is linearly isomorphic to the space $H^k(A)$.

Proof. For open $B \subset C$ and a k-form η on C, write $\eta|B = j^*\eta$ ($j : B \to C$ the inclusion), the **restriction** of η to B. Since j^* commutes with d, restrictions of closed (exact) forms are closed (exact); and $(\eta|B)|A = \eta|A$ for $A \subset B \subset C$.

Step 1. $H^0(X)$ is trivial. Let f be a closed 0-form on X. $f|U$ and $f|V$ are closed, and U, V are homologically trivial in dimension 0, so $f|U = c_1$, $f|V = c_2$ are constants; as $U \cap V \neq \emptyset$, $c_1 = c_2$ and f is constant.

Step 2. Let $\phi : X \to [0,1]$ be C^∞ with $\phi = 0$ in a neighborhood U' of $U - A$ and $1 - \phi = 0$ in a neighborhood V' of $V - A$ (Lemma 40.2). For $k \geq 0$ define $\delta : \Omega^k(A) \to \Omega^{k+1}(X)$ by

$$\delta(\omega) = \begin{cases} d\phi \wedge \omega & \text{on } A, \\ 0 & \text{on } U' \cup V'. \end{cases}$$

Since $d\phi = 0$ on $U' \cup V'$, $\delta(\omega)$ is well-defined, and since A and $U' \cup V'$ are open with union X, it is C^∞ on X. δ is linear and commutes with d up to sign: $d(\delta(\omega)) = -d\phi \wedge d\omega$ on A and 0 on $U' \cup V'$, i.e. $d\delta(\omega) = -\delta(d\omega)$. So δ carries closed forms to closed forms and exact forms to exact forms ($\delta(d\theta) = -d\delta(\theta)$), and induces a linear map $\tilde{\delta} : H^k(A) \to H^{k+1}(X)$. We show $\tilde{\delta}$ is an isomorphism.

Step 3. $\tilde{\delta}$ is one-to-one: if ω is a closed k-form on A with $\delta(\omega) = d\theta$ for a k-form θ on X, then ω is exact. Define k-forms

$$\omega_1 = \begin{cases} \phi\omega & \text{on } A, \\ 0 & \text{on } U', \end{cases} \qquad \omega_2 = \begin{cases} (1-\phi)\omega & \text{on } A, \\ 0 & \text{on } V', \end{cases}$$

well-defined and C^∞ on U and V. Since $d\omega = 0$, $d\omega_1 = d\phi \wedge \omega = \delta(\omega)$ on A and 0 on U', so $d\omega_1 = \delta(\omega)|U = d\theta|U$; thus $\omega_1 - \theta|U$ is closed on U. Similarly $d\omega_2 = -d\theta|V$ and $\omega_2 + \theta|V$ is closed on V. If $k > 0$, homological triviality gives $(k-1)$-forms η_1, η_2 on U, V with $\omega_1 - \theta|U = d\eta_1$, $\omega_2 + \theta|V = d\eta_2$. Restricting to A and adding,

$$\phi\omega + (1-\phi)\omega = \omega_1|A + \omega_2|A = d(\eta_1|A + \eta_2|A),$$

so ω is exact on A. If $k = 0$, there are constants c_1, c_2 with $\omega_1 - \theta|U = c_1$, $\omega_2 + \theta|V = c_2$, and $\phi\omega + (1-\phi)\omega = c_1 + c_2$ on A; so ω is constant, i.e. exact.

Step 4. $\tilde{\delta}$ is onto: given a closed $(k+1)$-form η on X, we find a closed k-form ω on A with $\eta - \delta(\omega)$ exact. $\eta|U$ and $\eta|V$ are closed, so there are k-forms θ_1, θ_2 on U, V with $d\theta_1 = \eta|U$, $d\theta_2 = \eta|V$. Let $\omega = \theta_1|A - \theta_2|A$; it is closed since $d\omega = \eta|A - \eta|A = 0$. Define a k-form θ on X by

$$\theta = \begin{cases} (1-\phi)\theta_1 + \phi\theta_2 & \text{on } A, \\ \theta_1 & \text{on } U', \\ \theta_2 & \text{on } V', \end{cases}$$

well-defined and C^∞. We show $d\theta = \eta - \delta(\omega)$. On A:

$$d\theta|A = [-d\phi \wedge (\theta_1|A) + (1-\phi)(d\theta_1|A)] + [d\phi \wedge (\theta_2|A) + \phi(d\theta_2|A)] = \eta|A + d\phi \wedge (-\omega) = \eta|A - \delta(\omega)|A.$$

On U' and V': $d\theta|U' = d\theta_1|U' = \eta|U' = \eta|U' - \delta(\omega)|U'$ and similarly on V', since $\delta(\omega)$ vanishes on $U' \cup V'$. Hence $d\theta = \eta - \delta(\omega)$. ∎

Theorem 40.4

Let $n \geq 1$. Then

$$\dim H^k(\mathbb{R}^n - \mathbf{0}) = \begin{cases} 0 & \text{for } k \neq n-1, \\ 1 & \text{for } k = n-1. \end{cases}$$

Proof.

Step 1. $n = 1$. Let $A = \mathbb{R}^1 - 0 = A_0 \cup A_1$, with A_0 the negative and A_1 the positive reals. If ω is a closed k-form on A with $k > 0$, $\omega|A_0$ and $\omega|A_1$ are closed; A_0, A_1 are star-convex, so there are

$(k-1)$-forms η_i on A_i with $d\eta_i = \omega|A_i$. Let $\eta = \eta_0$ on A_0 and η_1 on A_1; then η is C^∞ on A and $d\eta = \omega$. For $k = 0$: let $f_0 = 0$ on A_0 and $f_0 = 1$ on A_1. f_0 is a closed 0-form that is not constant, hence not exact. $\{f_0\}$ is a basis of $H^0(A)$: if f is a closed 0-form on A, $f|A_0 = c_0$ and $f|A_1 = c_1$ are constants, so $f = c_1 f_0 + c_0(1 - f_0)$, i.e. $f = (c_1 - c_0)f_0 + c_0$, and $\{f\} = (c_1 - c_0)\{f_0\}$.

Step 2. If B is open in $\mathbb{R}^n$, then $B \times \mathbb{R}$ is open in $\mathbb{R}^{n+1}$ and $\dim H^k(B) = \dim H^k(B \times \mathbb{R})$ for all k. Let $g : B \to B \times \mathbb{R}$, $g(\mathbf{x}) = (\mathbf{x}, 0)$, and $h : B \times \mathbb{R} \to B$, $h(\mathbf{x}, s) = \mathbf{x}$. Then $h \circ g$ is the identity of B, and $g \circ h$ is differentiably homotopic to the identity of $B \times \mathbb{R}$ via the straight-line homotopy $H((\mathbf{x}, s), t) = t(\mathbf{x}, s) + (1 - t)(\mathbf{x}, 0) = (\mathbf{x}, st)$. Apply Theorem 40.1.

Step 3. Assume the theorem for n $(n \geq 1)$ and prove it for $n + 1$. Let

$$U = \mathbb{R}^{n+1} - \{(0, \ldots, 0, t) : t \geq 0\}, \qquad V = \mathbb{R}^{n+1} - \{(0, \ldots, 0, t) : t \leq 0\},$$

so U is $\mathbb{R}^{n+1}$ minus the half-line $\mathbf{0} \times \mathbb{H}^1$ and V is $\mathbb{R}^{n+1}$ minus the half-line $\mathbf{0} \times \mathbb{L}^1$. Then $A = U \cap V = \mathbb{R}^n \times \mathbb{R}$ minus the line $\mathbf{0} \times \mathbb{R}$, i.e. $A = (\mathbb{R}^n - \mathbf{0}) \times \mathbb{R}$, and $X = U \cup V = \mathbb{R}^{n+1} - \mathbf{0}$. U is star-convex with respect to $\mathbf{p} = (0, \ldots, 0, -1)$ and V with respect to $\mathbf{q} = (0, \ldots, 0, 1)$. By Theorem 40.3, $H^0(X)$ is trivial and $\dim H^{k+1}(X) = \dim H^k(A)$ for $k \geq 0$. By Step 2, $\dim H^k(A) = \dim H^k(\mathbb{R}^n - \mathbf{0})$, which by the induction hypothesis is 0 if $k \neq n - 1$ and 1 if $k = n - 1$. So $\dim H^{k+1}(\mathbb{R}^{n+1} - \mathbf{0})$ is 0 if $k + 1 \neq n$ and 1 if $k + 1 = n$. ■

Restated in terms of forms:

Theorem 40.5

Let $A = \mathbb{R}^n - \mathbf{0}$, with $n \geq 1$.
(a) If $k \neq n - 1$, then every closed k-form on A is exact on A.
(b) There is a closed $n - 1$ form η_0 on A that is not exact. If η is any closed $n - 1$ form on A, then there is a unique scalar c such that $\eta - c\eta_0$ is exact.

Proof [Supplement]. (a) $H^k(A) = 0$.

(b) $H^{n-1}(A)$ has dimension 1; take η_0 with $\{\eta_0\}$ a basis. Then $\{\eta\} = c\{\eta_0\}$ for a unique c. ■

This guarantees a closed non-exact $(n-1)$-form on $\mathbb{R}^n - \mathbf{0}$ without giving a formula. An explicit one is

$$\eta_0 = \sum_{i=1}^{n} (-1)^{i-1} f_i \, dx_1 \wedge \cdots \wedge \widehat{dx_i} \wedge \cdots \wedge dx_n, \qquad f_i(\mathbf{x}) = x_i / \|\mathbf{x}\|^n.$$

Why η_0 is closed but not exact [Supplement]

$\eta_0 = \beta_{n-1} G$ for the vector field $G(\mathbf{x}) = (\mathbf{x}; \mathbf{x}/\|\mathbf{x}\|^n)$, so $d\eta_0 = (\operatorname{div} G)\, dx_1 \wedge \cdots \wedge dx_n$ (Theorem 31.1). With $r = \|\mathbf{x}\|$, $D_i(x_i/r^n) = 1/r^n - n x_i^2/r^{n+2}$, so $\operatorname{div} G = n/r^n - nr^2/r^{n+2} = 0$. On $S^{n-1} = \partial B^n$ with the induced orientation, the corresponding normal is the outward one $N(\mathbf{x}) = (\mathbf{x}; \mathbf{x})$ (Lemma 38.7), and $\langle G, N \rangle = \|\mathbf{x}\|^2/\|\mathbf{x}\|^n = 1$ there; by Lemma 38.5, $\int_{S^{n-1}} \eta_0 = \mathrm{v}(S^{n-1}) \neq 0$. If $\eta_0 = d\theta$, Stokes' theorem would give $\int_{S^{n-1}} \eta_0 = \int_{\partial S^{n-1}} \theta = 0$ (S^{n-1} has empty boundary).

Theorem 40.6

Let $A = \mathbb{R}^n - \mathbf{0}$, with $n > 1$. If η is a closed $n - 1$ form in A, then η is exact in A if and only if

$$\int_{S^{n-1}} \eta = 0.$$

Proof. If η is exact, its integral over S^{n-1} is 0 by Stokes' theorem. Conversely suppose the integral is zero. By Theorem 40.5 there is a unique c with $\eta - c\eta_0$ exact. Then, by Stokes' theorem, $\int_{S^{n-1}} \eta = c \int_{S^{n-1}} \eta_0$. Since $\int_{S^{n-1}} \eta_0 \neq 0$, $c = 0$, and η is exact. ■

Chapter 9. Epilogue—Life Outside $\mathbb{R}^n$

Overview

We have dealt with submanifolds of euclidean space and forms on open sets of euclidean space: conceptually simple, but important ideas are sometimes obscured by the familiar surroundings, and in mathematics and physics manifolds often occur as abstract spaces. This chapter sketches how to move outside $\mathbb{R}^n$.

§41. Differentiable Manifolds and Riemannian Manifolds

Motivation. So far manifolds sat inside $\mathbb{R}^n$. Abstract manifolds remove the ambient space and keep only coordinate charts. Everything proved for submanifolds, including forms and Stokes' theorem, carries over once tangent vectors are defined intrinsically.

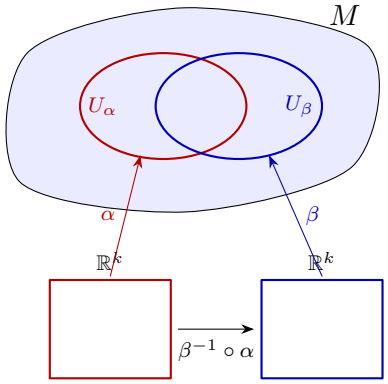


Figure. *Coordinate patches α, β of an abstract manifold overlap; the transition map $\beta^{-1} \circ \alpha$ between open sets of $\mathbb{R}^k$ must be C^r. Smoothness on M is defined through these charts, without any ambient $\mathbb{R}^n$.*

41.1 Differentiable manifolds

Definition 41.1 (differentiable manifold)

Let M be a metric space. Suppose there is a collection of homeomorphisms $\alpha_i : U_i \to V_i$, where U_i is open in $\mathbb{H}^k$ or $\mathbb{R}^k$ and V_i is open in M, such that the V_i cover M. (α_i is a **homeomorphism** if it carries U_i onto V_i one-to-one and both α_i, α_i^{-1} are continuous.) Suppose the α_i overlap with class C^∞: the **transition function** $\alpha_i^{-1} \circ \alpha_j$ is of class C^∞ whenever $V_i \cap V_j$ is non-empty. The α_i are **coordinate patches** on M, as is any other homeomorphism $\alpha : U \to V$ (U open in $\mathbb{H}^k$ or $\mathbb{R}^k$, V open in M) that overlaps the α_i with class C^∞. M with this collection of patches is a **differentiable k-manifold** (of class C^∞).

For $k = 1$ we again allow domains open in $\mathbb{L}^1$. If there is a patch $\alpha : U \to V$ about p with U open in $\mathbb{R}^k$, p is an **interior point**; otherwise a **boundary point**; the boundary points form ∂M. If $\alpha : U \to V$ is a patch about p, then $p \in \partial M$ iff U is open in $\mathbb{H}^k$ and $p = \alpha(\mathbf{x})$ for some $\mathbf{x} \in \mathbb{R}^{k-1} \times 0$ (proof as in Lemma 24.2). Throughout, M denotes a differentiable k-manifold.

Definition 41.2 (orientation)

Patches α_0, α_1 on M **overlap positively** if $\det D(\alpha_1^{-1} \circ \alpha_0) > 0$. If M can be covered by patches overlapping positively, M is **orientable**. An **orientation** of M consists of such a covering together with all other patches overlapping these positively; an **oriented manifold** is a manifold with an orientation. If $\{\alpha_i\}$ is an orientation, $\{\alpha_i \circ r\}$ (r the reflection of $\mathbb{R}^k$) is the **opposite** orientation.

If $\partial M \neq \emptyset$, then ∂M is a differentiable $(k-1)$-manifold without boundary, with patches $\alpha \circ b$ (α a patch about $p \in \partial M$, $b(x_1, \ldots, x_{k-1}) = (x_1, \ldots, x_{k-1}, 0)$), as in Theorem 24.3. If α_0, α_1 overlap positively, so do $\alpha_0 \circ b$ and $\alpha_1 \circ b$ (Theorem 34.1). So if M is oriented, ∂M can be oriented by restricting the patches of the orientation of M; for k even this is the **induced orientation** of ∂M, for k odd its opposite is.

Definition 41.3 (C^∞ maps between manifolds)

Let M, N be differentiable manifolds of dimensions k, n, $A \subset M$ and $f : A \to N$. f is **of class C^∞** if for each $x \in A$ there are a patch $\alpha : U \to V$ on M about x and a patch $\beta : W \to Y$ on N about $y = f(x)$ such that $\beta^{-1} \circ f \circ \alpha$ is of class C^∞ as a map of a subset of $\mathbb{R}^k$ into $\mathbb{R}^n$. (Independent of the patches, since transition functions are C^∞.) A one-to-one $f : M \to N$ onto N with f and f^{-1} of class C^∞ is a **diffeomorphism**.

41.2 Tangent vectors

Without a surrounding euclidean space, what is a tangent vector? For M in $\mathbb{R}^n$, a tangent vector at p is the velocity $(p; D\gamma(t_0))$ of a C^∞ curve $\gamma : [a, b] \to M$ with $\gamma(t_0) = p$. For abstract M we cannot differentiate γ, but for a patch $\alpha : U \to V$ about p, $\alpha^{-1} \circ \gamma$ maps a subset of $\mathbb{R}^1$ into $\mathbb{R}^k$, so we may think of the "derivative" of γ at t_0 as the function $\mathbf{v}$ assigning to each patch α about p the matrix $\mathbf{v}(\alpha) = D(\alpha^{-1} \circ \gamma)(t_0)$. If α_0, α_1 are two patches about p, the chain rule gives

$$\mathbf{v}(\alpha_1) = Dg(\mathbf{x}_0) \cdot \mathbf{v}(\alpha_0), \qquad g = \alpha_1^{-1} \circ \alpha_0, \quad \mathbf{x}_0 = \alpha_0^{-1}(p).$$

Definition 41.4 (tangent vector, tangent space)

Given $p \in M$, a **tangent vector** to M at p is a function $\mathbf{v}$ assigning to each coordinate patch $\alpha : U \to V$ about p a k by 1 column matrix $\mathbf{v}(\alpha)$, such that for any two patches α_0, α_1 about p,

$$\mathbf{v}(\alpha_1) = Dg(\mathbf{x}_0) \cdot \mathbf{v}(\alpha_0), \tag{$*$}$$

where $g = \alpha_1^{-1} \circ \alpha_0$ and $\mathbf{x}_0 = \alpha_0^{-1}(p)$. The entries of $\mathbf{v}(\alpha)$ are the **components** of $\mathbf{v}$ with respect to α.

By $(*)$, $\mathbf{v}$ is determined by its components in a single patch, and we may define $(a\mathbf{v} + b\mathbf{w})(\alpha) = a\mathbf{v}(\alpha) + b\mathbf{w}(\alpha)$. The tangent vectors at p form the **tangent space** $\mathcal{T}_p(M)$, a k-dimensional vector space: if $\alpha(\mathbf{x}) = p$, the map $\mathbf{v} \mapsto (\mathbf{x}; \mathbf{v}(\alpha))$ is a linear isomorphism of $\mathcal{T}_p(M)$ onto $\mathcal{T}_{\mathbf{x}}(\mathbb{R}^k)$, whose inverse is denoted $\alpha_* : \mathcal{T}_{\mathbf{x}}(\mathbb{R}^k) \to \mathcal{T}_p(M)$; so $\alpha_*(\mathbf{x}; \mathbf{v}(\alpha)) = \mathbf{v}$. For a C^∞ curve $\gamma : [a, b] \to M$ with $\gamma(t_0) = p$, the **velocity vector** is defined by $\mathbf{v}(\alpha) = D(\alpha^{-1} \circ \gamma)(t_0)$; every tangent vector at p is the velocity vector of some curve.

Tangent vectors as derivations

A common alternative: to $\mathbf{v} \in \mathcal{T}_p(M)$ associate an operator $X_{\mathbf{v}}$ on real-valued C^∞ functions near p, the **derivative with respect to $\mathbf{v}$**. If $\mathbf{v}$ is the velocity of γ at t_0 and $\alpha(\mathbf{x}) = p$, then

writing $f \circ \gamma = (f \circ \alpha) \circ (\alpha^{-1} \circ \gamma)$,

$$\frac{d(f \circ \gamma)}{dt}(t_0) = D(f \circ \alpha)(\mathbf{x}) \cdot D(\alpha^{-1} \circ \gamma)(t_0) = D(f \circ \alpha)(\mathbf{x}) \cdot \mathbf{v}(\alpha),$$

which depends only on f and $\mathbf{v}$. So define $X_{\mathbf{v}}(f) = D(f \circ \alpha)(\mathbf{x}) \cdot \mathbf{v}(\alpha)$; it is independent of α, $X_{\mathbf{v}+\mathbf{w}} = X_{\mathbf{v}} + X_{\mathbf{w}}$, $X_{c\mathbf{v}} = cX_{\mathbf{v}}$, and for $M = \mathbb{R}^k$ it is the directional derivative of f with respect to $\mathbf{v}$. It satisfies (1) *locality*: if f, g agree near p, $X_{\mathbf{v}}(f) = X_{\mathbf{v}}(g)$;
(2) *linearity*: $X_{\mathbf{v}}(af + bg) = aX_{\mathbf{v}}(f) + bX_{\mathbf{v}}(g)$;
(3) the *product rule*: $X_{\mathbf{v}}(f \cdot g) = X_{\mathbf{v}}(f)g(p) + f(p)X_{\mathbf{v}}(g)$. These characterize $X_{\mathbf{v}}$: every operator X on C^∞ functions near p satisfying (1)–(3) equals $X_{\mathbf{v}}$ for a unique $\mathbf{v} \in \mathcal{T}_p(M)$. So one can define a tangent vector at p to be such an operator; many authors prefer this "intrinsic" definition, which does not involve coordinate patches explicitly.

41.3 Forms on M

Definition 41.5 (forms on a manifold)

An **l-form** on M is a function ω assigning to each $p \in M$ an alternating l-tensor $\omega(p) \in \mathcal{A}^l(\mathcal{T}_p(M))$. If $\alpha : U \to V$ is a patch about p with $\alpha(\mathbf{x}) = p$, let $T = \alpha_* : \mathcal{T}_{\mathbf{x}}(\mathbb{R}^k) \to \mathcal{T}_p(M)$ and $T^* : \mathcal{A}^l(\mathcal{T}_p(M)) \to \mathcal{A}^l(\mathcal{T}_{\mathbf{x}}(\mathbb{R}^k))$; define $(\alpha^*\omega)(\mathbf{x}) = T^*(\omega(p))$. ω is **of class C^∞ near** p if $\alpha^*\omega$ is of class C^∞ near $\mathbf{x}$ (independent of α), and of class C^∞ if this holds for every patch.

Henceforth all forms are C^∞. $\Omega^l(M)$ is the space of l-forms on M. There are no elementary forms on M giving a canonical form, but $\alpha^*\omega = \sum_{[I]} f_I \, dx_I$ in $\mathbb{R}^k$; the f_I are the **components** of ω with respect to α.

Definition 41.6 (differential)

For an l-form ω on M, $d\omega$ is defined as follows: given $p \in M$ and tangent vectors $\mathbf{v}_1, \ldots, \mathbf{v}_{l+1}$ at p, choose a patch α about p with $\alpha(\mathbf{x}) = p$ and set

$$d\omega(p)(\mathbf{v}_1, \ldots, \mathbf{v}_{l+1}) = d(\alpha^*\omega)(\mathbf{x})((\mathbf{x}; \mathbf{v}_1(\alpha)), \ldots, (\mathbf{x}; \mathbf{v}_{l+1}(\alpha))).$$

That is, pull ω back to $\alpha^*\omega$ in $\mathbb{R}^k$, pull the $\mathbf{v}_i$ back to tangent vectors in $\mathbb{R}^k$, and apply d in $\mathbb{R}^k$. Equivalently (with $\mathbf{a}_i = \mathbf{v}_i(\alpha)$), $d\omega(p)(\alpha_*(\mathbf{x}; \mathbf{a}_1), \ldots, \alpha_*(\mathbf{x}; \mathbf{a}_{l+1})) = d(\alpha^*\omega)(\mathbf{x})((\mathbf{x}; \mathbf{a}_1), \ldots, (\mathbf{x}; \mathbf{a}_{l+1}))$, i.e. $d\omega$ is the unique $(l+1)$-form on M such that

$$\alpha^*(d\omega) = d(\alpha^*\omega)$$

for every coordinate patch α on M. (The d on the right is the operator in $\mathbb{R}^k$.)

This is independent of the patch, and $d\omega$ is C^∞.

Theorem 41.1

Let M be a compact differentiable manifold. Given a covering of M by coordinate patches, there exist functions $\phi_i : M \to \mathbb{R}$ of class C^∞, for $i = 1, \ldots, l$, such that:
(1) $\phi_i(p) \geq 0$ for each $p \in M$.
(2) For each i, the set Support ϕ_i is covered by one of the given coordinate patches.
(3) $\sum \phi_i(p) = 1$ for each $p \in M$.

Proof. Given $p \in M$, choose a patch $\alpha : U \to V$ of the covering about p, $\alpha(\mathbf{x}) = p$, and a non-negative C^∞ function $f : U \to \mathbb{R}$ whose support is compact and contained in U, with $f(\mathbf{x}) > 0$.

Define $\psi_p : M \to \mathbb{R}$ by $\psi_p(y) = f(\alpha^{-1}(y))$ for $y \in V$ and $\psi_p(y) = 0$ otherwise. Since $f \circ \alpha^{-1}$ vanishes outside a compact subset of V, ψ_p is C^∞ on M. ψ_p is positive on an open set U_p about p. Cover M by finitely many U_p, for $p = p_1, \dots, p_l$, and set $\lambda = \sum_{j=1}^{l} \psi_{p_j}$ and $\phi_i = (1/\lambda)\psi_{p_i}$. ■

Definition 41.7 (integral of a form)

Let M be a compact oriented differentiable k-manifold and ω a k-form on M. If Support ω lies in a single patch $\alpha : U \to V$ of the orientation, $\int_M \omega = \int_{\operatorname{Int} U} \alpha^*\omega$. In general, with $\phi_1, \dots, \phi_l$ as in Theorem 41.1,

$$\int_M \omega = \sum_{i=1}^{l} \Big[\int_M \phi_i \omega \Big].$$

The usual argument shows this is well-defined and linear.

Theorem 41.2 (Stokes' theorem)

Let M be a compact, oriented differentiable k-manifold. Let ω be a $k-1$ form on M. If ∂M is non-empty, give ∂M the induced orientation; then

$$\int_M d\omega = \int_{\partial M} \omega.$$

If ∂M is empty, then $\int_M d\omega = 0$.

Proof. The proof of Theorem 37.2 goes through verbatim: all computations were carried out within coordinate patches, so no changes are needed. The special conventions for $k = 1$ and ∂M a 0-manifold are handled as before (Theorem 37.4). ■

Chapter 8 also generalizes: the **deRham group** $H^k(M)$ is the quotient of the closed k-forms on M by the exact ones, and there are methods to compute it, including a general *Mayer–Vietoris theorem* relating the deRham groups of $M = U \cup V$, U, V and $U \cap V$ (see Bott–Tu). $H^k(M)$ is obviously a diffeomorphism invariant of M. Strikingly, it is also a *topological* invariant: if M and N are homeomorphic, $H^k(M)$ and $H^k(N)$ are linearly isomorphic. This follows from *deRham's theorem*: the algebra of closed forms modulo exact forms on M is isomorphic to an algebra defined in algebraic topology for any topological space, the "cohomology algebra of M with real coefficients".

41.4 Riemannian manifolds

Not everything generalizes so readily: the volume of M and $\int_M f\, dV$ do not make sense on an abstract differentiable manifold. By §36, the volume of a compact oriented k-manifold M in $\mathbb{R}^n$ is $\mathrm{v}(M) = \int_M \omega_V$, where the volume form ω_V has value 1 on *any orthonormal basis* of $\mathcal{T}_p(M)$ in its natural orientation. This uses the inner product that $\mathcal{T}_p(M) \subset \mathcal{T}_p(\mathbb{R}^n) = p \times \mathbb{R}^n$ inherits from the dot product of $\mathbb{R}^n$; an abstract $\mathcal{T}_p(M)$ has none, so "orthonormal" is meaningless. We need an inner product on each tangent space.

Definition 41.8 (Riemannian metric)

A **Riemannian metric** on a differentiable k-manifold M is an inner product $\langle \mathbf{v}, \mathbf{w} \rangle$ on each tangent space $\mathcal{T}_p(M)$, required to be of class C^∞ as a 2-tensor field on M. A **Riemannian manifold** is a differentiable manifold together with a Riemannian metric.

("Metric" here has nothing to do with "metric space".) Every differentiable manifold has a Riemannian metric (the proof uses a partition of unity), but it is not unique. A Riemannian metric gives a

volume function $V(\mathbf{v}_1, \dots, \mathbf{v}_k)$ on k-tuples in $\mathcal{T}_p(M)$ (as in the exercises of §21), so one defines, for continuous f supported in a single patch $\alpha : U \to V$,

$$\int_M f\, dV = \int_{\operatorname{Int} U} (f \circ \alpha)\, V(\alpha_*(\mathbf{x}; \mathbf{e}_1), \dots, \alpha_*(\mathbf{x}; \mathbf{e}_k)),$$

in general via a partition of unity as in §25, and $\mathrm{v}(M) = \int_M dV$. If M is compact, oriented and Riemannian, $\int_M \omega = \int_M \lambda\, dV$ where $\lambda(p)$ is the value of $\omega(p)$ on an orthonormal k-tuple in $\mathcal{T}_p(M)$ belonging to its natural orientation; if $\lambda \equiv 1$, ω is the **volume form** ω_V of M, and $\mathrm{v}(M) = \int_M \omega_V$.

Many questions arise: the *length* of a smooth curve $\gamma : [a, b] \to M$ is $\int_{t=a}^{t=b} \|\gamma_*(t; \mathbf{e}_1)\|$, the integral of the norm of its velocity vector (using the inner product on $\mathcal{T}_p(M)$); one studies *geodesics* (curves of minimal length joining two points), *curvature*, and so on—the subject of *Riemannian geometry*.

What does not generalize

Most of this book generalizes to abstract differentiable or Riemannian manifolds. The interpretation of Stokes' theorem in terms of scalar and vector fields (§38) does not: the "translation functions" of §31 depend on forms defined in $\mathbb{R}^n$, grad and div apply only to fields in $\mathbb{R}^n$, curl only in $\mathbb{R}^3$, and even the "normal vector" to a manifold depends on the surrounding space. Manifolds, differential forms and Stokes' theorem have meaning outside euclidean space; classical vector analysis does not.